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**Report prepared for the WHO annual consultation on
the composition of influenza vaccine for the Southern
Hemisphere 2018**

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The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and presented in Flu News Europe (<http://www.flunewseurope.org>) the joint ECDC-WHO Europe weekly influenza update, with the assistance of members of the Influenza and Other Respiratory Pathogens Group at the WHO European Office:

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WHO Vaccine Recommendation Meeting for the Southern Hemisphere 2018 influenza season.

WHO CC, London - Final Report for the meeting: 25-27 September 2017.

Influenza Activity in the WHO European Region, February - August 2017

Weekly reporting on influenza activity started in week 40/2016. The course of the 2016-2017 season was characterised by an early increase in levels of transmission compared to previous years (**Figure E1-A**) and no clear west to east pattern in the spread of influenza across the Region (**Figure E2**). Influenza A viruses predominated, accounting for 89% of all sentinel detections and the great majority (99%) of subtyped influenza A viruses from sentinel sites were A(H3N2). Confirmed cases of influenza virus type A infection reported from hospitals were predominantly in adults aged over 65 years. Substantial excess all-cause mortality was observed in people aged 15–64 years, and markedly so in people aged 65 years or older, in the majority of the 18 countries reporting these data. Two-thirds of the A(H3N2) viruses genetically characterised belong to genetic subclade 3C.2a1. Vaccine effectiveness estimates were similar to those in recent years. Only seven of 3,440 influenza viruses assessed for susceptibility to neuraminidase inhibitors showed reduced/highly reduced inhibition (RI/HRI) by oseltamivir and/or zanamivir.

The proportions of respiratory specimens from patients presenting with influenza-like illness (ILI) or acute respiratory infection (ARI) symptoms that tested positive for influenza virus increased between weeks 46/2016 and 52/2016. The overall positivity rate in specimens from sentinel sources reached approximately 50%, indicating increased influenza activity, until week 4/2017 followed by declining rates (but remaining above 10% until week 19/2017). The peak in activity was slightly earlier than those in recent years but the magnitude was similar to peaks observed in the 2012-2013, 2014-2015 and 2015-2016 seasons (**Figure E1-A**). During the peak period the greatest number of influenza virus detections was reported for week 3/2017. Increasing trends in ILI and ARI were reported by at least one country in each sub-region by week 49/2017 and each sub-region has experienced at least one week of high or very high intensity of influenza activity during the season. However, there have been no sustained periods of high or very high intensity in western or central Asian parts of the region. Southern Europe is the only sub-region to have reported sustained periods of very high intensity influenza activity this season (**Figure E2**).

For the whole WHO European Region between weeks 40/2016 and 20/2017, over 148,000 influenza detections from sentinel and non-sentinel sources were reported to the European Surveillance System (TESSy) – with type A detections (87%) predominating over type B (13%) (**Table E1**). This represents an increase of approximately 10,000 detections compared to the same period of the previous season. Within type A, the A(H3N2) subtype (99%) greatly predominated over A(H1N1)pdm09 viruses (1%) (**Table E1**) and this pattern was evident for both sentinel and non-sentinel influenza virus detections (**Figure E1-B/C**). Approximately two-thirds of the A(H3N2) viruses genetically characterised belonged to subclade 3C.2a1. Of the influenza type B detections, 72% of the viruses for which lineage-determination was performed were of the Yamagata-lineage (**Table E1**). Influenza B detections predominated in the latter part of the season but numbers were low and no significant peak was observed (**Figure E1-B/C**): for weeks 21-34/2017 a total of 1243 influenza detections across the Region were reported, 257 (21%) influenza A and 986 (79%) influenza B. Of the type A viruses subtyped (n = 120) 24 (20%) were A(H1N1)pdm09 and

96 (80%) A(H3N2), while of the 160 influenza B viruses ascribed to a lineage 14 (9%) were Victoria- and 146 (91%) were Yamagata-lineage.

All countries in the Region reported influenza type A virus detections to outnumber those of influenza B; however, the proportions of influenza B detections have been considerably greater in some countries in central Asia compared to those in other sub-regions with influenza B viruses accounting for 38% of all detections. It is of note that A(H3N2) detections greatly outnumbered A(H1N1)pdm09 detections across the Region; there were no A(H1N1)pdm09 detections in central Asia and no more than 2% of subtyped influenza A viruses were A(H1N1)pdm09 in other regions. For influenza type B, Yamagata-lineage viruses predominated (60% and over) in southern, western and northern European while in western and central Asia Victoria-lineage viruses were dominant (**Table E1 & Figure E3**).

Between weeks 40/2016 and 20/2017, nine countries (Czech Republic, Finland, France, Ireland, Romania, Slovakia, Spain, Sweden and the United Kingdom) reported a total of 7,484 laboratory-confirmed hospitalised influenza cases (3,708 cases from ICUs and 3,776 from other hospital wards). Cases from ICU peaked over the New Year period (**Figure E4-A**) and among cases admitted to ICUs, influenza type A detections (97%) have predominated over type B (3%). Of the influenza A viruses subtyped, A(H3N2) (91%) predominated over A(H1N1)pdm09 viruses (9%). Similarly, the observed proportion of influenza B detections (2%) was low in all other hospital wards and A(H3N2) was the dominant subtype (>99%) among influenza A detections. Among hospitalised patients with reported age, those aged 65 years and older accounted for the largest number of cases this season (**Figure E4-B**). A(H3N2) viruses were dominant in all age groups.

This season 16 countries, territories or areas with surveillance systems for severe acute respiratory infection (SARI; Albania, Armenia, Azerbaijan, Belarus, Belgium, Georgia, Kazakhstan, Kosovo*, Kyrgyzstan, Montenegro, Republic of Moldova, Russian Federation, Serbia, Tajikistan, Ukraine and Uzbekistan) reported a total of 39,713 cases. These cases were predominantly among children but in some countries, including Albania and Serbia, increases in those aged 65 years and older were also observed. Of 10,876 cases tested for influenza virus, 3,573 (33%) were positive: type A viruses (76%) predominated over type B (24%) and, of the influenza A viruses subtyped, the vast majority (>99%) were A(H3N2). The distribution of influenza types and subtypes by age group are not available for these data.

Neuraminidase inhibitor susceptibility was assessed for 3,440 influenza viruses (3,082 A(H3N2), 53 A(H1N1)pdm09 and 305 type B) with collection dates between week 40/2016 and 20/2017. All but seven A(H3N2) viruses showed no phenotypic or genotypic signs for reduced inhibition (RI) or highly reduced inhibition (HRI): four showed RI by oseltamivir in phenotypic assays, while the three also tested against zanamivir showed normal inhibition (NI); two showed HRI by oseltamivir and RI by zanamivir by genotypic analysis (both had NA R292K amino acid substitution); and one showed RI by zanamivir and NI by oseltamivir in phenotypic assays. In addition, one influenza A(H1N1)pdm09 virus showed RI by oseltamivir and NI by zanamivir in phenotypic assays and three influenza B/Victoria-lineage viruses showed RI: one by oseltamivir (not tested for zanamivir) and two by zanamivir (NI by oseltamivir) in phenotypic assays.

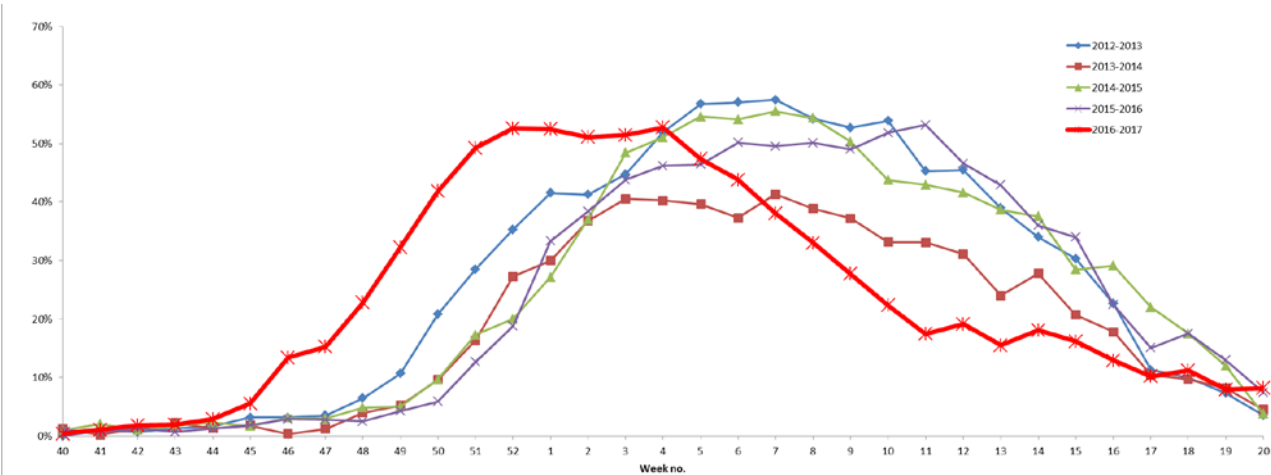
Using data reported to the European monitoring of excess mortality for public health action project (EuroMOMO – <http://www.euromomo.eu>), significant excess all-cause mortality was observed among people aged 15–64 years, and markedly so in people aged 65 years or older, in the majority of the 18 reporting countries (Belgium, Denmark, Estonia, Finland,

* in accordance with United Nations Security Council. Resolution 1244 (1999)

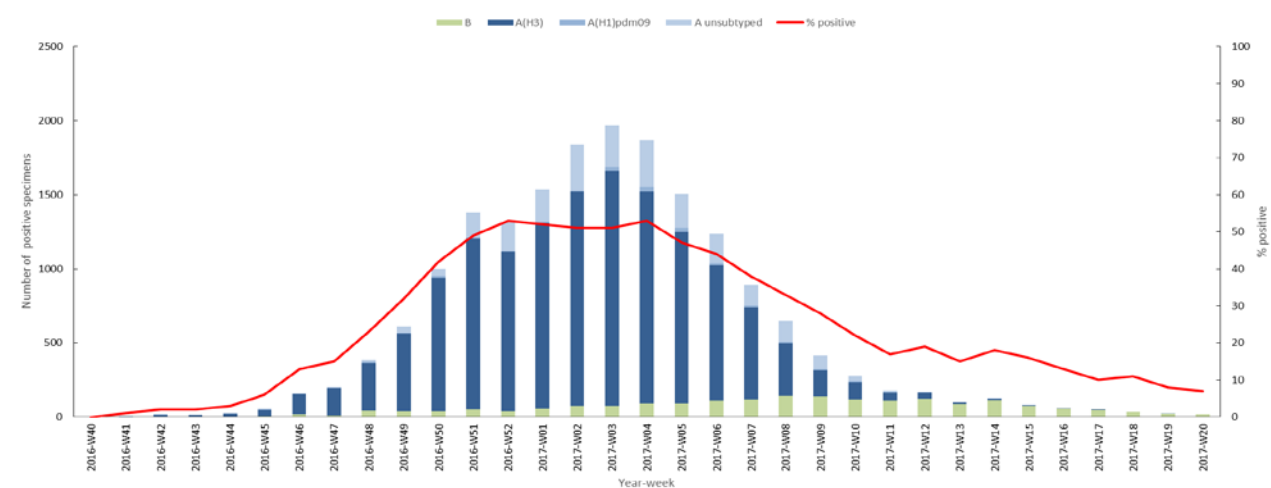
France, Germany (Berlin), Greece, Hungary, Ireland, Italy, Malta, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom (England, Scotland and Wales only)). This is commonly seen when the predominant viruses circulating are A(H3N2). In the last season where A(H3N2) viruses predominated (2014-2015), an excess all-cause mortality was also observed in those aged 65 years and older with a lower signal among the 15-64 years age group.

Figure E1. Influenza Activity in the WHO European Region 2016-2017 (weeks 40/2016-20/2017)

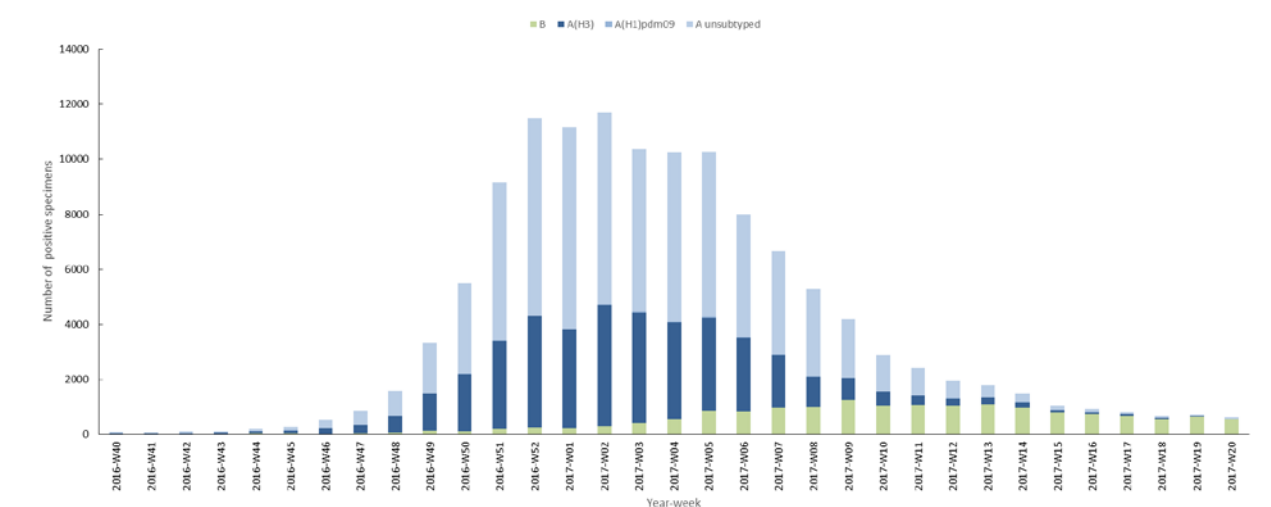
A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel Influenza Detections 2016-2017 season



C. Non-sentinel Influenza Detections 2016-2017 season



Intensity of Influenza Activity and Dominant Virus (sub)type

EAST

* Baseline influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.

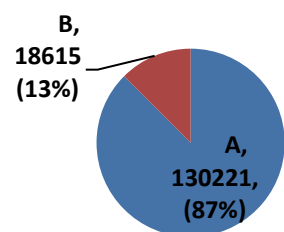
^a in accordance with. United Nations Security Council. Resolution 1244 (1999).

^a in accordance with. United Nations Security Council. Resolution 1244 (1999).

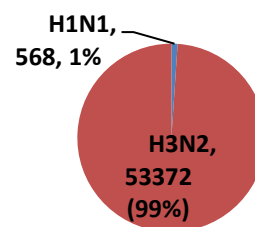
^a in accordance with. United Nations Security Council. Resolution 1244 (1999).

Table E1. Distribution of Influenza Detections in the WHO European Region², 2016-2017 Season (weeks 40/2016-20/2017)

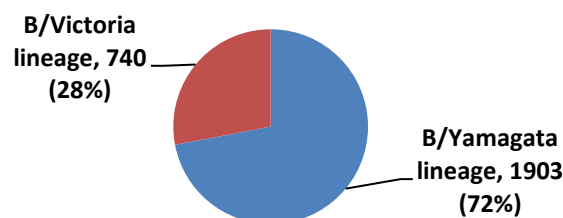
Proportions of Influenza Types/Subtypes/Lineages based on combined sentinel and non-sentinel detections



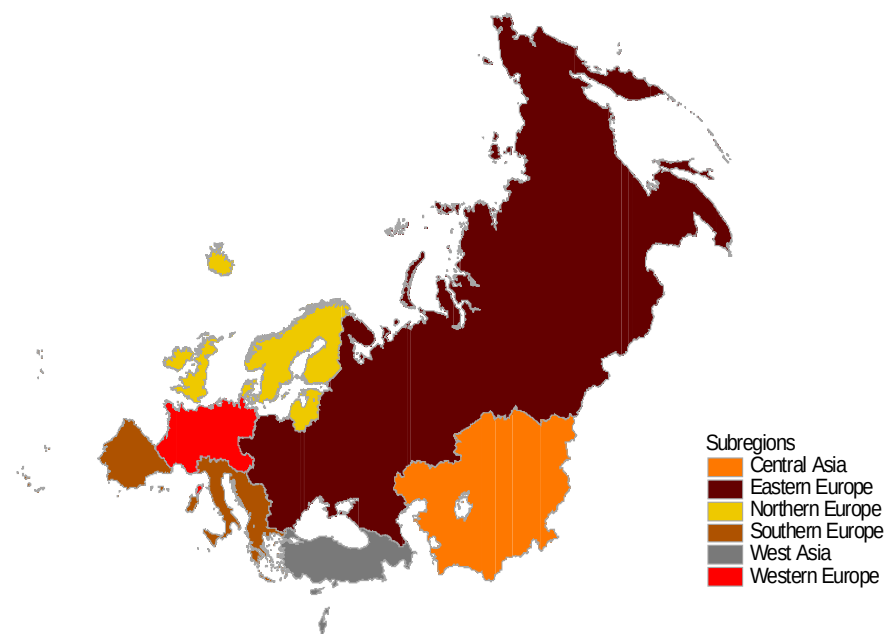
Influenza types (n=108,254 detections)



Influenza A subtypes (n=44,776; 43% of detections subtyped)



Influenza B lineages (n=882; 20% of influenza B detections ascribed to lineage)



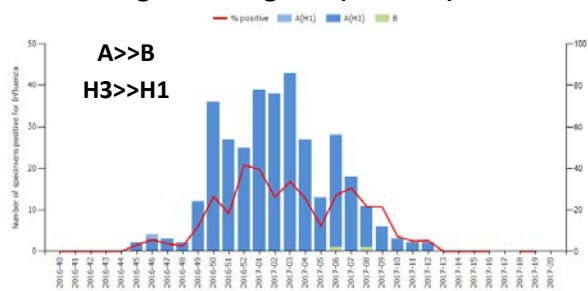
Influenza Detections*	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	22505	91%	33592	98%	17902	73%	52394	87%	2754	81%	1074	62%	130221	87%
Influenza A subtyped	12577		10358		15089		12953		1891		1072		53940	
<i>A (H1N1)pdm09</i>	101	1%	99	1%	265	2%	94	1%	9	0%	0	0%	568	1%
<i>A (H3N2)</i>	12476	99%	10259	99%	14824	98%	12859	99%	1882	100%	1072	100%	53372	99%
Influenza B	2120	9%	707	2%	6583	27%	7889	13%	664	19%	652	38%	18615	13%
Influenza B lineage determined	485		195		191		1484		115		173		2643	
<i>B/Yamagata-lineage</i>	402	83%	168	86%	99	52%	1203	81%	15	13%	16	9%	1903	72%
<i>B/Victoria-lineage</i>	83	17%	27	14%	92	48%	281	19%	100	87%	157	91%	740	28%
Total	24625		34299		24485		60283		3418		1726		148836	

*combined ILI, ARI sentinel and non-sentinel respiratory specimens positive for influenza

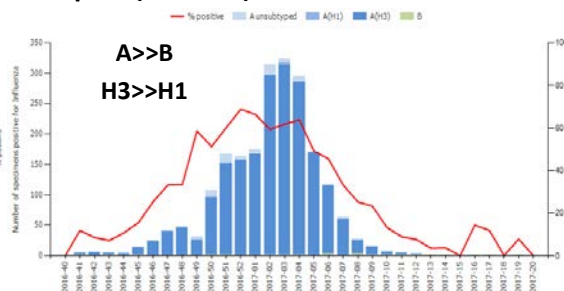
² WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: <http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>

Figure E3. Patterns of Influenza Type and Subtype Circulation in the WHO European Region*, weeks 40/2016-20/2017

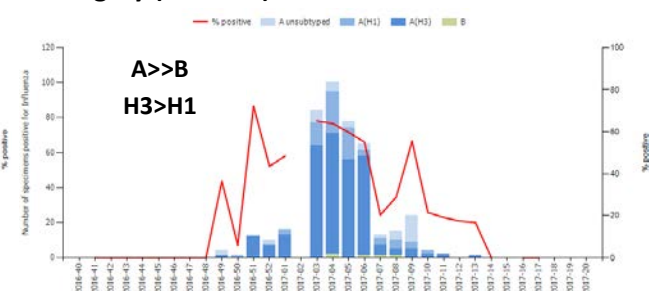
United Kingdom – England (Sentinel)



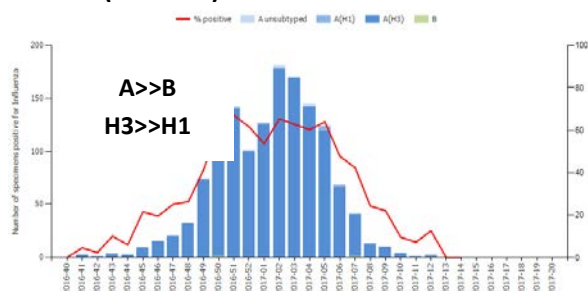
Spain (Sentinel)



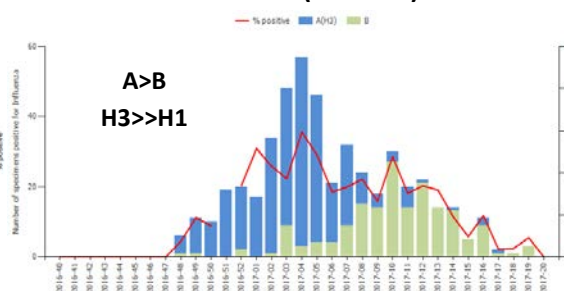
Hungary (Sentinel)



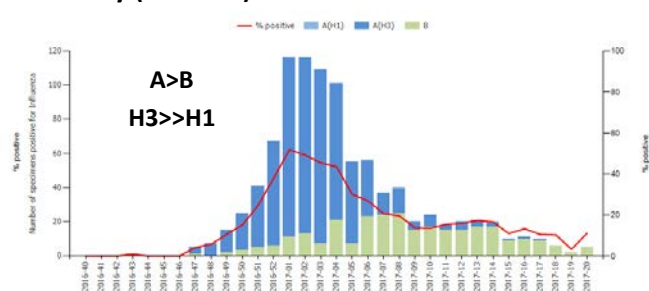
France (Sentinel)



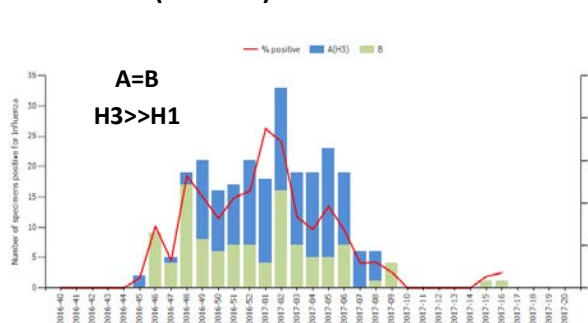
Russian federation (Sentinel)



Turkey (Sentinel)



Kazakhstan (Sentinel)

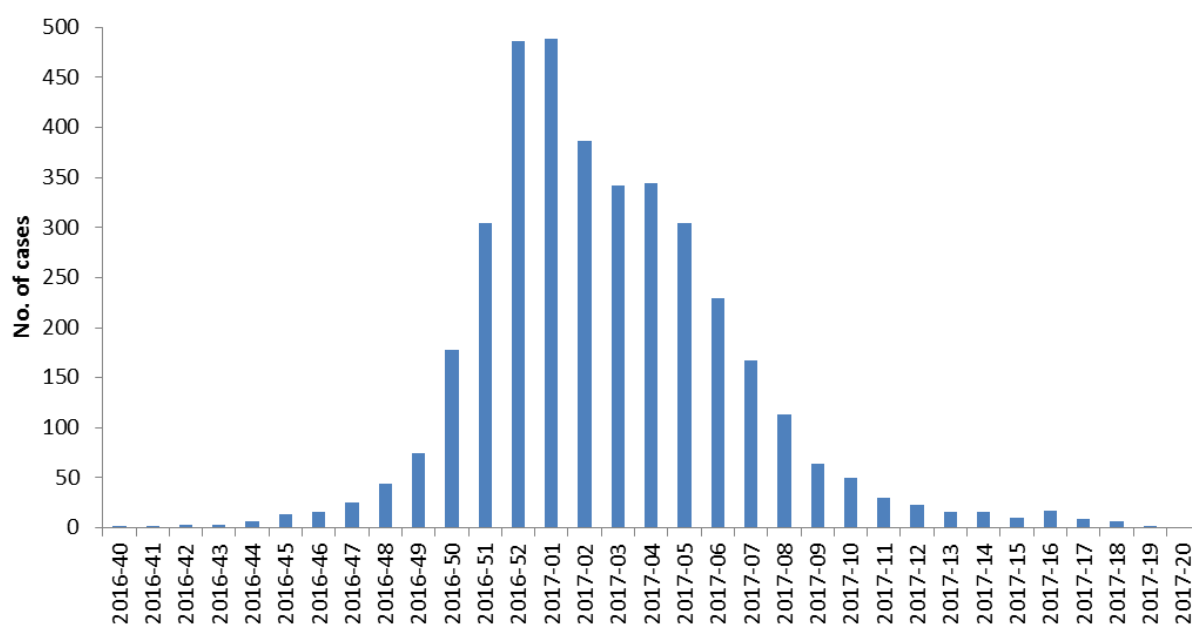


*= represents similar ratio; > represents a ratio of at least 1.5:1 but less than 5:1; >> represents a ratio of $\geq 5:1$

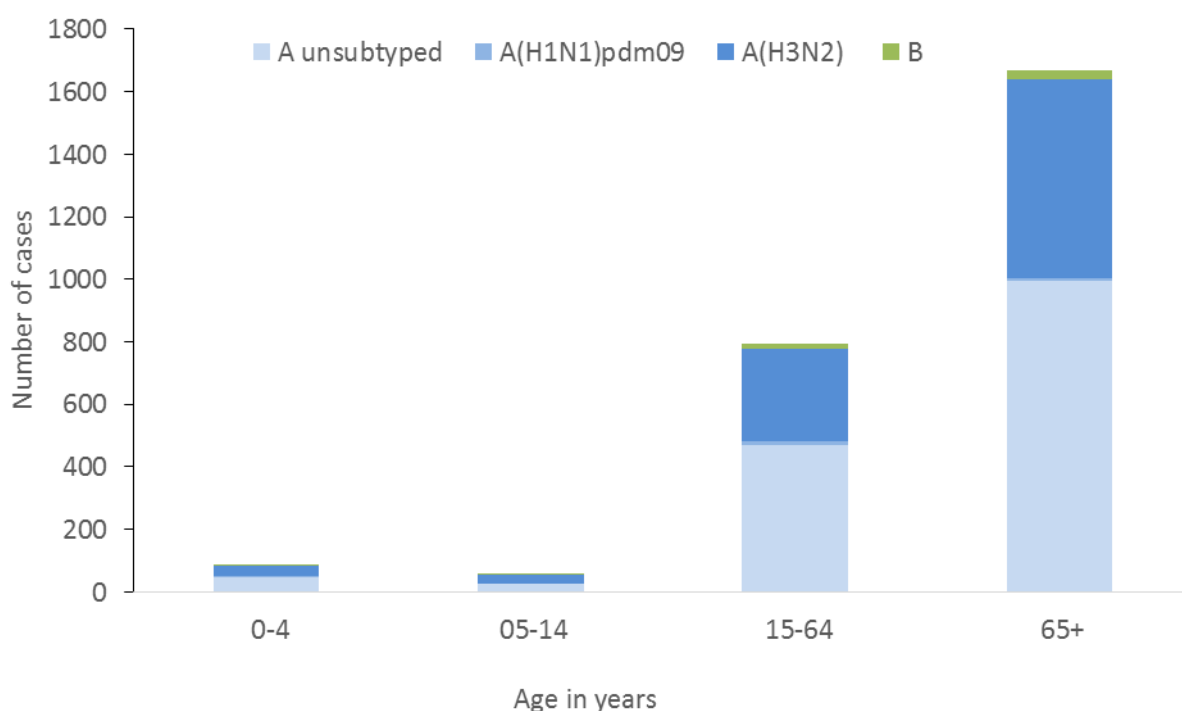
Source: European Centre for Disease Prevention and Control/WHO Regional Office for Europe. Flu News Europe, Joint ECDC–WHO/Europe weekly influenza update

Figure E4. Hospitalised influenza cases

A. Laboratory-confirmed cases admitted to ICU by week*, weeks 40/2016-20/2017



B. Distribution of virus (sub)type by age group in laboratory-confirmed hospitalised cases (ICU)*, weeks 40/2016-20/2017



*Reporting countries: Czech Republic, Finland, France, Ireland, Romania, Slovakia, Spain, Sweden, and the United Kingdom

Influenza specimens received at the Francis Crick Institute.

Samples, viruses and/or clinical samples, with collection dates after 2017-01-31 have been received from 37 countries in Europe, Africa, the Middle East, the Indian Ocean, Central Asia and the Far East. Approximately 73% of the samples received were type A viruses, with influenza A(H3N2) viruses outnumbering A(H1N1)pdm09 viruses at a ratio of 5.3:1 (**Table 1**). Of the type B viruses, viruses of the B/Victoria lineage predominated over those of the B/Yamagata lineage at a ratio of 1.7:1.

Table 2 shows the recovery/isolation rates from specimens received with collection dates after 2017-01-31. Viruses were recovered efficiently with recoveries of between 70% and 84% from samples sent as clinical specimens, depending on virus subtype or lineage, and of $\geq 95\%$ for samples sent as virus isolates, irrespective of subtype or virus lineage. For A(H3N2) viruses, efficient recovery was obtained as assessed by detection of virus neuraminidase activity using the MUNANA sialidase assay, but the minority (approximately 35%) of these were able to be assessed by HI assay as the majority of viruses had low HA titres (<4 HA units in the presence of 20nM oseltamivir).

Table 3 summarises the reactivity of A(H1N1)pdm09, A(H3N2) and viruses of the B/Victoria and B/Yamagata lineages with post-infection ferret antisera raised against vaccine viruses or reference viruses in HI assays. For A(H3N2) viruses, the reactivity of test viruses in plaque reduction neutralisation assays (PRNA) is also summarised. Numbers and percentages of viruses showing ≤ 2 -, 4- and ≥ 8 -fold reductions in titres relative to the titres of the antisera for their respective homologous reference/vaccine viruses are shown. For A(H1N1)pdm09 test viruses the HI titres of an antiserum raised against the vaccine virus, egg-propagated A/Michigan/45/2015, were compared with that with the vaccine virus. For A(H3N2) test viruses the HI titres of antisera raised against egg-propagated A/Hong Kong/4801/2014, tissue culture-propagated A/Hong Kong/5738/2014 and tissue culture-propagated A/Hong Kong/4801/2014 were compared to the titres of the antisera with their homologous viruses. For the A(H3N2) test viruses assessed by PRNA the 50% neutralising titres of antisera raised against egg-propagated A/Hong Kong/4801/2014, tissue culture-propagated A/Hong Kong/7295/2014 and tissue culture-propagated A/Oman2585/2016 were compared to the titres of the antisera with their homologous viruses. For Influenza B viruses of the B/Victoria lineage the HI titres of test viruses with antisera raised against at least one of three tissue culture-propagated B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016) were compared to the titres of the antisera with their homologous viruses. For Influenza B viruses of the B/Yamagata lineage the HI titres of test viruses with an antiserum raised against the egg-propagated vaccine virus, B/Phuket/3073/2013, were compared to the titre of the antiserum with the homologous virus.

Table 1. Summary of clinical samples and virus isolates received, with collection dates after 2017-01-31, by country*

MONTH*	TOTAL RECEIVED	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage		
Country		Number received	Number propagated	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated	Number received	Number propagated ¹	Number received	Number propagated ¹	
2017 FEBRUARY														
Austria	16			3	3	7	0	7				6	6	
Belgium	9					7	0	7				1	1	
Bosnia and Herzegovina	15	1	0			14	in process			1	1			
Bulgaria	12			2	2	7	1	6		3	2			
Egypt	16	4	0	1	0	1	0	1	9	0	1			
France	7					4	2	2		2	2	1	1	
Georgia	1					1	0	1						
Germany	14			4	4	4	2	2		2	2	4	4	
Ghana	1											1	1	
Greece	7			1	0	5	0	0		1	0			
Iceland	13					10	7	3				3	3	
Ireland	1									1	1			
Italy	9			2	1	7	2	3						
Jordan	1					1	0	1						
Kazakhstan	16					1	1	0	7	0	8			
Latvia	9					5	3	2		1	1	3	3	
Lithuania	5					4	3	1		1	1			
Madagascar	1									1	1			
Moldova	4					1	0	1		3	3			
Niger	21								9	in process	10	in process	2	in process
Norway	31			2	1	14	2	12		8	5	7	5	
Poland	10					10	0	5						
Portugal	2					2	1	1						
Russia	42			1	1	28	12	13		12	10	1	1	
Saudi Arabia	8	2	0	2	2	1	0	1		2	2	1	1	
Senegal	2									2	2			
Serbia	10			3	2	6	0	3				1	1	
Slovakia	9			2	2	3	1	2		3	3	1	1	
Slovenia	7					5	1	4				2	2	
Spain	1					1	1	0						
Sweden	4					3	3	0		1	1			
Turkey	10								10	0				
Ukraine	5					1	0	0		4	4			
United Kingdom	3					3	0	3						
2017 MARCH														
Austria	3					1	0	1				1	1	
Bosnia and Herzegovina	3					2	0	2	1	in process	1			
Bulgaria	1					1	0	1						
Egypt	3								3	0				
France	5			1	1	3	2	1				1	1	
Georgia	1													
Germany	13					8	7	1		1	1	2	2	
Iceland	6					4	2	2		3	3	2	2	
Ireland	1											1	1	
Italy	3											3	2	
Jordan	1											1	1	
Kazakhstan	8								7	0	1			
Latvia	2					1	0	1				1	1	
Lithuania	6					1	1	0				4	4	
Madagascar	4					1	0	1		1	1			
Moldova	4									3	2			
Niger	8					7	in process			4	4			
Norway	17			1	1	11	4	5		1	0	1	in process	
Poland	6					6	0	2				4	3	
Romania	7													
Russia	20					6	3	2		7	5			
Saudi Arabia	18	1	0	9	8	6	1	5		14	14	2	2	
Senegal	10			5	5					5	5			
Serbia	2								1	0		1	1	
Slovakia	1											1	1	
Slovenia	7					2	2	0		1	1	4	4	
Turkey	1								1	0				
Ukraine	18			2	1	1	0	0	5	0	10	9		
United Kingdom	1										1	1		

MONTH*	TOTAL RECEIVED	A		H1N1pdm09		H3N2			B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated	Number received	Number propagated ¹	Number received	Number propagated ²		Number received	Number propagated	Number received	Number propagated ¹	Number received	Number propagated ¹
2017 APRIL														
Austria	2												2	2
Georgia	7										7	7		
Germany	4					2	1	1			1	1	1	1
Hong Kong	7										7	sequenced		
Ireland	2												2	2
Jordan	1					1	1	0						
Latvia	2					1	1	0					1	1
Madagascar	7					4	0	3						
Moldova	5										3	2		
Niger	2										5	5		
Norway	24			3	3	2	in process							
Poland	1					4	1	3			9	8	8	8
Romania	8								1	0				
Russia	15					2	0	2			7	5	1	1
Senegal	7			6	6						13	13		
Slovakia	1										1	1		
Slovenia	1										1	1		
Ukraine	14										1	1	1	1
United Kingdom	3			1	1						13	13	1	1
											1	1	1	1
2017 MAY														
Bosnia and Herzegovina	1								1	in process				
France	2			2	2									
Georgia	3			1	1									
Hong Kong	120			30	sequenced	62	sequenced				2	1	24	sequenced
Madagascar	11					10	2	6			4	sequenced		
Moldova	2										1	1		
Niger	2					2	in process				2	2		
Norway	5					3	1	1					2	2
Russia	2													
Ukraine	1										2	2		
United Kingdom	1										1	1	1	1
2017 JUNE														
Ghana	8			4	in process	3	in process						1	0
Hong Kong	50					48	sequenced							
Madagascar	6					5	0	4			2	1		
Niger	1										1	1		
South Africa	15			1	1	14	3	11					1	in process
2017 JULY														
Ghana	10			6	6	4	0	2						
Hong Kong	125					125	sequenced							
South Africa	11			3	1	5	0	5					3	3
2017 AUGUST														
Ghana	1			1	1									
Hong Kong	9					9	sequenced							
	499	0	0	58	22	306	10	38	2	0	83	66	50	24
		0.0%		11.6%		61.3%			0.4%		16.6%		10.0%	
37 Countries					72.9%						27.1%			

* Month indicates the months in which the clinical specimens were collected

1. Propagated to sufficient titre to perform HI assay

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir; numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

311 clinical specimens received - all were assessed by NGS and selected samples chosen for virus isolation and antigenic characterisation

Table 2. Isolation rates for specimens collected after 2017-01-31

	Clinical sample			Virus isolate		
	Number processed	Number propagated ¹	Percent recovered	Number processed	Number propagated ¹	Percent recovered
H1	41	33	80%	41	39	95%
H3	94	12 ²	13%	216	84 ²	39%
		49 ³	70% (20%) ⁴ 53%		125 ³	97% (40%) ⁴ 58%
B Victoria	43	31	72%	140	136	97%
B Yamagata	25	21	84%	67	65	97%

1. Propagated to sufficient HA titre to perform HI assay

2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay

3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay

4. The percentages for viruses recovered that could be assayed by HI are shown in parentheses

Table 3. Summary of isolates with collection dates after 2017-01-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays (PRNA)

Virus type/subtype	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ¹	70	68 (97.1%)	2 (2.9%)	0 (0%)
H3 ^{2a}	82	69 (84.1%)	11 (13.4%)	2 (2.4%)
H3 ^{2b}	124	57 (46.0%)	34 (27.4%)	33 (26.6%)
H3 ^{2c}	124	63 (50.8%)	45 (36.3%)	16 (12.9%)
H3 ^{3a}	94	76 (80.8%)	12 (12.8%)	6 (6.4%)
H3 ^{3b}	140	8 (5.7%)	17 (12.1%)	115 (82.1%)
H3 ^{3c}	140	113 (80.7%)	18 (12.9%)	9 (6.4%)
B/Victoria ⁴	165	160 (97.0%)	2 (1.2%)	3 (1.8%)
B/Yamagata ⁵	83	30 (36.1%)	26 (31.3%)	27 (32.5%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titres for ferret antiserum raised against vaccine virus A/Michigan/45/2015

2a. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Hong Kong/5738/2014 (genetic subgroup 3C.2a)

2b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a)

2c. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a)

3a. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Oman2585/2016 (genetic subgroup 3C.2a1) in plaque-reduction neutralisation assays

3b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a [NYK]) in plaque-reduction neutralisation assays

3c. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Hong Kong/7295/2014 (genetic subgroup 3C.2a) in plaque-reduction neutralisation assays

4. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture propagated B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016)

5. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus B/Phuket/3073/2013

Influenza A(H1N1)pdm09 virus analyses.

Influenza A(H1N1)pdm09 viruses have been received from NICs in 19 countries or administrative regions in four WHO Regions: African (3: Ghana, Senegal and South Africa), Eastern Mediterranean (2: Egypt and Saudi Arabia), European (13: Austria, Bulgaria, France, Georgia, Germany, Greece, Italy, Norway, Russia, Serbia, Slovakia, Ukraine and United Kingdom) and Western Pacific (1: Hong Kong SAR). **Table 4** summarises the antigenic (HI) results for viruses, recovered from clinical samples with collection dates after 2017-01-31, that were propagated successfully at the Francis Crick Institute. The vast majority (~97%) were antigenically indistinguishable from the egg-propagated vaccine virus A/Michigan/45/2015.

Tables 5-1 to 5-12 show the results of HI assays of H1N1 viruses carried out since our February 2017 report. Test viruses in each table are sorted by date of collection and viruses with collection dates after 2017-01-31 are shown below red lines in the tables. A full summary of the HI results for viruses with collection dates after 2017-01-31 is shown in **Table 5-13**.

Approximately 97% of the test viruses were recognised by the antisera raised against A/Michigan/45/2015 vaccine viruses at titres within 2-fold of the titres of the antiserum with the homologous virus. For antisera raised against A/California/7/2009, A/Bayern/69/2009, A/St Petersburg/27/2011, A/Hong Kong/5659/2012, A/South Africa/3626/2013 and A/Slovenia/2903/2015 over 90% of the test viruses were recognised at titres within 2-fold, with all but one titre being within 4-fold, of the titres of the antisera with their respective homologous viruses (**Table 5-13**). The antiserum raised against A/Lviv/N6/2009 recognised only ~18% of test viruses at titres within 2-fold of the titre of the antiserum with the homologous virus and ~30% of the test viruses were recognised at titres $\geq$ 8-fold reduced compared to the homologous titre. The antisera raised against A/Astrakhan/1/2011 and A/St Petersburg/100/2011 recognised > 90% of the test viruses at titres within 4-fold of the homologous titres of the antisera. Antiserum raised against A/Israel/Q-504/2015, a genetic subgroup 6B.2 virus, recognised >85% of viruses at titres within 2-fold of the titre of the antiserum with the homologous virus and all but one virus (98%) at titres within 4-fold of the homologous titre of the antiserum. Of the 70 test viruses for which the genetic group or subgroup were known, all viruses with collection dates after 2017-01-31 fell into genetic subgroup 6B.1.

Figure 1 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. Over the last eight years the HA genes have evolved compared to A/California/7/2009 with viruses in genetic group 6 forming clusters designated 6A, 6B and 6C. Viruses collected after 2017-01-31 fell into subgroup 6B.1 (**Figure 1**). Group 6B and subgroup 6B.1 are defined by the following amino acid substitutions in **HA1** and **HA2**:

- **P83S, D97N, S185T, S203T K163Q, A256T, K283E and I321V** in **HA1**, and **E47K, S124N and E172K** in **HA2** compared with A/California/7/2009 define **genetic group 6B**, e.g. A/South Africa/3626/2013.

- **S84N, S162N (+CHO)** and **I216T** in **HA1** define **genetic subgroup 6B.1** within group 6B, e.g. A/Michigan/45/2015. Within this subgroup a number of virus clusters have emerged defined by **HA1** substitutions **A215G** or **I324V** sometimes with **D274N** or **S74R** with **I295V**, with the latter cluster having subclusters defined by **HA1** substitutions **R45K** or **S164T** or **H273Q** or **K308R**.

An amino acid sequence alignment of the HAs for representative H1N1 viruses is shown in **Figure 2**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 3 illustrates the location of amino acid residues on an H1-HA structure defining genetic group 6B and the two major subgroups 6B.1 and 6B.2.

Figure 4 shows a phylogenetic tree for NA genes of the same representative set of H1N1 viruses as used in **Figure 1**. The HA and NA phylogenetic trees are generally congruent.

Figure 5 shows a NA amino acid alignment for the same representative set of viruses of these viruses as used in **Figure 2**, annotated as **Figure 2** but lacking a proteolytic-processing site.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 47 viruses with collection dates after 2017-01-31. All but one of the viruses tested showed normal inhibition (NI) by both neuraminidase inhibitors. The exception was A/Paris/1227/2017 that encoded NA H275Y substitution known to confer highly reduced inhibition (HRI) (**Table 14**).

Conclusion.

The number of H1N1pdm09 viruses received in recent months has been low. The vast majority of A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus A/Michigan/45/2015 and the vast majority also retained susceptibility to oseltamivir and zanamivir.

Table 4. Summary of influenza A(H1N1)pdm09 viruses analysed, collected after 2017-01-31

MONTH	H1N1pdm09				
Country	Number received ¹	Number propagated ²	Fold reduction in HI titre		
			≤2 fold ³	4 fold ³	≥8 fold ³
2017 FEBRUARY					
Austria	3	3	3		
Bulgaria	2	2	2		
Egypt	1	0			
Germany	4	4	4		
Greece	1	0			
Italy	2	1	1		
Norway	2	1	1		
Russia	1	1	1		
Saudi Arabia	2	2	2		
Serbia	3	2	2		
Slovakia	2	2	2		
2017 MARCH					
France	1	1	1		
Norway	1	1	1		
Saudi Arabia	9	8	8		
Senegal	5	5	4	1	
Ukraine	2	1	1		
2017 APRIL					
Norway	3	3	3		
Senegal	6	6	5	1	
United Kingdom	1	1	1		
2017 MAY					
France	2	2	2		
Georgia	1	1	1		
Hong Kong	30	12*	12		
2017 JUNE					
Ghana	4	2	2		
South Africa	1	1	1		
2017 JULY					
Ghana	6	6	6		
South Africa	3	1	1		
2017 AUGUST					
Ghana	1	1	1		
	99	70	68	2	0
19 Countries			97.1%	2.9%	0.0%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assays

3. Reduction in HI titre with post-infection ferret antiserum raised against A/Michigan/45/2015 (egg grown vaccine virus)

* 12/30 selected for virus isolation after NGS of all 30 clinical specimens

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-03-01 + 2017-03-08)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre												
					Post-infection ferret antisera												
					NEW												
					A/Mich 45/15 Egg NIB F42/16 ¹ 6B.1	A/Cal 7/09 Egg F06/16 ¹	A/Bayern 69/09 MDCK F09/15 ¹	A/Lviv N6/09 MDCK F14/13 ¹	A/Astrak 1/11 MDCK F22/13 ¹ 5	A/St. P 27/11 Egg F26/14 ¹ 6	A/St. P 27/11 Egg F07/17 ¹ 6	A/St. P 100/11 Egg F24/11 ¹ 7	A/HK 5659/12 MDCK F30/12 ¹ 6A	A/Sth Afr 3626/13 Egg F03/14 ¹ 6B	A/Slov 2903/2015 Egg F02/16 ² 6B.1	A/Israel Q-504/15 MDCK F08/16 ² 6B.2	A/Israel Q-504/15 MDCK F08/17 6B.2
					Genetic group												
REFERENCE VIRUSES																	
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E2	640	640	320	160	640	640	1280	1280	1280	640	1280	1280	1280
A/California/7/2009			2009-04-09	E3/E4	640	640	320	160	640	320	1280	1280	640	640	1280	640	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	<	160	80	<	<	<	<	40	<	<	<	<
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	80	80	640	640	40	80	160	80	160	80	160	80	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	1280	640	640	320	640	640	2560	1280	1280	640	1280	640	1280
A/St. Petersburg/27/2011	6	2011-02-14	E1/E4	1280	640	640	320	640	320	1280	1280	640	640	1280	640	1280	1280
A/St. Petersburg/100/2011	7	2011-03-14	E1/E4	2560	1280	1280	640	2560	1280	5120	2560	2560	1280	2560	1280	2560	2560
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	80	40	160	160	640	320	320	160	320	160	320	320
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	320	320	320	320	320	640	640	320	640	640	320	320	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	1280	1280	640	320	1280	640	2560	2560	1280	640	2560	1280	1280
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	640	640	320	160	640	320	1280	1280	640	640	1280	1280	1280
TEST VIRUSES																	
A/Iran/18529/2016	6B.1	2016-01-19	MDCK2/MDCK1	320	320	160	160	320	160	ND	640	640	320	640	320	ND	ND
A/Iran/24101/2016	6B.1	2016-01-27	MDCK3/MDCK1	640	320	160	80	320	160	ND	640	640	320	640	640	ND	ND
A/Iran/24119/2016	6B.1	2016-01-30	MDCK3/MDCK1	640	320	160	80	320	160	ND	640	640	320	1280	640	ND	ND
A/Mayotte/2114/2016	6B.1	2016-02-23	MDCK2/MDCK1	1280	320	320	160	640	320	ND	640	640	640	1280	640	ND	ND
A/Madagascar/5599/2016	6B.1	2016-09-27	MDCK1	640	320	160	80	320	160	1280	640	640	320	640	640	640	640
A/Madagascar/5602/2016	6B.1	2016-09-27	MDCK1	640	640	320	160	640	320	1280	1280	640	640	1280	640	640	640
A/Madagascar/5768/2016	6B.1	2016-10-05	MDCK1	640	320	160	80	320	160	640	640	320	640	640	640	640	640
A/Madagascar/6112/2016	6B.1	2016-10-17	MDCK1	1280	640	320	160	640	320	1280	1280	640	640	1280	640	1280	1280
A/Madagascar/6380/2016	6B.1	2016-10-24	MDCK1	1280	640	320	160	320	320	1280	1280	640	640	1280	640	1280	1280
A/Cherkessk/2/2016		2016-11-09	MDCK2/MDCK1	160	160	320	160	160	160	ND	320	320	320	320	160	ND	ND
A/Yunnan-Simao/SWL1989/2016	6B.1	2016-11-23	C3/MDCK1	640	320	160	80	320	160	ND	640	320	320	640	640	ND	ND
A/Shanghai-Huangpu/SWL13439/2016	6B.1	2016-12-06	C3/MDCK1	640	320	320	160	640	320	ND	640	640	640	1280	640	ND	ND

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND = Not Done

Sequences in phylogenetic trees

Vaccine

G155E
G155E>G, D222G

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-03-15 + 2017-03-22)

Viruses	Other information	Passage history	Collection date	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Mich 45/15 Egg NIB F42/16 ⁻¹	A/Cal 7/09 Egg F06/16 ⁻¹	A/Bayern 69/09 MDCK F09/15 ⁻¹	A/Lviv N6/09 MDCK F14/13 ⁻¹	A/Astrak 1/11 MDCK F22/13 ⁻¹	A/St. P 27/11 Egg F26/14 ⁻¹	A/St. P 100/11 Egg F24/11 ⁻¹	A/HK 5659/12 MDCK F30/12 ⁻¹	A/Sth Afr 3626/13 Egg F03/14 ⁻¹	A/Slov 2903/2015 Egg F02/16 ⁻²	A/Israel Q-504/15 MDCK F08/16 ⁻²	
				6B.1			5	6	7	6A	6B	6B.1	6B.2		
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32		2015-09-07	E3/E2	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280
A/California/7/2009			2009-04-09	E3/E4	1280	640	640	320	1280	640	2560	1280	1280	2560	1280
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	<	160	80	<	<	<	40	<	<	<
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	80	80	640	640	40	80	80	160	80	160	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	1280	640	640	320	640	640	1280	1280	640	1280	640
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	1280	640	640	320	640	320	1280	640	640	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	2560	1280	1280	640	2560	1280	2560	2560	1280	2560	1280
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	160	80	40	160	160	320	320	160	320	160
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	640	320	320	320	320	320	640	320	640	640	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	1280	1280	640	320	1280	640	2560	1280	640	2560	1280
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	640	640	320	160	640	320	1280	640	640	1280	1280
TEST VIRUSES															
A/Oman/8208/2016		6B.1		MDCK1/MDCK1	2560	1280	640	320	1280	640	2560	1280	1280	2560	2560
A/Oman/8006/2016		6B.1		MDCK1/MDCK1	640	640	320	80	640	320	640	640	640	640	640
A/Sapporo/94/2016		6B.1	2016-11-14	MDCK2/MDCK2/MDCK1	1280	640	640	320	640	320	1280	1280	640	2560	1280
A/Montana/50/2016		6B.1	2016-10-10	C1/MDCK1	320	80	160	160	160	160	320	320	160	640	320
A/Kanagawa/206/2016		6B.1	2016-11-02	MDCK2/MDCK2/MDCK1	640	320	160	80	320	160	640	640	320	640	640
A/Qatar/16-VI-17-0000458/2017		6B.1	2017-01-03	MDCK1	1280	1280	640	320	640	320	1280	1280	1280	2560	1280
A/Qatar/10-VI-17-0000393/2017		6B.1	2017-01-03	MDCK1	640	320	160	80	320	160	640	640	320	1280	640
A/Qatar/16-VI-17-0001368/2017		6B.1	2017-01-08	MDCK1	1280	1280	640	320	1280	640	1280	1280	1280	2560	1280
A/Qatar/10-VI-17-0001391/2017		6B.1	2017-01-09	MDCK2	2560	1280	640	640	1280	640	2560	2560	1280	2560	2560
A/Qatar/16-VI-17-0001687/2017		6B.1	2017-01-10	MDCK1	2560	1280	640	640	1280	640	2560	2560	1280	2560	2560
A/Qatar/10-VI-17-0001917/2017		6B.1	2017-01-11	MDCK1	1280	640	320	160	640	320	1280	1280	640	1280	640
A/Qatar/34-VI-17-0001851/2017		6B.1	2017-01-11	MDCK1	2560	1280	640	320	1280	640	2560	1280	1280	2560	1280
A/Qatar/10-VI-17-0001960/2017		6B.1	2017-01-12	MDCK1	1280	1280	640	320	640	640	1280	1280	1280	2560	1280
A/Qatar/10-VI-17-0002160/2017		6B.1	2017-01-13	MDCK1	640	320	160	160	320	320	640	640	320	640	640
A/Qatar/10-VI-17-0002271/2017		6B.1	2017-01-14	MDCK1	2560	1280	640	320	1280	640	2560	2560	1280	2560	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine

G155E

G155E>G, D222G

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-04-05 + 2017-05-31)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				A/Mich 45/15 Egg NIB F42/16 ⁻¹ 6B.1	A/Cal 7/09 Egg F06/16 ⁻¹	A/Bayern 69/09 MDCK F09/15 ⁻¹	A/Lviv N6/09 MDCK F14/13 ⁻¹	A/Astrak 1/11 MDCK F22/13 ⁻¹ 5	A/St. P 27/11 Egg F26/14 ⁻¹ 6	A/St. P 100/11 Egg F24/11 ⁻¹ 7	A/HK 5659/12 MDCK F30/12 ⁻¹ 6A	A/Sth Afr 3626/13 Egg F03/14 ⁻¹ 6B	A/Slov 2903/2015 Egg F02/16 ⁻² 6B.1	A/Israel Q-504/15 MDCK F08/16 ⁻² 6B.2		
				Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	Passage history	
REFERENCE VIRUSES																
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E2	1280	2560	640	640	1280	640	2560	2560	2560	2560	2560	G155E G155E>G, D222G
A/California/7/2009			2009-04-09	E3/E4	1280	1280	640	640	1280	640	2560	2560	1280	2560	1280	
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	40	40	320	160	80	40	40	80	80	80	40	
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	160	160	1280	1280	80	80	160	320	80	160	160	
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	2560	1280	1280	640	1280	640	2560	2560	1280	2560	1280	
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	2560	1280	1280	640	2560	640	2560	2560	1280	2560	1280	
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	5120	5120	2560	1280	5120	1280	5120	5120	5120	5120	5120	
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	160	160	80	160	320	320	640	160	320	320	
A/South Africa/3626/2013	clone 37	6B	2013-06-06	E1/E3	1280	640	640	640	640	640	1280	1280	1280	1280	640	
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E1	2560	2560	640	640	1280	640	2560	2560	1280	2560	2560	
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	2560	1280	640	320	1280	640	2560	2560	1280	2560	2560	
TEST VIRUSES																
A/Dakar/13/2016		6B.1	2016-10-11	p1/M1	1280	1280	320	320	640	640	1280	1280	1280	2560	1280	
A/Dakar/14/2016		6B.1	2016-11-02	p1/M1	2560	2560	1280	640	2560	1280	2560	2560	2560	5120	2560	
A/Dakar/02/2016		6B.1	2016-11-07	p1/M1	2560	1280	640	320	1280	640	1280	1280	1280	2560	1280	
A/Serbia/2680/2017		6B.1	2017-02-03	M1	2560	1280	640	320	1280	640	2560	1280	1280	5120	2560	
A/Serbia/2712/2017		6B.1	2017-02-10	M1	2560	1280	640	320	1280	640	2560	2560	2560	5120	2560	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-05-03)

Haemagglutination inhibition titre															
Viruses	Other information	Passage history	Collection date	Post-infection ferret antisera											
				A/Mich 45/15 Egg NIB F42/16 ⁻¹ 6B.1	A/Cal 7/09 Egg F06/16 ⁻¹	A/Bayern 69/09 MDCK F09/15 ⁻¹	A/Lviv N6/09 MDCK F14/13 ⁻¹	A/Astrak 1/11 MDCK F22/13 ⁻¹ 5	A/St. P 27/11 Egg F26/14 ⁻¹ 6	A/St. P 100/11 Egg F24/11 ⁻¹ 7	A/HK 5659/12 MDCK F30/12 ⁻¹ 6A	A/Sth Afr 3626/13 Egg F03/14 ⁻¹ 6B	A/Slov 2903/2015 Egg F02/16 ⁻² 6B.1	A/Israel Q-504/15 MDCK F08/16 ⁻² 6B.2	
				Passage history	Genetic group										
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E2	640	640	320	160	640	320	1280	1280	640	1280	640
A/California/7/2009			2009-04-09	E3/E4	640	640	320	160	640	320	1280	640	640	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	40	40	320	160	80	40	40	80	80	80	40
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	160	160	1280	1280	80	80	160	320	80	160	160
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	2560	1280	1280	640	1280	640	2560	2560	1280	2560	1280
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	2560	1280	1280	640	2560	640	2560	2560	1280	2560	1280
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	5120	5120	2560	1280	5120	1280	5120	5120	5120	5120	5120
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	160	160	80	160	320	320	640	160	320	320
A/South Africa/3626/2013	clone 37	6B	2013-06-06	E1/E3	1280	640	640	640	640	640	1280	1280	1280	1280	640
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E1	2560	2560	640	640	1280	640	2560	2560	1280	2560	2560
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	2560	1280	640	320	1280	640	2560	2560	1280	2560	2560
TEST VIRUSES															
A/Bulgaria/58/2017		6B.1	2017-01-02	SIAT2/MDCK1	640	320	160	80	320	160	640	320	320	640	320
A/Bulgaria/26/2017		6B.1	2017-01-02	SIAT2/MDCK1	640	320	160	80	320	160	640	640	320	640	640
A/Bulgaria/312/2017		6B.1	2017-01-11	SIAT2/MDCK1	640	320	160	80	320	160	640	640	320	640	320
A/Sachsen/26/2017		6B.1	2017-01-26	C2/MDCK1	640	320	160	80	320	320	640	640	640	1280	640
A/Norway/786/2017		6B.1	2017-01-27	MDCK1/MDCK1	640	320	160	160	640	160	640	640	640	1280	640
A/Norway/807/2017		6B.1	2017-01-27	MDCK1/MDCK1	640	320	160	80	320	160	640	640	640	1280	320
A/Sachsen-Anhalt/3/2017		6B.1	2017-01-27	C1/MDCK1	320	320	80	40	320	160	640	640	320	640	640
A/Sachsen/41/2017		6B.1	2017-01-27	C1/MDCK1	1280	160	320	320	320	160	640	320	320	640	640
A/Niedersachsen/101/2017		6B.1	2017-02-01	C1/MDCK1	320	320	160	80	320	160	640	320	320	640	320
A/Bulgaria/430/2017		6B.1	2017-02-01	SIAT2/MDCK1	640	320	160	160	320	160	640	640	320	1280	640
A/Bulgaria/429/2017		6B.1	2017-02-01	SIAT2/MDCK1	640	320	320	160	320	160	1280	640	640	1280	640
A/Rheinland-Pfalz/42/2017		6B.1	2017-02-17	C2/MDCK1	640	320	160	80	320	160	640	640	640	1280	640
A/Bayern/81/2017		6B.1	2017-02-27	C1/MDCK1	640	640	160	80	320	320	640	640	640	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine

G155E
G155E>G, D222G

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-06-20 + 2017-06-28)

					Haemagglutination inhibition titre										
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera										
					A/Mich 45/15 Egg	A/Cal 7/09 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Astrak 1/11 MDCK	A/St. P 27/11 Egg	A/St. P 100/11 Egg	A/HK 5659/12 MDCK	A/Sth Afr 3626/13 Egg	A/Slov 2903/2015 Egg	A/Israel Q-504/15 MDCK
					NIB F42/16 ¹	F06/16 ¹	F09/15 ¹	F14/13 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²
					6B.1				5	6	7	6A	6B	6B.1	6B.2
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E2	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280
A/California/7/2009			2009-04-09	E3/E4	640	640	640	320	640	320	1280	640	640	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	40	320	160	<	<	40	<	40	40	<
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	80	80	1280	640	80	80	80	160	80	160	160
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	640	320	640	320	1280	1280	640	1280	640
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	640	640	640	320	640	320	1280	1280	640	2560	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E5	640	640	640	320	640	320	1280	640	640	1280	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	320	160	80	320	160	640	320	640	640	320
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	640	320	320	320	320	320	640	640	640	640	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	640	640	320	320	640	320	1280	640	640	1280	1280
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	1280	1280	640	320	640	320	1280	1280	1280	2560	1280
TEST VIRUSES															
A/Sassari/1/2016		6B.1	2016-12-30	MDCK3/MDCK1	640	640	320	160	640	320	1280	1280	640	2560	1280
A/Norway/1019/2017			2017-02-06	MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Kosice/355/2017		6B.1	2017-02-15	MDCKx/MDCK1	640	160	320	160	320	160	640	640	640	1280	640
A/Kosice/354/2017		6B.1	2017-02-16	MDCKx/MDCK1	640	320	320	160	640	320	1280	640	640	1280	640
A/Torino/93/2017		6B.1	2017-02-17	MDCK3/MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Norway/1888/2017		6B.1	2017-03-20	MDCK1/MDCK1	1280	640	640	320	640	320	1280	640	1280	2560	1280
A/Norway/2146/2017		6B.1	2017-04-03	MDCK1/MDCK1	640	640	640	160	1280	320	1280	640	1280	2560	1280
A/Norway/2147/2017		6B.1	2017-04-04	MDCK1	1280	1280	640	320	1280	320	2560	1280	1280	2560	1280
A/Norway/2339/2017		6B.1	2017-04-19	MDCK1	640	1280	640	320	640	320	2560	1280	1280	2560	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Vaccine

Sequences in phylogenetic trees

G155E

G155E>G, D222G

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-07-12)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					A/Mich 45/15 Egg NIB F42/16 ^{†1} 6B.1	A/Cal 7/09 Egg F06/16 ^{†1}	A/Bayern 69/09 MDCK F09/15 ^{†1}	A/Lviv N6/09 MDCK F14/13 ^{†1}	A/Astrak 1/11 MDCK F22/13 ^{†1} 5	A/St. P 27/11 Egg F26/14 ^{†1} 6	A/St. P 100/11 Egg F24/11 ^{†1} 7	A/HK 5659/12 MDCK F30/12 ^{†1} 6A	A/Sth Afr 3626/13 Egg F03/14 ^{†1} 6B	A/Slov 2903/2015 Egg F02/16 ^{†2} 6B.1	A/Israel Q-504/15 MDCK F08/16 ^{†2} 6B.2
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E2	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280
A/California/7/2009			2009-04-09	E3/E4	640	1280	640	320	1280	640	2560	1280	1280	2560	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	40	320	320	40	40	40	40	80	80	40
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	40	80	1280	1280	80	80	80	160	80	160	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	1280	640	320	1280	640	2560	1280	1280	2560	1280
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	640	640	640	320	640	320	1280	640	1280	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E5	640	640	640	320	640	320	2560	640	640	2560	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	160	160	160	80	320	160	640	320	320	640	160
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	320	320	320	320	320	320	640	320	640	640	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280
TEST VIRUSES															
A/Thuringen/130/2017		6B.1	2017-02-18	C1/MDCK1	1280	640	640	320	640	320	1280	1280	1280	2560	1280
A/Dakar/10/2017		6B.1	2017-03-01	P0/MDCK1	640	640	320	160	640	160	1280	640	640	1280	640
A/Dakar/03/2017		6B.1	2017-03-14	P0/MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Dakar/04/2017		6B.1	2017-03-14	P0/MDCK1	640	640	320	80	320	320	640	320	640	1280	640
A/Ukraine/7763/2017			2017-03-23	C1/MDCK2	1280	1280	1280	640	640	640	2560	1280	1280	2560	1280
A/Dakar/11/2017		6B.1	2017-03-29	P1/MDCK1	320	320	160	80	320	160	640	320	320	640	640
A/Dakar/02/2017		6B.1	2017-03-31	P1/MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Dakar/06/2017		6B.1	2017-04-01	P0/MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Dakar/09/2017		6B.1	2017-04-01	P0/MDCK1	320	320	320	160	320	160	640	320	320	1280	640
A/Dakar/05/2017		6B.1	2017-04-02	P0/MDCK1	640	640	320	160	640	160	1280	640	640	1280	640
A/Dakar/01/2017		6B.1	2017-04-04	P0/MDCK1	1280	1280	640	320	1280	320	2560	1280	1280	2560	1280
A/Dakar/07/2017		6B.1	2017-04-05	P0/MDCK1	640	640	320	160	640	160	1280	640	640	1280	640
A/Dakar/08/2017		6B.1	2017-04-05	P0/MDCK1	640	320	160	80	320	160	640	320	640	640	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine

G155E

G155E>G, D222G

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-07-19)

Viruses	Other information	Passage history	Collection date	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Mich 45/15 Egg	A/Cal 7/09 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Astrak 1/11 MDCK	A/St. P 27/11 Egg	A/St. P 100/11 Egg	A/HK 5659/12 MDCK	A/Sth Afr 3626/13 Egg	A/Slov 2903/2015 Egg	A/Israel Q-504/15 MDCK	
				NIB F42/16 ¹	F06/16 ¹	F09/15 ¹	F14/13 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
Genetic group		6B.1	5	6	7	6A	6B	6B.1	6B.2						
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E4	1280	2560	5120	640	2560	1280	5120	2560	2560	5120	2560
A/California/7/2009			2009-04-09	E3/E4	1280	1280	320	640	1280	640	2560	1280	1280	2560	1280
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	40	320	320	40	40	40	80	80	40	40
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	40	80	320	640	80	80	80	80	80	160	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	320	320	1280	320	2560	1280	640	1280	640
A/St. Petersburg/27/2011		6	2011-02-14	E1/E4	640	640	640	320	640	320	2560	1280	640	2560	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E5	640	640	640	320	640	320	2560	640	640	2560	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	320	160	80	320	160	1280	640	640	640	320
A/South Africa/3626/2013	clone 37	6B	2013-06-06	E1/E3	640	640	320	640	640	640	1280	640	1280	640	640
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E1	1280	1280	640	640	1280	640	2560	1280	1280	2560	2560
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	640	640	640	320	640	320	1280	640	640	1280	1280
TEST VIRUSES															
A/Belgium/G198/2017		6B.1	2017-01-23	SIATx/MDCK1	640	640	320	160	320	160	640	640	640	1280	640
A/Sachsen/41/2017		6B.1	2017-01-27	C1/MDCK1	320	40	320	160	320	80	320	320	160	640	640
A/Georgia/617/2017		6B.1	2017-05-25	MDCK1	1280	1280	640	320	1280	640	2560	1280	1280	2560	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine

G155E
G155E>G, D222G

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-08-02 + 2017-08-08)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Mich 45/15 Egg NIB F42/16 ¹	A/Cal 7/09 Egg F06/16 ¹	A/Bayern 69/09 MDCK F09/15 ¹	A/Lviv N6/09 MDCK F14/13 ¹	A/Astrak 1/11 MDCK F22/13 ¹	A/St. P 27/11 Egg F26/14 ¹	A/St. P 100/11 Egg F24/11 ¹	A/HK 5659/12 MDCK F30/12 ¹	A/Sth Afr 3626/13 Egg F03/14 ¹	A/Slov 2903/2015 Egg F02/16 ²	A/Israel Q-504/15 MDCK F08/16 ²	
				6B.1				5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E2	1280	2560	5120	640	2560	1280	5120	2560	2560	5120	2560
A/California/7/2009		2009-04-09	E3/E4	1280	1280	320	640	1280	640	2560	1280	1280	2560	1280	
A/Bayern/69/2009		2009-07-01	MDCK5	<	40	320	320	40	40	40	80	80	40		
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	40	80	320	640	80	80	80	80	160	80		
A/Astrakhan/1/2011	clone 37	5	2011-02-28	MDCK1/MDCK6	640	640	320	320	1280	320	2560	1280	640	1280	640
A/St. Petersburg/27/2011		2011-02-14	E1/E3	640	640	640	320	640	320	2560	1280	640	2560	640	
A/St. Petersburg/100/2011		2011-03-14	E1/E4	640	640	640	320	640	320	2560	640	640	2560	640	
A/Hong Kong/5659/2012		2012-05-21	MDCK4/MDCK2	320	320	160	80	320	160	1280	640	640	640	320	
A/South Africa/3626/2013		2013-06-06	E1/E3	640	640	320	640	640	640	1280	640	1280	640	640	
A/Slovenia/2903/2015		2015-10-26	E4/E1	1280	1280	640	640	1280	640	2560	1280	1280	2560	2560	
A/Israel/Q-504/2015		2015-12-15	C1/MDCK2	640	640	640	320	640	320	1280	640	640	1280	1280	
TEST VIRUSES															
A/Irkutsk/670/2017		6B.1	2017-02-01	MDCK1/MDCK1	640	640	320	160	640	160	1280	640	640	1280	640
A/Austria/971790/2017		6B.1	2017-02-20	C2/MDCK1	1280	640	640	320	640	640	1280	1280	1280	2560	1280
A/Austria/971917/2017		6B.1	2017-02-13	C2/MDCK1	640	640	320	160	640	320	1280	1280	1280	1280	1280
A/Austria/971919/2017		6B.1	2017-02-13	C1/MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Bretagne/1184/2017		6B.1	2017-03-22	MDCK1/MDCK1	640	640	320	160	640	320	1280	640	640	1280	1280
A/England/41/2017		6B.1	2017-04-12	MDCK1/MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Paris/1226/2017		6B.1	2017-05-03	MDCK1/MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Paris/1227/2017		6B.1	2017-05-03	MDCK1/MDCK1	640	640	320	160	640	320	1280	640	640	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine

G155E
G155E>G, D222G

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-08-16)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Mich 45/15	A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Astrak 1/11	A/St. P 27/11	A/St. P 100/11	A/HK 5659/12	A/Sth Afr 3626/13	A/Slov 2903/2015	A/Israel Q-504/15	
				Egg	Egg	MDCK	MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
				NIB F42/16 ¹	F06/16 ¹	F09/15 ¹	F14/13 ¹	F22/13 ¹	F26/14 ¹	F24/11 ¹	F30/12 ¹	F03/14 ¹	F02/16 ²	F08/16 ²	
	Passage history			6B.1				5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/Michigan/45/2015		6B.1	2015-09-07	E3/E2	640	640	320	320	1280	320	2560	640	640	1280	1280
A/California/7/2009	clone 38-32		2009-04-09	E3/E4	640	640	320	320	640	320	2560	640	640	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5	<	40	160	40	<	80	<	40	40	<	
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	40	80	640	640	80	80	160	80	80	160	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK6	640	640	640	160	1280	320	2560	640	640	1280	640
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	640	640	640	160	640	640	2560	640	640	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	640	640	640	320	640	320	2560	640	640	2560	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	160	160	160	80	320	160	1280	320	320	320	320
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	320	320	320	320	640	320	1280	320	640	640	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	640	1280	640	320	640	640	2560	1280	640	2560	1280
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	640	640	640	320	640	320	2560	640	640	2560	1280
TEST VIRUSES															
A/Montana/50/2016		6B.1	2016-10-10	E3/E1	160	160	160	80	160	80	640	160	160	320	320
NYMC X-299 (A/Montana/50/2016)		6B.1	2016-10-10	Ex	640	640	320	160	640	320	2560	640	640	1280	640
A/Shanghai-Huangpu/SWL13439/2016		6B.1	2016-12-06	E3/E1	640	640	320	160	1280	320	2560	640	640	1280	1280

Vaccine

G155E

G155E>G, D222G

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-08-29)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Mich 45/15 Egg NIB F42/16 ⁻¹	A/Cal 7/09 Egg F06/16 ⁻¹	A/Bayern 69/09 MDCK F09/15 ⁻¹	A/Lviv N6/09 MDCK F14/13 ⁻¹	A/Astrak 1/11 MDCK F22/13 ⁻¹	A/St. P 27/11 Egg F26/14 ⁻¹	A/St. P 100/11 Egg F24/11 ⁻¹	A/HK 5659/12 MDCK F30/12 ⁻¹	A/Sth Afr 3626/13 Egg F03/14 ⁻¹	A/Slov 2903/2015 Egg F02/16 ⁻²	A/Israel Q-504/15 MDCK F08/16 ⁻²	
				6B.1				5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E3	640	640	320	320	1280	320	2560	640	640	1280	1280
A/California/7/2009			2009-04-09	E3/E5	640	640	320	320	640	320	2560	640	640	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	40	160	160	40	<	80	<	40	80	<
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	<	80	640	640	40	40	80	80	40	160	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	320	320	320	160	640	160	1280	640	640	640	320
A/St. Petersburg/27/2011	clone 37	6	2011-02-14	E1/E4	640	640	640	320	640	320	2560	640	640	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	1280	1280	1280	640	1280	640	5120	1280	1280	2560	1280
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	160	160	160	80	320	80	640	320	160	320	160
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	320	320	320	320	640	320	1280	320	640	640	320
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E1	640	640	320	160	640	320	1280	640	640	2560	640
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280
TEST VIRUSES															
A/Saudi Arabia/1017089598/2017		6B.1	2017-01-21	MDCK1	640	640	320	160	640	160	1280	640	640	1280	1280
A/Saudi Arabia/1045481031/2017		6B.1	2017-01-27	MDCK2	640	640	320	160	320	320	1280	640	640	1280	640
A/Saudi Arabia/246644/2017		6B.1	2017-02-25	SIAT1	320	320	160	80	320	160	1280	320	320	1280	320
A/Saudi Arabia/1133704245/2017		6B.1	2017-02-26	MDCK1	320	640	320	160	640	160	1280	640	640	1280	640
A/Saudi Arabia/247711/2017		6B.1	2017-03-01	MDCK1	320	320	320	160	320	160	1280	320	640	1280	640
A/Saudi Arabia/2362806503/2017		6B.1	2017-03-07	MDCK1	320	640	320	160	640	160	1280	640	640	1280	640
A/Saudi Arabia/253107/2017		6B.1	2017-03-15	SIAT1	320	640	160	80	640	160	1280	640	640	1280	640
A/Saudi Arabia/253340/2017		6B.1	2017-03-15	MDCK1	320	640	160	80	320	160	1280	640	640	1280	640
A/Saudi Arabia/NK3106091/2017		6B.1	2017-03-15	MDCK1	320	320	160	80	320	160	1280	320	320	1280	640
A/Saudi Arabia/106153704/2017		6B.1	2017-03-18	MDCK2	640	640	160	80	640	160	1280	640	640	1280	640
A/Saudi Arabia/254177/2017		6B.1	2017-03-19	MDCK1	640	640	320	160	640	160	1280	640	640	2560	640
A/Saudi Arabia/1034579050/2017		6B.1	2017-03-20	MDCK1	640	640	320	160	640	320	2560	640	640	1280	1280
A/Ghana/1762/2107		6B.1	2017-06-16	C1/MDCK1	640	640	320	320	640	320	2560	640	640	2560	1280
A/Ghana/1777/2107		6B.1	2017-06-23	C2/MDCK1	640	640	320	160	640	320	2560	640	640	2560	640
A/South Africa/R05819/2017		6B.1	2017-06-30	MDCK2	640	640	160	160	640	160	1280	640	640	2560	640
A/Ghana/1901/2107		6B.1	2017-07-06	C1/MDCK1	640	1280	320	320	640	320	2560	640	640	2560	640
A/Ghana/2032/2107		6B.1	2017-07-14	C3/MDCK1	1280	1280	640	320	1280	320	2560	1280	1280	2560	1280
A/South Africa/R06693/2017		6B.1	2017-07-17	MDCK1/MDCK1	320	320	160	160	320	160	1280	640	320	1280	640
A/Ghana/2145/2107		6B.1	2017-07-25	MDCK1	640	640	320	160	640	320	2560	640	640	2560	1280
A/Ghana/2237/2107		6B.1	2017-08-03	MDCK1	640	640	320	160	640	320	1280	640	640	2560	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Sequences in phylogenetic trees

Vaccine

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-09-06 + 2017-09-12)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Mich	A/Cal	A/Bayern	A/Lviv	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
				45/15	7/09	69/09	N6/09	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
				Egg	Egg	MDCK	MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	
	Passage history														
	Ferret number														
	Genetic group														
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E3	640	640	320	160	640	320	2560	640	640	2560	640
A/California/7/2009			2009-04-09	E3/E3	320	640	320	160	640	320	1280	640	640	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	40	320	160	40	<	80	<	40	80	40
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	40	80	640	640	80	80	160	80	80	320	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	640	320	1280	320	2560	1280	1280	1280	640
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	640	320	640	320	2560	640	640	2560	640	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E5	640	640	640	320	640	320	2560	640	640	1280	640	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	320	160	80	640	160	1280	640	320	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	320	320	320	160	320	320	1280	320	640	1280	320	
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	320	640	320	160	640	320	1280	640	640	1280	640
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	640	640	320	160	640	320	2560	640	1280	2560	1280
TEST VIRUSES															
A/Yunnan-Wuhua/SWL1938/2016		6B.1	2016-12-14	C2/M3	640	640	320	160	640	320	1280	640	640	1280	640
A/Yunnan-Wuhua/SWL1938/2016		6B.1	2016-12-14	E4/E1	320	320	320	160	640	160	1280	320	320	1280	640
A/Guizhou-Qingzhen/SWL1331/2017		6B.1	2017-04-24	C3/MDCK1	320	40	160	320	320	80	320	160	160	1280	640
A/Guizhou-Qingzhen/SWL1331/2017		6B.1	2017-04-24	E3/E1	320	640	320	160	640	320	1280	640	640	1280	640
A/Ghana/1883/2017		6B.1	2017-07-04	C1/MDCK3	320	320	160	80	320	160	640	320	320	640	320
A/Ghana/475/2017		6B.1	2017-07-09	MDCK2	320	320	160	80	320	160	640	320	320	640	320
A/Ghana/482/2017		6B.1	2017-07-12	MDCK2	640	640	320	160	640	320	1280	640	640	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

Vaccine

G155E

G155E>G, D222G

Table 5-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2017-09-19)

				Haemagglutination inhibition titre											
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				A/Mich	A/Cal	A/Bayern	A/Lviv	A/Astrak	A/St. P	A/St. P	A/HK	A/Sth Afr	A/Slov	A/Israel	
				45/15	7/09	69/09	N6/09	1/11	27/11	100/11	5659/12	3626/13	2903/2015	Q-504/15	
				Egg	Egg	MDCK	MDCK	MDCK	Egg	MDCK	Egg	MDCK	Egg	MDCK	
	Passage history	Ferret number	Genetic group	NIB F42/16 ⁻¹	F06/16 ⁻¹	F09/15 ⁻¹	F14/13 ⁻¹	F22/13 ⁻¹	F26/14 ⁻¹	F24/11 ⁻¹	F30/12 ⁻¹	F03/14 ⁻¹	F02/16 ⁻²	F08/16 ⁻²	
				6B.1				5	6	7	6A	6B	6B.1	6B.2	
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E2	1280	1280	640	320	1280	640	2560	1280	1280	2560	1280
A/California/7/2009			2009-04-09	E3/E3	320	320	320	160	640	320	1280	640	640	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5/MDCK1	<	40	320	160	40	<	80	40	40	80	40
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	40	80	640	640	160	80	160	160	80	320	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	640	640	320	2560	640	2560	2560	1280	2560	1280
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	640	640	640	320	1280	320	2560	1280	640	1280	1280
A/St. Petersburg/100/2011		7	2011-03-14	E1/E5	640	640	640	320	1280	320	2560	1280	640	2560	1280
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	160	320	160	80	320	160	1280	640	320	640	320
A/South Africa/3626/2013	clone 37	6B	2013-06-06	E1/E3	640	320	320	320	640	320	1280	640	1280	1280	640
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E1	640	640	320	320	1280	640	2560	1280	1280	2560	1280
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	1280	640	320	320	1280	320	2560	1280	1280	2560	2560
TEST VIRUSES															
A/Hong Kong/2106/2017		6B.1	2017-05-10	MDCK2	640	640	640	160	640	320	2560	1280	1280	2560	1280
A/Hong Kong/2109/2017		6B.1	2017-05-12	MDCK2	640	640	640	160	640	320	1280	1280	640	2560	1280
A/Hong Kong/2111/2017		6B.1	2017-05-15	MDCK2	640	640	640	160	640	320	2560	1280	640	2560	1280
A/Hong Kong/2113/2017		6B.1	2017-05-13	MDCK2	640	640	320	160	640	320	1280	640	640	2560	1280
A/Hong Kong/2199/2017		6B.1	2017-05-16	MDCK2	640	640	320	160	640	320	2560	1280	640	2560	1280
A/Hong Kong/2201/2017		6B.1	2017-05-18	MDCK2	640	640	640	320	640	320	2560	1280	640	2560	1280
A/Hong Kong/2239/2017		6B.1	2017-05-19	MDCK1	640	640	320	160	640	320	2560	1280	640	2560	1280
A/Hong Kong/2240/2017		6B.1	2017-05-20	MDCK1	640	640	320	160	640	320	1280	640	640	1280	1280
A/Hong Kong/2242/2017		6B.1	2017-05-20	MDCK1	1280	1280	640	320	1280	640	2560	1280	1280	2560	2560
A/Hong Kong/2243/2017		6B.1	2017-05-21	MDCK1	640	640	320	160	640	320	1280	640	640	1280	640
A/Hong Kong/2245/2017		6B.1	2017-05-21	MDCK1	640	640	320	160	640	320	2560	1280	640	2560	1280
A/Hong Kong/2311/2017		6B.1	2017-05-24	MDCK1	640	640	320	320	640	320	2560	1280	640	2560	1280

G155E

G155E>G, D222G

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Sequences in phylogenetic trees

G155E
G155E>G, D222G

Table 5-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses with collection dates after 2017-01-31 - Summary

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Mich 45/15 Egg	A/Cal 7/09 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Astrak 1/11 MDCK	A/St. P 27/11 Egg	A/St. P 100/11 Egg	A/HK 5659/12 MDCK	A/Sth Afr 3626/13 Egg	A/Slov 2903/2015 Egg	A/Israel Q-504/15 MDCK	
				NIB F42/16 ^{*1} 6B.1	F06/16 ^{*1}	F09/15 ^{*1}	F14/13 ^{*1}	F22/13 ^{*1} 5	F26/14 ^{*1} 6	F24/11 ^{*1} 7	F30/12 ^{*1} 6A	F03/14 ^{*1} 6B	F02/16 ^{*2} 6B.1	F08/16 ^{*2} 6B.2	
REFERENCE VIRUSES															
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E2	640	640	320	320	1280	320	2560	640	640	1280	1280
A/California/7/2009			2009-04-09	E3/E4	640	640	320	320	640	320	2560	640	640	1280	640
A/Bayern/69/2009			2009-07-01	MDCK5	<	40	160	160	40	<	80	<	40	40	<
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	40	80	640	640	80	80	160	80	80	160	80
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK6	640	640	640	160	1280	320	2560	640	640	1280	640
A/St. Petersburg/27/2011		6	2011-02-14	E1/E3	640	640	640	160	640	640	2560	640	640	1280	640
A/St. Petersburg/100/2011		7	2011-03-14	E1/E4	640	640	640	320	640	320	2560	640	640	2560	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	160	160	160	80	320	160	1280	320	320	320	320
A/South Africa/3626/2013		6B	2013-06-06	E1/E3	320	320	320	320	640	320	1280	320	640	640	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E1	640	1280	640	320	640	640	2560	1280	640	2560	1280
A/Israel/Q-504/2015		6B.2	2015-12-15	C1/MDCK2	640	640	640	320	640	320	2560	640	640	2560	1280
TEST VIRUSES															
Number of viruses tested*					72	72	72	72	72	72	72	72	72	72	72
No with titre reduction ≤2-fold					70	67	72	13	49	67	48	71	69	69	63
%					97.2	93.1	100	18.0	68.1	93.1	66.7	98.6	95.8	95.8	87.5
No with titre reduction =4-fold					2	4		37	23	5	19	1	3	3	8
%					2.8	5.6		51.4	31.9	6.9	26.4	1.4	4.2	4.2	11.1
No with titre reduction ≥8-fold						1		22			5				1
%						1.3		30.6			6.9				1.4

* Of those with available HA sequence (X), all were clade 6B.1

Vaccine

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 1. Phylogenetic comparison of A(H1N1)pdm09 HA genes

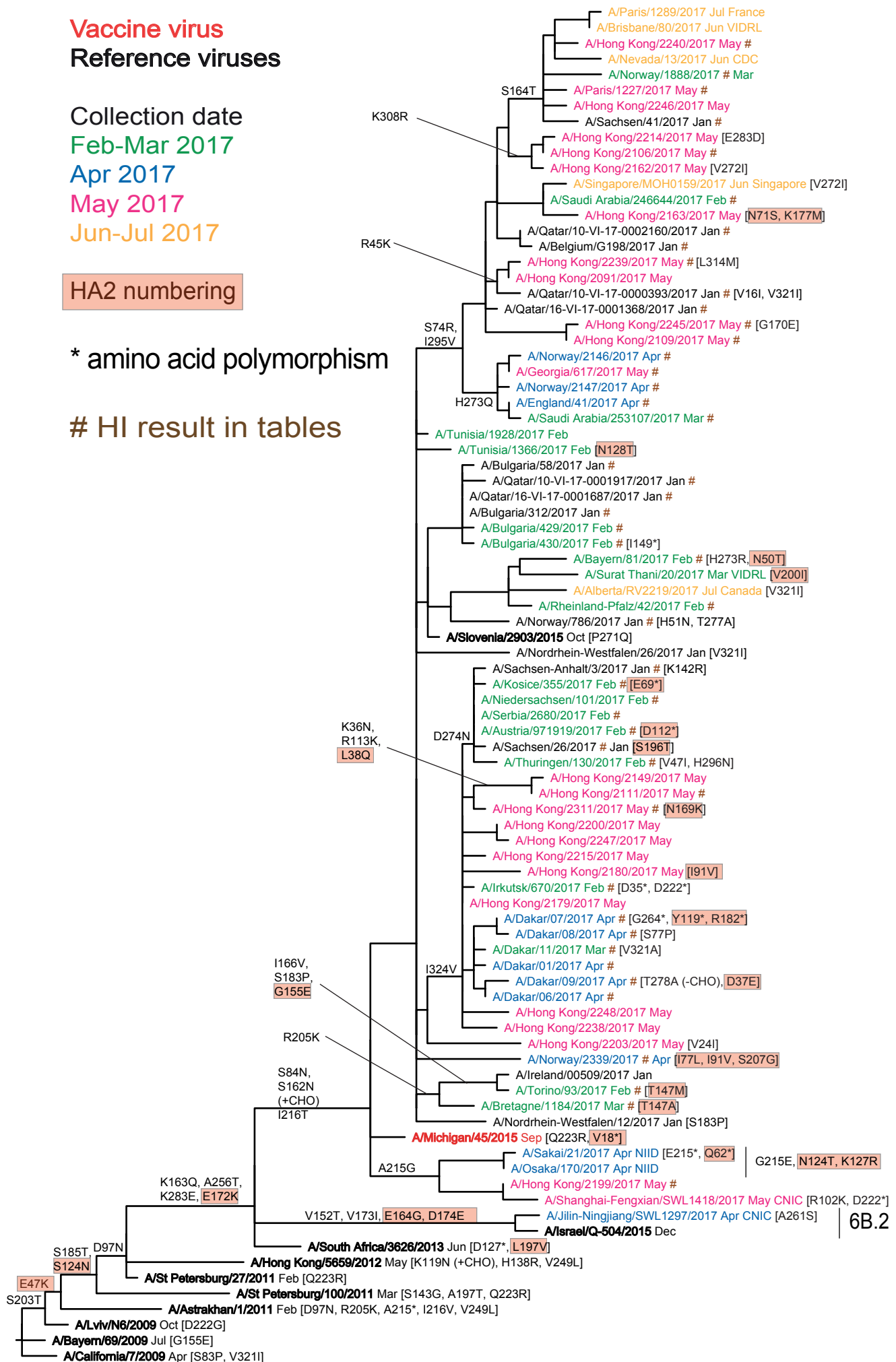


Figure 2. HA protein alignment for influenza A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100	
A/California/7/2009	DTLCIGYHANNST	DTVDTVLEKNVT	VTHTSVNLL	EDKHNGKLC	LRGVAPLHLG	KCNIA	GWILGNPE	CESLSTASS	WSYIVETP	SSDNGT	CYPGDFIDYEE
A/Bayern/69/2009									S		
A/Lviv/N6/2009									S		
A/Astrakhan/1/2011									S		N 5
A/St Petersburg/27/2011									S		N 6
A/St Petersburg/100/2011									S		N 7
A/Hong Kong/5659/2012									S		N 6A
A/South Africa/3626/2013									S		N 6B
A/Israel/Q-504/2015									S		N 6B.2
A/Jilin-Ningjiang/SWL1297/2017									S		N 6B.2
A/Hong Kong/2199/2017									SN		N 6B.1
A/Michigan/45/2015									SN		N 6B.1
A/Bretagne/1184/2017									SN		N 6B.1
A/Torino/93/2017									SN		N 6B.1
A/Hong Kong/2248/2017									SN		N 6B.1
A/Dakar/11/2017									SN		N 6B.1
A/Hong Kong/2149/2017					N				SN		N 6B.1
A/Kosice/355/2017									SN		N 6B.1
A/Slovenia/2903/2015									SN		N 6B.1
A/Bayern/81/2017									SN		N 6B.1
A/Bulgaria/429/2017									SN		N 6B.1
A/Saudi Arabia/253107/2017								R	SN		N 6B.1
A/Georgia/617/2017								R	SN		N 6B.1
A/Hong Kong/2245/2017								R	SN		N 6B.1
A/Hong Kong/2239/2017						K		R	SN		N 6B.1
A/Saudi Arabia/246644/2017								R	SN		N 6B.1
A/Hong Kong/2214/2017								R	SN		N 6B.1
A/Hong Kong/2246/2017								R	SN		N 6B.1
A/Norway/1888/2017								R	SN		N 6B.1
A/Paris/1289/2017								R	SN		N 6B.1

	110	120	130	140	150	160	170	180	190	200		
A/California/7/2009	LREQLSSVSS	FERFEIFPKT	SSWPNHDS	NKGVTAAC	PHAGAKSF	YKNLIWL	VKKGNSY	PKLSKSY	INDKGEV	LVVLWG	IHPHST	SADQQLSYQNADAYVF
A/Bayern/69/2009							E					
A/Lviv/N6/2009												
A/Astrakhan/1/2011											T	
A/St Petersburg/27/2011											T	
A/St Petersburg/100/2011						G					T	
A/Hong Kong/5659/2012		N			R						T	
A/South Africa/3626/2013			X					Q			T	
A/Israel/Q-504/2015								Q		I	T	
A/Jilin-Ningjiang/SWL1297/2017						T		Q		I	T	
A/Hong Kong/2199/2017								NQ			T	
A/Michigan/45/2015								NQ			T	
A/Bretagne/1184/2017								NQ			T	
A/Torino/93/2017								NQ	V		P	T
A/Hong Kong/2248/2017								NQ			T	
A/Dakar/11/2017								NQ			T	
A/Hong Kong/2149/2017			K					NQ			T	
A/Kosice/355/2017								NQ			T	
A/Slovenia/2903/2015								NQ			T	
A/Bayern/81/2017								NQ			T	
A/Bulgaria/429/2017								NQ			T	
A/Saudi Arabia/253107/2017								NQ			T	
A/Georgia/617/2017								NQ			T	
A/Hong Kong/2245/2017								NQ	E		T	
A/Hong Kong/2239/2017								NQ			T	
A/Saudi Arabia/246644/2017								NQ			T	
A/Hong Kong/2214/2017								NQ			T	
A/Hong Kong/2246/2017								NQT			T	
A/Norway/1888/2017								NQT			T	
A/Paris/1289/2017								NQT			T	

	210	220	230	240	250	260	270	280	290	300	
A/California/7/2009	VGSSRYSKFK	PEIARPKVRD	QEGRMNYWT	LVPEGDKIT	FEATGNLV	VPRYAFAM	ERNAGSGII	ISDTPVHDC	NTTCQTPK	GAIN	TSLPFQNIHPITI
A/Bayern/69/2009											
A/Lviv/N6/2009				G							
A/Astrakhan/1/2011	T	K	XV			L					
A/St Petersburg/27/2011				R							
A/St Petersburg/100/2011				R							
A/Hong Kong/5659/2012						L					
A/South Africa/3626/2013											
A/Israel/Q-504/2015											
A/Jilin-Ningjiang/SWL1297/2017											
A/Hong Kong/2199/2017											
A/Michigan/45/2015											
A/Bretagne/1184/2017											
A/Torino/93/2017											
A/Hong Kong/2248/2017											
A/Dakar/11/2017											
A/Hong Kong/2149/2017											
A/Kosice/355/2017											
A/Slovenia/2903/2015											
A/Bayern/81/2017											
A/Bulgaria/429/2017											
A/Saudi Arabia/253107/2017											
A/Georgia/617/2017											
A/Hong Kong/2245/2017											
A/Hong Kong/2239/2017											
A/Saudi Arabia/246644/2017											
A/Hong Kong/2214/2017											
A/Hong Kong/2246/2017											
A/Norway/1888/2017											
A/Paris/1289/2017											

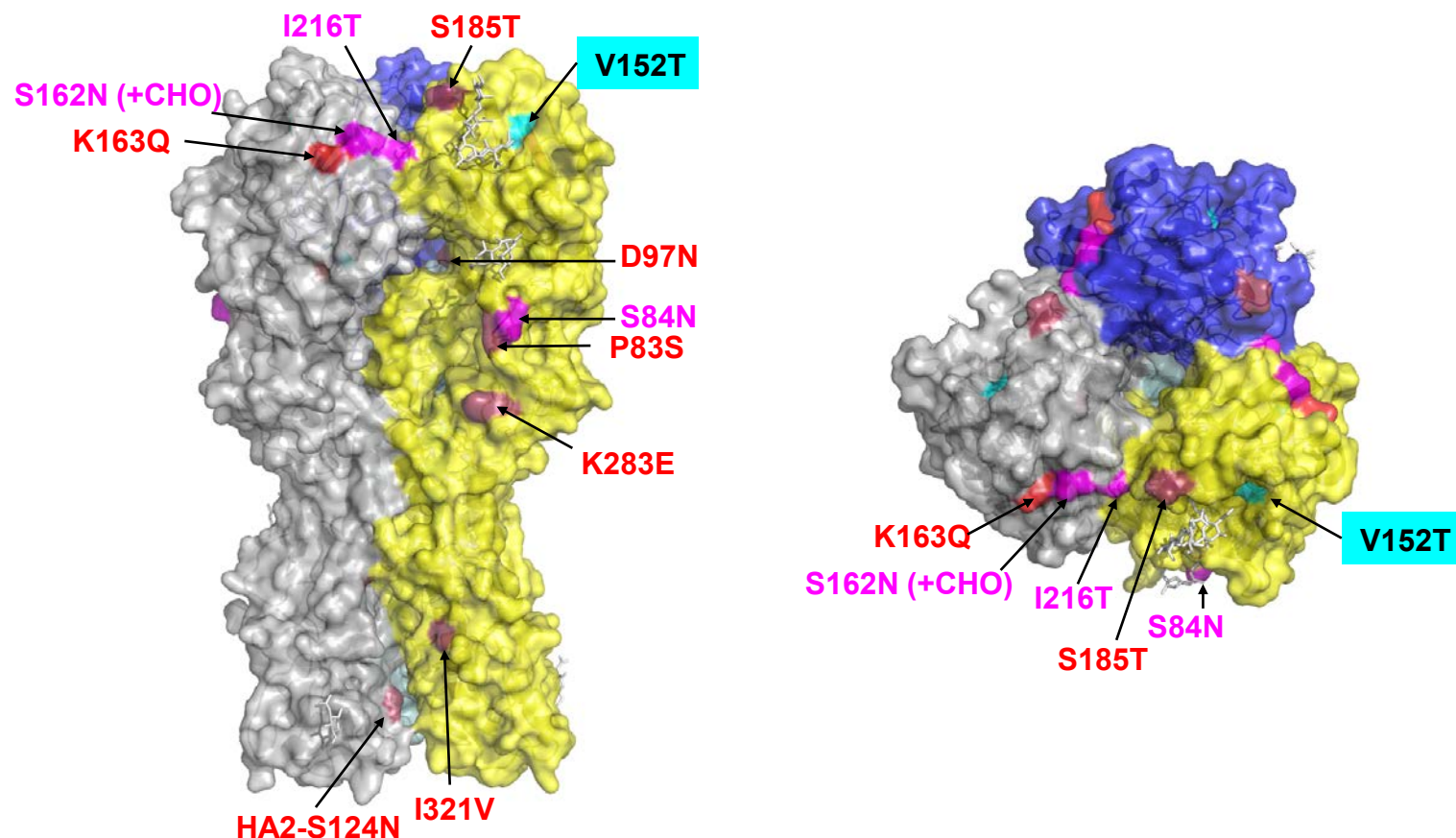
	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	GKCPKYVKSTKLR	LATGLRNIPSIQSR	RG	GLFGA	IAGFIEGGWTGMVDG	WYGYHHQNEQSG	GYAADLKSTQNA	IDEITNKVNSVIEK	MNTQFTAVG	KEFNHL
A/Bayern/69/2009		V								
A/Lviv/N6/2009		V								
A/Astrakhan/1/2011		V						K		
A/St Petersburg/27/2011		V						K		
A/St Petersburg/100/2011		V						K		
A/Hong Kong/5659/2012		V						K		
A/South Africa/3626/2013		V						K		
A/Israel/Q-504/2015		V						K		
A/Jilin-Ningjiang/SWL1297/2017		V						K		
A/Hong Kong/2199/2017		V						K		
A/Michigan/45/2015		V			X			K		
A/Bretagne/1184/2017		V						K		
A/Torino/93/2017		V						K		
A/Hong Kong/2248/2017		V						K		
A/Dakar/11/2017		A	V					K		
A/Hong Kong/2149/2017		V	V				Q	K		
A/Kosice/355/2017		V	V					K		X
A/Slovenia/2903/2015		V						K		
A/Bayern/81/2017		V						K	T	
A/Bulgaria/429/2017		V						K		
A/Saudi Arabia/253107/2017		V						K		
A/Georgia/617/2017		V						K		
A/Hong Kong/2245/2017		V						K		
A/Hong Kong/2239/2017	M	V						K		
A/Saudi Arabia/246644/2017		V						K		
A/Hong Kong/2214/2017	R	V						K		
A/Hong Kong/2246/2017		V						K		
A/Norway/1888/2017		V						K		
A/Paris/1289/2017		V						K		

	410	420	430	440	450	460	470	480	490	500
A/California/7/2009	EKRIENLNKKVDDG	FLDIWTYN	AEALLVLL	ENERTLDYHDS	NVKNLYEKVRS	QLKNNAKEI	GNCGFEFYHK	CNDTCMESV	KNGTYDYPKY	SEEAALNREEI
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Astrakhan/1/2011					N					
A/St Petersburg/27/2011					N					
A/St Petersburg/100/2011					N					
A/Hong Kong/5659/2012					N					
A/South Africa/3626/2013					N					K
A/Israel/Q-504/2015					N				G	K
A/Jilin-Ningjiang/SWL1297/2017					N				G	K
A/Hong Kong/2199/2017					N					K
A/Michigan/45/2015					N					K
A/Bretagne/1184/2017					N		A			K
A/Torino/93/2017					N		M	E		K
A/Hong Kong/2248/2017					N					K
A/Dakar/11/2017					N					K
A/Hong Kong/2149/2017					N					K
A/Kosice/355/2017					N					K
A/Slovenia/2903/2015					N					K
A/Bayern/81/2017					N					K
A/Bulgaria/429/2017					N					K
A/Saudi Arabia/253107/2017					N					K
A/Georgia/617/2017					N					K
A/Hong Kong/2245/2017					N					K
A/Hong Kong/2239/2017					N					K
A/Saudi Arabia/246644/2017					N					K
A/Hong Kong/2214/2017					N					K
A/Hong Kong/2246/2017					N					K
A/Norway/1888/2017					N					K
A/Paris/1289/2017					N					K

	510	520	530	540
A/California/7/2009	DGVKLESTRIYQIL	AIYSTVASSLVL	VSLGAISFWMCS	NGSLQCRICI
A/Bayern/69/2009				
A/Lviv/N6/2009				
A/Astrakhan/1/2011				
A/St Petersburg/27/2011				
A/St Petersburg/100/2011				
A/Hong Kong/5659/2012				
A/South Africa/3626/2013			V	
A/Israel/Q-504/2015	E			
A/Jilin-Ningjiang/SWL1297/2017	E			
A/Hong Kong/2199/2017				
A/Michigan/45/2015				
A/Bretagne/1184/2017				
A/Torino/93/2017				
A/Hong Kong/2248/2017				
A/Dakar/11/2017				
A/Hong Kong/2149/2017				
A/Kosice/355/2017				
A/Slovenia/2903/2015				
A/Bayern/81/2017				
A/Bulgaria/429/2017				
A/Saudi Arabia/253107/2017				
A/Georgia/617/2017				
A/Hong Kong/2245/2017				
A/Hong Kong/2239/2017				
A/Saudi Arabia/246644/2017				
A/Hong Kong/2214/2017				
A/Hong Kong/2246/2017				
A/Norway/1888/2017				
A/Paris/1289/2017				

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 3. H1 HA locations of amino acids substitutions in genetic subgroups 6B.1 and 6B.2



Amino acid substitutions compared with A/California/7/2009 that define genetic group 6B are in **red** (**S203T**, **A256T** in HA1 with **E47K**, **E172K** in HA2 not visible on this structure), those that define subgroup 6B.1 in **magenta**, those that define subgroup 6B.2 in **cyan** (**V173I** in HA1 with **E164G**, **D174E** in HA2 not visible on this structure). A number of viruses in subgroup 6B.2 carry additional substitutions (e.g. **R113K**, **D127E** in HA1 with **K47Q** in HA2).

6B.1 viruses have circulated virtually exclusively and a number of genetic subclusters have emerged defined by various HA1 amino acid substitutions: e.g. A215G, R205K, I166V + S183P, I324V, D274N + I324V, K36N + R133K + I324V, S74R + I295V, S74R + H273Q + I295V, R45K + S74R + I295V, S74R + I295V + K308R, S74R + S164T + I295V

Figure 4. Phylogenetic comparison of A(H1N1)pdm09 NA genes

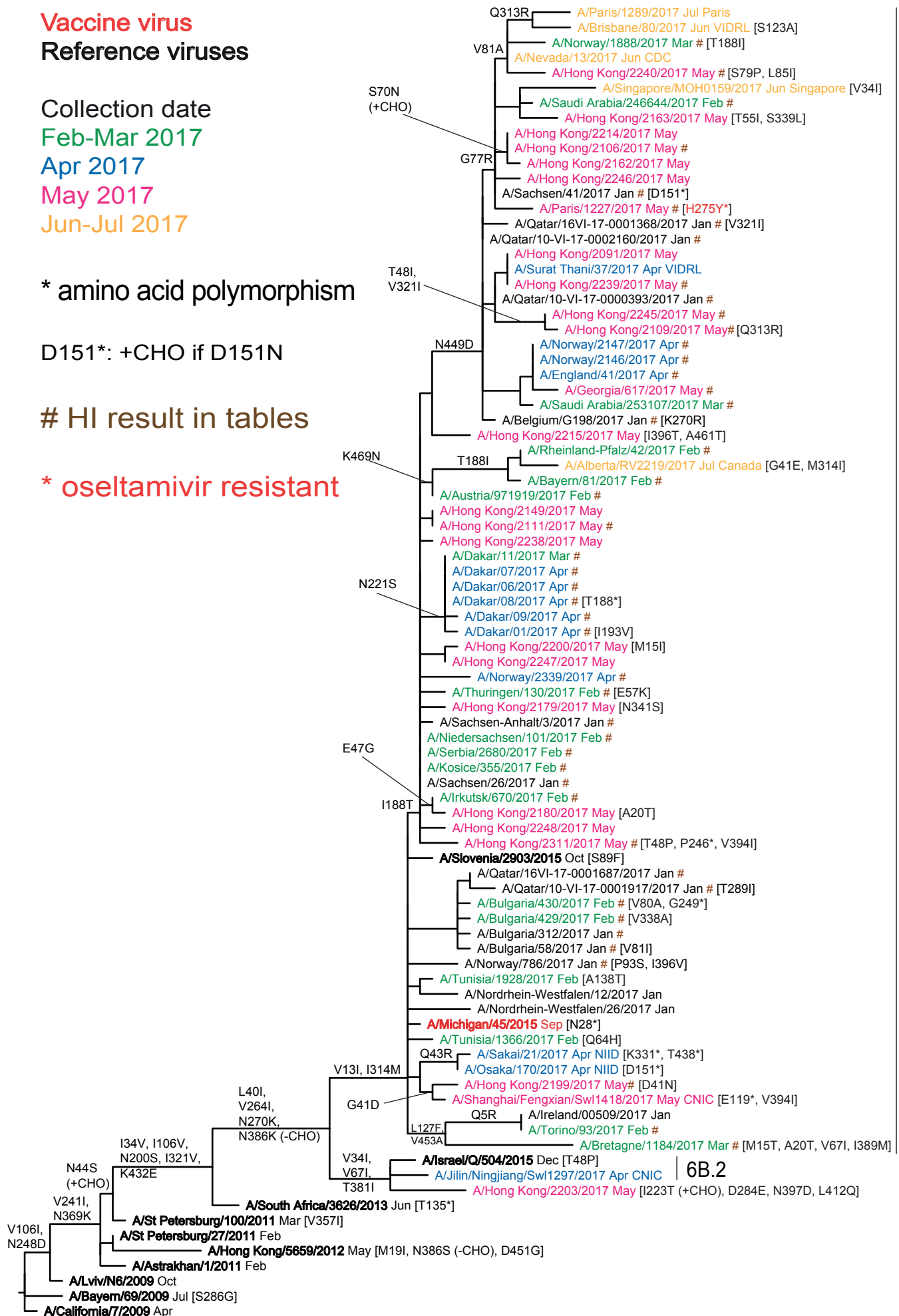
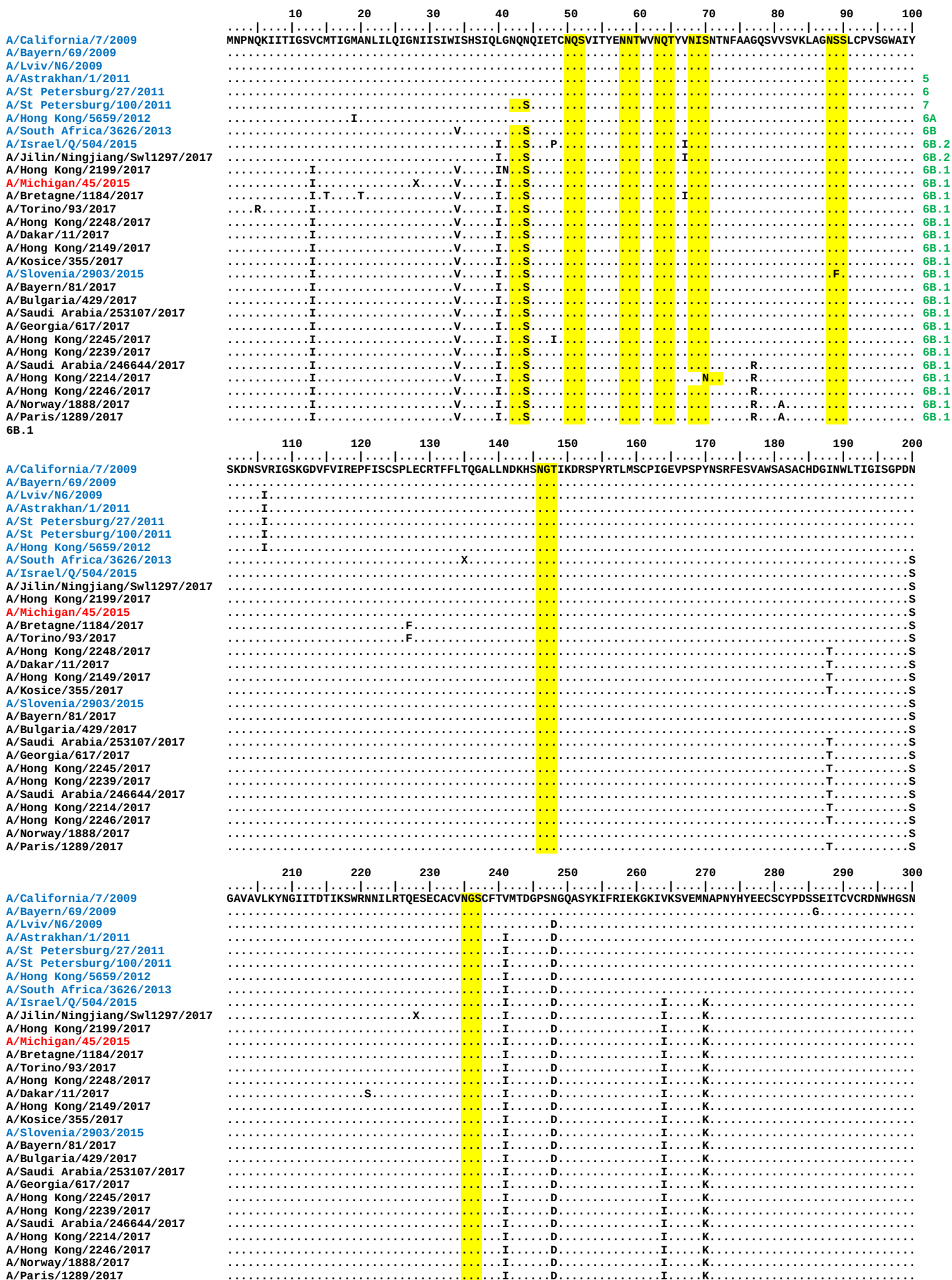


Figure 5. NA protein alignment for influenza A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2017-01-31 have been received from NICs in 33 countries or administrative regions in four WHO Regions: African (4: Ghana, Madagascar, Niger and South Africa), Eastern Mediterranean (3: Egypt, Jordan, and Saudi Arabia), European (25 countries) and Western Pacific (1: Hong Kong SAR).

Influenza A(H3N2) viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pig, turkey and humans or the loss of the ability of viruses to agglutinate any of these RBCs. Of the successfully propagated viruses collected since 2017-01-31, just over 40% had sufficient HA titre to be analysed by HI assays conducted using guinea pig RBCs in the presence of 20nM oseltamivir, added to circumvent NA-mediated binding to the RBCs (**Tables 1, 2, 6 & 7**), this being an increase over the proportion seen in February 2017. The remainder were unable to agglutinate guinea pig RBCs at all (**Tables 2 & 6**), or were unable to agglutinate RBCs in the presence of 20nM oseltamivir.

HI analyses.

A total of 124 viruses with collection dates after 2017-01-31 could be analysed by HI assay and summaries of the HI results are shown in **Tables 2 and 7**. Individual HI results are shown in **Tables 8-1 to 8-21**. Test viruses in each table are sorted by genetic subclade and by date of clinical specimen collection and those below the red horizontal lines in the tables (**Tables 8-7 and 8-9 to 8-21**) are viruses with collection dates after 2017-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. Of the viruses sequenced with collection dates after 2017-01-31 only viruses from Iceland (**Table 8-10**) fall into genetic subclade 3C.3a.

A summary of the HI results for viruses collected since 2017-01-31 with antisera raised against the cell culture-propagated cultivars of A/Hong Kong/5738/2014 and A/Hong Kong/4801/2014 and the egg-propagated cultivar of A/Hong Kong/4801/2014 is shown in **Table 7**. A more detailed summary of the recognition of the viruses by the full panel of antisera together with a breakdown based on gene sequencing for viruses in subclades 3C.2a and 3C.2a1 are shown in **Table 8-22**. The antiserum raised against A/Hong Kong/5738/2014 recognised >97% of the test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus, 84% within 2-fold of the homologous titre; viruses in genetic subgroups 3C.2a and 3C.2a1 were recognised in similar proportions (**Table 8-22**). Likewise, the antiserum raised against cell culture-propagated A/Hong Kong/4801/2014 recognised 87% of test viruses with collection dates after 2017-01-31 at titres within 4-fold, and 50% at titres within 2-fold, of the titre with the homologous virus; viruses in genetic

subgroups 3C.2a and 3C.2a1 were recognised in generally similar proportions (**Table 8-22**).

The antiserum raised against egg-propagated A/Hong Kong/4801/2014 recognised 73% of test viruses with collection dates after 2017-01-31 at titres within 4-fold, and 46% at titres within 2-fold, of the titre with the homologous virus (**Table 7**). A somewhat higher proportion of viruses in genetic subgroup 3C.2a1 were poorly recognised by this antiserum (at titres $\geq$ 8-fold lower than the titre of the antiserum with the homologous virus) compared with test viruses from the 3C.2a subclade.

For test viruses with collection dates after 2017-01-31 characterised both genetically and antigenically (**Table 8-22**):

- an antiserum raised against the egg-propagated cultivar of A/Hong Kong/4801/2014 (3C.2a) recognised 90% of those viruses in subclade 3C.2a but 60% of the viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/4801/2014 (3C.2a) recognised 96% of the test viruses in subclade 3C.2a and 81% of the viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/5738/2014 (3C.2a) recognised all of the test viruses in subclade 3C.2a and all but one (98%) of the test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against a cell culture-propagated cultivar of A/Georgia/532/2015 (3C.2a), a reference virus that carries the **HA1** amino acid substitutions **R142K** and **Q197R** and shows polymorphism at the 158-160 glycosylation motif, recognised only 38.5% of those viruses in subclade 3C.2a and only two (6.7%) of the viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;
- an antiserum raised against the previously recommended vaccine virus, egg-propagated A/Switzerland/9715293/2013 (3C.3a), recognised only three of the test viruses at a titre within 4-fold of the titre of the antiserum with the homologous virus, all three were viruses were from Iceland and two for which the HA gene sequences were determined fell into genetic subclade 3C.3a;
- the antiserum raised against a cell culture-propagated cultivar of A/Switzerland/9715293/2013 (3C.3a) had a homologous titre of 80 or 160 and, in contrast, recognised 96% of all the test viruses, 90% in subclade 3C.2a and 84% in subclade 3C.2a1, at titres within 4-fold of the homologous titre;
- an antiserum raised against a cell culture-propagated cultivar of A/Stockholm/6/2014 (3C.3a) had a higher homologous titre of 320 and recognised 85% of all the test viruses, 84% of test viruses in subclade 3C.2a

and 83% of test viruses in subclade 3C.2a1 at titres within 4-fold of the homologous titre of the antiserum;

- an antiserum raised against an egg-propagated cultivar of A/Stockholm/6/2014 (3C.3a) had a homologous titre of 320 and recognised only 66% of all the test viruses, 85% of viruses in subclade 3C.2a and 50% in subclade 3C.2a1, at titres within 4-fold of the homologous titre of the antiserum.

Antisera have been included for which no homologous titre is available because the reference viruses for these antisera were not able to agglutinate RBCs. Nevertheless, these three antisera, raised against the cell culture-propagated cultivars of A/Oman/2585/2016, A/Norway/4436/2016 and A/Greece/4/2017 – all 3C.2a1 viruses, were used and recognised the test viruses at titres generally similar to the titres of the antisera for the majority of the reference viruses at titres of 160 or 320, but some test viruses were recognised at lower titres: specifically the antiserum raised against A/Oman/2585/2016 recognised 68% of the tested viruses at titres ≥ 160 , the antiserum raised against A/Norway/4436/2016 recognised 73% of the tested viruses at titres ≥ 160 , and the antiserum raised against A/Greece/4/2017 recognised 87% of the tested viruses at titres ≥ 160 .

Antisera raised against viruses in genetic groups no longer in circulation generally recognised the test viruses poorly; these were raised against A/Texas/50/2012 (a former vaccine virus in genetic group 3C.1) and reference viruses in other genetic groups - A/Hong Kong/146/2013 (3C.2), A/Samara/73/2013 (3C.3) and A/Netherlands/525/2014 (3C.3b).

Antisera raised against recent egg-propagated viruses and several high yield (growth) reassortant (HY(G)R) viruses have been generated. The antiserum raised against A/Norway/3806/2016, a 3C.2a1 virus, recognised the panel of test viruses generally very well but the antisera raised against HY(G)Rs derived from A/Norway/3806/2016, NIB-103 and CBER-08 C1.4, had higher titres for the homologous viruses and recognised the test viruses poorly (**Table 8-22**). An antiserum raised against egg-propagated A/Norway/4849/2016, a 3C.2a virus, recognised the cell culture-propagated test viruses poorly and, likewise, an antiserum raised against egg-propagated A/Greece/4/2017, a 3C.2a1 virus, recognised the cell culture-propagated test viruses very poorly (**Table 8-22**).

Virus Neutralisation.

Plaque reduction neutralisation assays (PRNA) were used to complement HI tests and allowed antigenic characterisation of representative viruses of the many that could not be assayed by HI. **Tables 9-1 to 9-13** show the results of the plaque reduction assays performed since the February 2017 VCM. All test viruses with results shown in **Tables 9-6 to 9-13** have collection dates after 2017-01-31 with a total of 117 viruses analysed, the genetic subclade to which the viruses belonged was determined for all but three viruses. A blue line on each of the tables separates

viruses in the 3C.2a genetic subgroup from those in the 3C.2a1 subgroup. The final column shows the specific amino acid substitutions in HA1 of the reference and test viruses. The results in **Tables 9-1 to 9-13** show the neutralisation titres as estimated by the software used to score the neutralisation titres and these numbers are shown 'rounded' to the nearest value in a 2-fold dilution series, analogous to the HI assay.

Selected antisera were raised against egg-propagated A/Hong Kong/4801/2014 (subclade 3C.2a), the currently recommended vaccine virus, and egg-propagated cultivars of A/Norway/3806/2016 (a 3C.2a1 virus with **HA1** substitutions of **N121K** and **I140M**) and A/Greece/4/2017 (a 3C.2a1 virus with **HA1** substitutions of **G78D**, **N122D** resulting in the loss of a glycosylation motif at residue **122** and **T135K** resulting in the loss of glycosylation at residue **133**). All egg-propagated viruses had lost the glycosylation motif at residues **158 – 160** of **HA1** and had acquired a substitution **L194P** as a result of egg adaptation, A/Hong Kong/4801/2014 also carried the substitution **N96S** in **HA1**. Antisera raised against five cell culture-propagated viruses from genetic subclades 3C.2a or 3C.2a1 were also used: A/Hong Kong/7295/2014 (3C.2a), A/Cote d'Ivoire/544/2016 (3C.2a with the substitutions **N121K**, **S144K** and **S198P** in **HA1**), A/Oman/2585/2016 (3C.2a1 with the substitution **N121K** in **HA1**), A/Norway/4436/2016 (3C.2a1 with substitutions **N121K** and **I140M** in **HA1**) and A/Greece/4/2017.

Test viruses were selected to represent genetic groups within subclades 3C.2a and 3C.2a1.

The PRNA results are summarised in **Table 9-14** in which the numbers and percentages of viruses, with the specified amino acid substitutions, recognised by the antisera raised against the reference viruses at titres $\leq$ 4-fold of the titres of the antisera for the homologous viruses are shown.

The summary shows that antisera raised against the reference viruses propagated in cell culture (A/Hong Kong/7295/2014, A/Cote d'Ivoire/544/2014, A/Oman/2585/2016, A/Norway/4436/2016 and A/Greece/4/2017) recognise the test viruses propagated in cell culture generally well. When assessed by genetic subgroup, with the exception of the antiserum raised against A/Cote d'Ivoire/544/2016, the antisera recognised approximately 90% or more of the test viruses in the 3C.2a and 3C.2a1 subgroups at titres within 4-fold of the titres of the antisera for the homologous viruses. Viruses grouped by specific amino acid substitutions showed some fluctuation in the proportion recognised by the antisera at titres within 4-fold of the homologous titres of the antisera.

Antisera raised against reference viruses propagated in eggs (A/Hong Kong/4801/2014, A/Norway/3806/2016 and A/Greece/4/2017) all recognised the test viruses poorly. This gives no evidence for antisera raised against A/Norway/3806/2016 and A/Greece/4/2017 recognising viruses in any subclade better than the antiserum raised against egg-propagated A/Hong Kong/4801/2014.

Statistical analysis showed that antisera raised against the cell culture-propagated reference viruses in subclade 3C.2a recognised cell culture-propagated test viruses in subclade 3C.2a somewhat better than they recognised the test viruses belonging to subclade 3C.2a1. Antisera raised against the subclade 3C.2a1 cell culture-propagated viruses, A/Oman/2585/2016 and A/Greece/4/2017, but not the antiserum raised against cell culture-propagated A/Norway/4436/2016 recognised cell culture-propagated test viruses from subclade 3C.2a1 somewhat better than those in subclade 3C.2a. The antiserum raised against cell culture-propagated A/Norway/4436/2016 recognised test viruses from subclade 3C.2a no differently than those in subclade 3C.2a1.

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 6**. The HA genes fell within subclades 3C.3a, 3C.2a and 3C.2a1.

Viruses in subclades 3C.2a and 3C.2a1 have predominated recently.

- **3C.2a** viruses are defined by **L3I**, **N128T** (resulting in the gain of a potential glycosylation site), **N144S** (resulting in the loss of a potential glycosylation site), **N145S**, **F159Y**, **K160T** (resulting in the gain of a potential glycosylation site at residue **158**), **P198S**, **F219S**, **N225D** and **Q311H** in **HA1**, e.g. A/Hong Kong/5738/2014. Several subgroups have emerged within 3C.2a subclades with HA substitutions:
 - **N121K** and **S144K** in **HA1**, with further subgroups of viruses that carry HA genes that encode **N122D** (resulting in the loss of a glycosylation motif at residues **122-124**) and **S262N** in **HA1**, or the substitution **R261Q** in **HA1**,
 - **T131K** and **R261Q** in **HA1** many also having HA genes encoding **R142K** in **HA1**,
 - **N31S**, **D53N**, **R142G**, **S144R**, **N171K**, **I192T**, **Q197H** in **HA1**.
- **3C.2a1** viruses are defined by the same amino acid substitutions as **3C.2a** viruses but additionally encode **N171K** in **HA1** and **I77V** and **G155E** in **HA2**. Several subgroups have emerged within the 3C.2a1 subclade with HA substitutions; the majority carry an **N121K** substitution in **HA1** other groups are defined by:
 - **K92R** and **H311Q** in **HA1**, with further subgroups of viruses that carry HA genes that encode **I58V** in **HA1**, or **R142G** in **HA1** with further substitutions **E62G** and **T135K** (resulting in the loss of a potential glycosylation site) in **HA1**
 - **G150E** in **HA2** many with **T135K** (resulting in the loss of a potential glycosylation site) or **I140M** in **HA1** many also having HA genes encoding **R142K** in **HA1**. Some viruses with the **T135K** substitution

also have **HA1 N122D** substitution which results in the further loss of a glycosylation site.

Figure 7 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny. The vaccine virus A/Hong Kong/4801/2014 is shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The group/subclade to which each of the sequences belongs is also shown.

Figure 8 shows the location on a 2005 H3 HA structure of amino acid substitutions that define the main genetic subgroups of 3C.2a and 3C.2a1 subclade viruses defined in **Table 9-14**, each compared with the vaccine virus A/Hong Kong/4801/2014.

Figure 9 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated. **Figure 10** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine virus is shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the group/subclade to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of recent viruses with HA genes in genetic groups within 3C.2a (including 3C.2a1) generally cluster similarly to the HA sequences. Most viruses fall into genetic groups that differ from the NA gene sequence of A/Hong Kong/4801/2014 at several positions: **S245N**, resulting in the formation of a glycosylation motif at residues **245-247**, **S247T** maintaining the glycosylation motif, **T267K**, **I380V** and **T392I**. There are several subgroups defined by amino acid substitutions:

- **G93D**, **D339N**, **P468L**, associated with 3C.2a viruses,
- **P468H**, many with the additional substitution **D339N** and many of these have the substitution **N329S**, resulting in the loss of a glycosylation motif at **329 - 331**, associated with viruses in both 3C.2a and 3C.2a1 subclades.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 194 viruses with collection dates after 2017-01-31. Nine viruses showed reduced inhibition (RI) by oseltamivir, with seven of these also showing RI by zanamivir (**Table 14**). NA substitutions were found at residues 329 or 331 in the NA which results in the loss of a glycosylation motif in the NA in these viruses. Loss of this glycosylation motif, commonly associated with N329S substitution in the NA, was also seen in some other viruses classed as showing normal inhibition (NI) by both antivirals. The location of the substitutions is illustrated in **Figure 20**.

Conclusion.

Viruses in subclades 3C.2a and 3C.2a1 were predominant and just under 60% of viruses in these subclades did not agglutinate RBCs efficiently such that they could not be characterised by HI assay.

In HI assays that could be carried out the antiserum raised against a cell culture-propagated 3C.2a virus (genetically similar to the vaccine virus A/Hong Kong/4801/2014) recognised test viruses in subclades 3C.2a and 3C.2a1 very well. An antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/4801/2014, polymorphic at the HA1 158-160 glycosylation motif, also recognised the test viruses well. An antiserum raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 recognised the majority (73%) of viruses at titres within 4-fold of its homologous titre.

In PRNA, subclade 3C.2a and 3C.2a1 viruses were recognised very well by antisera raised against three cell culture-propagated viruses in genetic subclade 3C.2a. These were antisera raised against a cell culture-propagated virus genetically similar to the vaccine virus A/Hong Kong/4801/2014, A/Hong Kong/7295/2014, and three recent cell culture-propagated reference viruses in subclade 3C.2a1 (A/Oman/2585/2016, A/Norway/4436/2016 and A/Greece/4/2017). An antiserum raised against another recent 3C.2a virus, A/Cote d'Ivoire/544/2016, showed more variable recognition of the test viruses in each genetic subclade.

Antisera raised against the egg-propagated vaccine virus A/Hong Kong/4801/2014 (3C.2a) and other egg-propagated viruses (A/Norway/3806/2016 and A/Greece/4/2017) recognised the test viruses poorly.

Table 6. A(H3N2) viruses recovered from specimens collected after 2017-01-31), as assessed by NA activity (MUNANA-based assay)

Situation at the time of the September 2017 VCM (viruses with collection dates Jan-Aug 2017)

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3C.2a1
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir						
+	+	+	Yes	49	43 (88%)	5	22	16
-	+	+	Yes	75	46 (61%)	0	17	29
-	-	+	Yes	0	0	0	0	0
+	-	+	Yes	0	0	0	0	0
			Totals	124	89 (72%)	5	39	45
+	+	-	No	21	17 (81%)	0	9	8
+	-	-	No	0	0	0	0	0
-	+	-	No	23	13 (57%)	0	6	7
-	-	-	No	131	104 (79%)	0	58	46
			Totals	175	134 (77%)	0	73	61
			Totals	299	223	5	112	106

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2017-09-13: a number still in process

Situation at the time of the February 2017 VCM (viruses with collection dates Sep 2016-Jan 2017)

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3C.2a1
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir						
+	+	+	Yes	33	24 (73%)	1	11	12
-	+	+	Yes	23	18 (78%)	0	7	11
-	-	+	Yes	0	0	0	0	0
+	-	+	Yes	0	0	0	0	0
			Totals	56	42 (75%)	1	18	23
+	+	-	No	52	46 (88%)	0	19	27
+	-	-	No	0	0	0	0	0
-	+	-	No	7	5 (71%)	0	4	1
-	-	-	No	407	324 (80%)	0	144	180
			Totals	466	375 (80%)	0	167	208
			Totals	522	417 (80%)	1	185	231

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2017-02-17: a number still in process

Table 7. Summary of influenza A(H3N2) viruses analysed, collected after 2017-01-31

MONTH	H3N2											
			Egg-propagated A/Hong Kong/4801/2014 (3C.2a)			Cell-propagated A/Hong Kong/4801/2014 (3C.2a)			Cell-propagated A/Hong Kong/5738/2014 (3C.2a)			
	Country	Number received	Number propagated ¹	≤2 fold ²	4 fold ²	≥8 fold ²	≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴
2017 FEBRUARY												
Austria	7	0										
Belgium	7	0										
Bosnia and Herzegovina	14	in process	1	1	2	1	1	2	2	1		
Bulgaria	7	1		1			1					
Egypt	1	0										
France	4	2			2		1	1				
Georgia	1	0										
Germany	4	2		1	1	1	1		1	1		
Greece	5	0										
Iceland	10	7	3	4		1	2	4	3	4		
Italy	7	2			2		1	1	2			
Jordan	1	0										
Kazakhstan	1	1	1			1						
Latvia	5	3	3			3			3			
Lithuania	4	3	3			3			3			
Moldova	1	0										
Norway	14	2	1	1			2		1	1		
Poland	10	0										
Portugal	2	1		1			1					
Russia	28	12	2	7	3	7	5					
Saudi Arabia	1	0										
Serbia	6	0										
Slovakia	3	1	1			1			1			
Slovenia	5	1	1			1			1			
Spain	1	1	1				1			1		
Sweden	3	3		3		3						
Ukraine	1	0										
United Kingdom	3	0										
2017 MARCH												
Austria	1	0										
Bosnia and Herzegovina	2	0										
Bulgaria	1	0										
France	3	2			2			2				
Germany	8	7		1	6		7		6			
Iceland	4	2	1	1			2			1	1	
Latvia	1	0										
Lithuania	1	1	1			1			1			
Madagascar	1	0										
Niger	7	in process										
Norway	11	4	1	1	2	1	3		3	1		
Poland	6	0										
Russia	6	3	2	1		3						
Saudi Arabia	6	1		1			1					
Slovenia	2	2		1	1		1	1	1		1	
Ukraine	1	0										
2017 APRIL												
Germany	2	1			1		1		1			
Jordan	1	1	1			1			1			
Latvia	1	1	1			1			1			
Madagascar	4	0										
Niger	2	in process										
Norway	4	1	1			1			1			
Russia	2	0										
2017 MAY												
Hong Kong	62	sequenced	3	4	8	5	7	3	5			
Madagascar	10	2	1	1		1	1					
Niger	2	in process										
Norway	3	1	1			1			1			
2017 JUNE												
Ghana	3	3	2	1		3			3			
Hong Kong	48	sequenced		1		1			1			
Madagascar	5	0										
South Africa	14	3	3			3			3			
2017 JULY												
Ghana	4	0										
Hong Kong	125	sequenced	22	2	3	19	6	2	23	1		
South Africa	5	0										
2017 AUGUST												
Hong Kong	9	sequenced										
	523	77	57	34	33	63	45	16	69	11	2	
33 Countries			46.0%	27.4%	26.6%	50.8%	36.3%	12.9%	84.1%	13.4%	2.4%	

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

2. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/4801/2014 (3C.2a)

3. Compared to homologous titres for ferret antisera raised against tissue culture-propagated virus A/Hong Kong/4801/2014 (3C.2a)

4. Compared to homologous titres for ferret antisera raised against tissue culture-propagated virus A/Hong Kong/5738/2014 (3C.2a) - 42 isolates not tested.

Table 8-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-02-21

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas 50/12 Egg F36/12 [†]	A/Samara 73/13 SIAT F35/15 [†]	A/Stock 6/14 SIAT F14/14 [†]	A/Stock 6/14 Egg F20/14 [†]	A/Switz 9715293/13 SIAT F18/15 [†]	A/Switz 9715293/13 Egg F32/14 [†]	A/HK 5738/14 MDCK F30/14 [†]	A/HK 4801/14 MDCK F43/15 [†]	A/HK 4801/14 Egg F12/15 [†]	A/Georgia 532/15 SIAT F33/15 [†]	A/Oman 2585/16 SIAT NIB F50/16 [†]	A/Nor 4436/16 SIAT F03/17 [†]
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES															
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	5120	640	160	640	80	640	160	640	80	160	160	320
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	1280	640	320	320	80	320	320	640	80	320	320	640
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	40	40	320	160	80	80	160	160	40	80	160	160
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	640	80	160	320	40	640	160	320	40	80	160	320
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	40	40	160	160	160	80	80	160	40	80	160	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	320	160	160	320	40	640	160	160	40	160	160	320
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	80	80	160	160	80	80	160	320	160	160	320	320
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK2	160	80	320	160	160	160	320	640	160	320	640	320
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	40	80	40	40	40	40	160	320	160	320	320	160
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	160	80	320	160	80	80	320	320	160	1280	320	320
TEST VIRUSES															
A/Kurgan/24/2016	3C.2a		MDCK1/SIAT1	40	40	80	80	40	40	80	160	80	160	80	160
A/Yaroslavl/20170/2016	3C.2a	2016-12-05	MDCK1/SIAT1	40	40	160	80	80	40	160	160	80	160	160	160
A/Novosibirsk/5288/2016	3C.2a	2016-12-03	MDCK1/SIAT1	160	160	320	160	160	160	160	640	320	320	320	640
A/Gorno-Altaysk/1/2016	3C.2a	2016-12-13	MDCK3/SIAT1	160	160	320	160	160	80	320	320	320	320	320	640
A/Luxembourg/63722/2016	3C.2a1	2016-12-16	SIAT2	80	40	320	80	160	40	160	160	80	160	320	320
A/Omsk/4174/2016	3C.2a1	2016-12-19	MDCK2/SIAT1	40	40	160	80	80	40	80	160	80	80	160	160
A/Armenia/2363/2016	3C.2a1	2016-12-28	MDCK1/SIAT1	<	<	160	80	80	<	80	160	80	80	160	160
A/Ireland/00098/2016	3C.3a	2016-12-29	C1/SIAT1	40	40	320	160	80	80	80	80	40	80	80	80
A/Ireland/86110/2016	3C.2a1	2016-12-29	C1/SIAT1	<	<	40	40	<	<	40	160	40	40	80	80
A/Ireland/00492/2017	3C.2a1	2017-01-02	C1/SIAT1	40	40	80	80	40	40	80	160	80	160	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [†] < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-02-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas 50/12 Egg	A/Samara 73/13 SIAT	A/Stock 6/14 SIAT	A/Stock 6/14 Egg	A/Switz 9715293/13 SIAT	A/Switz 9715293/13 Egg	A/HK 5738/14 MDCK	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Georgia 532/15 SIAT	A/Oman 2585/16 SIAT	A/Nor 4436/16 SIAT
				F36/12 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹	NIB F50/16 ¹	F03/17 ¹
	Passage history Ferret number Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	5120	640	160	640	80	640	160	640	80	160	160	320
A/Samara/73/2013		2013-03-12	C1/SIAT4	1280	640	320	320	80	320	320	640	80	320	320	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT3	40	40	320	160	80	80	160	160	40	80	160	160
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	640	80	160	320	40	640	160	320	40	80	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT2	40	40	160	160	160	80	80	160	40	80	160	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	320	160	160	320	40	640	160	160	40	160	160	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT2	80	80	160	160	80	80	160	320	160	160	320	320
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK2	160	80	320	160	160	160	320	640	160	320	640	320
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	40	80	40	40	40	40	160	320	160	320	640	160
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	160	80	320	160	80	80	320	320	160	1280	320	320
TEST VIRUSES															
A/Stockholm/48/2016			MDCK1/SIAT1	40	40	160	80	40	40	80	320	80	80	160	160
A/Czech Republic/178/2017			MDCKx/SIAT1	<	<	40	40	40	<	40	80	40	40	160	80
A/Sachsen/74/2016		2016-11-07	C1/SIAT1	40	<	160	80	80	<	80	160	80	40	160	160
A/Stockholm/42/2016		2016-11-10	MDCK1/SIAT1	40	40	160	80	40	40	80	320	80	80	160	160
A/Latvia/11-072125/2016		2016-11-28	C2/SIAT1	40	40	80	80	40	80	80	160	80	80	80	80
A/Austria/956928/2016		2016-11-30	C1/SIAT1	80	80	160	160	80	80	80	320	80	160	160	160
A/Austria/957344/2016		2016-12-04	C1/SIAT1	<	<	160	40	40	<	40	160	40	40	80	160
A/Austria/957860/2016		2016-12-06	C2/SIAT1	40	40	160	80	80	40	80	160	80	80	160	160
A/Latvia/12-020474/2016		2016-12-08	C2/SIAT1	40	40	80	80	40	40	80	320	80	80	160	80
A/Czech Republic/169/2016		2016-12-14	MDCK2/SIAT1	40	<	160	80	40	40	160	160	80	80	160	160
A/Sweden/153/2016		2016-12-16	MDCK1/SIAT1	<	<	160	40	40	<	40	160	40	80	160	80
A/Sweden/157/2016		2016-12-17	MDCK1/SIAT1	40	<	160	80	80	<	80	160	40	40	160	160
A/Czech Republic/176/2016		2016-12-19	MDCK2/SIAT1	40	<	160	40	40	40	160	160	40	160	160	160
A/Greece/726/2016		2016-12-19	SIAT1	40	40	160	80	40	<	160	160	80	80	160	160
A/Greece/35/2017		2017-01-10	SIAT2	40	<	80	40	40	<	80	160	80	40	160	160
A/Belarus/1136/2017		2017-01-13	SIAT1	<	<	40	40	<	<	80	80	<	<	160	80
A/Belarus/1146/2017		2017-01-14	SIAT1	40	40	160	80	80	40	160	160	160	80	320	320
A/Belarus/1147/2017		2017-01-14	SIAT1	40	<	160	40	40	<	80	160	40	80	160	120
A/Belarus/1140/2017		2017-01-15	SIAT1	40	<	80	40	40	<	80	160	40	160	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-03-03 + 2017-03-10

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas 50/12 Egg	A/Samara 73/13 SIAT	A/Stock 6/14 SIAT	A/Stock 6/14 Egg	A/Switz 9715293/13 SIAT	A/Switz 9715293/13 Egg	A/HK 5738/14 MDCK	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Georgia 532/15 SIAT	A/Oman 2585/16 SIAT	A/Nor 4436/16 SIAT
				F36/12 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F32/14 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹	NIB F50/16 ¹	F03/17 ¹
	Passage history Ferret number Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	5120	640	80	640	80	640	160	320	80	160	160	320
A/Samara/73/2013		2013-03-12	C1/SIAT4	1280	640	320	320	80	640	320	640	80	320	320	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT3	40	40	320	160	80	80	80	160	40	80	160	160
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	320	80	80	320	40	640	80	320	40	80	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	<	<	80	80	160	40	40	80	40	80	80	80
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	320	160	80	320	<	640	160	160	40	160	160	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT2	40	40	80	80	40	80	160	320	80	160	160	160
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK2	80	40	160	160	80	80	160	320	160	160	160	160
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	40	80	40	40	40	<	160	640	160	320	320	160
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	160	320	160	640	160	320
TEST VIRUSES															
A/Mayotte/2107/2016		2016-03-03	MDCK3/SIAT1	80	80	160	160	80	80	80	320	160	160	160	320
A/Mayotte/2100/2016		2016-03-18	MDCK4/SIAT1	80	40	320	80	80	40	80	160	80	80	160	160
A/Mayotte/2097/2016		2016-03-21	MDCK4/SIAT1	40	40	80	40	80	40	160	160	80	160	160	160
A/Iran/55457/2016		2016-10-01	MDCK2/SIAT1	80	80	320	160	160	80	160	320	160	160	320	320
A/Lyon/2207/2016		2016-10-24	MDCK3/SIAT1	40	40	160	80	80	40	80	320	80	160	160	160
A/Poitiers/2238/2016		2016-11-01	MDCK3/SIAT1	80	80	320	160	160	80	160	320	160	160	320	320
A/Clermont Ferrand/2227/2016		2016-11-02	MDCK2/SIAT1	<	40	160	80	40	40	80	160	40	80	80	160
A/Poitiers/2239/2016		2016-11-02	MDCK3/SIAT1	40	40	160	80	40	40	80	160	80	160	160	160
A/Fes/67/2016		2016-11-08	SIAT1	40	40	160	80	80	40	80	160	80	80	160	160
A/Hungary/175/2016		2016-12-02	MDCK1/SIAT1	40	<	80	80	40	<	80	80	40	40	160	160
A/Bayern/76/2016		2016-12-13	C1/SIAT1	40	40	160	80	160	40	160	160	80	80	320	320
A/Hungary/188/2016		2016-12-13	MDCK1/SIAT1	40	<	40	40	40	<	40	40	<	40	80	80
A/Hungary/191/2016		2016-12-13	MDCK1/SIAT1	40	40	160	80	40	40	80	160	80	80	160	160
A/Bosnia and Herzegovina/12/2016		2016-12-14	SIAT2	40	40	320	80	160	40	160	160	80	80	160	160
A/Hungary/202/2016		2016-12-14	MDCK1/SIAT1	40	<	80	40	40	<	80	80	<	40	80	80
A/Slovenia/53/2017		2017-01-04	MDCK1/SIAT1	40	40	160	80	40	40	80	160	40	80	80	160
A/Tripoli GR/475/2017		2017-01-17	SIAT1	40	40	160	80	80	80	160	320	80	160	320	320
A/Athens GR/496/2017		2017-01-17	SIAT1	40	40	160	80	80	40	160	160	80	80	160	320
A/Athens GR/499/2017		2017-01-17	SIAT1	40	40	160	80	80	40	160	320	80	80	160	160
A/Athens GR/510/2017		2017-01-18	SIAT1	80	40	320	160	160	80	160	320	80	160	320	320
A/Athens GR/566/2017		2017-01-19	SIAT1	40	40	160	80	40	40	80	160	80	80	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-03-17 + 2017-03-23 + 2017-03-29

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/HK	A/HK	A/HK	A/Georgia	A/Oman	A/Nor
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	5738/14	4801/14	4801/14	532/15	2585/16	4436/16
				Egg	SIAT	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	SIAT	SIAT
Passage history	Ferret number	Genetic group													

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-04-04 + 2017-05-03

Haemagglutination inhibition titre																	NEW	NEW
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera													
					A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/HK	A/HK	A/HK	A/Georgia	A/Oman	A/Nor	A/Nor	A/Nor
					50/12	73/13	6/14	6/14	9715293/13	9715293/13	5738/14	4801/14	4801/14	532/15	2585/16	4436/16	4465/16	4465/16
					Egg	SIAT	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	SIAT	SIAT	SIAT	EGG
Ferret number					F08/14 ^{†1}	F35/15 ^{†1}	F14/14 ^{†1}	F20/14 ^{†1}	F18/15 ^{†1}	F32/14 ^{†1}	F30/14 ^{†1}	F43/15 ^{†1}	F12/15 ^{†1}	F33/15 ^{†1}	NIB F50/16 ^{†1}	F03/17 ^{†1}	F11/17 ^{†1}	F12/17 ^{†1}
Genetic group					3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a	3C.2a
REFERENCE VIRUSES																		
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	2560	640	160	640	40	640	160	320	80	320	160	320	160	160
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT4	1280	1280	640	640	160	640	320	1280	160	640	640	1280	320	80
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	80	80	320	320	160	160	160	320	80	160	320	320	160	40
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	160	80	80	320	40	640	80	320	40	80	160	320	160	40
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT3	40	40	320	160	160	80	80	160	40	160	160	320	80	40
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	160	160	160	320	80	640	160	320	40	160	320	320	160	40
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	40	40	320	80	160	80	320	640	160	160	320	640	320	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK2	80	80	160	160	80	80	160	320	160	160	160	320	160	40
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	40	40	40	40	160	640	160	640	320	160	320	1280
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT5	40	80	320	320	160	160	320	640	80	1280	320	640	1280	320
A/Norway/4465/2016	isolate 2 clone 27	3C.2a		E6	<	<	<	40	40	<	160	160	80	1280	320	160	1280	1280
TEST VIRUSES																		
A/Kyrgyzstan/29/2016		3C.2a	2016-11-14	MDCKx/SIAT1	80	80	160	160	80	80	160	160	80	160	320	320	80	40
A/Kyrgyzstan/28/2016		3C.2a	2016-11-14	MDCKx/SIAT1	40	80	160	80	80	80	80	320	80	80	160	320	40	40
A/Iceland/65/2016		3C.2a1	2016-12-12	MDCK1/SIAT1	40	40	160	80	40	40	160	160	80	160	160	160	ND	ND
A/Iceland/66/2016		3C.2a	2016-12-14	MDCK1/SIAT1	40	40	160	80	40	40	160	160	80	160	160	160	320	80
A/Iceland/68/2016		3C.3a	2016-12-15	MDCK1/SIAT1	80	80	640	320	160	160	160	320	160	160	320	320	80	<
A/Iceland/70/2016		3C.2a	2016-12-19	MDCK1/SIAT1	40	40	160	80	40	40	160	160	80	320	160	160	160	80
A/Iceland/72/2016		3C.2a	2016-12-19	MDCK1/SIAT1	40	40	160	80	40	40	160	320	80	160	160	160	80	40
A/Iceland/74/2016		3C.3a	2016-12-21	MDCK1/SIAT1	40	40	320	160	80	80	160	160	80	160	160	160	80	40
A/Iceland/76/2016		3C.2a1	2016-12-24	MDCK1/SIAT1	40	40	160	80	40	40	80	160	80	160	160	160	160	40
A/Iceland/78/2016		3C.2a	2016-12-26	MDCK1/SIAT1	40	40	160	80	40	40	160	160	80	160	160	320	80	40
A/Israel/R-742/2016		3C.3a	2016-12-29	C1/SIAT1	<	<	80	80	80	40	40	40	<	80	40	80	ND	ND

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done

Sequences in phylogenetic trees

Vaccine

Table 8-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-05-12 + 2017-05-19

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/HK	A/HK	A/HK	A/Georgia	A/Oman	A/Nor	
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	5738/14	4801/14	4801/14	532/15	2585/16	4436/16	
				Egg	SIAT	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	SIAT	SIAT	
F08/14 ⁻¹	F35/15 ⁻¹	F14/14 ⁻¹	F20/14 ⁻¹	F18/15 ⁻¹	F73/16 ⁻¹	F30/14 ⁻¹	F43/15 ⁻¹	F12/15 ⁻¹	F33/15 ⁻¹	NIB	F50/16 ⁻¹	F03/17 ⁻¹				
3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a1				
REFERENCE VIRUSES																
A/Texas/50/2012		2012-04-15	E5/E2	2560	640	160	640	80	320	160	320	80	320	160	640	
A/Samara/73/2013		2013-03-12	C1/SIAT4	640	640	320	320	160	320	320	640	160	640	320	640	
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT3	40	40	320	160	160	80	160	160	80	160	160	160	
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	160	160	160	320	80	320	160	320	40	80	160	320	
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	<	40	160	80	160	80	80	80	40	80	160	160	
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	80	160	160	320	80	640	160	320	40	160	160	320	
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	80	40	160	160	80	40	160	640	160	320	320	320	
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK1	80	80	160	160	80	80	160	640	160	320	320	320	
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	40	80	80	80	40	<	320	640	160	640	640	320	
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	160	320	80	640	320	320	
TEST VIRUSES																
A/Belgium/G0687/2016		2016-11-30	SIAT1	40	40	160	80	40	40	160	160	80	80	320	320	
A/Khmelnitsky/1990/2016		2016-12-02	MDCK1/SIAT1	40	<	160	80	80	40	160	160	80	80	160	160	
A/Khmelnitsky/1958/2016		2016-12-05	MDCK1/SIAT1	40	<	160	80	80	40	160	160	80	80	320	160	
A/Khmelnitsky/1981/2016		2016-12-05	MDCK1/SIAT1	80	160	160	160	80	80	320	640	160	640	320	640	
A/Kyiv/920/2016		2016-12-06	SIAT2/SIAT1	40	<	160	80	80	40	160	160	80	80	160	160	
A/Kyiv/923/2016		2016-12-08		40	<	160	80	80	40	160	160	80	80	160	160	
A/Kyiv/926/2016		2016-12-10	SIAT1/SIAT1	40	<	160	80	40	40	160	160	80	80	160	160	
A/Montenegro/78/2016		2016-12-12	SIAT1	40	40	160	80	80	40	160	160	80	80	320	160	
A/Armenia/2298/2016		2016-12-22	SIAT1	40	40	160	80	80	40	160	160	80	80	160	160	
A/Belgium/G0024/2017		2017-01-03	SIAT1	40	40	160	80	80	40	160	320	80	160	160	160	
A/Belgium/G0042/2017		2017-01-05	SIAT1	40	<	160	80	40	40	160	160	80	80	160	160	
A/Athens GR/563/2017		2017-01-19	SIAT1	40	<	160	40	80	40	80	160	80	80	160	160	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done

Sequences in phylogenetic trees

Vaccine

Table 8-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-05-23

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas 50/12 Egg F08/14 ⁻¹	A/Samara 73/13 SIAT F35/15 ⁻¹	A/Stock 6/14 SIAT F14/14 ⁻¹	A/Stock 6/14 Egg F20/14 ⁻¹	A/Switz 9715293/13 SIAT F18/15 ⁻¹	A/Switz 9715293/13 Egg F73/16 ⁻¹	A/HK 5738/14 MDCK F30/14 ⁻¹	A/HK 4801/14 MDCK F43/15 ⁻¹	A/HK 4801/14 Egg F12/15 ⁻¹	A/Georgia 532/15 SIAT F33/15 ⁻¹	A/Oman 2585/16 SIAT NIB F50/16 ⁻¹	A/Nor 4436/16 SIAT F03/17 ⁻¹
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	1280	640	160	640	40	640	160	320	80	160	160	320
A/Samara/73/2013		2013-03-12	C1/SIAT4	640	640	640	640	160	320	320	1280	160	320	320	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	80	80	640	160	160	160	160	320	160	160	320	320
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	160	80	160	320	40	640	160	640	40	80	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	40	40	320	160	160	160	160	160	80	80	160	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	160	160	320	320	80	640	320	320	80	160	320	640
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	80	80	320	160	80	80	160	320	160	160	320	320
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK1	80	80	320	160	80	80	160	640	160	160	320	320
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	40	80	80	80	40	<	320	640	160	320	640	320
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	80	80	320	160	80	80	160	320	160	640	320	320
TEST VIRUSES															
A/Montenegro/41/2016		2016-11-28	SIAT3	80	40	320	80	80	80	160	320	160	80	320	160
A/Ireland/80971/2016		2016-12-15	C1/SIAT1	40	40	80	160	40	40	80	160	80	80	160	160
A/Ireland/81104/2016		2016-12-15	C1/SIAT1	80	80	320	640	160	320	160	320	160	160	320	320
A/Ireland/81354/2016		2016-12-17	C2/SIAT1	40	<	80	160	40	40	80	160	80	160	160	160
A/Ireland/81180/2016		2016-12-19	C1/SIAT1	40	<	80	160	40	40	80	320	40	80	160	160
A/Ireland/81920/2016		2016-12-21	C1/SIAT1	40	40	320	640	160	160	160	160	80	80	160	160
A/Ireland/82326/2016		2016-12-21	C1/SIAT1	40	40	320	640	160	160	160	160	80	80	160	160
A/Ireland/82745/2016		2016-12-22	C1/SIAT1	80	80	640	320	160	160	160	160	80	80	160	160
A/Latvia/01-024575/2017		2017-01-10	C2/SIAT1	80	80	160	320	160	80	160	320	80	160	320	320
A/Latvia/02-057392/2017		2017-02-17	C1/SIAT1	80	40	160	320	160	80	160	320	160	80	320	320
A/Latvia/02-056748/2017		2017-02-20	C2/SIAT1	80	40	160	160	80	80	160	320	80	320	160	160
A/Latvia/02-062187/2017		2017-02-22	C2/SIAT1	80	80	160	320	80	80	160	320	80	160	320	320
A/Latvia/04-003626/2017		2017-04-03	C1/SIAT1	80	40	160	160	80	80	160	320	80	80	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-05-26

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/HK	A/HK	A/HK	A/Georgia	A/Oman	A/Nor
				50/12	73/13	6/14	6/14	9715293/13	9715293/13	5738/14	4801/14	4801/14	532/15	2585/16	4436/16
	Passage history			Egg	SIAT	SIAT	Egg	SIAT	Egg	MDCK	MDCK	Egg	SIAT	SIAT	SIAT
	Ferret number			F08/14 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F73/16 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹	NIB F50/16 ¹	F03/17 ¹
	Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	1280	640	160	640	40	640	160	320	80	160	160	320
A/Samara/73/2013		2013-03-12	C1/SIAT4	640	640	640	640	160	320	320	1280	160	320	320	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	80	80	640	160	160	160	160	320	160	160	320	320
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	160	80	160	320	40	640	160	640	40	80	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	40	40	320	160	160	160	160	160	80	80	160	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	160	160	160	320	80	640	320	320	80	160	320	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	80	80	320	160	80	80	160	320	160	160	320	320
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK1	80	80	320	160	80	80	160	640	160	160	320	320
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	40	80	80	80	80	<	320	640	160	320	640	320
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	80	80	320	160	80	80	160	320	160	640	320	320
TEST VIRUSES															
A/Ireland/72254/2016		2016-11-12	C1/SIAT1	40	40	320	160	160	160	80	160	80	80	160	160
A/Ireland/72703/2016		2016-11-14	C1/SIAT1	<	<	<	<	<	<	40	160	40	40	80	40
A/Diyarbakir/1847/2016		2016-11-23	SIAT1/SIAT1	40	40	160	80	40	40	160	160	80	40	320	320
A/Ireland/76255/2016		2016-11-29	C2/SIAT1	<	<	<	<	<	<	<	40	<	<	40	40
A/Ireland/78717/2016		2016-12-06	C1/SIAT1	80	80	640	320	160	160	160	160	80	80	160	160
A/Ankara/1907/2016		2016-12-06	MDCK1/SIAT1	40	40	160	80	80	40	80	160	80	80	160	320
A/Diyarbakir/1905/2016		2016-12-07	SIAT1/SIAT1	40	40	160	80	80	80	160	160	80	40	160	160
A/Diyarbakir/1941/2016		2016-12-14	SIAT1/SIAT1	40	<	80	40	40	<	160	80	<	<	160	160
A/Gaziantep/1950/2016		2016-12-14	SIAT1/SIAT1	80	80	320	160	160	80	320	320	160	80	320	320
A/Denizli/1984/2016		2016-12-19	SIAT1/SIAT1	40	40	160	80	40	40	80	160	40	80	160	160
A/Canakkale/7/2017		2016-12-29	SIAT1/SIAT1	40	40	160	80	80	80	160	160	80	40	160	160
A/Canakkale/10/2017		2016-12-29	SIAT1/SIAT1	40	40	320	80	80	40	160	160	80	40	320	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-06-09

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas	A/Samara	A/Stock	A/Stock	A/Switz	A/Switz	A/HK	A/HK	A/HK	A/Georgia	A/Oman	A/Nor
				50/12 Egg F08/14 ¹	73/13 SIAT F35/15 ¹	6/14 SIAT F14/14 ¹	6/14 Egg F20/14 ¹	9715293/13 SIAT F18/15 ¹	9715293/13 Egg F73/16 ¹	5738/14 MDCK F30/14 ¹	4801/14 MDCK F43/15 ¹	4801/14 Egg F12/15 ¹	532/15 SIAT F33/15 ¹	2585/16 SIAT NIB F50/16 ¹	4436/16 SIAT F03/17 ¹
	Passage history Ferret number Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	1280	640	160	640	40	320	160	320	80	160	160	320
A/Samara/73/2013		2013-03-12	C1/SIAT4	640	640	640	640	160	320	320	1280	160	320	320	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	80	40	320	160	160	160	160	160	80	80	160	160
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	160	80	160	320	80	320	160	320	40	40	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	<	<	320	80	80	80	80	80	40	40	80	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	80	160	160	320	40	640	160	320	40	80	160	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	80	160	320	160	80	80	320	640	160	160	320	320
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK1	80	80	160	160	80	80	160	640	160	160	160	160
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	40	80	80	80	40	40	320	640	160	320	640	160
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	160	320	160	640	320	320
TEST VIRUSES															
A/Catalonia/9305S/2016		2016-11-24	C0/SIAT1	40	<	160	80	40	40	80	160	80	40	160	160
A/Catalonia/9415S/2017		2016-12-21	C0/SIAT1	40	40	160	80	40	40	80	160	80	40	160	160
A/Serbia/2252/2016		2016-12-23	SIAT1	40	40	160	80	40	40	80	160	80	80	160	160
A/Cyprus/F49/2017		2017-01-12	SIAT1	40	<	160	80	40	40	80	160	80	40	160	160
A/Norway/484/2017		2017-01-22	SIAT1/SIAT1	<	<	160	80	<	<	40	80	40	160	160	160
A/Cyprus/F148/2017		2017-01-26	SIAT1	40	40	80	80	40	40	80	320	80	80	160	160
A/Catalonia/3509691NS/2017		2017-02-02	C0/SIAT1	40	40	160	80	40	40	80	160	80	80	160	160
A/Norway/1120/2017		2017-02-11	SIAT1/SIAT1	<	<	160	80	40	40	80	160	40	160	160	160
A/Iceland/40/2017		2017-02-22	MDCK1/SIAT1	<	<	160	40	40	<	160	80	40	80	80	160
A/Iceland/64/2017		2017-03-08	MDCK1/SIAT1	<	<	160	40	40	<	40	160	40	40	80	160
A/Iceland/69/2017		2017-03-09	MDCK1/SIAT1	40	40	320	160	80	160	80	160	80	80	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-06-16

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas 50/12 Egg F08/14 ⁻¹	A/Samara 73/13 SIAT F35/15 ⁻¹	A/Stock 6/14 SIAT F14/14 ⁻¹	A/Stock 6/14 Egg F20/14 ⁻¹	A/Switz 9715293/13 SIAT F18/15 ⁻¹	A/Switz 9715293/13 Egg F73/16 ⁻¹	A/HK 5738/14 MDCK F30/14 ⁻¹	A/HK 4801/14 MDCK F43/15 ⁻¹	A/HK 4801/14 Egg F12/15 ⁻¹	A/Georgia 532/15 SIAT F33/15 ⁻¹	A/Oman 2585/16 SIAT NIB F50/16 ⁻¹	A/Nor 4436/16 SIAT F03/17 ⁻¹
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
				Passage history	Ferret number	Genetic group									
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	1280	640	160	640	40	320	160	320	80	160	160	320
A/Samara/73/2013		2013-03-12	C1/SIAT4	640	640	640	640	160	320	320	1280	160	320	320	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	80	40	320	160	160	160	160	160	80	80	160	160
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	160	80	160	320	80	320	160	320	40	40	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	<	<	320	80	80	80	80	80	40	40	80	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	160	160	160	320	40	640	160	320	40	80	160	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	80	160	320	160	80	80	320	640	160	160	320	320
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK1	80	80	160	160	80	80	160	640	160	160	160	160
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	40	80	40	40	40	40	320	640	160	320	640	160
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	160	320	160	640	320	320
TEST VIRUSES															
A/Egypt/8070/2016		2016-12-05	SIAT1	40	<	160	80	80	40	160	160	40	40	160	160
A/Egypt/8123/2016		2016-12-08	SIAT1	40	<	160	80	80	40	80	160	40	40	160	160
A/Egypt/8200/2016		2016-12-08	SIAT2	40	<	40	80	40	<	80	160	40	40	40	80
A/Egypt/8437/2016		2016-12-12	SIAT1	40	40	160	80	80	40	160	160	80	40	160	320
A/Egypt/8375/2016		2016-12-16	SIAT1	<	40	160	80	80	40	160	160	80	40	160	160
A/Egypt/8619/2016		2016-12-26	SIAT1	<	<	80	40	40	40	80	40	40	<	80	80
A/Jordan/20160020743/2016		2016-12-28	SIAT1	40	40	160	80	40	80	160	160	80	40	160	160
A/Jordan/20170020054/2017		2017-01-03	SIAT1	<	40	160	40	40	40	80	80	80	<	160	160
A/Sachsen/4/2017		2017-01-10	C1/SIAT1	40	40	80	80	40	40	160	160	80	80	160	160
A/Jordan/20170020076/2017		2017-01-12	SIAT1	40	80	160	80	40	40	160	320	80	160	160	160
A/Berlin/9/2017		2017-01-16	C1/SIAT1	<	<	160	40	<	<	80	80	80	<	160	160
A/Poland/412/2017		2017-01-18	SIAT1	40	40	160	80	40	40	160	320	80	80	160	160
A/Poland/555/2017		2017-01-24	SIAT1	40	40	80	80	40	40	160	160	80	160	160	160
A/Hessen/14/2017		2017-01-26	C1/SIAT1	40	40	320	80	40	80	160	160	80	40	160	160
A/Iceland/27/2017		2017-02-16	MDCK1/SIAT1	<	40	160	160	40	80	80	80	40	40	80	80
A/Iceland/23/2017		2017-02-16	MDCK1/SIAT1	40	40	320	160	80	80	80	80	40	40	80	80
A/Iceland/31/2017		2017-02-19	MDCK1/SIAT1	<	40	160	160	40	80	80	80	40	40	80	80
A/Iceland/37/2017		2017-02-21	MDCK1/SIAT1	40	40	320	160	80	160	80	160	80	80	160	160
A/Iceland/36/2017		2017-02-21	MDCK1/SIAT1	40	<	160	80	40	40	160	160	80	40	160	160
A/Norway/1481/2017		2017-02-27	SIAT1	40	40	160	80	40	40	160	160	80	80	160	160
A/Norway/1477/2017		2017-03-04	SIAT1	40	<	160	80	40	40	80	160	40	80	160	160
A/Norway/2400/2017		2017-04-28	SIAT1/SIAT1	80	80	320	160	80	80	160	640	160	80	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-06-21 + 2017-06-23

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				A/Texas 50/12 Egg	A/Samara 73/13 SIAT	A/Stock 6/14 SIAT	A/Stock 6/14 Egg	A/Switz 9715293/13 SIAT	A/Switz 9715293/13 Egg	A/HK 5738/14 MDCK	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Georgia 532/15 SIAT	A/Oman 2585/16 SIAT	A/Nor 4436/16 SIAT	
				F08/14 ^{*1}	F35/15 ^{*1}	F14/14 ^{*1}	F20/14 ^{*1}	F18/15 ^{*1}	F73/16 ^{*1}	F30/14 ^{*1}	F43/15 ^{*1}	F12/15 ^{*1}	F33/15 ^{*1}	NIB F50/16 ^{*1}	F03/17 ^{*1}	
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	
REFERENCE VIRUSES																
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	2560	640	160	640	80	320	160	320	80	160	160	320	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	640	640	640	640	160	320	320	640	160	320	320	640	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	80	80	640	320	160	160	160	320	160	160	320	320	
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	160	160	160	320	80	640	160	640	40	80	160	320	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	40	40	320	160	80	80	80	160	80	80	160	160	
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	160	160	160	320	80	640	320	320	80	160	320	320	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT3	80	80	160	160	80	80	160	320	160	160	160	160	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK1	80	80	160	160	80	80	160	320	160	160	160	160	
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	80	80	80	80	80	<	320	640	160	320	640	160	
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	160	320	160	640	320	320	
TEST VIRUSES																
A/Lithuania/6735/2017	3C.2a1		MDCK2/SIAT1	<	<	160	40	<	<	40	40	40	40	80	160	
A/Lithuania/7612/2017	3C.2a		MDCK2/SIAT1	<	<	80	40	40	<	80	160	40	40	80	80	
A/Bulgaria/2/2017	3C.2a1	2017-01-02	SIAT2/SIAT1	40	40	320	80	80	80	160	160	160	80	320	320	
A/Bulgaria/56/2017	3C.2a	2017-01-04	SIAT2/SIAT1	40	<	160	80	<	<	160	160	40	40	80	160	
A/Jordan/J-16-4294/2017	3C.2a	2017-01-04	SIAT1	40	40	80	80	40	<	160	160	80	160	160	160	
A/Bulgaria/132/2017	3C.2a	2017-01-06	SIAT2/SIAT1	40	80	320	160	80	80	160	320	160	80	320	320	
A/Bulgaria/301/2017	3C.2a	2017-01-09	SIAT2/SIAT1	40	80	320	160	80	80	160	320	160	80	320	320	
A/Bulgaria/182/2017	3C.2a1	2017-01-10	SIAT2/SIAT1	80	40	320	160	80	80	160	320	160	80	320	320	
A/Bulgaria/252/2017	3C.2a1	2017-01-12	SIAT2/SIAT1	40	40	320	80	80	40	160	160	80	80	320	320	
A/Bulgaria/373/2017	3C.2a	2017-01-24	SIAT2/SIAT1	40	40	320	160	80	80	320	160	80	80	320	320	
A/Bulgaria/427/2017	3C.2a	2017-01-30	SIAT2/SIAT1	<	<	160	80	40	<	80	80	40	40	160	160	
A/Bulgaria/414/2017	3C.2a	2017-02-01	SIAT2/SIAT1	<	<	80	80	40	<	80	80	40	80	80	160	
A/Iceland/18/2017		2017-02-13	MDCK1/MDCK2	40	80	640	320	160	160	160	160	80	80	160	160	
A/Lithuania/6165/2017	3C.2a	2017-02-20	MDCK2/SIAT1	80	80	160	80	80	80	160	320	80	640	320	320	
A/Lithuania/6583/2017	3C.2a	2017-02-21	MDCK2/SIAT1	40	80	80	80	40	40	160	320	80	160	160	320	
A/Lithuania/7703/2017	3C.2a1	2017-02-27	MDCK2/SIAT1	40	40	80	80	80	<	80	160	80	40	80	160	
A/Lithuania/7906/2017	3C.2a	2017-03-06	MDCK2/SIAT1	40	40	160	80	80	80	160	160	160	320	320	320	
A/Jordan/20157/2017	3C.2a	2017-04-22	SIAT2	80	80	160	160	80	80	80	160	80	80	320	320	
A/Norway/2629/2017	3C.2a1	2017-05-04	SIAT1	160	80	640	160	160	80	320	320	320	160	320	320	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-07-07

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas 50/12 Egg F08/14 ¹	A/Samara 73/13 SIAT F35/15 ¹	A/Stock 6/14 SIAT F14/14 ¹	A/Stock 6/14 Egg F20/14 ¹	A/Switz 9715293/13 SIAT F18/15 ¹	A/Switz 9715293/13 Egg F73/16 ¹	A/HK 5738/14 MDCK F30/14 ¹	A/HK 4801/14 MDCK F43/15 ¹	A/HK 4801/14 Egg F12/15 ¹	A/Georgia 532/15 SIAT F33/15 ¹	A/Oman 2585/16 SIAT NIB F50/16 ¹	A/Nor 4436/16 SIAT F03/17 ¹
				3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	2560	1280	160	1280	80	640	320	640	80	160	160	640
A/Samara/73/2013		2013-03-12	C1/SIAT4	640	640	320	640	160	320	640	1280	160	320	640	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	80	40	320	160	160	160	160	320	80	160	320	320
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	160	80	160	320	40	640	160	320	40	40	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	40	40	320	160	80	80	80	80	40	80	160	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	160	320	320	640	80	1280	320	320	80	160	320	640
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	160	160	320	320	160	80	160	640	160	160	320	640
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK1	160	80	160	160	160	80	160	640	160	160	160	320
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	80	160	80	80	80	40	320	640	320	640	640	320
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	160	160	320	320	160	160	320	640	320	1280	640	640
TEST VIRUSES															
A/Priedviza/77/2016		2016-12-19	MDCK1/SIAT1	80	80	160	160	80	80	160	640	160	160	320	320
A/Komarno/93/2016		2016-12-20	MDCK1/SIAT1	80	80	320	160	160	80	320	320	160	160	320	320
A/Trnava/97/2016		2016-12-21	MDCKx/SIAT1	80	80	320	160	160	80	160	320	160	320	320	320
A/Trnava/90/2016		2016-12-21	MDCK2/SIAT1	40	40	160	160	80	40	320	320	160	160	160	320
A/Trencin/94/2016		2016-12-22	MDCK1/SIAT1	160	160	320	160	160	160	320	320	160	640	640	640
A/Bratislava/112/2016		2016-12-30	MDCKx/SIAT1	80	80	160	160	80	80	160	320	160	80	160	160
A/Lubica/132/2017		2017-01-03	MDCKx/SIAT1	<	<	<	<	<	<	40	80	<	80	80	80
A/Levice/133/2017		2017-01-04	MDCKx/SIAT1	160	80	320	160	160	80	160	640	160	160	320	320
A/Kosice/182/2017		2017-01-05	MDCKx/SIAT1	80	80	160	160	80	40	160	320	160	160	160	160
A/Povazska Bystrica/151/2017		2017-01-10	MDCKx/SIAT1	80	40	160	80	80	80	160	320	160	80	320	320
A/Nitra/157/2017		2017-01-11	MDCKx/SIAT1	80	80	320	160	80	80	160	320	160	160	320	320
A/Nove Zamky/184/2017		2017-01-17	MDCKx/SIAT1	80	80	320	160	160	80	160	320	160	320	320	320
A/Piestany/205/2017		2017-01-18	MDCKx/SIAT1	80	80	320	160	160	80	160	320	160	160	320	320
A/Dunajska Streda/232/2017		2017-01-24	MDCKx/SIAT1	80	80	320	160	80	80	160	320	80	160	320	320
A/Slovenia/625/2017		2017-01-25	MDCKx/SIAT1	80	80	320	160	160	80	320	320	160	80	640	320
A/Slovenia/1159/2017		2017-02-15	MDCKx/SIAT1	160	40	320	160	160	80	160	320	160	80	320	320
A/Bratislava/322/2017		2017-02-15	MDCKx/SIAT1	80	80	160	160	80	40	160	320	160	320	160	320
A/Slovenia/1391/2017		2017-03-08	MDCKx/SIAT1	40	40	320	80	80	80	160	160	80	80	320	320
A/Slovenia/1406/2017		2017-03-08	MDCKx/SIAT1	<	<	<	<	<	<	40	40	<	<	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-07-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas 50/12 Egg	A/Samara 73/13 SIAT	A/Stock 6/14 SIAT	A/Stock 6/14 Egg	A/Switz 9715293/13 SIAT	A/Switz 9715293/13 Egg	A/HK 5738/14 MDCK	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Georgia 532/15 SIAT	A/Oman 2585/16 SIAT	A/Nor 4436/16 SIAT
				F08/14 [†]	F35/15 [†]	F14/14 [†]	F20/14 [†]	F18/15 [†]	F73/16 [†]	F30/14 [†]	F43/15 [†]	F12/15 [†]	F33/15 [†]	NIB F50/16 [†]	F03/17 [†]
	Passage history			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
	Genetic group														
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	2560	1280	160	1280	80	640	320	640	80	160	160	640
A/Samara/73/2013		2013-03-12	C1/SIAT4	640	640	320	640	160	320	640	1280	160	320	640	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	80	40	320	160	160	160	160	320	80	160	320	320
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	160	80	160	320	40	640	160	320	40	40	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	40	40	320	160	80	80	80	80	40	80	160	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	160	160	160	640	80	640	320	320	80	160	320	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	160	160	320	320	160	80	160	640	160	160	320	640
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK1	160	80	160	160	160	80	160	640	160	160	160	320
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	80	160	80	80	80	40	320	640	320	320	640	320
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	160	160	320	320	160	160	320	640	320	1280	640	640
TEST VIRUSES															
A/Trencin/46/2016		2016-12-01	MDCK1/SIAT1	80	80	160	160	80	80	160	320	80	80	160	160
A/Piestany/80/2016		2016-12-12	MDCK1/SIAT1	40	40	160	80	80	40	160	160	80	80	160	160
A/Zaporizka/157/2017		2017-01-01	MDCK2/SIAT1	40	40	320	80	80	80	160	320	160	80	320	160
A/Odesa/62/2017		2017-01-02	MDCK/SIAT1/SIAT1	80	80	320	160	160	80	160	320	160	80	320	320
A/Khmelnitsky/138/2017		2017-01-02	MDCK/SIAT1/SIAT1	40	<	160	80	80	40	160	160	40	40	160	160
A/Dnipro/244/2017		2017-01-02	MDCK1/SIAT1	40	40	160	80	80	40	160	160	80	40	320	160
A/Odesa/66/2017		2017-01-05	MDCK/SIAT1/SIAT1	40	<	160	40	40	40	80	160	40	<	160	160
A/Odesa/70/2017		2017-01-05	MDCK/SIAT1/SIAT1	40	<	160	40	40	40	80	160	<	<	160	160
A/Odesa/75/2017		2017-01-05	MDCK/SIAT1/SIAT1	80	80	640	160	160	160	320	640	320	160	640	640
A/Khmelnitsky/143/2017		2017-01-06	MDCK/SIAT1/SIAT1	<	<	80	40	40	<	80	80	<	<	160	80
A/Khmelnitsky/145/2017		2017-01-06	MDCK/SIAT1/SIAT2	40	40	320	80	80	80	160	160	80	80	320	320
A/Odesa/69/2017		2017-01-08	MDCK/SIAT1/SIAT1	160	80	320	160	160	160	320	320	160	160	640	320
A/Odesa/73/2017		2017-01-09	MDCK/SIAT1/SIAT1	40	40	160	80	80	40	80	160	80	40	160	160
A/Odesa/72/2017		2017-01-10	MDCK/SIAT1/SIAT1	40	40	160	80	80	40	160	160	80	40	160	320
A/Khmelnitsky/147/2017		2017-01-11	MDCK/SIAT1/SIAT1	40	40	160	80	80	80	160	160	80	40	320	320
A/Khmelnitsky/148/2017		2017-01-12	MDCK/SIAT1/SIAT1	40	40	160	80	80	80	160	160	80	80	320	320
A/Ukraine/7730/2017		2017-01-12	C1/SIAT1	40	<	160	80	40	<	80	80	40	<	160	160
A/Ukraine/7610/2017		2017-01-12	C1/SIAT1	40	<	160	80	40	40	160	160	80	<	160	160
A/Ukraine/7552/2017		2017-01-16	Cx/SIAT1	40	<	160	80	40	<	80	80	80	<	160	160
A/Ukraine/7767/2017		2017-01-17	C2/SIAT1	40	40	320	160	80	80	160	320	160	80	320	320
A/Ukraine/7611/2017		2017-01-17	C1/SIAT1	<	<	160	40	40	<	80	80	<	<	160	160
A/Perugia/26/2017		2017-01-20	SIAT3/SIAT2	80	80	320	160	160	80	160	320	160	160	320	320
A/Zaporizka/160/2017		2017-01-21	MDCK/SIAT1/SIAT1	40	40	160	80	40	40	80	160	80	40	160	160
A/Kyiv/84/2017		2017-01-22	SIAT1/SIAT1	80	40	320	160	80	80	320	320	160	160	320	320
A/Khmelnitsky/150/2017		2017-01-26	MDCK/SIAT1/SIAT1	40	40	320	160	160	40	160	320	160	40	320	320
A/Sassari/9/2017		2017-01-30	SIAT1/SIAT1	80	80	320	160	160	80	320	320	160	80	320	320
A/Friuli-Venezia Giulia/59/2017		2017-02-01	SIAT1/SIAT1	40	40	80	80	40	<	80	160	40	40	80	80
A/Perugia/39/2017		2017-02-07	SIAT1/SIAT1	<	<	80	40	<	<	80	80	<	<	80	80
A/Norway/1940/2017		2017-03-20	SIAT1/SIAT1	80	80	320	160	160	80	160	320	160	160	320	80
A/Norway/1934/2017		2017-03-21	SIAT1/SIAT1	40	40	160	80	40	<	160	160	40	40	160	160
A/Norway/2113/2017		2017-03-30	SIAT1/SIAT1	40	<	160	80	80	<	160	160	40	<	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [†] < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-07-21

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Texas 50/12 Egg	A/Samara 73/13 SIAT	A/Stock 6/14 SIAT	A/Stock 6/14 Egg	A/Switz 9715293/13 SIAT	A/Switz 9715293/13 Egg	A/HK 5738/14 MDCK	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Georgia 532/15 SIAT	A/Oman 2585/16 SIAT	A/Nor 4436/16 SIAT
				F08/14 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F73/16 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹	NIB F50/16 ¹	F03/17 ¹
	Passage history Ferret number Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES															
A/Texas/50/2012		2012-04-15	E5/E2	2560	640	160	640	160	320	160	320	80	160	160	640
A/Samara/73/2013		2013-03-12	C1/SIAT4	640	640	320	320	160	320	320	640	80	160	320	640
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	160	80	320	160	160	160	160	320	80	160	320	320
A/Stockholm/6/2014	isolate 2	2014-02-06	E4/E1	320	80	160	320	40	320	160	320	40	40	160	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	80	40	320	160	80	160	80	160	40	40	160	160
A/Switzerland/9715293/2013	clone 123	2013-12-06	E4/E1	160	160	160	320	40	640	160	160	40	80	160	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	80	40	160	80	80	80	160	320	80	160	160	320
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK1	80	40	160	80	80	80	80	320	80	80	160	160
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	80	80	80	80	40	40	160	640	320	320	320	320
A/Georgia/532/2015	plaq 20	2015-03-09	SIAT1/SIAT5	80	80	320	160	80	80	160	320	80	640	160	320
TEST VIRUSES															
A/Stockholm/59/2016		2016-12-25	MDCK2/SIAT1	40	<	160	80	40	40	160	160	40	40	160	160
A/Ukraine/7768/2016		2016-12-27	C1/SIAT1	<	<	80	40	40	<	40	80	<	<	40	80
A/Ukraine/7646/2017		2017-01-03	C1/SIAT1	40	<	160	40	40	<	80	80	40	<	40	80
A/Stockholm/6/2017		2017-01-03	MDCK2/SIAT1	40	40	160	80	40	40	80	160	40	40	80	80
A/Ukraine/7650/2017		2017-01-07	C1/SIAT1	40	40	160	40	40	40	80	160	40	<	160	160
A/Ukraine/7651/2017		2017-01-11	C1/SIAT1	<	<	80	<	<	<	40	40	40	<	40	160
A/Stockholm/8/2017		2017-01-12	MDCK0/SIAT1	40	<	80	40	40	<	80	160	40	40	80	80
A/Poitiers/767/2017		2017-01-18	MDCK2/SIAT1	80	80	320	160	80	80	160	320	80	320	320	320
A/Stockholm/26/2017		2017-01-21	MDCK0/SIAT1	40	40	40	40	<	<	80	160	40	40	40	80
A/Sachsen/107/2017		2017-02-07	C1/SIAT1	<	<	160	<	<	<	40	80	<	<	80	40
A/Bosnia and Herzegovina/107/2017		2017-02-13	SIAT1	<	<	40	<	<	<	40	40	<	<	80	40
A/Bosnia and Herzegovina/110/2017		2017-02-15	SIAT1	40	<	80	40	<	<	80	80	<	40	80	80
A/Sachsen/99/2017		2017-02-23	C2/SIAT1	80	40	80	80	40	40	80	160	80	40	80	80
A/Mecklenburg-Vorpommern/17/2017		2017-03-08	C1/SIAT1	40	<	160	40	40	<	160	80	40	<	160	80
A/Brandenburg/40/2017		2017-03-10	C1/SIAT1	40	<	160	40	40	<	80	80	40	<	160	160
A/Thuringen/52/2017		2017-03-13	C1/SIAT1	40	<	160	40	40	<	160	80	40	<	160	80
A/Nordrhein-Westfalen/106/2017		2017-03-17	C1/SIAT1	40	<	160	<	40	<	160	80	40	<	160	160
A/Sachsen/71/2017		2017-03-20	C1/SIAT1	40	<	160	40	40	<	80	80	40	<	40	80
A/Niedersachsen/185/2017		2017-03-20	C2/SIAT1	40	<	160	40	40	40	80	80	40	40	160	160
A/Bremen/27/2017		2017-04-19	C1/SIAT1	40	<	160	80	40	<	80	80	40	<	40	160
A/Hong Kong/2277/2017		2017-05-19	SIAT1	40	<	160	40	40	40	80	80	40	<	160	80
A/Hong Kong/2274/2017		2017-05-20	SIAT1	40	<	160	80	40	40	160	160	40	40	160	160
A/Hong Kong/2279/2017		2017-05-21	SIAT1	40	<	80	40	40	40	80	80	40	40	160	80
A/Hong Kong/2285/2017		2017-05-23	SIAT1	<	<	80	<	<	<	80	80	<	<	80	80
A/Hong Kong/2286/2017		2017-05-23	SIAT1	<	<	80	<	<	<	80	40	<	<	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-07-28

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre					
				Post-infection ferret antisera					
				A/Stock 6/14	A/Switz 9715293/13	A/HK 4801/14	A/HK 4801/14	A/Oman 2585/16	A/Nor 4436/16
				SIAT F14/14 ^{*1}	SIAT F18/15 ¹	MDCK F43/15 ^{*1}	Egg F12/15 ^{*1}	SIAT NIB F50/16 ^{*1}	SIAT F03/17 ^{*1}
	Passage history Ferret number Genetic group			3C.3a	3C.3a	3C.2a	3C.2a	3C.2a1	3C.2a1
REFERENCE VIRUSES									
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	320	160	320	80	320	320
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	320	80	160	40	160	160
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK1	160	80	320	80	160	160
A/Hong Kong/4801/2014	isolate 1 3C.2a	2014-02-26	E6/E2	80	40	640	320	320	320
TEST VIRUSES									
A/Stockholm/58/2016	3C.2a	2016-12-28	MDCK0/SIAT1	160	40	160	80	160	160
A/Sundsvall/1/2017	3C.2a1	2017-01-17	MDCK1/SIAT1	320	80	320	80	160	160
A/Belgium/H25/2017	3C.2a1	2017-01-25	SIATx/SIAT1	160	40	320	80	160	160
A/Belgium/H23/2017	3C.2a	2017-01-29	SIATx/SIAT1	40	<	80	40	40	80
A/St Petersburg/314/2017	3C.2a	2017-02-06	MDCK2/SIAT2	160	40	160	40	80	160
A/St Petersburg/302/2017	3C.2a	2017-02-15	MDCK2/SIAT1	160	40	160	80	80	160
A/Krasnoyarsk/R1107/2017	3C.2a	2017-02-15	MDCK2/SIAT1	160	40	160	80	80	160
A/Linköping/3/2017	3C.2a	2017-02-17	MDCK1/SIAT1	160	80	320	80	160	160
A/St Petersburg/301/2017	3C.2a	2017-02-17	MDCK2/SIAT1	160	40	160	80	80	160
A/Novosibirsk/34/2017	3C.2a	2017-02-18	MDCK2/SIAT1	80	40	80	40	80	80
A/Novosibirsk/39/2017	3C.2a	2017-02-18	MDCK2/SIAT1	160	40	160	80	160	160
A/Linköping/2/2017	3C.2a1	2017-02-27	MDCK3/SIAT1	160	40	160	80	80	80
A/Novosibirsk/47/2017	3C.2a	2017-03-07	MDCK2/SIAT1	80	40	320	80	160	160
A/Omsk/06/2017	3C.2a	2017-03-08	MDCK2/SIAT1	160	80	320	160	320	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-16. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-08-04

					Haemagglutination inhibition titre				NEW	NEW	NEW	NEW		
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera									
					A/Stock 6/14 SIAT	A/Switz 9715293/13 SIAT	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Oman 2585/16 SIAT	A/Nor 4436/16 SIAT	A/Nor 4849/16 Egg	A/Greece 4/17 T/C	A/Greece 4/17 Egg	CBER08 C1.4 Egg
					F14/14 ¹	F18/15 ¹	F43/15 ¹	F12/15 ¹	NIB F50/16 ¹	F03/17 ¹	F23/17 ¹	F27/17 ¹	F28/17 ¹	CBER FE18 ¹
					3C.3a	3C.3a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a	3C.2a1	3C.2a1	3C.2a1
REFERENCE VIRUSES														
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	640	160	320	160	320	320	80	320	160	160	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	640	80	160	40	160	160	<	160	80	40	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK1	320	160	640	160	320	320	80	320	160	160	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E2	80	80	640	320	640	320	640	640	1280	2560	
A/Norway/4849/2016	3C.2a	2016-12-02	E7	80	80	640	160	320	320	640	320	1280	1280	
A/Greece/4/2017	3C.2a1	2017-01-02	E9	160	80	640	160	640	320	320	1280	2560	1280	
A/Norway/3806/2016	CBER 08 C1.4	3C.2a1	2016-06-13	E7/E8	1280	160	>5120	640	1280	2560	1280	640	1280	
TEST VIRUSES														
A/Serbia/7921/2017	3C.2a	2016-12-09	C3/SIAT1	160	80	160	160	320	320	80	ND	160	160	
A/Serbia/7997/2017	3C.2a	2016-12-12	C3/SIAT1	160	80	160	80	160	320	80	ND	80	160	
A/Serbia/8255/2017	3C.2a	2016-12-16	C3/SIAT1	320	80	320	80	320	320	80	ND	160	160	
A/Serbia/8283/2017	3C.2a	2016-12-21	C3/SIAT1	160	80	160	80	160	160	80	ND	80	160	
A/Serbia/8374/2017	3C.2a	2016-12-21	C3/SIAT1	160	40	160	80	160	160	80	ND	80	160	
A/Serbia/8352/2017	3C.2a	2016-12-22	C3/SIAT1	160	80	320	160	160	320	80	ND	160	160	
A/Serbia/8375/2017	3C.2a	2016-12-22	C3/SIAT1	160	80	160	160	320	320	80	ND	80	160	
A/Serbia/8411/2017	3C.2a	2016-12-26	C3/SIAT1	160	80	320	80	160	320	80	ND	80	160	
A/Serbia/8499/2017	3C.2a	2016-12-27	C2/SIAT1	160	80	320	80	160	320	40	ND	40	80	
A/Stockholm/60/2016	3C.2a1	2016-12-28	MDCK1/SIAT2	160	40	160	80	160	160	<	ND	80	40	
A/Serbia/8501/2017	3C.2a	2016-12-28	C3/SIAT1	320	80	320	160	320	320	80	ND	160	160	
A/Serbia/8554/2017	3C.2a	2016-12-30	C1/SIAT1	160	40	320	80	80	160	40	ND	40	80	
A/Ukraine/7647/2017	3C.2a1	2017-01-04	C1/SIAT2	320	80	320	80	320	320	40	640	160	80	
A/Thuringen/165/2017	3C.2a	2017-01-13	C1/SIAT2	320	80	320	80	160	320	<	320	40	40	
A/Stockholm/11/2017	3C.2a1	2017-01-14	MDCK0/SIAT2	160	40	160	80	160	160	<	320	80	40	
A/England/19/2017	3C.2a1	2017-01-31	SIAT1/SIAT1	160	80	320	80	80	160	40	320	320	80	
A/Orenburg/70/2017	3C.2a	2017-02-07	SIAT1/SIAT1	160	40	160	80	160	160	<	ND	40	40	
A/Moscow/34/2017	3C.2a	2017-02-07	SIAT1/SIAT1	160	80	160	80	320	320	40	ND	80	40	
A/Orenburg/72/2017	3C.2a	2017-02-13	SIAT1/SIAT1	160	80	640	160	160	320	160	ND	160	160	
A/Paris/1023/2017	3C.2a1	2017-02-16	MDCK1/SIAT1	80	40	80	40	80	160	<	ND	80	<	
A/Linkoping/1/2017	3C.2a1	2017-02-17	MDCK2/SIAT2	320	80	320	80	320	320	40	640	160	80	
A/Moscow/41/2017	3C.2a	2017-02-18	SIAT1/SIAT1	160	40	160	40	320	160	40	ND	80	40	
A/Basse Normandie/1055/2017	3C.2a1	2017-02-20	MDCK1/SIAT1	80	40	160	40	160	160	<	ND	40	40	
A/Bosnia and Herzegovina/119/2017	3C.2a1	2017-02-24	SIAT2	160	40	80	80	160	160	<	ND	80	40	
A/Moscow/47/2017	3C.2a1	2017-02-25	SIAT1/SIAT1	160	80	160	80	160	160	40	ND	40	40	
A/Paris/1145/2017	3C.2a1	2017-03-06	MDCK1/SIAT1	160	<	80	40	80	160	<	ND	40	40	
A/Dijon/1176/2017	3C.2a1	2017-03-14	MDCK1/SIAT1	80	<	80	40	80	40	<	ND	40	<	
A/Niedersachsen/186/2017	3C.2a1	2017-03-21	C2/SIAT2	320	80	160	80	320	320	<	640	160	40	
A/St Petersburg/320/2017	3C.2a1	2017-03-21	MDCK2/SIAT2	320	160	320	160	320	320	80	ND	160	160	
A/Hong Kong/2267/2017	3C.2a	2017-05-19	Cx/SIAT1	160	80	320	160	320	320	80	160	80	160	
A/Hong Kong/2277/2017	3C.2a1	2017-05-19	Cx/SIAT1	80	40	160	40	80	80	<	160	80	80	
A/Hong Kong/2274/2017	3C.2a1	2017-05-20	Cx/SIAT1	160	40	320	80	80	160	40	320	160	80	
A/Hong Kong/2273/2017	3C.2a	2017-05-21	Cx/SIAT1	320	40	320	160	320	320	80	320	160	160	
A/Hong Kong/2279/2017	3C.2a1	2017-05-21	Cx/SIAT1	160	40	160	80	160	160	<	320	80	40	
A/Hong Kong/2283/2017	3C.2a1	2017-05-23	Cx/SIAT1	80	40	160	80	80	80	<	160	40	80	
A/Hong Kong/2284/2017	3C.2a	2017-05-23	Cx/SIAT1	160	80	160	160	160	160	40	160	80	160	
A/Hong Kong/2285/2017	3C.2a1	2017-05-23	Cx/SIAT1	80	<	40	40	80	40	<	80	40	40	
A/Hong Kong/2291/2017	3C.2a	2017-05-23	Cx/SIAT1	<	<	<	<	40	40	<	40	<	<	
A/Hong Kong/2293/2017	3C.2a1	2017-05-23	Cx/SIAT1	160	160	320	80	160	80	<	160	80	80	
A/Hong Kong/3159/2017	3C.2a	2017-07-03	Cx/SIAT1	160	40	160	80	160	160	40	ND	80	40	
A/Hong Kong/3160/2017	3C.2a	2017-07-04	Cx/SIAT1	80	40	160	40	80	160	<	160	40	80	
A/Hong Kong/3162/2017	3C.2a	2017-07-04	Cx/SIAT1	80	<	80	<	80	80	<	80	<	<	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = 40

ND = Not Done

Sequences in phylogenetic trees

Vaccine

Table 8-17. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-08-11

Viruses	Other information	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre					
					Post-infection ferret antisera					
					A/Stock 6/14 SIAT F14/14 ^{*1} 3C.3a	A/Switz 9715293/13 SIAT F18/15 ^{*1} 3C.3a	A/HK 4801/14 MDCK F43/15 ^{*1} 3C.2a	A/HK 4801/14 Egg F12/15 ^{*1} 3C.2a	A/Oman 2585/16 SIAT NIB F50/16 ^{*1} 3C.2a1	A/Greece 4/17 SIAT F27/17 ^{*1} 3C.2a1
REFERENCE VIRUSES										
A/Stockholm/6/2014	isolate 1	3C.3a	2014-02-06	SIAT1/SIAT2	1280	320	320	160	320	320
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT3	640	160	160	80	160	160
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK1	640	160	640	160	320	640
A/Hong Kong/4801/2014		3C.2a	2014-02-26	E6/E2	160	160	640	320	640	320
TEST VIRUSES										
A/Serbia/8418/2017		3C.2a		C3/SIAT1	320	80	160	80	160	160
A/St Etienne/951/2017		3C.2a1	2017-01-05	MDCK2/SIAT1	640	160	640	160	320	320
A/Lisboa/2/2017		3C.2a1	2017-01-05	SIAT1/SIAT1	320	80	160	80	320	320
A/Lisboa/4/2017		3C.2a	2017-01-05	SIAT1/SIAT1	320	80	160	80	320	160
A/Lisboa/12/2017		3C.2a	2017-01-05	SIAT1/SIAT1	320	80	160	80	320	320
A/Lisboa/5/2017		3C.2a1	2017-01-12	SIAT1/SIAT1	160	40	160	80	160	320
A/Lisboa/17/2017		3C.2a1	2017-01-22	SIAT2/SIAT2	320	160	320	160	320	320
A/Lisboa/16/2017		3C.2a	2017-02-10	SIAT1/SIAT1	160	40	160	80	160	320
A/Antsirabe/2096/2017		3C.2a	2017-05-02	MDCK2/SIAT1	160	40	160	80	160	160
A/Toamasina/2190/2017		3C.2a	2017-05-03	MDCK2/SIAT1	320	160	320	160	320	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine

Sequences in phylogenetic trees

Table 8-18. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-08-18

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre						NEW
				Post-infection ferret antisera						
				A/Stock 6/14 SIAT	A/Switz 9715293/13 SIAT	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Oman 2585/16 SIAT	A/Greece 4/17 SIAT	A/Oman 5466/16 SIAT
				F14/14* ¹	F18/15* ¹	F43/15* ¹	F12/15* ¹	NIB F50/16* ¹	F27/17* ¹	CDC F2017-010* ¹
				3C.3a	3C.3a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a1
REFERENCE VIRUSES										
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	640	320	320	160	320	320	320
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	320	160	160	80	160	160	160
A/Hong Kong/4801/2014	3C.2a	2014-02-26	MDCK4/MDCK3	320	160	640	160	320	320	320
A/Hong Kong/4801/2014	isolate 1 3C.2a	2014-02-26	E6/E2	80	80	640	320	640	640	640
A/Oman/5466/2016	3C.2a1	2016-07-06	SIAT2/SIAT3/SIAT1	160	160	320	160	160	320	160
TEST VIRUSES										
A/Washington/16/2017	3C.2a	2017-01-25	E4/E1	40	40	160	160	320	160	320
A/Barnaul/139/2017	3C.2a	2017-01-31	MDCK2/SIAT2	160	80	160	80	160	160	160
A/Magadan/187/2017	3C.2a	2017-02-06	MDCK2/SIAT3	160	160	320	160	160	320	160
A/Saudi Arabia/252925/2017	3C.2a	2017-03-15	SIAT1	160	160	160	80	160	320	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-19. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-08-23

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre					
				Post-infection ferret antisera					
				A/Stock 6/14 SIAT F14/14* ¹ 3C.3a	A/Switz 9715293/13 SIAT F18/15* ¹ 3C.3a	A/HK 4801/14 MDCK F43/15* ¹ 3C.2a	A/HK 4801/14 Egg F12/15* ¹ 3C.2a	A/Oman 2585/16 SIAT NIB F50/16* ¹ 3C.2a1	A/Greece 4/17 SIAT F27/17* ¹ 3C.2a1
				Passage history Ferret number Genetic group					
REFERENCE VIRUSES									
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	320	160	160	80	320	320
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	320	80	160	40	160	80
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK3	160	80	320	80	160	160
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	80	40	640	160	320	640
TEST VIRUSES									
A/Singapore/INFTT-16-0612/2016		2016-06-02	E5/E2	80	40	320	160	320	320
A/Singapore/INFIMH-16-0019/2016		2016-06-14	E5/E1	<	<	160	<	40	80
A/South Australia /135/2016		2016-10-01	E5/E1	40	<	160	40	80	40
A/Karaganda/112/2017		2017-02-07	MDCK x/MDCK1	80	40	160	80	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-20. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-08-25

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre							
				Post-infection ferret antisera							
				A/Stock 6/14	A/Switz 9715293/13	A/HK 5738/14	A/HK 4801/14	A/HK 4801/14	A/Oman 2585/16	A/Nor 4436/16	A/Greece 4/17
				SIAT	SIAT	MDCK	MDCK	Egg	SIAT	SIAT	SIAT
	Passage history										
	Ferret number			F14/14 ^{*1}	F18/15 ¹	F30/14 ^{*1}	F43/15 ^{*1}	F12/15 ^{*1}	NIB F50/16 ^{*1}	F03/17 ^{*1}	F27/17 ^{*1}
	Genetic group			3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a1
REFERENCE VIRUSES											
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	320	160	160	160	80	160	160	160
A/Switzerland/9715293/2013		2013-12-06	SIAT1/SIAT3	320	160	160	160	80	160	320	160
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT3	320	160	160	640	160	320	320	320
A/Hong Kong/4801/2014		2014-02-26	MDCK4/MDCK3	320	160	160	640	160	160	160	320
A/Hong Kong/4801/2014	isolate 1	2014-02-26	E6/E2	160	80	320	640	320	320	320	640
TEST VIRUSES											
A/South Africa/R04468/2017		2017-06-13	MDCK1/SIAT1	320	160	160	320	160	160	320	640
A/South Africa/R04939/2017		2017-06-20	MDCK1/SIAT1	640	320	320	640	160	320	320	640
A/South Africa/R05728/2017		2017-06-30	MDCK1/SIAT1	640	160	320	320	160	320	320	640
A/Hong Kong/3115/2017		2017-07-01	Cx/SIAT1	320	160	160	320	160	160	320	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-21. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2017-09-08

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Stock 6/14 SIAT	A/Switz 9715293/13 SIAT	A/HK 5738/14 MDCK	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Oman 2585/16 SIAT	A/Greece 4/2017 SIAT	A/Nor 4436/16 SIAT	A/Nor 3806/16 Egg	NIB-103 Egg
					F14/14 ¹	F18/15 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F50/16 ¹	F27/17 ¹	F03/17 ¹	F45/16 ¹	F13/17 ¹
		Genetic group			3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a1	3C.2a1	3C.2a1
REFERENCE VIRUSES														
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	320	320	160	320	160	320	320	320	320	80
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT3	320	160	160	160	80	160	160	160	160	40
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT3	160	80	160	320	160	160	160	320	160	80
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK3	160	80	160	320	160	320	320	320	320	80
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	80	80	320	640	160	640	640	320	320	1280
A/Norway/3806/2016	isolate 1 clone 25	3C.2a1	2016-06-13	E9	80	40	160	320	160	320	320	160	320	640
NIB-103 (A/Norway/3806/2016)		3C.2a1	2016-06-13	Ex	40	40	80	320	160	320	320	160	320	1280
TEST VIRUSES														
A/Antananarivo/1067/2016		3C.2a1	2016-04-05	E5/E1	40	40	320	1280	160	320	160	320	320	80
A/South Australia/54/2016		3C.2a	2016-07-12	E4/E1	320	80	80	160	40	80	80	160	80	<
A/Singapore/TT1374/2016		3C.2a	2016-12-01	E6/E1	80	80	160	640	160	320	320	320	160	160
A/Guizhou-Qingzhen/1968/2016		3C.2a1	2016-12-06	C2/SIAT2/SIAT1	80	40	80	160	80	160	160	160	160	<
A/Singapore/GP2646/2016		3C.2a1	2016-12-08	E6/E1	40	40	80	320	80	160	80	160	80	40
A/Brisbane/321/2016		3C.2a	2016-12-19	E6/E1	40	40	160	320	80	320	320	160	320	640
A/Bosnia & Herzegovina/18294/2017			2017-02-01	SIAT1	160	80	160	160	80	160	320	160	160	40
A/Sydney/52/2017		3C.2a1	2017-03-06	E5/E1	40	<	80	320	80	160	160	80	160	160
A/Brisbane/29/2017		3C.2a	2017-03-08	E4/E1	<	<	40	160	40	160	80	40	80	320
A/Hong Kong/2473/2017		3C.2a	2017-06-04	Cx/SIAT1	160	80	160	160	40	160	160	160	160	40
A/Shanghai-Pudongxin/1987/2017		3C.2a1	2017-06-13	C2/SIAT1/SIAT1	80	40	80	160	80	160	160	160	80	<
A/Ghana/1841/2017			2017-06-24	SIAT1	160	<	80	160	80	160	160	160	160	<
A/Ghana/1797/2017			2017-06-27	MDCK1	40	160	80	320	80	160	160	160	160	40
A/Ghana/1782/2017			2017-06-28	SIAT1	80	<	80	160	40	160	160	160	160	<
A/Hong Kong/3730/2017		3C.2a	2017-07-11	Cx/SIAT1	160	80	160	160	80	160	320	160	160	40
A/Hong Kong/3815/2017		3C.2a1	2017-07-13	Cx/SIAT1	160	80	160	160	80	320	320	160	160	<
A/Hong Kong/3861/2017		3C.2a	2017-07-14	Cx/SIAT1	320	80	160	160	80	320	320	160	160	<
A/Hong Kong/3819/2017		3C.2a1	2017-07-15	Cx/SIAT1	160	80	160	160	80	160	320	160	160	40
A/Hong Kong/3820/2017			2017-07-17	Cx/SIAT1	160	80	160	80	80	160	320	160	160	<
A/Hong Kong/3946/2017			2017-07-17	Cx/SIAT1	160	80	160	160	80	160	160	320	160	40
A/Hong Kong/3823/2017		3C.2a	2017-07-18	Cx/SIAT1	160	80	160	160	80	160	160	160	160	<
A/Hong Kong/3800/2017		3C.2a1	2017-07-20	Cx/SIAT1	160	80	160	160	80	160	320	160	160	<
A/Hong Kong/3824/2017		3C.2a	2017-07-21	Cx/SIAT1	160	80	160	160	80	160	320	160	160	<
A/Hong Kong/3854/2017		3C.2a	2017-07-21	Cx/SIAT1	160	80	80	160	80	160	160	160	160	<
A/Hong Kong/3860/2017		3C.2a	2017-07-21	Cx/SIAT1	160	80	160	160	80	160	160	160	160	<
A/Hong Kong/3928/2017		3C.2a	2017-07-21	Cx/SIAT1	160	80	80	160	80	160	160	160	160	40
A/Hong Kong/4103/2017			2017-07-21	Cx/SIAT1	80	40	80	80	80	80	320	80	80	40
A/Hong Kong/4082/2017			2017-07-22	Cx/SIAT1	160	80	80	160	80	160	320	160	160	<
A/Hong Kong/3855/2017		3C.2a1	2017-07-23	Cx/SIAT1	160	80	80	160	80	160	160	160	160	<
A/Hong Kong/3856/2017		3C.2a1	2017-07-23	Cx/SIAT1	80	<	40	40	<	40	40	40	40	<
A/Hong Kong/4104/2017			2017-07-23	Cx/SIAT1	80	40	80	160	80	80	80	80	160	<
A/Hong Kong/3853/2017		3C.2a	2017-07-24	Cx/SIAT1	80	80	80	160	80	160	160	160	160	<
A/Hong Kong/4105/2017			2017-07-25	Cx/SIAT1	80	40	80	40	40	80	160	80	80	<
A/Hong Kong/4108/2017			2017-07-25	Cx/SIAT1	160	40	160	160	80	80	160	160	160	<
A/Hong Kong/3978/2017		3C.2a1	2017-07-28	Cx/SIAT1	160	40	80	80	80	160	160	80	80	40
A/Hong Kong/4018/2017		3C.2a1	2017-07-28	Cx/SIAT1	160	80	80	160	80	160	160	160	160	<
A/Hong Kong/4212/2017			2017-07-31	Cx/SIAT1	160	40	80	160	80	160	160	160	160	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Sequences in phylogenetic trees

Vaccine

Table 8-22. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) with collection dates after 2017-01-31 - Summary

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre																			
					Post-infection ferret antisera																			
					A/Texas 50/12 Egg	A/Samara 73/13 SIAT	A/Stock 6/14 SIAT	A/Stock 6/14 Egg	A/Switz 9715293/13 SIAT	A/Switz 9715293/13 Egg	A/HK 5738/14 MDCK	A/HK 4801/14 MDCK	A/HK 4801/14 Egg	A/Georgia 532/15 SIAT	A/Oman 2585/16 SIAT	A/Nor 4436/16 SIAT	A/Nor 4849/16 Egg	A/Greece 4/17 SIAT	A/Greece 4/17 Egg	A/Nor 3806/16 Egg	NIB-103 Egg	CBER08 C1.4 Egg	A/Oman 5466/16 SIAT	
					F08/14 ¹	F35/15 ¹	F14/14 ¹	F20/14 ¹	F18/15 ¹	F73/16 ¹	F30/14 ¹	F43/15 ¹	F12/15 ¹	F33/15 ¹	NIB F50/16 ¹	F03/17 ¹	F23/17 ¹	F27/17 ¹	F28/17 ¹	F45/16 ¹	F13/17 ¹	FE18 ¹	F2017-010 ¹	
		Genetic group			3C.1	3C.3	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a	3C.2a1	3C.2a1	3C.2a1	3C.2a1	3C.2a1		
REFERENCE VIRUSES																								
A/Texas/50/2012		3C.1	2012-04-15	E5/E2	1280	640	160	640	40	320	160	320	80	160	160	320								
A/Samara/73/2013		3C.3	2013-03-12	C1/SIAT4	640	640	640	640	160	320	320	1280	160	320	320	640								
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT2	80	40	320	160	160	160	160	160	80	80	160	160	80	320	160	320	80	160	320	
A/Stockholm/6/2014	isolate 2	3C.3a	2014-02-06	E4/E1	160	80	160	320	80	320	160	320	40	40	160	320								
A/Switzerland/9715293/2013		3C.3a	2013-12-06	SIAT1/SIAT3	<	<	320	80	80	80	80	80	40	40	80	160	<	160	80	160	40	40	160	
A/Switzerland/9715293/2013	clone 123	3C.3a	2013-12-06	E4/E1	80	160	160	320	40	640	160	320	40	80	160	320								
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT3	80	160	320	160	80	80	320	640	160	160	320	320				160	80			
A/Hong Kong/4801/2014		3C.2a	2014-02-26	MDCK4/MDCK1	80	80	160	160	80	80	160	640	160	160	160	160	80	320	160	320	80	160	320	
A/Hong Kong/4801/2014	isolate 1	3C.2a	2014-02-26	E6/E2	40	80	80	80	40	40	320	640	160	320	640	160	640	640	1280	320	1280	2560	640	
A/Georgia/532/2015	plaq 20	3C.2a	2015-03-09	SIAT1/SIAT5	80	80	160	160	80	80	160	320	160	640	320	320								
A/Norway/4849/2016		3C.2a	2016-12-02	E7			80		80				160		320	320	640	320	1280			1280		
A/Greece/4/2017		3C.2a1	2017-01-02	E9			160		80				640	160	640	320	320	1280	2560			1280		
A/Norway/3806/2016	CBER 08 C1.4	3C.2a1	2016-06-13	E7/E8			1280		160			>5120	640		1280	2560	1280	640	640			1280		
A/Oman/5466/2016		3C.2a1	2016-07-06	SIAT2/SIAT3/SIAT1			160		160			320	160		160	320						160		
A/Norway/3806/2016	Isolate 1 clone 25	3C.2a1	2016-06-13	E9			80		40		160	320	160		320	160		320		320	640			
NIB-103 (A/Norway/3806/2016)		3C.2a1	2016-06-13	Ex			40		40		80	320	160		320	160		320		320	1280			
TEST VIRUSES																								
Number of viruses tested					50	50	127	50	127	50	86	127	127	50	127*	121*	26	55*	26	31	31	26	2	
No with titre reduction ≤2-fold							68	16	101		71	66	59	3						25			2	
%							53.5	32.0	79.5		82.6	52.0	46.5	6.0						80.6			100	
No with titre reduction =4-fold							40	17	21	3	13	44	36	5			1			5	1			
%							31.5	34.00	16.5	6.0	15.1	34.6	28.3	10.0			3.8			16.1	3.2			
No with titre reduction ≥8-fold					50	50	19	17	5	47	2	17	32	42			25		26	1	30	26		
%					100	100	15.00	34.00	4.0	94.0	2.3	13.4	25.2	84.0			96.2		100	3.3	96.8	100		
Number of clade 3C.2a viruses tested					13	13	49	13	49	13	24	49	49	13	49*	44*	11	22*	11	10	10	11	2	
No with titre reduction ≤2-fold							23	2	42		19	29	22	3						9			2	
%							46.9	15.4	85.7		79.2	59.2	44.9	23.1						90.0			100	
No with titre reduction =4-fold							18	9	2		5	18	17	2			1			1	1			
%							36.7	69.2	4.1		20.8	36.7	34.7	15.4			9.1			10.0	10.00			
No with titre reduction ≥8-fold					13	13	8	2	5	13		2	10	8			10		11		9	11		
%					100	100	16.4	15.4	10.2	100		4.1	20.4	61.5			90.9		100		90.0	100		
Number of clade 3C.2a1 viruses tested					30	30	58	30	58	30	42	58	58	30	58*	58*	15	20*	15	9	9	15		
No with titre reduction ≤2-fold							32	6	46		38	25	22							6				
%							55.2	20.0	79.3		90.5	43.1	37.9							66.7				
No with titre reduction =4-fold							16	9	2		3	22	13	2						2				
%							27.6	30.0	3.5		7.1	37.9	22.4	6.7						22.2				
No with titre reduction ≥8-fold					30	30	10	15	10	30	1	11	23	28			15		15	1	9	15		
%					100	100	17.2	50.0	17.2	100	2.4	19.0	39.7	93.3			100		100	11.1	100	100		

* Homologous HI titres not available

Reference virus results are taken from individual tables as exampleS. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Vaccine

Table 9-1 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-03-06

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹												HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera												
				A/HK 4801/14 Egg NIB F53/16 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a	A/Oman 2585/16 SIAT NIB F50/16 3C.2a1	A/Norway 4436/16 SIAT F03/17 3C.2a1	A/Norway 3806/16 Egg NIB F45/16 3C.2a1							
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	320	320	431	N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, N171K N121K, I140M, N171K N121K, I140M, T160K (-CHO), N171K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	320	251	80	71	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	1280	993	160	168	
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	1280	993	80	119	
A/Norway/4436/2016	3C.2a1	2016-04-06	SIAT1	160	122	640	775	640	481	1280	1406	1280	992	160	120	
A/Norway/3806/2016 Is2 cl25	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	727	640	506	640	647	
TEST VIRUSES																
A/Oman/7840/2016	3C.2a	2016-11-15	SIAT1/SIAT1	<	0	640	538	1280	983	640	832	640	693	ND	ND	N121K, S144K
A/Denmark/57/2017	3C.2a	2016-11-23	SIAT4/SIAT1	<	0	160	198	320	427	640	533	320	320	ND	ND	N121K, S144K
A/Tula/10/2016	3C.2a	2016-11-24	MDCK3/SIAT1	40	40	640	541	1280	1036	640	873	640	737	ND	ND	N121K, S144K
A/Irkutsk/1/2016	3C.2a	2016-12-09	MDCK2/SIAT1	<	0	320	247	640	533	320	431	320	308	ND	ND	N121K, S144K
A/Portugal/SU203/2016	3C.2a	2016-12-21	SIAT1/SIAT1	<	1	640	599	1280	1067	1280	1009	640	583	ND	ND	N121K, S144K, R222K
A/Oman/7695/2016	3C.2a	2016-11-06	SIAT1/SIAT1	40	53	640	574	1280	1034	1280	1050	640	720	ND	ND	N121K, S144K, D291N
A/Bretagne/2898/2016	3C.2a	2016-10-28	MDCK2/SIAT2	<	1	80	112	320	380	320	358	160	145	ND	ND	N121K, T135A (-CHO), S144K, R261Q
A/Kazakhstan/4984/2016	3C.2a	2016-12-03	SIAT1	<	0	320	448	320	427	640	696	640	743	ND	ND	N121K, N122D (-CHO), S144K, S262N
A/Kazakhstan/4986/2016	3C.2a	2016-12-05	SIAT1	40	55	640	579	640	788	640	890	640	711	ND	ND	N121K, N122D (-CHO), S144K, S262N
A/Kazakhstan/4982/2016	3C.2a	2016-12-12	SIAT1	40	40	640	520	640	604	1280	1045	640	725	ND	ND	N121K, N122D (-CHO), S144K, S262N
A/Portugal/SU237/2016	3C.2a1	2016-12-27	SIAT1/SIAT1	<	3	320	389	160	220	1280	1076	640	608	ND	ND	R142G, N171K
A/Georgia/10/2017	3C.2a1	2017-01-03	SIAT1	<	1	320	335	160	155	640	731	320	414	ND	ND	R142G, N171K
A/Georgia/1815/2016	3C.2a1	2016-12-23	SIAT1	<	0	320	295	320	293	640	837	640	594	ND	ND	K92R, N121K, N171K, H311Q
A/Finland/658/2016	3C.2a1	2016-11-22	SIAT1	40	20	320	287	160	148	640	754	640	487	ND	ND	N121K, R142G, N171K
A/Portugal/SU227/2016	3C.2a1	2016-12-25	SIAT1/SIAT1	40	40	320	333	160	155	1280	987	640	576	ND	ND	N121K, R142G, N171K
A/Finland/672/2016	3C.2a1	2016-12-15	SIAT1	40	49	320	288	160	171	1280	1001	640	492	40	56	N121K, T135K (-CHO), N171K
A/Finland/659/2016	3C.2a1	2016-11-25	SIAT1	40	20	160	216	80	80	640	918	320	395	160	145	N121K, T135K (-CHO), N171K
A/Finland/664/2016	3C.2a1	2016-12-12	SIAT1	40	46	320	243	160	151	640	947	640	640	160	128	N121K, T135K (-CHO), N171K
A/Ukraine/7458/2016	3C.2a1	2016-12-21	P1/SIAT1	40	20	160	204	160	130	640	674	320	299	80	63	N121K, T135K (-CHO), N171K, S279Y
A/Khmelnitsky/2471/2016	3C.2a1	2016-12-21	MDCK1/SIAT1	<	0	80	98	60	69	640	549	320	250	40	49	N121K, T135K (-CHO), N171K, S279Y
A/Kyiv/956/2016	3C.2a1	2016-12-14	SIAT1/SIAT1	320	240	320	411	640	480	1280	1085	640	707	ND	ND	N121K, T135K (-CHO), N171K, S279Y
A/Iceland/82/2016	3C.2a1	2016-12-28	MDCK1/SIAT1	<	0	320	411	320	267	1280	1280	640	617	80	62	A106V, N121K, T135K (-CHO), N171K
A/Portugal/EVA164/2016	3C.2a1	2016-12-26	SIAT1/SIAT1	40	40	640	539	320	427	1280	1396	1280	960	ND	ND	K92R, N121K, N171K, E172D, S198P, H311Q

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation
ND = Not Done

Table 9-2 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-03-09

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				A/HK 4801/14 Egg NIB F53/16 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a	A/Oman 2585/16 SIAT NIB F50/16 3C.2a1	A/Norway 4436/16 SIAT F03/17 3C.2a1						
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, N171K N121K, I140M, N171K N121K, I140M, T160K (-CHO), N171K, L194P
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	465	320	320	
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	311	320	251	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	1880	1280	993	
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT2	160	203	320	386	640	508	2560	2115	1280	993	
A/Norway/4436/2016	3C.2a1	2016-04-06	SIAT1	160	122	640	775	640	481	1280	1406	1280	992	
A/Norway/3806/2016 Is2 cl25	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	727	640	506	
TEST VIRUSES														N121K, S144K I58V, N121K, S144K, S219Y T131K, R142K T131K, R142K, R261Q T131K, N133S, R142K, R261Q
A/Finland/649/2016	3C.2a	2016-10-20	SIAT1/SIAT1	80	100	640	794	1280	1129	1280	1131	1280	981	
A/Montenegro/152/2016	3C.2a	2016-12-20	MDCK1/SIAT1	<	20	160	153	640	569	640	480	320	404	
A/Denmark/69/2016	3C.2a	2016-12-06	SIAT2/SIAT1	80	100	320	354	640	480	1280	1024	640	870	
A/Finland/656/2016	3C.2a	2016-11-15	SIAT1/SIAT1	80	109	640	493	640	508	640	880	640	815	
A/Finland/670/2016	3C.2a	2016-12-12	SIAT1	40	49	160	142	30	91	640	683	320	455	
A/Kyiv/953/2016	3C.2a1	2016-12-12	SIAT1/SIAT1	40	45	640	628	640	488	1280	1222	640	868	
A/Austria/960155/2016	3C.2a1	2016-12-19	C1/SIAT1	<	20	640	566	320	337	1280	1143	640	727	D53N, R142G, N171K D53N, R142G, N171K D53N, R142G, N171K
A/Ukraine/7489/2016	3C.2a1	2016-12-20	P1/SIAT1	40	40	640	626	320	240	1280	1196	1280	815	
A/Ukraine/7490/2016	3C.2a1	2016-12-23	P1/SIAT1	40	20	640	562	320	448	1280	1493	1280	1029	
				Vaccine										

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-3 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-03-16

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Norway 4436/16 SIAT F03/17 3C.2a1		A/Norway 3806/16 Egg NIB F45/16 3C.2a1		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	320	320	431	N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, I140M, N171K N121K, I140M, T160K (-CHO), N171K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	251	80	71	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	993	160	168	
A/Norway/4436/2016	3C.2a1	2016-04-06	SIAT1	160	122	640	775	640	481	1280	992	160	120	
A/Norway/3806/2016 Is2 cl25	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	506	640	647	
TEST VIRUSES														
A/Alba/206712/2017	3C.2a	2017-01-10	SIAT1/SIAT1	80	67	160	220	160	144	640	570	40	40	Q57H, T131K, R142K
A/Macedonia/8/2017	3C.2a	2017-01-04	SIAT1	<	1	640	512	1280	1066	1280	978	40	40	N121K, N122D (-CHO), S144K, S262N,
A/Israel/C-470/2017	3C.2a	2017-01-08	C1/SIAT1	<	1	1280	979	1280	1208	1280	1054	80	68	N121K, N122D (-CHO), S144K, S262N
A/Israel/C-758/2017	3C.2a	2017-01-12	C1/SIAT1	<	15	320	304	1280	1600	640	815	80	73	N121K, S124I (-CHO), T135A (-CHO), S144K,
A/Poland/30539/2017	3C.2a1	2017-01-09	SIAT1	40	40	640	500	320	443	640	819	40	49	R142G, N171K
A/Sweden/154/2016	3C.2a1	2016-12-17	MDCK1/SIAT1	40	40	320	299	160	156	640	941	80	104	N121K, N171K
A/Hungary/197/2016	3C.2a1	2016-12-19	MDCK2/SIAT1	40	48	640	512	1280	1386	1280	1610	160	144	N121K, N171K, E280G
A/Poland/30540/2017	3C.2a1	2017-01-09	SIAT1	<	0	320	308	320	245	640	880	40	40	K92R, N121K, N171K, H310Q
A/Luxembourg/65905/2016	3C.2a1	2016-12-29	SIAT1	<	0	320	345	320	272	640	612	40	20	K92R, N121K, N171K, H310Q
A/Luxembourg/204/2017	3C.2a1	2017-01-03	MDCK1/SIAT1	<	0	160	153	80	80	320	457	40	40	N121K, I140M, N171K
A/Bucuresti/206677/2017	3C.2a1	2017-01-09	SIAT1/SIAT1	<	1	320	255	160	143	640	560	80	52	K2Q, N121K, I140M, N171K
A/Stockholm/47/2016	3C.2a1	2016-11-17	MDCK1/SIAT1	<	0	80	114	80	66	320	354	<	10	K2Q, R90X, N121K, I140M, N171K
A/Luxembourg/64296/2016	3C.2a1	2016-12-19	SIAT1	40	40	160	170	320	246	640	736	80	60	N121K, I140M, N171K, F193S
A/Netherlands/3793/2016	3C.2a1	2016-12-27	MDCK2/SIAT1	<	0	160	177	160	151	640	492	80	63	N121K, T135K (-CHO), N171K, Q197K
A/Edirne/38/2017	3C.2a1	2017-01-06	MDCK1/SIAT1	<	3	320	262	640	520	640	787	40	51	N121K, T135K (-CHO), N171K, Q197K
A/Athens GR/554/2017	3C.2a1	2017-01-19	SIAT1	40	20	320	248	160	146	640	640	40	58	N121K, N122D (-CHO), T135K (-CHO), N171K
A/Athens GR/558/2017	3C.2a1	2017-01-19	SIAT1	<	0	160	169	160	139	320	394	40	40	N121K, N122X (-CHO), T135K (-CHO), N171K
				Vaccine										

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-04-12

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹																		HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera																		
				A/HK 4801/14		A/HK 7295/14		A/Cote d'Ivoire 544/16		A/Norway 4436/16		A/Norway 3806/16		New 4465/16		New 4465/16		New 659/16		New 1943909/16		
				Egg	NIB F53/16	MDCK	F02/15	SIAT	NIB F54/16	SIAT	F03/17	Egg	NIB F45/16	SIAT	F11/17	F12/17	EGG	SIAT	F09/17	SIAT	F10/17	
				3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a	3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a	3C.2a	3C.2a1	3C.2a1			
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																						
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	320	320	431	320	462	2560	3584	160	167	160	219	N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, I140M, N171K N121K, I140M, T160K (-CHO), N171K, L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	251	80	71	320	444	40	44	160	171	160	133	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	993	160	168	640	946	80	77	640	520	640	503	
A/Norway/4436/2016	3C.2a1	2016-04-06	SIAT1	160	122	640	775	640	481	1280	992	160	120	1280	1185	160	147	640	516	640	845	
A/Norway/3806/2016 Is2 cl25	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	506	640	647	320	320	2560	2711	320	406	320	300	
TEST VIRUSES																						
A/Norway/4465/2016	3C.2a	2016-11-07	SIAT2	160	131	640	522	640	807	1280	1138	80	90	10240	7758	2560	2382	320	249	320	278	T131K, R142K, R261Q
A/Norway/4465/2016	3C.2a	2016-11-07	E6	320	283	320	298	160	151	160	140	160	172	2560	2217	2560	2627	160	133	80	83	T131K, R142K, T160K (-CHO), L194P,T203I, R261Q
A/Finland/659/2016	3C.2a1	2016-11-25	SIAT2	80	116	640	659	320	460	640	930	160	127	1280	1792	80	75	5120	3901	5120	4511	N121K, T135K (-CHO), N171K
A/Switzerland/1943909/2016	3C.2a1	2016-12-16	SIAT1	80	70	640	516	320	267	640	624	80	72	640	576	40	42	1280	1526	2560	2405	N121K, T135K (-CHO), N171K
A/Qatar/10-VI-17-0001178/2017	3C.2a1	2017-01-08	SIAT1	40	49	320	403	320	260	640	682	80	64	ND	ND	ND	ND	ND	ND	ND	ND	N121K, T135K (-CHO), N171K, Q197K
A/Qatar/10-VI-17-0001469/2017	3C.2a1	2017-01-09	SIAT1	80	62	320	434	640	542	1280	991	160	184	ND	ND	ND	ND	ND	ND	ND	ND	N121K, T135K (-CHO), N171K, Q197K
				Vaccine																		

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

ND = Not Done

Table 9-5 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-05-22

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹														HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera														
				A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Norway 4436/16 SIAT F03/17 3C.2a1		A/Norway 3806/16 Egg NIB F45/16 3C.2a1		New A/Qatar 10-VI-17-0001178/17 SIAT F16/17 3C.2a1		New A/Qatar 10-VI-17-0001469/17 SIAT F15/17 3C.2a1		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	780	640	601	320	272	320	320	320	431	640	483	80	99	N96S, T160K (-CHO), L194P
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80	61	640	570	160	144	320	251	80	71	640	508	80	72	none
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT2	80	118	1280	1768	2560	2384	1280	993	160	168	640	612	80	71	N121K, S144K, S198P
A/Norway/4436/2016	3C.2a1	2016-04-06	SIAT1	160	122	640	775	640	481	1280	992	160	120	640	518	80	106	N121K, I140M, N171K
A/Norway/3806/2016	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	506	640	647	ND	ND	ND	ND	N121K, I140M, T160K (-CHO), N171K, L194P
TEST VIRUSES																		
A/Estonia/103188/2016	3C.2a	2016-11-07	SIAT2	<	0	80	92	40	20	160	154	<	0	ND	ND	ND	ND	T131K, R142K, R261Q
A/Latvia/11-072125/2016	3C.2a	2016-11-07	E6	<	0	40	47	40	20	80	66	<	0	ND	ND	ND	ND	T131K, R142K, T160K (-CHO), L194P, T203I, R261Q
A/Qatar/10-VI-17-0001469/2017	3C.2a1	2017-01-09	SIAT1	80	62	320	434	640	542	1280	991	160	184	1280	960	160	152	N121K, T135K (-CHO), N171K
A/Spain/104138/2016	3C.2a1	2016-11-25	SIAT2	<	0	80	66	40	20	80	73	<	0	ND	ND	ND	ND	N121K, T135K (-CHO), N171K
A/Qatar/10-VI-17-0001178/2017	3C.2a1	2017-01-08	SIAT1	40	49	320	403	320	260	640	682	80	64	1280	974	160	233	N121K, T135K (-CHO), N171K, Q197K
				Vaccine														

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

ND = Not Done

Repeat assays

Table 9-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-08-02

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹												HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera												
				A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Oman 2585/16 SIAT NIB F50/16 3C.2a1		A/Greece 4/17 SIAT F27/17 3C.2a1		A/Greece 4/17 Egg F28/17 3C.2a1		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, N171K G78D, N121K, N122D(-CHO), T135K (-CHO), N171K G78D, N121K, N122D(-CHO), T135K(-CHO), T160K(-CHO), L194P, N171K, T203I
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	160	182	320	332	640	555	2560	2997	
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	640	680	1280	1457	2560	2150	160	147	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT3	160	150	640	908	1280	1234	1280	1335	2560	2090	80	77	
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT3	160	128	640	582	1280	1015	1280	1491	1280	1593	80	70	
A/Greece/4/2017	3C.2a1	2017-01-02	SIAT3	40	70	160	153	80	114	640	880	1280	1536	320	264	
A/Greece/4/2017	3C.2a1	2017-01-02	E9	160	236	160	165	80	105	160	218	640	728	5120	4129	
TEST VIRUSES																
A/Norway/2603/2017	3C.2a	2017-05-02	SIAT1	80	102	640	680	1280	1051	1280	1140	2560	2106	80	98	N121K, S144K
A/Bulgaria/589/2017	3C.2a	2017-03-23	SIAT1	80	91	1280	1214	1280	1646	1280	1280	2560	1920	80	118	T10M (-CHO), N121K, S144K
A/Bosnia and Herzegovina/122/2017	3C.2a	2017-03-02	SIAT1	320	270	1280	1801	2560	2095	2560	2040	2560	2093	320	274	N121K, S144K, R261Q
A/Jordan/20157/2017	3C.2a	2017-04-22	SIAT2	160	230	1280	1126	2560	2133	2560	2816	5120	4096	160	151	N121K, S144K, R261Q
A/Norway/2149/2017	3C.2a	2017-04-06	SIAT1	320	373	1280	1093	1280	975	1280	975	2560	2780	80	90	N121K, N122D (-CHO), S144K, S262N
A/Hong Kong/2291/2017	3C.2a	2017-05-23	SIAT1	160	128	2560	2453	1280	1457	2560	1991	1280	1474	80	110	N31S, D53N, R142G,S144R, N171K, I192T, Q197H
A/Bosnia and Herzegovina/127/2017	3C.2a1	2017-03-06	SIAT1	160	205	1280	1707	640	736	2560	3000	2560	2804	320	258	K92R, N121K, N171K, H311Q
A/Hong Kong/2279/2017	3C.2a1	2017-05-21	SIAT1	80	67	640	664	320	311	2560	1950	2560	2450	320	384	N121K, T135K (-CHO), N171K
A/Norway/2629/2017	3C.2a1	2017-05-04	SIAT1	40	53	320	345	320	320	640	396	2560	2364	640	762	K2Q, N121K, T135K (-CHO), N171K
A/Hong Kong/2274/2016	3C.2a1	2017-05-20	SIAT1	160	178	640	747	320	311	1280	1493	5120	4352	1280	1440	N121K, T135K (-CHO), N171K, K264E
A/Hong Kong/2277/2017	3C.2a1	2017-05-19	SIAT1	80	73	640	693	320	312	1280	1205	2560	2076	320	384	E62G, K92R, N121K, N133S/T135K (-CHO), R142G, N171K, H311Q
A/Hong Kong/2285/2017	3C.2a1	2017-05-23	SIAT1	80	62	320	431	160	145	1280	1006	2560	2304	320	285	E62G, K92R, N121K, T135K (-CHO), R142G, N171K, H311Q
				Vaccine												

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-7 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-08-09

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹ Post-infection ferret antisera												HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				A/HK 4801/14	A/HK 7295/14	A/Cote d'Ivoire 544/16	A/Oman 2585/16	A/Greece 4/17	A/Greece 4/17							
				Egg	MDCK	SIAT	SIAT	SIAT	Egg							
				NIB F53/16	F02/15	NIB F54/16	NIB F50/16	F27/17	F28/17							
				3C.2a	3C.2a	3C.2a	3C.2a1	3C.2a1	3C.2a1							
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	160	182	320	332	640	555	2560	2997	N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, N171K G78D, N121K, N122D (-CHO), T135K (-CHO), N171K G78D, N121K, N122D(-CHO), T135K(-CHO), T160K(-CHO), L194P, N171K, T203I
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	640	680	1280	1457	2560	2150	160	147	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT3	160	150	640	908	1280	1234	1280	1335	2560	2090	80	77	
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT3	160	128	640	582	1280	1015	1280	1491	1280	1593	80	70	
A/Greece/4/2017	3C.2a1	2017-01-02	SIAT3	40	70	160	153	80	114	640	880	1280	1536	320	264	
A/Greece/4/2017	3C.2a1	2017-01-02	E9	160	236	160	165	80	105	160	218	640	728	5120	4129	
TEST VIRUSES																
A/Iceland/69/2017	3C.3a	2017-03-09	MDCK1/SIAT1	40	40	40	55	40	20	80	67	40	52	40	40	I3L, K92R, T128A (-CHO), A138S, R142G, S144N (+CHO), Y159S, T160K (-CHO), Q197K, S198P, A271T, H311Q, S312N, K326R K92R, T131K, T160X (-CHO), R261Q A106T, T131K, R142G, R261Q T131K, R142K, A212T, R261Q T131K, R142K, A212T, R261Q S91N, N121K, N122D (-CHO), S144K, S262N
A/Egypt/3562/2017		2017-02-16	SIAT1	320	384	640	782	640	756	1280	1205	2560	3641	1280	1008	
A/Catalonia/3509691NS/2017	3C.2a	2017-02-02	C0/SIAT1	80	67	320	256	160	160	320	388	640	518	160	124	
A/Hong Kong/2230/2017	3C.2a	2017-05-19	Cx/SIAT1	160	213	640	564	320	290	640	739	640	825	80	62	
A/Hong Kong/2227/2017	3C.2a	2017-05-19	Cx/SIAT1	160	125	640	613	320	269	640	815	640	807	160	132	
A/Hong Kong/2261/2017	3C.2a	2017-05-19	Cx/SIAT1	320	280	640	624	320	296	640	773	640	845	80	67	
A/Nosy Be/2866/2017	3C.2a	2017-06-19	SIAT1	80	107	1280	989	1280	1280	1280	1454	2560	2188	80	106	
A/Serbia/2672/2017	3C.2a1	2017-02-03	SIAT1	80	64	160	237	160	181	320	455	1280	1017	640	548	N121K, T135K (-CHO), T160X (-CHO), N171K G78D, Y94H, N121K, N122D (-CHO), T135K (-CHO), N171K N121K, N122D (-CHO), T135K (-CHO), T160X (-CHO), N171K N121K, T135K (-CHO), N171K, Q197K N121K, T135K (-CHO), N171K, S198P K92R, N121K, T135K (-CHO), N171K, H311Q V43I, K92R, N121K, T135N (-/+CHO), N171K, H311Q
A/Thuringen/52/2017	3C.2a1	2017-03-13	C1/SIAT1	160	168	640	506	320	272	1280	1105	2560	2520	1280	1224	
A/Iceland/64/2017	3C.2a1	2017-03-08	MDCK1/SIAT1	40	45	80	90	80	88	320	268	640	747	320	301	
A/Austria/971918/2017	3C.2a1	2017-02-13	C1/SIAT1	80	94	160	218	160	227	640	614	1280	1778	1280	992	
A/Nordrhein-Westfalen/106/2017	3C.2a1	2017-03-17	C1/SIAT1	40	40	320	304	160	179	640	547	1280	1664	640	890	
A/Hong Kong/3161/2017	3C.2a1	2017-07-03	Cx/SIAT1	40	40	80	78	160	138	160	207	320	436	80	113	
A/Hong Kong/3163/2017	3C.2a1	2017-07-03	Cx/SIAT1	40	40	80	56	160	132	160	218	320	420	160	167	

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-8. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-08-17

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera										
				A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Oman 2585/16 SIAT NIB F50/16 3C.2a1		A/Greece 4/17 SIAT F27/17 3C.2a1		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, N171K G78D, N121K, N122D (-CHO), T135K (-CHO), N171K
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	160	182	320	332	640	555	
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	640	680	1280	1457	2560	2150	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT3	160	150	640	908	1280	1234	1280	1335	2560	2090	
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT3	160	128	640	582	1280	1015	1280	1491	1280	1593	
A/Greece/4/2017	3C.2a1	2017-01-02	SIAT3	40	70	160	153	80	114	640	880	1280	1536	
TEST VIRUSES														
A/Hong Kong/2263/2017	3C.2a	2017-05-19	Cx/SIAT1	80	63	160	197	160	120	320	438	640	499	S124R (-CHO), T131K, R142K, R261Q
A/Analavory/2821/2017	3C.2a	2017-06-20	SIAT1	40	40	640	699	640	627	1280	987	1280	1792	Q1K, D94N, N121K, N122D (-CHO), S144K, S262N
A/Antsirabe/2961/2017	3C.2a	2017-06-27	SIAT1	40	40	320	468	640	727	1280	960	1280	1009	S91N, N121K, N122D (-CHO), S144K, S262N
A/Austria/974918/2017	3C.2a1	2017-03-06	C1/SIAT1	40	20	320	299	320	292	1280	1024	1280	1368	K92R, N121K, R142G, N171K, H311Q
A/Hong Kong/2260/2017	3C.2a1	2017-05-19	Cx/SIAT1	<	0	320	303	160	142	640	790	640	901	R142G, N171K
				Vaccine										

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

ND = Not Done

Table 9-9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-08-21

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera										
				A/HK 4801/14 Egg NIB F53/16 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a	A/Oman 2585/16 SIAT NIB F50/16 3C.2a1	A/Greece 4/17 SIAT F27/17 3C.2a1						
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, N171K G78D, N121K, N122D(-CHO), T135K (-CHO), N171K
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	160	182	320	332	640	555	
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	640	680	1280	1457	2560	2150	
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT3	160	150	640	908	1280	1234	1280	1335	2560	2090	
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT3	160	128	640	582	1280	1015	1280	1491	1280	1593	
A/Greece/4/2017	3C.2a1	2017-01-02	SIAT3	40	70	160	153	80	114	640	880	1280	1536	
TEST VIRUSES														
A/Moldova/140.06/2017	3C.2a	2017-02-06	SIAT1	80	80	640	854	1280	1160	1280	1737	1280	1778	N121K
A/Poland/17C/2017	3C.2a	2017-03-01	SIAT1	40	40	320	359	640	500	640	563	1280	1040	V112X, N121K, S144K
A/Bulgaria/414/2017	3C.2a	2017-02-01	SIAT2/SIAT1	<	10	320	301	320	474	320	320	640	608	N121K, S144K, N158X (-CHO)
A/Slovenia/1126/2017	3C.2a	2017-02-14	MDCKx/SIAT1	40	44	640	549	640	823	640	815	1280	1280	N121K, S144K, I260V, S262G
A/Bulgaria/545/2017	3C.2a	2017-02-24	SIAT2/SIAT1	80	75	640	663	640	840	1280	1024	2560	1978	N121K, N122D (-CHO), S144K, S262N
A/Bulgaria/524/2017	3C.2a	2017-02-24	SIAT2/SIAT1	80	61	640	602	640	902	1280	972	2560	2133	G49S, N121K, N122D (-CHO), S144K, S262N
A/Lithuania/6583/2017	3C.2a	2017-02-21	MDCK2/SIAT1	<	3	320	349	320	427	320	263	160	180	N121K, S144K, N158X/T160X (-CHO), N216I
A/Hong Kong/2271/2017	3C.2a	2017-05-21	Cx/SIAT1	40	43	160	160	640	666	320	470	1280	1141	N121K, T135K (-CHO), S144K, R150K, R261Q, S312G
A/Hong Kong/2288/2017	3C.2a	2017-05-23	Cx/SIAT1	160	192	640	611	320	320	640	931	1280	1103	T131K, R142K, A212T
A/Hong Kong/2266/2017	3C.2a	2017-05-20	Cx/SIAT1	80	96	640	510	160	160	640	629	640	902	T131K, R142K, R261Q
A/Hong Kong/2272/2017	3C.2a	2017-05-20	Cx/SIAT1	40	45	160	149	80	77	320	414	640	544	T131K, R142K, A212T, R261Q, R307K
A/Hong Kong/2280/2017	3C.2a	2017-05-20	Cx/SIAT1	80	80	320	260	80	77	320	289	640	640	T131K, R142K, S145G, S198P, R261Q
A/Lithuania/6165/2017	3C.2a	2017-02-20	MDCK2/SIAT1	80	64	160	156	80	112	160	194	160	184	T131K, R142K, T160X (-CHO), R261Q, A304T
A/Poland/1963/2017	3C.2a1	2017-03-23	SIAT1	40	47	320	435	320	251	640	800	1280	1233	K92R, N121K, N171K, H311Q
A/Poland/19B/2017	3C.2a1	2017-02-23	SIAT1	80	64	640	499	320	320	1280	1193	1280	1840	K92R, N121K, N171K, H311Q
A/Bulgaria/435/2017	3C.2a1	2017-02-01	SIAT2/SIAT1	80	65	640	675	640	580	1280	1148	2560	2210	S9N, K92R, N121K, N171K, S198P, H311Q
A/Bulgaria/415/2017	3C.2a1	2017-02-01	SIAT2/SIAT1	80	80	320	437	160	151	640	640	1280	960	N121K, I140M, T160X (-CHO), N171K
A/Poland/16E/2017	3C.2a1	2017-02-22	SIAT1	80	80	320	369	320	345	1280	986	2560	3707	N121K, T135K (-CHO), N171K
A/Slovenia/1207/2017	3C.2a1	2017-02-19	SIATx/SIAT1	80	71	320	352	320	343	1280	1083	2560	2655	N121K, T135K (-CHO), N171K
A/Iceland/59/2017	3C.2a1	2017-03-04	MDCK1/SIAT1	40	53	320	252	160	160	640	875	2560	2120	N121K, T135K (-CHO), N171K
A/Bulgaria/471/2017	3C.2a1	2017-02-09	SIAT2/SIAT1	40	55	640	563	640	890	1280	1216	2560	2560	N121K, T135K (-CHO), N171K, K310R
A/Iceland/54/2017	3C.2a1	2017-03-01	MDCK1/SIAT1	40	40	160	126	80	76	640	544	1280	1242	N121K, N122D (-CHO), T135K (-CHO), N171K
A/Hong Kong/2286/2017	3C.2a1	2017-05-23	Cx/SIAT1	40	20	160	196	80	73	640	611	1280	1280	E62G, G78D, K92R, N121K, T135K (-CHO), R142G, N171K, R255K, H311Q
				Vaccine										

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-10. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-08-23

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
				Post-infection ferret antisera											
				A/HK 4801/14 Egg NIB F53/16 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a	A/Oman 2585/16 SIAT NIB F50/16 3C.2a1	A/Greece 4/17 SIAT F27/17 3C.2a1							
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES															
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	160	182	320	332	640	555	N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, N171K G78D, N121K, N122D (-CHO), T135K (-CHO), N171K	
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	640	680	1280	1457	2560	2150		
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT3	160	150	640	908	1280	1234	1280	1335	2560	2090		
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT3	160	128	640	582	1280	1015	1280	1491	1280	1593		
A/Greece/4/2017	3C.2a1	2017-01-02	SIAT3	40	70	160	153	80	114	640	880	1280	1536		
TEST VIRUSES															
A/Belgium/G448/2017	3C.2a	2017-02-16	SIATx/SIAT1	320	375	1280	1024	640	557	1280	1246	2560	1920	R142K, M168V, R261Q	
A/Khabarovsk/18/2017	3C.2a	not available	MDCK3/SIAT1	80	64	640	603	640	640	640	928	1280	1147	N121K, S144K	
A/Bremen/8/2017	3C.2a	2017-02-02	C1/SIAT1	80	74	640	853	1280	1120	1280	989	1280	1396	N121K, S144K, I260V	
A/Austria/971753/2017	3C.2a	2017-02-18	C1/SIAT1	80	67	640	493	640	812	640	947	1280	1006	N121K, S144K, S312N	
A/Belgium/G451/2017	3C.2a	2017-02-20	SIATx/SIAT1	160	146	640	747	1280	1259	1280	1280	1280	1396	I58V, N121K, S144K, S219Y, M268I	
A/Norway/1969/2017	3C.2a	2017-03-23	SIAT1/SIAT1	160	220	1280	1355	1280	1280	2560	2480	2560	2640	S9G, N121K, S144K, R261Q, V309I	
A/Sachsen/107/2017	3C.2a	2017-02-07	C1/SIAT1	40	40	160	207	320	272	320	280	640	488	E62G, N121K, N122D (-CHO), S144K, S262N, K276R	
A/Pays de Loire/1045/2017	3C.2a1	2017-02-15	MDCK1/SIAT1	160	221	640	876	640	488	1280	1821	2560	2504	N171K	
A/Belgium/G432/2017	3C.2a1	2017-02-16	SIATx/SIAT1	160	187	1280	1203	640	538	2560	2150	2560	2508	N171K	
A/Austria/971745/2017	3C.2a1	2017-02-18	C1/SIAT1	160	173	1280	1024	320	304	1280	1882	1280	1184	K92R, N121K, N171K, H311Q	
A/Paris/1141/2017	3C.2a1	2017-03-06	MDCK1/SIAT1	160	196	640	640	640	564	1280	1880	2560	2643	N121K, I140M, N171K	
A/Belgium/G444/2017	3C.2a1	2017-02-15	SIATx/SIAT1	80	80	640	604	640	480	1280	1656	5120	4352	N121K, T135K (-CHO), N171K	
A/Slovenia/994/2017	3C.2a1	2017-02-08	SIATx/SIAT1	320	250	640	630	640	500	1280	1455	5120	4985	N121K, T135K (-CHO), N171K	
A/St Petersburg/274/2017	3C.2a1	not available	MDCK1/SIAT1	40	40	80	74	80	68	160	237	640	501	N121K, T135K (-CHO), T160X (-CHO), N171K	
A/Austria/973209/2017	3C.2a1	2017-02-24	C1/SIAT1	80	80	640	540	320	312	1280	1196	2560	3047	N121K, T135K (-CHO), N171K, L177X	
A/Norway/1849/2017	3C.2a1	2017-03-16	SIAT1/SIAT1	320	420	640	640	640	645	2560	2418	5120	4211	T10K (-CHO), N121K, T135K (-CHO), N171K	
A/Norway/2171/2017	3C.2a1	2017-04-02	SIAT1/SIAT1	160	139	640	787	640	741	2560	1991	5120	4608	G78D, Y94H, N121K, N122D (-CHO), N171K	
A/Slovenia/1105/2017	3C.2a1	2017-02-13	MDCKx/SIAT1	160	197	1280	1043	640	756	1280	1669	2560	2337	I58V, K92R, N121K, N171K, Q197R, H311Q	
A/Bayern/87/2017	3C.2a1	2017-03-09	C1/SIAT1	80	99	640	827	320	249	640	938	1280	1778	R33Q, D53E, Q75H, K92R, N121K, N171K, H311Q	
				Vaccine											

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-11. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-08-31

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				A/HK 4801/14 Egg NIB F53/16 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a	A/Oman 2585/16 SIAT NIB F50/16 3C.2a1	A/Greece 4/17 SIAT F27/17 3C.2a1						
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES													N96S, T160K (-CHO), L194P none N121K, S144K, S198P N121K, N171K G78D, N121K, N122D (-CHO), T135K (-CHO), N171K	
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	160	182	320	332	640		555
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	640	680	1280	1457	2560		2150
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT3	160	150	640	908	1280	1234	1280	1335	2560		2090
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT3	160	128	640	582	1280	1015	1280	1491	1280		1593
A/Greece/4/2017	3C.2a1	2017-01-02	SIAT3	40	50	160	145	80	101	640	688	1280	1440	
TEST VIRUSES														
A/Novgorog Velikiy/67/2017		2017-02-17	SIAT1/SIAT1	80	64	160	210	160	160	640	533	640	782	
A/Antananarivo/2531/2017		2017-05-30	SIAT1	<	10	320	463	640	566	640	861	1280	1076	
A/Moscow/33/2017	3C.2a	2017-02-06	MDCK1/SIAT1	<	5	320	417	640	514	640	568	640	853	N121K, N122D (-CHO), S144K, G240R, S262N
A/Orenburg/71/2017	3C.2a	2017-02-07	SIAT1/SIAT1	40	20	320	433	320	451	640	579	640	909	N121K, N122D (-CHO), S144K, S262N, Y308F
A/Orenburg/72/2017	3C.2a	2017-02-13	SIAT1/SIAT1	<	5	80	80	320	279	40	55	20	20	K83Q, N121K, S144K, T160K (-CHO)
A/England/27/2017	3C.2a	2017-02-15	SIAT1/SIAT1	40	20	320	410	640	582	640	840	1280	1031	V112F, N121K, N122D (-CHO), S144K, S262N
A/Moscow/40/2017	3C.2a	2017-02-17	SIAT1/SIAT1	40	46	320	356	320	411	640	522	640	554	N121K, S144K
A/Antsirabe/1076/2017	3C.2a	2017-03-21	SIAT1	40	20	320	444	320	414	640	667	1280	980	I3L, S91N, N121K, N122D (-CHO), S144K, S262N
A/Antananarivo/1563/2017	3C.2a	2017-04-10	SIAT1	40	46	160	129	80	88	320	312	320	427	T131K, R142K, R261Q
A/Antananarivo/1621/2017	3C.2a	2017-04-13	SIAT1	40	20	640	527	640	487	640	788	1280	1040	N121K, N122D (-CHO), S144K, S262N, I282V
A/Moramanga/1907/2017	3C.2a	2017-04-26	SIAT1	<	0	640	487	1280	1006	640	835	1280	1074	S91N, N121K, N122D (-CHO), S144K, S262N
A/Antananarivo/2103/2017	3C.2a	2017-05-03	MDCK2/SIAT1	<	0	160	220	320	302	320	368	1280	1006	S9R, N121K, N122D (-CHO), S144K, S262N
A/Antananarivo/2396/2017	3C.2a	2017-05-15	SIAT1	<	3	320	315	640	501	640	573	640	782	I34V, N121K, N122D (-CHO), S144K, S262N
A/Fianarantsoa/2442/2017	3C.2a	2017-05-19	SIAT1	40	40	320	415	640	487	640	640	640	920	N121K, N122D (-CHO), S144K, S262N
A/Nosy Be/2610/2017	3C.2a	2017-05-24	SIAT1	<	10	640	528	640	600	1280	1351	1280	1509	I34V, N121K, S144K, S262N
A/Toamasina/2757/2017	3C.2a	2017-06-13	SIAT1	40	40	640	498	640	556	640	689	1280	1019	N121K, N122D (-CHO), S144K, S262N
A/penza/77/2017	3C.2a1	2017-02-01	SIAT1/SIAT1	<	1	160	146	80	62	640	469	640	563	E50K, K92R, N121K, N171K, H311Q
A/Lisboa/15/2017	3C.2a1	2017-02-01	SIAT1/SIAT1	40	40	320	306	160	146	640	683	1280	1227	N171K
A/England/30/2017	3C.2a1	2017-02-13	MDCK1/SIAT1	40	40	320	260	160	209	640	743	640	862	I58V, K92R, N121K, N171K, Q197R, H311Q
A/England/32/2017	3C.2a1	2017-02-28	SIAT1/SIAT1	40	20	80	80	80	71	640	668	1280	1214	N121K, N122D (-CHO), T135K (-CHO), N171K, I192T
A/Antananarivo/2398/2017	3C.2a1	2017-05-16	SIAT1	40	20	320	251	80	112	640	514	640	779	S47T, G78S, N171K
				Vaccine										

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-12. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-09-04

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera										
				A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Norway 3806/16 Egg NIB F45/16 3C.2a1		A/Norway 4436/16 SIAT F03/17 3C.2a1		A/Greece 4/17 Egg F28/17 3C.2a1		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	320	431	320	320	2560	2997	N96S, T160K (-CHO), L194P none N121K, I140M, T160K (-CHO), N171K, L194P N121K, I140M, N171K G78D, N121K, N122D, T135K, T160K, L194P, N171K, T203I
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	80	71	320	251	160	147	
A/Norway/3806/2016 Is2 cl25	3C.2a1	2016-06-13	E7	640	733	640	512	640	647	160	120	2560	2440	
A/Norway/4436/2016	3C.2a1	2016-11-03	SIAT1	160	122	320	775	640	506	1280	992	80	67	
A/Greece/4/2017	3C.2a1	2017-01-02	E9	160	236	160	165	320	300	160	130	5120	4129	
TEST VIRUSES														
A/Bulgaria/545/2017	3C.2a	2017-02-24	SIAT2/SIAT1	160	154	640	886	<	10	640	840	40	40	N121K, N122D (-CHO), S144K, S262N
A/Poland/17C/2017	3C.2a	2017-03-01	SIAT1	40	40	320	465	40	55	640	709	40	56	V112X, N121K, S144K
A/Saudi Arabia/1127975090/2017	3C.2a	2017-03-11	SIAT1	40	20	640	681	40	20	640	951	40	40	N31S, D53N, R142G, S144R, N171K, I192T, Q197H
A/Norway/2149/2017	3C.2a	2017-04-06	SIAT1	80	80	640	629	80	80	1280	1862	80	67	N121K, N122D (-CHO), S144K, S262N
A/Jordan/20157/2017	3C.2a	2017-04-22	SIAT2	80	60	640	547	80	69	640	608	80	78	N121K, S144K, R261Q
A/Norway/2603/2017	3C.2a	2017-05-02	SIAT1	40	40	640	640	80	62	640	866	80	64	N121K, S144K
A/Antananarivo/2396/2017	3C.2a	2017-05-15	SIAT1	40	20	640	597	80	66	640	779	80	62	I34V, N121K, N122D (-CHO), S144K, S262N
A/Hong Kong/2266/2017	3C.2a	2017-05-20	Cx/SIAT1	80	80	320	397	40	42	640	568	40	40	T131K, R142K, R261Q
A/Hong Kong/2272/2017	3C.2a	2017-05-20	Cx/SIAT1	40	43	160	119	40	47	320	265	80	61	T131K, R142K, A212T, R261Q, R307K
A/Hong Kong/2280/2017	3C.2a	2017-05-20	Cx/SIAT1	80	80	320	259	40	58	640	514	80	64	T131K, R142K, S145G, S198P, R261Q
A/Hong Kong/2291/2017	3C.2a	2017-05-23	SIAT1	40	40	640	837	40	20	640	626	40	40	N31S, D53N, R142G,S144R, N171K, I192T, Q197H
A/Antsirabe/2961/2017	3C.2a	2017-06-27	SIAT1	40	40	640	519	80	60	640	614	40	56	S91N, N121K, N122D (-CHO), S144K, S262N
A/Iceland/59/2017	3C.2a1	2017-03-04	MDCK1/SIAT1	40	20	320	280	40	40	640	640	640	587	N121K, T135K (-CHO), N171K
A/Paris/1141/2017	3C.2a1	2017-03-06	MDCK1/SIAT1	40	40	320	416	40	58	640	680	40	59	N121K, I140M, N171K
A/Bayern/87/2017	3C.2a1	2017-03-09	C1/SIAT1	40	48	320	286	40	40	320	311	40	41	R33Q, D53E, Q75H, K92R, N121K, N171K, H311Q
A/Thuringen/52/2017	3C.2a1	2017-03-13	C1/SIAT1	80	60	320	300	80	75	640	480	640	551	G78D, Y94H, N121K, N122D (-CHO), T135K (-CHO), N171K
A/Nordrhein-Westfalen/106/2017	3C.2a1	2017-03-17	C1/SIAT1	40	20	160	152	40	20	320	347	320	280	N121K, T135K (-CHO), N171K, S198P
A/Norway/2171/2017	3C.2a1	2017-04-02	SIAT1/SIAT1	80	60	320	265	80	104	640	570	640	714	G78D, Y94H, N121K, N122D (-CHO), N171K
A/Norway/2629/2017	3C.2a1	2017-05-04	SIAT1	40	20	320	253	40	49	320	265	320	340	K2Q, N121K, T135K (-CHO), N171K
A/Antananarivo/2398/2017	3C.2a1	2017-05-16	SIAT1	80	66	320	421	80	63	640	889	80	69	S47T, G78S, N171K
A/Hong Kong/2277/2017	3C.2a1	2017-05-19	SIAT1	40	40	320	368	80	60	640	690	160	160	E62G, K92R, N121K, N133S/T135K (-CHO), R142G, N171K, H311Q
A/Hong Kong/2260/2017	3C.2a1	2017-05-19	Cx/SIAT1	40	52	320	287	40	56	320	443	40	52	R142G, N171K
A/Hong Kong/2274/2016	3C.2a1	2017-05-20	SIAT1	80	60	320	392	80	80	640	616	640	614	N121K, T135K (-CHO), N171K, K264E
A/Hong Kong/3163/2017	3C.2a1	2017-07-03	Cx/SIAT1	40	40	80	90	80	60	160	228	160	120	V43I, K92R, N121K, T135N (-/+CHO), N171K, H311Q
				Vaccine										

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-13. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2017-09-07

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
				A/HK 4801/14 Egg NIB F53/16 3C.2a	A/HK 7295/14 MDCK F02/15 3C.2a	A/Norway 3806/16 Egg NIB F45/16 3C.2a1	A/Norway 4436/16 SIAT F03/17 3C.2a1	A/Greece 4/17 Egg F28/17 3C.2a1							
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES															
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	320	431	320	320	2560	2997	N96S, T160K (-CHO), L194P	
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	80	71	320	251	160	147	none	
A/Norway/3806/2016 Is2 cl25	3C.2a1	2016-06-13	E7	640	733	640	512	640	647	160	120	2560	2440	N121K, I140M, T160K (-CHO), N171K, L194P	
A/Norway/4436/2016	3C.2a1	2016-11-03	SIAT1	160	122	320	775	640	506	1280	992	80	67	N121K, I140M, N171K	
A/Greece/4/2017	3C.2a1	2017-01-02	E9	160	236	160	165	320	300	160	130	5120	4129	G78D, N121K, N122D, T135K, T160K, L194P, N171K, T203I	
TEST VIRUSES															
A/Saudi Arabia/256339/2017	3C.2a	2017-03-24	SIAT1	40	40	640	892	160	130	1280	1076	80	93	N121K, N122D (-CHO), S144K, S262N	
A/Saudi Arabia/1001498961/2017	3C.2a	2017-03-22	SIAT1	40	48	640	837	40	40	1280	1158	40	57	N121K, S144K	
A/Omsk/06/2017	3C.2a	2017-03-08	MDCK2/SIAT1	80	60	160	160	40	46	320	241	40	53	T131K, R142K, T160I (-CHO), R261Q	
A/Saudi Arabia/252773/2017	3C.2a	2017-03-14	SIAT1	<	5	1280	1129	<	1	1280	1338	40	20	N31S, D53N, R142G, S144R, N171K, I192T, Q197H, R261Q	
A/Saudi Arabia/252925/2017	3C.2a	2017-03-15	SIAT1	<	5	320	286	40	32	640	593	640	714	S54N, N121K, N122D (-CHO), T135K (-CHO), S144K, S262N	
A/Hong Kong/2273/2017	3C.2a	2017-05-21	Cx/SIAT1	40	56	160	229	40	50	640	486	40	56	T131K, R142G, A212T, R261Q	
A/South Africa/R05624/2017	3C.2a	2017-07-01	MDCK1/SIAT1	40	40	640	591	40	40	640	640	40	40	Q57R, N121K, S144K, I236V	
A/Hong Kong/3115/2017	3C.2a	2017-07-01	Cx/SIAT1	80	116	1280	1120	80	113	1280	1813	80	91	T131K, R142G, S198P, N246K (-CHO), R261Q	
A/Saudi Arabia/255031/2017	3C.2a1	2017-03-21	SIAT1	40	47	320	457	40	53	640	941	40	59	K92R, N121K, N171K, H311Q	
A/Dijon/1176/2017	3C.2a1	2017-03-14	MDCK1/SIAT1	40	20	320	301	40	45	640	747	640	782	S46T, N121K, T135K (-CHO), N171K	
A/Paris/1145/2017	3C.2a1	2017-03-06	MDCK1/SIAT1	40	59	640	491	80	68	640	907	40	50	N171K	
A/Basse Normandie/1055/2017	3C.2a1	2017-02-20	MDCK1/SIAT1	40	46	640	533	80	68	1280	1184	80	111	N121K, I140M, N171K	
A/South Africa/R06830/2017	3C.2a1	2017-07-18	MDCK1/SIAT1	<	5	320	420	40	20	640	600	40	52	E62G, K92R, N121K, R142G, N171K, H311Q	
A/South Africa/R06645/2017	3C.2a1	2017-07-17	MDCK1/SIAT1	40	20	320	260	<	10	640	556	40	55	I34V, E62G, K92R, N121K, I140M, R142G, N171K, H311Q	
A/South Africa/R06622/2017	3C.2a1	2017-07-14	MDCK1/SIAT1	40	42	320	260	40	40	640	500	80	70	I34V, E62G, K92R, N121K, I140M, R142G, T160X (-CHO), N171K, H311Q	
A/South Africa/R06488/2017	3C.2a1	2017-07-13	MDCK1/SIAT1	<	5	640	547	<	3	1280	987	40	40	I34V, E62G, K92R, N121K, R142G, N171K, H311Q	
A/South Africa/R05502/2017	3C.2a1	2017-06-29	MDCK1/SIAT1	40	41	320	410	40	51	640	536	320	448	N121K, T135K (-CHO), N171K, Q197K	
A/South Africa/R05477/2017	3C.2a1	2017-06-28	MDCK1/SIAT1	40	40	640	608	40	40	640	818	80	60	E62G, K92R, N121K, R142G, N171K, H311Q	
A/South Africa/R05138/2017	3C.2a1	2017-06-26	MDCK1/SIAT1	80	67	640	614	80	68	1280	1203	1280	1066	N121K, T135K (-CHO), N171K, Q197K	
A/South Africa/R04468/2017	3C.2a1	2017-06-13	MDCK1/SIAT1	40	56	80	98	80	62	160	137	320	336	N121K, T135K (-CHO), T160A (-CHO), N171K, Q197K	
A/South Africa/R04939/2017	3C.2a1	2017-06-20	MDCK1/SIAT1	40	40	160	157	40	58	320	313	640	560	N121K, T135K (-CHO), N171K, Q197K	
A/South Africa/R05728/2017	3C.2a1	2017-06-30	MDCK1/SIAT1	40	40	320	253	80	65	320	473	640	480	N121K, T135K (-CHO), N171K, Q197K	
				Vaccine											

Vaccine

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-14. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary

Neutralisation titre ¹																					
Post-infection ferret antisera																					
Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	A/HK 4801/14 Egg NIB F53/16 3C.2a		A/HK 7295/14 MDCK F02/15 3C.2a		A/Cote d'Ivoire 544/16 SIAT NIB F54/16 3C.2a		A/Oman 2585/16 SIAT NIB F50/16 3C.2a1		A/Norway 3806/16 Egg NIB F45/16 3C.2a1		A/Norway 4436/16 SIAT F03/17 3C.2a1		A/Greece 4/17 SIAT F27/17 3C.2a1		A/Greece 4/17 Egg F28/17 3C.2a1			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read
REFERENCE VIRUSES																					
A/Hong Kong/4801/2014 Is1 cl36	3C.2a	2014-02-26	E7	640	686	320	410	160	182	320	332	320	431	320	320	640	555	2560	2997		
A/Hong Kong/7295/2014	3C.2a	2014-08-07	SIAT2	80	111	640	636	640	680	1280	1457	80	71	320	251	2560	2150	160	147		
A/Cote d'Ivoire/544/2016	3C.2a	2016-04-06	P1/SIAT3	160	150	640	908	1280	1234	1280	1335	ND	ND	ND	ND	2560	2090	80	77		
A/Oman/2585/2016	3C.2a1	2016-03-04	SIAT3	160	128	640	582	1280	1015	1280	1491	ND	ND	ND	ND	1280	1593	80	70		
A/Norway/3806/2016 Is2 cl25	3C.2a1	2016-06-13	E7	640	733	640	512	320	365	640	727	640	647	160	120	640	560	2560	2440		
A/Norway/4436/2016	3C.2a1	2016-11-03	SIAT1	160	122	320	775	640	481	1280	1406	640	506	1280	992	1280	1280	80	67		
A/Greece/4/2017	3C.2a1	2017-01-02	SIAT3	40	70	160	153	80	114	640	880	ND	ND	ND	ND	1280	1536	320	264		
A/Greece/4/2017	3C.2a1	2017-01-02	E9	160	236	160	165	80	105	160	218	320	300	160	130	640	728	5120	4129		
TEST VIRUS SUBCLADE	Number tested			57	56	57	56	48	42	48	42	20	26	20	26	48	42	27	33		
Number and % of viruses reacting ≤4-fold reduced compared to the homologous titre																					
3C.2a				11	19.3%	56	98.2%	41	85.4%	46	95.8%	1	5.0%	20	100.0%	45	93.8%	0			
3C.2a1				11	19.6%	50	89.3%	24	57.1%	39	92.9%	0		24	92.3%	42	100%	4	12.1%		
HA1 amino acid substitutions																		Number ≤4-fold reduced compared to the homologous titre/ number tested & % neutralised			
																		No sequenced	Comment		
																		5	3C.3a		
3C.2a	N121K, S144K			4/18	22.2%	17/18	94.4%	16/16	100%	15/16	93.8%	0/5		5/5	100%	14/16	87.5%	0/7	74		
3C.2a	N121K, N122D (-CHO), S144K, S262N			1/18	5.6%	18/18	100%	17/17	100%	17/17	100%	1/5	20.0%	5/5	100%	17/17	100%	0/6	40		
3C.2a	T131K, R142G/K, R261Q			4/13	30.8%	13/13	100%	4/10	40.0%	9/10	90.0%	0/6		6/6	100%	9/10	90.0%	0/9	79		
3C.2a	N31S, D53N, R142G,S144R, N171K, I192T, Q197H			1/3	33.3%	3/3	100%	1/1	100%	1/1	100%	0/3		3/3	100%	1/1	100%	0/3	10		
3C.2a	Others (e.g. N121K only, +T135K)			1/5	20.0%	5/5	100%	3/4	75.0%	4/4	100%	0/1		1/1	100%	4/4	100%	0/2	8		
																		211			
3C.2a1	N171K			2/5	40.0%	5/5	100%	2/4	50.0%	4/4	100%	0/2		2/2	100%	4/4	100%	0/2	11		
3C.2a1	N121K, N171K			3/10	30.0%	10/10	100%	7/9	77.8%	9/9	100%	0/2		2/2	100%	9/9	100%	0/3	22		
3C.2a1	N121K, I140M, N171K			1/3	33.3%	3/3	100%	1/2	50.0%	2/2	100%	0/2		2/2	100%	2/2	100%	0/2	9		
3C.2a1	N121K, T135K (-CHO), N171K			3/22	13.6%	19/22	86.4%	10/16	62.5%	14/16	87.5%	0/10		9/10	90.0%	16/16	100%	3/14	59		
3C.2a1	N121K, N122D (-CHO), T135K (-CHO), N171K			1/4	25.0%	2/4	50.0%	1/4	25.0%	4/4	100%	0/1		1/1	100%	4/4	100%	1/2	50.0%		
3C.2a1	above five with R142G			0/10		10/10	100%	2/5	40.0%	5/5	100%	0/7		7/7	100%	5/5	100%	0/8	49		
3C.2a1	Others (e.g. T135N)			1/2	50.0%	1/2	50.0%	1/2	50.0%	1/2	50.0%	0/2		1/2	50%	2/2	100%	0/2	9		
																		386	170		
Vaccine																					

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

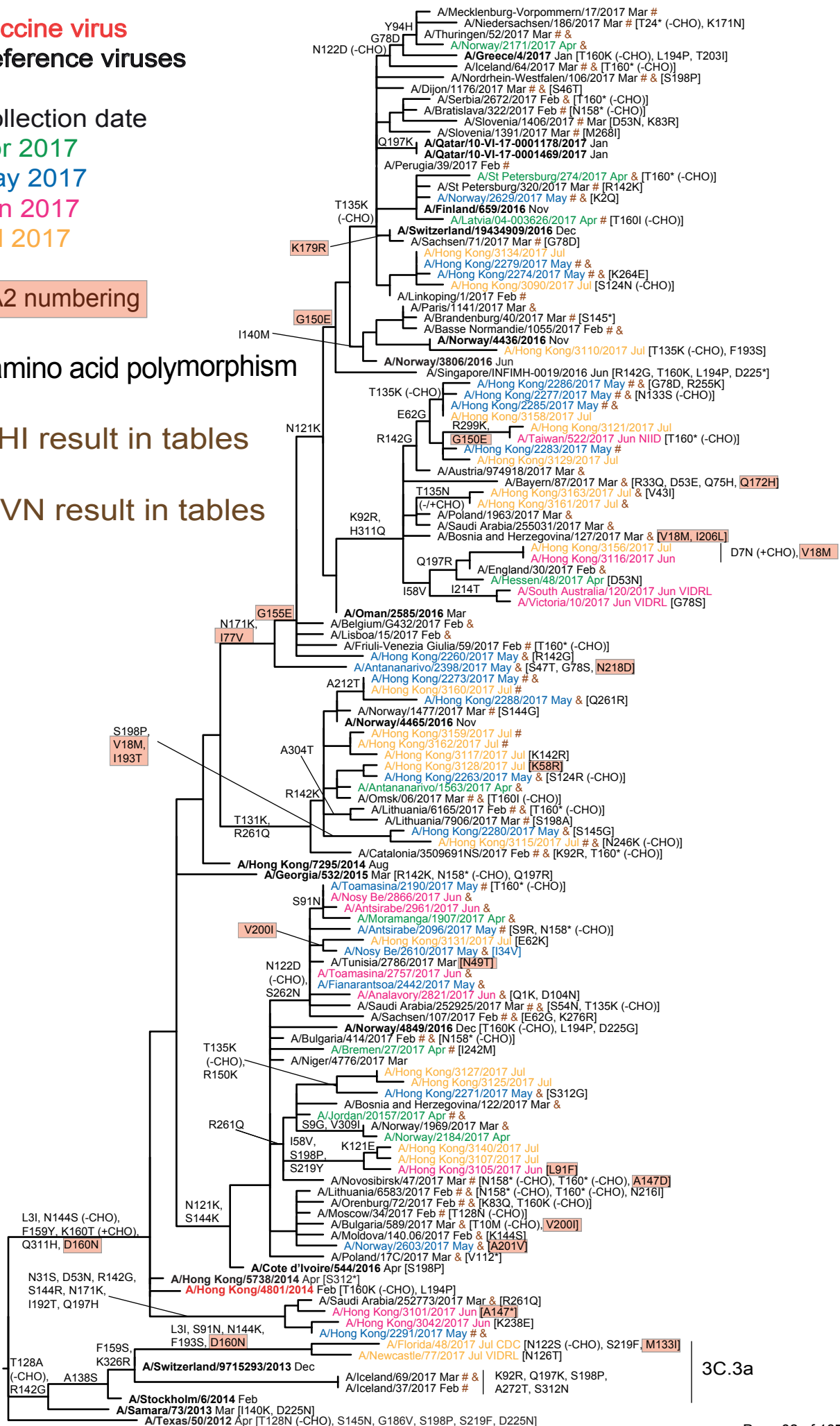
Vaccine virus

Reference viruses

Apr 2017
May 2017
Jun 2017
Jul 2017

* amino acid polymorphism

& VN result in tables



3C.2a

3C.3a

Figure 7. HA protein alignment for influenza A(H3N2) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100		
A/Texas/50/2012	Q	K	L	P	G	N	D	N	S	T	A	3C.1
A/Samara/73/2013	Q	K	L	P	G	N	D	N	S	T	A	3C.3
A/Stockholm/6/2014	Q	K	L	P	G	N	D	N	S	T	A	3C.3a
A/Iceland/69/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.3a
A/Switzerland/9715293/2013	Q	K	L	P	G	N	D	N	S	T	A	3C.3a
A/Florida/48/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.3a
A/Hong Kong/3042/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Saudi Arabia/252773/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Hong Kong/4801/2014	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Hong Kong/5738/2014	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Cote D'Ivoire/544/2016	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Norway/2603/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Hong Kong/3140/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Norway/2184/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Hong Kong/3125/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Analavory/2821/2017	K	I	.	.	.	.	.	.	.	.	.	3C.2a
A/Hong Kong/3131/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Antsirabe/2961/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Hong Kong/3115/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Hong Kong/3160/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a
A/Antananarivo/2398/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Oman/2585/2016	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/South Australia/120/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Hong Kong/3156/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Hong Kong/3163/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Hong Kong/3121/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Hong Kong/2286/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Hong Kong/3110/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Hong Kong/3090/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Latvia/04-003626/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Greece/4/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1
A/Norway/2171/2017	Q	K	L	P	G	N	D	N	S	T	A	3C.2a1

	110	120	130	140	150	160	170	180	190	200		
A/Texas/50/2012	D	V	P	D	Y	A	S	L	R	S	L	
A/Samara/73/2013	D	V	P	D	Y	A	S	L	R	S	L	
A/Stockholm/6/2014	D	V	P	D	Y	A	S	L	R	S	L	
A/Iceland/69/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Switzerland/9715293/2013	D	V	P	D	Y	A	S	L	R	S	L	
A/Florida/48/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3042/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Saudi Arabia/252773/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/4801/2014	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/5738/2014	D	V	P	D	Y	A	S	L	R	S	L	
A/Cote D'Ivoire/544/2016	D	V	P	D	Y	A	S	L	R	S	L	
A/Norway/2603/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3140/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Norway/2184/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3125/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Analavory/2821/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3131/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Antsirabe/2961/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3115/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3160/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Antananarivo/2398/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Oman/2585/2016	D	V	P	D	Y	A	S	L	R	S	L	
A/South Australia/120/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3156/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3163/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3121/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/2286/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3110/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Hong Kong/3090/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Latvia/04-003626/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Greece/4/2017	D	V	P	D	Y	A	S	L	R	S	L	
A/Norway/2171/2017	D	V	P	D	Y	A	S	L	R	S	L	

	210	220	230	240	250	260	270	280	290	300		
A/Texas/50/2012	R	I	T	V	S	T	K	R	S	Q	A	
A/Samara/73/2013	R	I	T	V	S	T	K	R	S	Q	A	
A/Stockholm/6/2014	R	I	T	V	S	T	K	R	S	Q	A	
A/Iceland/69/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Switzerland/9715293/2013	R	I	T	V	S	T	K	R	S	Q	A	
A/Florida/48/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3042/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Saudi Arabia/252773/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/4801/2014	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/5738/2014	R	I	T	V	S	T	K	R	S	Q	A	
A/Cote D'Ivoire/544/2016	R	I	T	V	S	T	K	R	S	Q	A	
A/Norway/2603/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3140/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Norway/2184/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3125/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Analavory/2821/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3131/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Antsirabe/2961/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3115/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3160/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Antananarivo/2398/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Oman/2585/2016	R	I	T	V	S	T	K	R	S	Q	A	
A/South Australia/120/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3156/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3163/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3121/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/2286/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3110/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Hong Kong/3090/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Latvia/04-003626/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Greece/4/2017	R	I	T	V	S	T	K	R	S	Q	A	
A/Norway/2171/2017	R	I	T	V	S	T	K	R	S	Q	A	

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	310	320	330	340	350	360	370	380	390	400
A/Texas/50/2012	TYGACPRYVKQSTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNKFEHFQIEKEFS									
A/Samara/73/2013										
A/Stockholm/6/2014										
A/Iceland/69/2017		N	R							
A/Switzerland/9715293/2013			R							
A/Florida/48/2017			R							
A/Hong Kong/3042/2017		H								
A/Saudi Arabia/252773/2017		H								
A/Hong Kong/4801/2014		H								
A/Hong Kong/5738/2014		HX								
A/Cote D'Ivoire/544/2016		H								
A/Norway/2603/2017		H								
A/Hong Kong/3140/2017		H								
A/Norway/2184/2017		I H								
A/Hong Kong/3125/2017		H								
A/Analavory/2821/2017		H								
A/Hong Kong/3131/2017		H								
A/Antsirabe/2961/2017		H								
A/Hong Kong/3115/2017		H			M					
A/Hong Kong/3160/2017		H								
A/Antananarivo/2398/2017		H								
A/Oman/2585/2016		H								
A/South Australia/120/2017										
A/Hong Kong/3156/2017					M					
A/Hong Kong/3163/2017										
A/Hong Kong/3121/2017										
A/Hong Kong/2286/2017										
A/Hong Kong/3110/2017		H								
A/Hong Kong/3090/2017		H								
A/Latvia/04-003626/2017		H								
A/Greece/4/2017		H								
A/Norway/2171/2017		H								

	410	420	430	440	450	460	470	480	490	500
A/Texas/50/2012	EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENGHTIDLTSEMKNLFKTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF									
A/Samara/73/2013										
A/Stockholm/6/2014										
A/Iceland/69/2017										
A/Switzerland/9715293/2013										
A/Florida/48/2017						I			N	
A/Hong Kong/3042/2017									N	
A/Saudi Arabia/252773/2017									N	
A/Hong Kong/4801/2014									N	
A/Hong Kong/5738/2014									N	
A/Cote D'Ivoire/544/2016									N	
A/Norway/2603/2017									N	
A/Hong Kong/3140/2017									N	
A/Norway/2184/2017									N	
A/Hong Kong/3125/2017									N	
A/Analavory/2821/2017									N	
A/Hong Kong/3131/2017									N	
A/Antsirabe/2961/2017									N	
A/Hong Kong/3115/2017									N	
A/Hong Kong/3160/2017									N	
A/Antananarivo/2398/2017		V							N	
A/Oman/2585/2016		V						E	N	
A/South Australia/120/2017		V						E	N	
A/Hong Kong/3156/2017		V						E	N	
A/Hong Kong/3163/2017		V						E	N	
A/Hong Kong/3121/2017		V					E	E	N	
A/Hong Kong/2286/2017		V						E	N	
A/Hong Kong/3110/2017		V					E	E	N	
A/Hong Kong/3090/2017		V					E	E	N	
A/Latvia/04-003626/2017		V					E	E	N	
A/Greece/4/2017		V					E	E	N	
A/Norway/2171/2017		V					E	E	N	

	510	520	530	540	550
A/Texas/50/2012	QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI				
A/Samara/73/2013					
A/Stockholm/6/2014					
A/Iceland/69/2017					
A/Switzerland/9715293/2013					
A/Florida/48/2017					
A/Hong Kong/3042/2017					
A/Saudi Arabia/252773/2017					
A/Hong Kong/4801/2014					
A/Hong Kong/5738/2014					
A/Cote D'Ivoire/544/2016					
A/Norway/2603/2017			V		
A/Hong Kong/3140/2017					
A/Norway/2184/2017		X			
A/Hong Kong/3125/2017					
A/Analavory/2821/2017					
A/Hong Kong/3131/2017			I		
A/Antsirabe/2961/2017					
A/Hong Kong/3115/2017			T		
A/Hong Kong/3160/2017					
A/Antananarivo/2398/2017				D	
A/Oman/2585/2016					
A/South Australia/120/2017					
A/Hong Kong/3156/2017					
A/Hong Kong/3163/2017					
A/Hong Kong/3121/2017					
A/Hong Kong/2286/2017					
A/Hong Kong/3110/2017					
A/Hong Kong/3090/2017					
A/Latvia/04-003626/2017					
A/Greece/4/2017					
A/Norway/2171/2017					

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 8. Locations of the main groups of amino acid substitutions in clade 3C.2a viruses

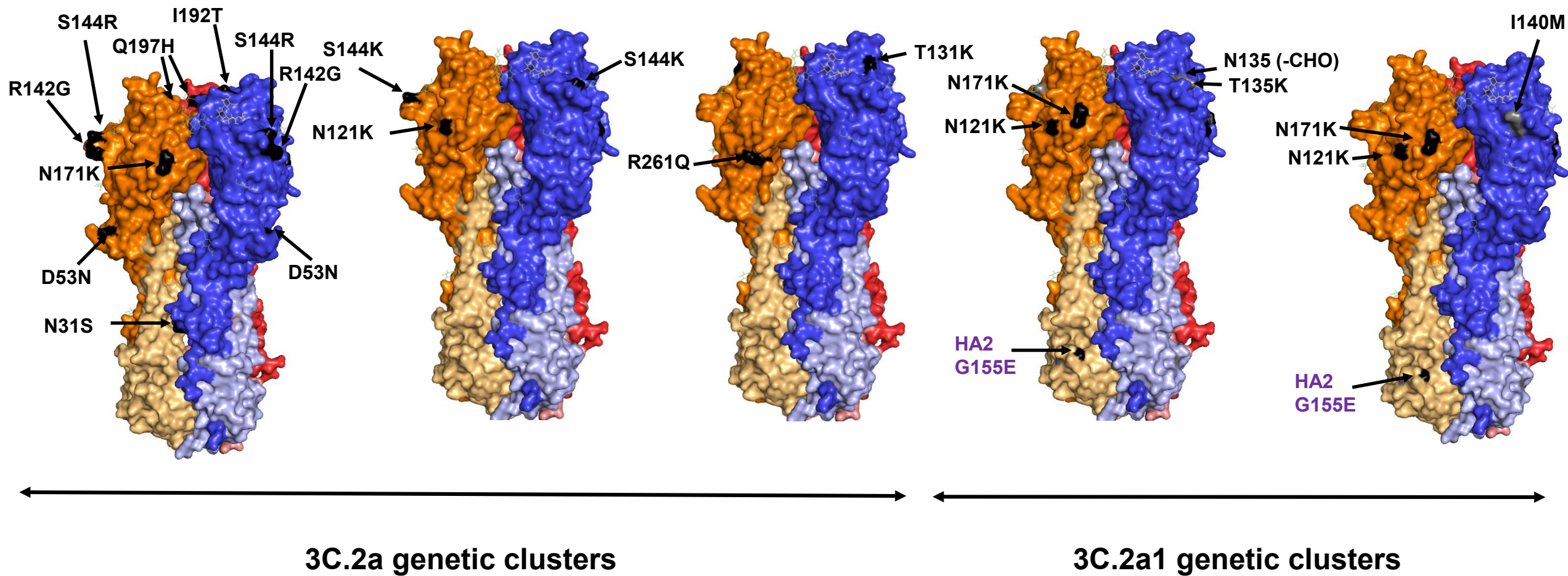


Figure 9. Phylogenetic comparison of A(H3N2) NA genes

Vaccine virus
Reference viruses

Collection date

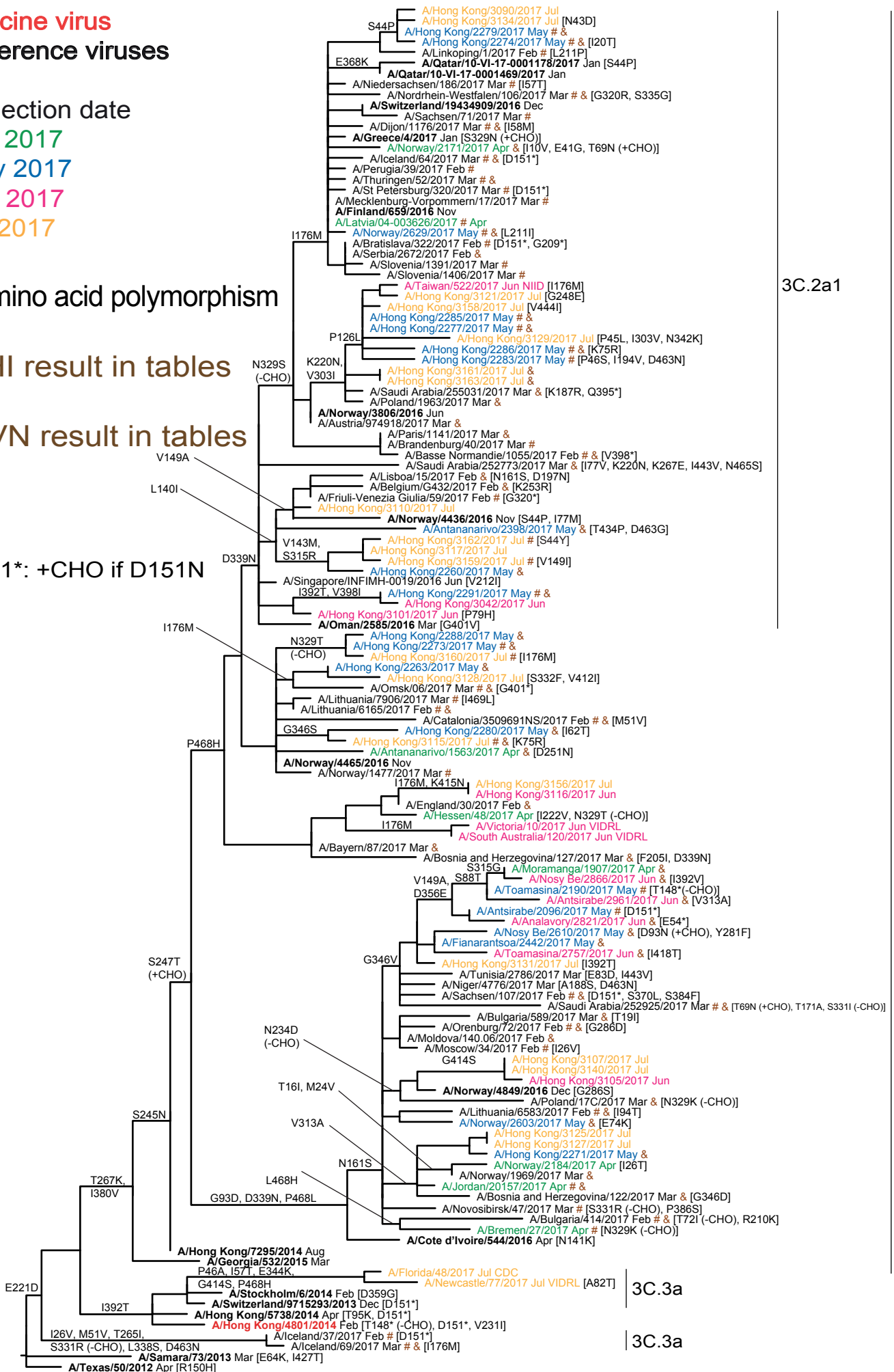
Apr 2017
May 2017
Jun 2017
Jul 2017

* amino acid polymorphism

HI result in tables

& VN result in tables

D151*: +CHO if D151N



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**

Influenza B virus analyses.

Influenza B viruses represented 27% of the samples received with collection dates after 2017-01-31 (**Table 1**). B/Victoria lineage viruses predominated over those of the B/Yamagata lineage at a ratio of approximately 2:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from 26 countries or administrative regions in four WHO Regions (**Table 10**): African (3: Madagascar, Niger and Senegal), European (20), Eastern Mediterranean (2: Egypt and Saudi Arabia) and Western Pacific (1: Hong Kong SAR). The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 10**.

HI results for the propagated viruses are shown in **Tables 11-1 to 11-16**: test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are those with collection dates after 2017-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. A summary of the recognition of the test viruses collected after 2017-01-31 is shown in **Table 11-17**. Only 12 (6.9%) of the 174 test viruses were recognised at titres within 2-fold of the homologous titre with the post-infection ferret antiserum raised against egg-propagated vaccine virus, B/Brisbane/60/2008, and only a total of 14 (8%) test viruses were recognised at titres within 4-fold of the homologous titre of this antiserum. Test viruses were very poorly recognised by post-infection ferret antisera raised against other egg-propagated reference viruses (B/Malta/636714/2011, B/Johannesburg/3964/2012, B/South Australia/81/2012 and B/Maryland/15/2016) with less than ~10% of the tested viruses being recognised at titres within 4-fold of the homologous titres of these antisera. In contrast, over 80% of the test viruses were recognised at titres within 2-fold of the titres with the corresponding homologous viruses for antisera raised against three of four reference viruses propagated in tissue culture; these four viruses are considered to be surrogate cell culture-propagated antigens representing the egg-propagated B/Brisbane/60/2008 prototype virus. These antisera were raised against B/Hong Kong/514/2009, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016; these antisera recognised 96%, 95% and 94%, respectively, of test viruses at titres within 2-fold, and 97%, 98% and 98%, respectively, at titres within 4-fold of the titres of the antisera with their corresponding homologous viruses.

An antiserum raised against cell culture-propagated B/Formosa/V2367/2012 recognised test viruses less well. Only 58% of the test viruses were recognised at titres within 2-fold of the titre of the antiserum with the homologous virus but >90% of the test viruses were recognised at titres within 4-fold of the homologous titre. It is noteworthy that B/Formosa/V2367/2012 had become polymorphic for an egg-adaptive substitution resulting in loss of the HA1 197-199 glycosylation site, despite isolation and propagation exclusively in tissue culture.

B/Maryland/15/2016 and B/Norway/2409/2017 are viruses with a deletion in HA1 at residues 162 and 163 (Δ 162-163) in HA1 (**Figure 13**) and antisera raised against cell

culture-propagated cultivars of B/Maryland/15/2016 and B/Norway/2409/2017 were used in a number of the tests (**Tables 11-13 to 11-16**). These antisera recognised the Δ 162-163 viruses well but the antiserum raised against the cell culture-propagated cultivar of B/Maryland/15/2016 showed very poor recognition of test viruses (no test viruses were recognised at titres within 4-fold of the homologous titre of the antiserum) and other reference viruses. The antiserum raised against cell culture-propagated B/Norway/2409/2017 recognised only 38% of the test viruses at titres within 4-fold of the homologous titre of the antiserum. The antiserum raised against the egg-propagated cultivar of B/Maryland/15/2016 recognised only 10% of cell culture-propagated test viruses at titres within 4-fold of the homologous titre, but it recognised all five of the egg-propagated reference viruses and cell culture-propagated B/Formosa/V2367/2012 at titres within 2-fold of the homologous titre. In addition, the antiserum raised against the egg-propagated cultivar of B/Maryland/15/2016 recognised two cell culture-propagated test viruses from Russia (B/St Petersburg/282/2017 and B/Khakassia/A1/2017) at titres 2-fold or 4-fold lower, respectively, than the homologous titre of the antiserum. B/Khakassia/A1/2017 had lost, and B/St Petersburg/282/2017 was polymorphic for an egg-adaptive substitution resulting in the complete or incomplete loss, respectively, of the HA1 197-199 glycosylation site.

B/Hong Kong/269/2017 is a virus with a three amino acid deletion in HA1 at residues 162, 163 and 164 (Δ 162-164) in HA1 (**Figure 13**). Antisera raised against the cell culture-propagated and egg-propagated cultivars of B/Hong Kong/269/2017 were used in **Table 11-16**. The antiserum raised against the cell culture-propagated cultivar recognised only the homologous virus and the egg-propagated cultivar of B/Hong Kong/269/2017. The antiserum raised against the egg-propagated cultivar of B/Hong Kong/269/2017 recognised the cell culture-propagated cultivar at a titre within 2-fold of the titre of the antiserum with the homologous virus but this antiserum also recognised egg-propagated B/Johannesburg/3964/2012 at a titre within 4-fold of the homologous titre of the antiserum. The egg-propagated cultivar of B/Hong Kong/269/2017 is polymorphic for an egg-adaptive substitution resulting in an incomplete loss of the HA1 197-199 glycosylation site.

Phylogenetic analysis of the HA genes of representative B/Victoria lineage viruses is shown in **Figure 11** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 12**. All recently circulating viruses analysed carried HA genes that fell into genetic group 1A, the B/Brisbane/60/2008 genetic group, the vast majority falling into a genetic group defined by two HA1 amino acid substitutions compared with B/Brisbane/60/2008: I117V and N129D. Two new virus clusters have emerged within clade 1A both with deletions in HA1. Viruses from the USA and Norway (B/Norway/2409/2017) have a two amino acid deletion in **HA1 Δ 162-163** in combination with a substitution **I180V** in **HA1** and **R151K** in **HA2**. The other cluster, viruses from Hong Kong, are viruses that have a three amino acid deletion in **HA1 Δ 162-164** in combination with a substitution **I180T** and **K209N** in **HA1**.

Figure 14 shows a phylogenetic tree for the NA gene and **Figure 15** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins. The NA gene phylogeny is largely congruent with the HA phylogeny.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from 23 countries or administrative regions in four WHO Regions (**Table 12**): African (3: Ghana, Niger and South Africa), Eastern Mediterranean (2: Jordan and Saudi Arabia), European (17) and Western Pacific (1: Hong Kong SAR). The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 12**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 13-1 to 13-10**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are viruses with collection dates after 2017-01-31. The clade into which the HA gene falls is shown, and viruses for which gene sequences are included in phylogenetic trees are highlighted. A summary of the results for all of the post-infection ferret antisera is shown in **Table 13-11**.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in quadrivalent vaccines for the Southern Hemisphere 2017 and the Northern Hemisphere 2017/2018 seasons, recognised 67% of the test viruses collected after 2017-01-31 at titres within 4-fold of the titre with the homologous virus, 37% within 2-fold.

Antisera raised against other viruses from the B/Phuket/3073/2013 clade (clade 3), egg-propagated B/Wisconsin/1/2010, B/Stockholm/12/2011 and B/Hong Kong/3417/2014 recognised 89%, 94% and 100%, respectively, of the test viruses collected after 2017-01-31 at titres within 4-fold of the titres of the antisera with their homologous viruses. The antisera raised against egg-propagated B/Wisconsin/1/2010 and B/Stockholm/12/2011, recognised, respectively 67%, and 58% of the test viruses collected after 2017-01-31 at titres within 2-fold of the homologous titres of the antisera. The antiserum raised against egg-propagated B/Hong Kong/3417/2014 recognised all the test viruses at titres within 2-fold of the titre of the antiserum with the homologous virus.

A post-infection ferret antiserum raised against a cell culture-propagated cultivar of B/Phuket/3073/2013 recognised only 56% of the recently collected test viruses at titres within 4-fold, and only 32% within 2-fold, of its titre with the homologous virus.

An antiserum raised against egg-propagated B/Massachusetts/02/2012 recognised only 37% of recently collected test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus. Antiserum raised against the cell culture-propagated cultivar of B/Massachusetts/02/2012 recognised 47% of recently collected test viruses at titres within 4-fold of, and 19% within 2-fold, of its homologous titres. Test viruses were recognised no better by an antiserum raised against cell culture-propagated B/Estonia/55669/2011, another virus from the B/Massachusetts/02/2012 clade (clade 2), with 22% of the test viruses collected after 2017-01-31 being recognised at titres within 4-fold, and <10% within 2-fold, of the titres of the antiserum with the homologous virus.

Figure 16 shows a phylogenetic analysis of the HA genes of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 17**. All tested viruses and the vast majority of recently circulating viruses fall in clade 3, the B/Phuket/3073/2013 clade, with most falling into a genetic group defined by two **HA1** amino acid substitutions compared with B/Phuket/3073/2013: **L172Q** and **M251V**.

Figure 18 shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 19** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. As for the HA gene, the vast majority of NA genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3) with a majority of viruses carrying **NA D342N** and **K373Q** amino acid substitutions compared to the B/Phuket/3073/2013 sequence.

Antiviral Analysis.

Susceptibility to the neuraminidase inhibitors zanamivir and oseltamivir was assessed phenotypically, backed up by full NA gene sequencing, for **207** influenza B viruses with collection dates after 2017-01-31 (**Table 14**). All viruses but one virus of the B/Victoria lineage showed normal inhibition (no reduced susceptibility) by both antivirals. The exception was that B/Moscow/107/2017 showed borderline RI by oseltamivir that was associated with NA H273Y amino acid substitution.

Conclusion.

Viruses of the B/Victoria lineage and the B/Yamagata lineage have been received in similar proportions but in small numbers. Most viruses of the B/Victoria lineage were antigenically similar to viruses genetically closely related to B/Brisbane/60/2008 but two distinct clusters of viruses have emerged with deletions in HA1, Δ 162-163 or Δ 162-164, of two or three amino acid residues. Viruses with these deletions are antigenically distinct from the majority of viruses in clade 1A and the Δ 162-163 and Δ 162-164 variants are antigenically distinct from each other. Viruses of the B/Yamagata lineage fell within genetic clade 3, the B/Phuket/3073/2013 clade, and the majority were recognised well by antisera raised against egg-propagated clade 3 vaccine/reference viruses.

Table 10. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2017-01-31

MONTH	B Victoria lineage				
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³
2017 FEBRUARY					
Belgium	1	1	1		
Bulgaria	3	2	2		
Egypt	1	1	1		
France	2	2	2		
Germany	2	2	2		
Greece	1	0			
Ireland	1	1	1		
Kazakhstan	8	8	8		
Latvia	1	1	1		
Lithuania	1	1	1		
Madagascar	1	1	1		
Moldova	3	3	3		
Niger	10	in process			
Norway	8	5	5		
Russia	12	10	10		
Saudi Arabia	2	2	2		
Senegal	2	2	2		
Slovakia	3	3	3		
Sweden	1	1	1		
Ukraine	4	4	4		
2017 MARCH					
Austria	1	1	1		
Georgia	1	1	1		
Germany	3	3	3		
Kazakhstan	1	1	1		
Lithuania	1	1	1		
Madagascar	3	2	2		
Moldova	4	4	4		
Norway	1	0			
Romania	7	5	5		
Russia	14	14	12	2	
Senegal	5	5	5		
Slovenia	1	1	1		
Ukraine	10	9	9		
United Kingdom	1	1	1		
2017 APRIL					
Georgia	7	7	7		
Germany	1	1	1		
Hong Kong	7	sequenced			
Madagascar	3	2	2		
Moldova	5	5	5		
Norway	9	8	7		1
Romania	7	5	5		
Russia	13	13	13		
Senegal	1	1	1		
Slovakia	1	1	1		
Ukraine	13	13	13		
United Kingdom	1	1	1		
2017 MAY					
Georgia	2	1	1		
Hong Kong	4	sequenced			1
Madagascar	1	1	1		
Moldova	2	2	2		
Russia	2	2	2		
Ukraine	1	1	1		
2017 JUNE					
Hong Kong	2	1			1
Madagascar	1	1	1		
	203	164	160	2	3
26 Countries			97.0%	1.2%	1.8%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Hong Kong/514/2009, B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016)

Deletion 162-163

Deletion 162-164

Table 11-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-02-28

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK		
				Sh 539, 540, 543, 544, 570, 571, 574* ^{1,3}	F41/14* ²	F26/13* ²	F29/13* ⁴	F01/13* ⁴	F04/13* ²	F41/13* ²	F09/13* ²	F15/16* ⁴	F16/16* ⁴		
				Genetic group	1A	1A	1A	1A	1A	1A	1B	1A	1A		
REFERENCE VIRUSES															
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	160	40	40	80	40	80	10	<	<		
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	640	320	320	160	640	80	40	20		
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	640	320	320	160	640	40	40	20		
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	160	1280	1280	1280	1280	1280	160	80	80		
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	640	160	160	160	320	40	20	20		
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	20	320	160	80	80	320	20	<	10		
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	1280	<	40	<	<	20	20	20	40	20		
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	40	<	<	40	20	20	40	20		
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	2560	10	80	<	20	40	40	40	40	40		
TEST VIRUSES															
B/Madagascar/5014/2016	1A	2016-09-05	MDCK1	2560	<	80	<	20	40	40	20	40	20		
B/Madagascar/5226/2016	1A	2016-09-19	MDCK1	2560	<	80	<	<	40	40	20	40	20		
B/Ghana/FIPI-16-0035/2016	1A	2016-10-03	Cx/MDCK1	1280	<	40	<	<	20	20	20	40	20		
B/Ghana/FS-16-1474/2016	1A	2016-10-04	Cx/MDCK1	1280	<	40	<	<	20	40	20	40	20		
B/Ghana/FS-16-1483/2016	1A	2016-10-07	Cx/MDCK1	1280	<	40	<	<	20	40	20	40	20		
B/Ghana/GH-GAR-SOD-16-0014/2016	1A	2016-10-07	MDCK1	2560	<	80	<	20	40	80	40	40	40		
B/Madagascar/6005/2016	1A	2016-10-10	MDCK1/MDCK1	2560	10	40	<	<	40	40	20	40	40		
B/Ghana/DILI-16-1070/2016		2016-10-21	Cx/MDCK1	2560	<	40	<	<	20	40	20	40	20		
B/Madagascar/6379/2016	1A	2016-10-26	MDCKx/MDCK1	2560	<	40	<	<	40	40	20	40	20		
B/Ghana/FS-16-1664/2016	1A	2016-10-31	MDCK1	2560	<	80	<	20	40	40	40	40	40		
B/Ghana/FIPI-16-040/2016	1A	2016-10-31	MDCK1	2560	<	80	<	20	40	40	40	40	40		
B/Rheinland-Pfalz/43/2016	Vaccine related	1A	2016-11-21	C2/MDCK1	2560	40	640	<	160	160	640	40	20		
B/Moscow/92/2016		2016-12-20	MDCK1/MDCK2	2560	<	40	<	<	40	20	20	40	20		
B/Samara/680/2016	1A	2016-12-27	MDCK1/MDCK1	2560	<	80	<	<	40	40	40	40	20		
B/Portugal/SU275/2017	1A	2017-01-02	MDCK1	2560	<	80	<	<	40	40	20	40	20		
B/Athens GR/78/2017	1A	2017-01-03	MDCK1	2560	20	80	<	20	40	40	20	40	40		
B/Ankara/29/2017	1A	2017-01-03	SIAT1/MDCK1	1280	<	40	<	<	40	20	20	40	20		
B/Ankara/41/2017	1A	2017-01-04	SIAT1/MDCK1	2560	<	80	<	<	40	40	20	40	20		
B/Ankara/90/2017	1A	2017-01-11	SIAT1/MDCK1	2560	<	80	<	<	40	40	20	40	20		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table11-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-03-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	F41/14 ^{*2}	F26/13 ^{*2}	F29/13 ^{*2}	F01/13 ^{*2}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*4}	F16/16 ^{*4}
	Passage history												
	Ferret number												
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	160	40	40	40	20	40	<	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	640	160	160	160	640	80	20	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	40	640	320	160	160	320	40	20	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	1280	1280	1280	1280	320	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	640	80	160	160	320	40	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	20	320	80	160	80	320	20	20	10
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	1280	<	80	10	10	20	40	20	40	20
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	<	40	<	<	20	20	20	40	20
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	2560	10	80	<	10	40	40	20	40	20
TEST VIRUSES													
B/Iran/48810/2016	1A	2016-03-22	MDCK2/MDCK1	2560	<	80	20	20	40	40	20	40	20
B/Zhejiang-Haishu/11897/2016	1A	2016-11-01	C5/MDCK1	2560	<	80	20	20	40	40	20	40	20
B/Kyrgyzstan/2/2016	1A	2016-11-11	MDCKx/MDCK2	1280	<	20	<	<	10	<	10	20	10
B/Kyrgyzstan/22/2016	1A	2016-11-21	MDCKx/MDCK1	2560	<	80	10	10	40	40	20	40	20
B/Kyrgyzstan/8/2016	1A	2016-11-21	MDCKx/MDCK1	2560	<	80	10	20	40	40	20	40	20
B/Kyrgyzstan/9/2016	1A	2016-11-22	MDCKx/MDCK1	2560	<	80	10	20	40	40	20	40	20
B/Kyrgyzstan/6/2016	1A	2016-11-23	MDCKx/MDCK1	2560	<	80	10	20	40	40	20	40	20
B/Kyrgyzstan/10/2016	1A	2016-11-24	MDCKx/MDCK1	1280	<	40	10	10	20	40	20	40	20
B/Yunnan-Zhaoyang/11160/2016	1A	2016-11-26	C4/MDCK1	1280	<	80	20	20	40	40	20	40	20
B/Algeria/23/2016	1A	2016-11-28	MDCK1	1280	<	80	20	10	20	40	40	20	20
B/Algeria/18/2016	1A	2016-11-30	MDCK1	2560	<	40	20	20	20	40	20	40	10
B/Algeria/19/2016	1A	2016-11-30	MDCK1	2560	<	40	20	10	20	40	40	40	10
B/Algeria/24/2016	1A	2016-12-04	MDCK1	2560	<	80	20	20	40	40	20	40	20
B/Algeria/98/2016	1A	2016-12-05	MDCK1	2560	<	80	20	20	40	40	20	40	20
B/Yunnan-Zhaoyang/11210/2016	1A	2016-12-07	C2/MDCK1	2560	<	80	20	10	40	40	40	40	20
B/Algeria/43/2016	1A	2016-12-10	MDCK1	1280	<	80	20	20	40	40	20	40	20
B/Algeria/61/2017	1A	2016-12-11	MDCK1	2560	<	80	10	20	40	40	20	40	20
B/Algeria/70/2017	1A	2016-12-11	MDCK1	2560	<	80	20	20	40	40	20	40	20
B/Ankara/5579/2016	1A	2016-12-22	MDCK1/MDCK1	1280	<	80	20	20	20	40	20	40	20
B/Ankara/5630/2016	1A	2016-12-26	SIAT1/MDCK1	2560	<	80	20	20	40	40	20	40	20
B/Algeria/137/2017	1A	2017-01-03	MDCK1	2560	<	80	20	20	40	80	20	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

Vaccine

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Table 11-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-03-15

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F41/14 ²	F37/14 ²	F29/13 ⁴	F01/13 ⁴	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
	Passage history												
	Ferret number												
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	160	40	40	40	20	40	<	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	320	160	320	160	640	40	20	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	40	640	320	160	160	320	40	20	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	1280	1280	1280	1280	320	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	640	80	160	160	320	40	40	20
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	20	320	80	160	80	320	20	20	10
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	1280	<	80	10	10	20	40	20	40	20
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	<	40	<	<	20	20	20	40	20
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	2560	10	80	<	10	40	40	20	40	20
TEST VIRUSES													
B/Yokohama/109/2016	1A	2016-09-25	MDCK1/MDCK2/MDCK1	2560	10	<	20	40	40	80	10	10	10
B/Laos/F3032/2016	1A	2016-09-28	MDCK1/MDCK2/MDCK1	640	<	<	<	<	10	<	<	<	<
B/Kazakhstan/4976/2016	1A	2016-11-22	MDCK1	2560	<	<	<	<	10	20	10	10	10
B/Kazakhstan/4933/2016	1A	2016-11-23	MDCK1	1280	<	<	<	<	20	40	10	10	10
B/Kazakhstan/4934/2016	1A	2016-11-23	MDCK1	2560	20	<	<	<	40	40	20	40	20
B/Cameroon/10372/2016	1A	2016-12-01	MDCK1/MDCK1	1280	<	<	<	<	40	40	20	40	20
B/Cameroon/10374/2016	1A	2016-12-02	MDCK1/MDCK1	1280	<	<	<	<	40	40	20	40	20
B/Kazakhstan/4941/2016	1A	2016-12-02	MDCK1	2560	<	<	<	<	40	40	20	40	20
B/Cameroon/10586/2016	1A	2016-12-09	MDCK2/MDCK1	2560	<	<	<	<	40	40	20	40	40
B/Cameroon/10887/2016	1A	2016-12-22	MDCK1/MDCK1	1280	<	<	<	<	20	40	20	40	20
B/Ankara/5608/2016	1A	2016-12-23	SIAT1/MDCK2	1280	<	<	<	<	20	20	10	20	20
B/Qatar/47-VI-17-0000937/2017	1A	2017-01-05	MDCK1	1280	<	<	<	<	10	10	10	10	10
B/Qatar/10-VI-17-0001141/2017	1A	2017-01-07	MDCK1	1280	<	<	<	<	10	40	10	10	10
B/Qatar/47-VI-17-0001088/2017	1A	2017-01-07	MDCK1	1280	<	<	<	<	40	40	20	40	20
B/Qatar/10-VI-17-0001268/2017	1A	2017-01-08	MDCK1	640	<	<	<	<	10	<	<	<	<
B/Qatar/39-VI-17-0001241/2017	1A	2017-01-08	MDCK1	1280	<	<	<	<	20	40	10	20	10
B/Qatar/10-VI-17-0001338/2017	1A	2017-01-08	MDCK1	640	<	<	<	<	10	<	<	10	<
B/Qatar/11-VI-17-0001490/2017	1A	2017-01-09	MDCK1	1280	<	<	<	<	20	40	20	20	20
B/Qatar/11-VI-17-0001667/2017	1A	2017-01-10	MDCK1	1280	<	<	<	<	20	40	20	20	20
B/Qatar/32-VI-17-0001834/2017	1A	2017-01-11	MDCK1	2560	<	<	<	<	40	40	20	40	20
B/Qatar/15-VI-17-0002411/2017	1A	2017-01-15	MDCK1	2560	<	<	<	<	40	40	20	40	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-03-22 + 2017-04-05

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574* ^{1,3}	F41/14* ²	F37/14* ²	F29/13* ⁴	F01/13* ⁴	F04/13* ²	F41/13* ²	F09/13* ²	F15/16* ²	F16/16* ²	
				Genetic group	1A	1A	1A	1A	1A	1A	1B	1A	1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	160	40	40	40	20	40	<	<	<	
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	80	160	80	160	80	320	20	10	10	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	40	640	320	160	160	320	40	20	20	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	1280	1280	1280	1280	320	160	160	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	640	80	160	160	320	40	40	20	
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	20	320	80	160	80	320	20	20	10	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	1280	<	80	10	10	20	40	20	40	20	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	<	40	<	<	20	20	20	40	20	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK1	2560	10	80	<	10	40	40	20	40	20	
TEST VIRUSES														
B/St Petersburg/293/2016		2016-04-20	MDCK1/MDCK1/MDCK1	2560	<	10	20	<	40	40	20	40	20	
B/Cameroon/8427/2016	1A	2016-09-28	MDCK1/MDCK1	2560	<	10	20	<	40	40	20	40	40	
B/Kazakhstan/4911/2016	1A	2016-11-08	MDCK2	320	<	<	<	<	<	<	<	<	<	
B/Kazakhstan/4925/2016	1A	2016-11-10	MDCK2	320	<	<	<	<	<	<	<	<	<	
B/Kazakhstan/4970/2016	1A	2016-11-10	MDCK1	2560	<	10	20	40	40	80	40	40	40	
B/Kazakhstan/4944/2016	1A	2016-11-12	MDCK1	320	<	<	<	<	<	<	<	<	<	
B/Kazakhstan/4975/2016	1A	2016-11-20	MDCK2	2560	<	10	20	20	40	40	20	40	20	
B/Kazakhstan/4936/2016	1A	2016-11-25	MDCK2	2560	<	10	20	20	40	40	20	40	20	
B/Cameroon/10149/2016	1A	2016-11-25	MDCK1	5120	<	40	80	<	80	160	80	80	80	
B/Kazakhstan/4939/2016	1A	2016-11-29	MDCK1	320	<	<	<	<	<	<	<	<	<	
B/Cameroon/10593/2016	1A	2016-12-01	MDCK1	1280	<	10	10	<	20	20	20	40	20	
B/Cameroon/10358/2016	1A	2016-12-01	MDCK1	2560	<	10	20	20	40	40	20	40	20	
B/Cameroon/10363/2016	1A	2016-12-02	MDCK2	1280	<	10	20	<	40	40	40	40	20	
B/Dakar/04/2016	1A	2016-12-05	P1/MDCK1	2560	<	10	20	<	40	40	40	40	20	
B/Cameroon/10600/2016	1A	2016-12-13	MDCK2	2560	<	10	20	20	40	40	20	40	20	
B/Cameroon/10357/2016	1A	2016-12-25	MDCK2	2560	<	10	20	20	40	40	20	40	20	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-05-04

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	F41/14 ^{*2}	F37/14 ^{*2}	F29/13 ^{*4}	F01/13 ^{*4}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}
	Passage history												
	Ferret number												
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	320	40	40	40	40	80	10	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	80	160	160	160	80	320	40	20	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	320	320	160	160	640	40	20	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	160	640	640	640	640	640	160	80	80
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	160	80	160	160	160	40	20	10
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	40	160	80	80	80	160	20	<	10
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	1280	<	<	<	<	20	20	20	40	20
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	<	<	<	20	20	20	40	20
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	2560	<	<	<	<	40	40	40	40	40
TEST VIRUSES													
B/Niedersachsen/4/2017	1A	2017-01-30	C1/MDCK1	2560	<	<	<	<	40	20	20	40	10
B/Bayern/10/2017	1A	2017-02-02	C1/MDCK1	2560	<	<	<	<	40	40	20	40	20
B/Hessen/1/2017	1A	2017-02-02	C1/MDCK1	2560	<	<	<	<	40	40	20	40	20
B/Norway/998/2017	1A	2017-02-08	MDCK1/MDCK1	2560	<	<	<	<	20	40	20	40	20
B/Norway/1102/2017	1A	2017-02-13	MDCK1/MDCK1	2560	<	<	<	<	40	40	20	20	20
B/Norway/1040/2017	1A	2017-02-15	MDCK1/MDCK1	2560	<	<	<	<	40	40	20	40	40
B/Norway/1076/2017	1A	2017-02-16	MDCK1/MDCK1	5120	<	<	<	<	40	80	40	40	40
B/Bulgaria/513/2017	1A	2017-02-23	SIAT2/MDCK1	1280	<	<	<	<	20	40	20	20	20
B/Bulgaria/516/2017	1A	2017-02-24	SIAT2/MDCK1	2560	<	<	<	<	40	40	20	40	20
B/Nordrhein-Westfalen/8/2017	1A	2017-03-07	C1/MDCK1	2560	<	<	<	<	40	40	40	40	20
B/Bremen/4/2017	1A	2017-03-13	C1/MDCK1	2560	<	<	<	<	40	40	20	40	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-05-24 + 2017-05-31 + 2017-06-06

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
					Sh 539, 540, 543, 544, 570, 571, 574*1,3	F41/14*2	NIB F52/16*4	F29/13*4	F01/13*4	F04/13*2	F41/13*2	F09/13*2	F15/16*2	F16/16*2
Genetic group					1A		1A	1A	1A	1A	1B	1A	1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004			2004-12-06	E3/E7	1280	320	80	40	80	40	80	<	<	<
B/Brisbane/60/2008	1A		2008-08-04	E4/E4	2560	80	640	160	160	80	320	40	40	20
B/Malta/636714/2011	1A		2011-03-07	E4/E1	2560	80	640	320	160	160	320	40	40	40
B/Johannesburg/3964/2012	1A		2012-08-03	E1/E2	5120	320	1280	1280	1280	640	1280	160	160	160
B/Formosa/V2367/2012	1A		2012-08-06	MDCK1/MDCK3	5120	40	320	80	160	160	320	40	40	40
B/South Australia/81/2012	1A		2012-11-28	E4/E1	1280	80	320	80	80	80	320	20	20	20
B/Hong Kong/514/2009	1B		2009-10-11	MDCK3	2560	<	20	20	<	20	40	40	80	80
B/Ireland/3154/2016	1A		2016-01-14	MDCK1/MDCK4	2560	<	20	20	<	40	40	40	80	80
B/Nordrhein-Westfalen/1/2016	1A		2016-01-04	C2/MDCK3	5120	<	20	20	<	80	40	40	80	160
TEST VIRUSES														
B/Cameroon/9000/2016	1A		2016-10-12	MDCK1	2560	20	40	20	20	40	40	40	80	80
B/Cameroon/9004/2016	1A		2016-10-25	MDCK2	2560	<	20	10	20	40	40	20	80	80
B/Cameroon/9010/2016	1A		2016-10-26	MDCK1	2560	<	40	20	20	80	40	40	80	80
B/Cameroon/9159/2016	1A		2016-10-31	MDCK1	5120	<	40	20	20	80	40	40	160	160
B/Cameroon/9431/2016	1A		2016-11-03	MDCK2	2560	<	20	20	20	40	40	40	80	80
B/Cameroon/9439/2016	1A		2016-11-05	MDCK1	5120	<	40	40	40	80	80	40	80	80
B/Cameroon/9434/2016	1A		2016-11-08	MDCK2	2560	<	20	20	20	40	40	40	80	80
B/Cameroon/9648/2016	1A		2016-11-10	MDCK1	5120	<	160	80	80	320	160	160	640	320
B/Cameroon/9637/2016	1A		2016-11-16	MDCK2	2560	<	20	20	20	40	40	40	80	80
B/Cameroon/10152/2016			2016-11-16	MDCK1	2560	<	40	20	20	40	40	40	160	80
B/Cameroon/10153/2016			2016-11-16	MDCK1	5120	<	40	40	40	80	80	40	160	160
B/Cameroon/10165/2016			2016-11-16	MDCK1	5120	<	40	20	20	80	80	80	160	160
B/Cameroon/9986/2016			2016-11-18	MDCK1	2560	<	20	20	20	40	40	40	80	80
B/Cameroon/9989/2016			2016-11-21	MDCK1	2560	<	20	10	<	40	40	40	80	80
B/Rheinland-Pfalz/43/2016	Vaccine related	1A	2016-11-21	MDCK3	5120	160	1280	160	160	160	640	40	40	40
A/Serbia/8058/2016			2016-12-12	SIAT3	2560	<	20	10	<	40	40	40	80	40
A/Serbia/8068/2016			2016-12-12	SIAT3	1280	<	20	20	<	40	40	40	80	40
A/Serbia/8057/2016			2016-12-13	SIAT3	2560	<	20	10	<	40	40	20	80	40
B/Egypt/159/2017	1A		2017-01-04	MDCK1	5120	20	80	20	20	80	80	80	160	160
B/Egypt/174/2017	1A		2017-01-04	MDCK1	2560	<	80	20	40	80	80	80	160	80
B/Latvia/02-04598/2017	1A		2017-02-02	C2/MDCK1	5120	<	20	10	<	40	40	40	80	80
B/Norway/1167/2017	1A		2017-02-16	MDCK2	5120	<	80	20	20	80	80	40	160	80
B/Lithuania/6295/2017	1A		2017-02-20	MDCK2/MDCK1	2560	20	20	10	<	40	40	20	80	80
B/Slovenia/1496/2017	1A		2017-03-20	MDCK1	5120	<	40	20	20	40	80	40	160	80
B/Lithuania/10137/2017	1A		2017-03-22	MDCK2/MDCK1	2560	<	20	20	20	40	40	40	80	80
B/Norway/2409/2017	1A		2017-04-27	MDCK1/MDCK1	80	<	<	<	<	<	<	<	<	<

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-06-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574* ^{1,3}	F41/14 ²	NIB F52/16 ²	F29/13 ²	F01/13 ⁴	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
				1A	1A	1A	1A	1A	1A	1B	1A	1A	
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	320	160	40	80	40	80	<	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	1280	160	160	160	1280	40	40	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	640	320	160	160	320	40	40	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	640	640	640	1280	160	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	320	80	80	160	160	20	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E1	2560	80	640	160	160	160	320	40	40	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	40	20	20	40	40	40	80	80
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	5120	<	40	10	<	40	40	40	80	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	5120	<	40	10	<	80	40	40	80	80
TEST VIRUSES													
B/Egypt/124/2017	1A	2017-01-04	MDCK2	2560	<	40	20	<	80	80	40	80	80
B/Povazska Bystrica/175/2017	1A	2017-01-16	MDCKx/MDCK1	2560	<	40	10	<	80	80	20	80	40
B/Bratislava/275/2017	1A	2017-02-01	MDCKx/MDCK1	2560	<	40	10	<	80	80	20	80	40
B/Sastin-Straze/332/2017	1A	2017-02-13	MDCKx/MDCK1	2560	<	40	10	<	80	80	20	80	40
B/Poprad/324/2017	1A	2017-02-13	MDCKx/MDCK1	5120	<	40	20	<	160	160	40	80	80
B/Egypt/1428/2017	1A	2017-02-27	MDCK1	5120	<	40	20	<	80	80	40	80	80
B/Moldova/204.09/2017	1A	2017-02-27	MDCK1	2560	<	40	20	<	80	80	40	80	80
B/Moldova/243.1/2017	1A	2017-03-09	MDCK1	5120	<	40	20	<	80	160	40	80	80
B/Moldova/254.11/2017	1A	2017-03-14	MDCK1	5120	<	40	20	<	160	160	40	80	80
B/Moldova/267.11/2017	1A	2017-03-15	MDCK1	2560	<	40	20	<	80	80	40	80	40
B/Moldova/278.12/2017	1A	2017-03-21	MDCK1	2560	<	20	20	<	80	80	40	80	40
B/Moldova/334.14/2017	1A	2017-04-01	MDCK1	2560	<	40	20	20	160	80	40	80	80
B/Kosice/392/2017	1A	2017-04-06	MDCKx/MDCK1	5120	<	40	10	<	80	80	40	80	80
B/Moldova/347.15/2017	1A	2017-04-10	MDCK1	2560	<	40	20	<	80	80	40	80	80
B/Moldova/362.15/2017	1A	2017-04-10	MDCK1	5120	<	80	40	20	160	160	40	80	80
B/Moldova/379.16/2017	1A	2017-04-17	MDCK1	5120	<	40	20	20	160	160	40	80	80
B/Moldova/371.16/2017	1A	2017-04-18	MDCK1	2560	<	40	20	<	80	80	40	80	80
B/Norway/2302/2017	1A	2017-04-21	MDCK1/MDCK1	2560	<	40	20	<	80	80	40	80	80
B/Norway/2424/2017	1A	2017-04-26	MDCK1/MDCK1	2560	<	20	20	<	80	80	20	80	40
B/Moldova/406.18/2017	1A	2017-05-02	MDCK1	2560	<	40	20	<	80	80	40	80	40
B/Moldova/422.18/2017	1A	2017-05-02	MDCK1	2560	<	40	20	<	160	80	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-8. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-06-27

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 6367/14/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F41/14 ²	NIB F52/16 ²	F29/13 ²	F01/13 ⁴	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
	Passage history												
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	320	160	40	80	40	80	<	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	80	640	160	160	160	640	40	40	20
B/Malta/6367/14/2011	1A	2011-03-07	E4/E1	2560	80	640	320	160	160	320	40	40	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	640	640	640	1280	160	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	320	80	80	160	160	20	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E1	2560	80	640	160	160	160	320	40	40	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	40	20	20	40	40	40	80	80
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	5120	<	40	10	<	40	40	40	80	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	5120	<	40	10	<	80	40	40	80	80
TEST VIRUSES													
B/Moldova/98.04/2017	1A	2017-01-23	MDCK1	2560	<	40	20	20	80	80	20	80	40
B/Moldova/168.07/2017	1A	2017-02-14	MDCK1	2560	<	40	20	<	80	80	20	40	40
B/Khmelnitsky/151/2017	1A	2017-02-14	MDCK/SIAT1/MDCK1	5120	<	40	20	20	80	80	40	80	80
B/Moldova/199.08/2017	1A	2017-02-23	MDCK1	2560	<	40	20	20	80	80	20	80	40
B/Zaporizha/164/2017	1A	2017-02-24	MDCK/SIAT1/MDCK1	5120	40	640	160	160	160	640	40	40	40
B/Kharkiv/191/2017	1A	2017-02-28	SIAT2/MDCK1	5120	<	40	20	<	80	80	20	80	80
B/Kharkiv/192/2017	1A	2017-03-01	SIAT1/MDCK1	2560	<	40	10	<	40	80	20	80	80
B/Odesa/169/2017	1A	2017-03-10	MDCK/SIAT1/MDCK1	2560	<	20	10	<	40	40	20	80	80
B/Kharkiv/184/2017	1A	2017-03-11	SIAT1/MDCK1	2560	<	40	10	<	80	80	20	80	80
B/Odesa/168/2017	1A	2017-03-11	MDCK/SIAT1/MDCK1	5120	10	80	20	20	80	160	20	40	40
B/Kharkiv/182/2017	1A	2017-03-13	SIAT1/MDCK1	2560	<	40	20	<	80	80	40	80	80
B/Khmelnitsky/154/2017	1A	2017-03-15	MDCK/SIAT1/MDCK1	5120	<	40	20	<	80	80	40	80	80
B/Kharkiv/178/2017	1A	2017-03-17	SIAT1/MDCK1	2560	<	20	10	<	40	80	20	40	40
B/Odesa/172/2017	1A	2017-03-24	MDCK/SIAT1/MDCK1	2560	<	20	10	<	40	40	20	40	80
B/Odesa/171/2017	1A	2017-03-28	MDCK/SIAT1/MDCK1	5120	<	40	10	<	80	80	20	80	80
B/Norway/2077/2017	1A	2017-04-01	MDCK1/MDCK1	2560	<	40	10	<	80	80	40	80	40
B/Kyiv/130/2017	1A	2017-04-02	SIAT1/MDCK1	5120	<	40	20	40	80	80	20	80	80
B/Dnipro/241/2017	1A	2017-04-03	MDCK/SIAT1	5120	<	40	10	<	80	80	40	80	80
B/Kyiv/128/2017	1A	2017-04-03	SIAT1/MDCK1	5120	<	40	20	20	80	160	40	160	80
B/Norway/2213/2017	1A	2017-04-10	MDCK1	5120	<	40	20	20	80	80	40	80	80
B/Sachsen-Anhalt/7/2017	1A	2017-04-18	C2/MDCK1	5120	<	40	20	20	80	80	40	80	80
B/Ukraine/7761/2017	1A	2017-04-20	C1/MDCK1	2560	<	40	10	<	80	80	20	80	80
B/Ukraine/7760/2017	1A	2017-04-20	C1/MDCK1	5120	<	40	20	<	80	80	40	80	80
B/Ukraine/7759/2017	1A	2017-04-20	C1/MDCK1	2560	<	40	10	<	80	80	20	80	80
B/Ukraine/7757/2017	1A	2017-04-20	C1/MDCK1	2560	<	40	10	<	40	80	20	80	80
B/Ukraine/7756/2017	1A	2017-04-20	C1/MDCK1	2560	<	40	20	20	80	80	40	80	80
B/Ukraine/7755/2017	1A	2017-04-20	C1/MDCK1	2560	<	20	10	<	40	40	20	80	80
B/Ukraine/7754/2017	1A	2017-04-20	C1/MDCK1	2560	<	20	10	<	40	80	20	80	40
B/Norway/2309/2017	1A	2017-04-21	MDCK1/MDCK1	2560	<	40	10	<	40	80	20	80	40
B/Kyiv/204/2017	1A	2017-04-21	SIAT1/MDCK1	2560	<	40	20	20	80	80	40	80	40
B/Kyiv/202/2017	1A	2017-04-23	SIAT2/MDCK1	2560	<	40	20	20	80	80	20	80	40
B/Norway/2615/2017	1A	2017-04-30	MDCK1	2560	<	40	20	20	80	80	20	80	40
B/Kyiv/197/2017	1A	2017-04-30	SIAT1/MDCK1	2560	<	40	10	<	80	80	20	80	40
B/Kyiv/303/2017	1A	2017-05-04	SIAT1/MDCK1	2560	<	40	10	<	80	80	20	40	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-9. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-07-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
Passage history	Ferret number	Genetic group	Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	F41/14 ^{*2}	NIB F52/16 ^{*2}	F29/13 ^{*2}	F01/13 ^{*4}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}	
1A	1A	1A	1A	1A	1A	1A	1A	1B	1A	1A			
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	320	160	40	80	40	80	<	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	160	1280	320	320	160	1280	80	40	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	640	320	160	160	320	40	40	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	640	640	640	1280	160	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	20	320	80	80	160	160	20	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E1	2560	80	640	160	160	160	320	40	40	40
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	40	20	20	40	40	40	80	80
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	5120	<	40	10	<	40	40	40	80	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	5120	<	40	10	<	80	40	40	80	80
TEST VIRUSES													
B/Dakar/02/2017	1A	2017-02-16	P0/MDCK1	2560	<	40	20	20	80	80	40	80	40
B/Dakar/03/2017	1A	2017-02-22	P0/MDCK1	2560	<	40	20	20	80	80	40	80	40
B/Dakar/04/2017	1A	2017-03-06	P0/MDCK1	2560	<	20	10	<	80	40	20	80	80
B/Niedersachsen/20/2017		2017-03-10	C1/MDCK1	2560	<	40	10	<	80	80	40	80	80
B/Dakar/05/2017	1A	2017-03-13	P0/MDCK1	2560	<	20	20	<	80	80	40	80	40
B/Dakar/07/2017	1A	2017-03-13	P0/MDCK1	2560	<	40	10	<	40	80	40	80	80
B/Dakar/06/2017	1A	2017-03-22	P0/MDCK1	2560	<	40	20	20	80	80	40	80	80
B/Dakar/08/2017	1A	2017-03-22	P0/MDCK1	2560	<	40	20	20	80	80	40	80	80
B/Norway/2076/2017		2017-04-01	MDCK2	2560	<	40	20	<	80	80	40	80	40
B/Dakar/09/2017	1A	2017-04-15	P0/MDCK1	2560	<	40	20	20	80	80	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-10. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-07-18

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F41/14 ²	NIB F52/16 ²	F29/13 ²	F01/13 ⁴	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
				1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E6	5120	320	80	40	80	40	80	10	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	5120	160	640	320	320	160	1280	80	40	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	5120	80	320	160	160	80	320	40	20	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	640	640	640	1280	160	80	80
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	320	160	80	160	320	40	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E1	5120	80	160	80	80	80	320	20	20	10
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	5120	<	20	20	20	40	80	40	40	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	5120	<	20	40	<	40	40	20	40	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	5120	<	40	80	<	80	80	40	80	80
TEST VIRUSES													
B/Mayotte/2139/2016	1A	2016-02-20	MDCK2/MDCK1	5120	20	20	10	<	40	80	20	40	40
B/Lyon/CHU/12.1R308/2017	1A	2017-01-12	MDCK2/MDCK1	5120	40	20	10	20	40	80	40	40	40
B/Astrakhan/02/2017	1A	2017-01-17	MDCK3/MDCK1	5120	160	160	80	80	80	320	40	40	20
B/St Etienne/931/2017	1A	2017-01-22	MDCK2/MDCK1	5120	10	40	20	40	80	80	40	80	40
B/St Etienne/932/2017	1A	2017-01-22	MDCK2/MDCK1	5120	40	40	20	40	80	80	40	80	40
B/Lyon/CHU/25.1R258/2017	1A	2017-01-25	MDCK2/MDCK1	5120	<	20	10	<	40	80	40	80	40
B/Lyon/CHU6.2R80/2017	1A	2017-02-05	MDCK2/MDCK1	5120	<	20	10	<	40	80	40	80	40
B/St Petersburg/148/2017	1A	2017-02-15	C2/MDCK1	5120	40	40	20	20	80	80	40	80	40
B/Lyon/1010/2017	1A	2017-02-17	MDCK2/MDCK1	5120	<	20	40	20	80	80	40	40	40
B/Linköping/4/2017	1A	2017-02-18	MDCK1/MDCK1	5120	<	20	80	20	80	40	40	40	40
B/Ukraine/7731/2017	1A	2017-02-25	C1/MDCK1	5120	40	320	160	80	160	320	40	40	40
B/Belgium/S1204/2017	1A	2017-02-28	SIATx/MDCK1	5120	<	20	10	<	40	80	20	40	40
B/Austria/975422/2017	1A	2017-03-07	C1/MDCK1	5120	<	40	20	20	80	80	40	80	320
B/Georgia/425/2017	1A	2017-03-18	C1/MDCK1	5120	<	20	10	20	80	80	20	40	80
B/Georgia/501/2017	1A	2017-04-04	C1/MDCK1	5120	<	20	20	20	80	80	40	80	80
B/Novosibirsk/68/2017	1A	2017-04-07	C1/MDCK1	5120	<	20	10	20	40	40	20	40	40
B/Georgia/515/2017	1A	2017-04-08	C1/MDCK1	5120	<	20	20	<	80	40	40	80	40
B/Georgia/506/2017	1A	2017-04-10	C2/MDCK1	5120	80	80	40	80	80	160	40	40	40
B/Novosibirsk/75/2017	1A	2017-04-10	MDCK1/MDCK1	5120	<	20	20	20	80	80	40	80	40
B/Yekaterinburg/28/2017	1A	2017-04-10	MDCK2/MDCK1	5120	<	20	20	20	40	40	40	40	40
B/Georgia/523/2017	1A	2017-04-10	C1/MDCK1	5120	40	20	20	20	40	80	40	80	40
B/Georgia/518/2017	1A	2017-04-11	C1/MDCK1	5120	40	40	20	40	40	80	40	40	40
B/St Petersburg/278/2017	1A	2017-04-17	MDCK1/MDCK1	5120	<	20	20	40	80	80	40	80	40
B/St Petersburg/282/2017	1A	2017-04-20	MDCK1/MDCK1	5120	160	320	80	80	80	320	40	20	20
B/St Petersburg/283/2017	1A	2017-04-21	MDCK1/MDCK1	5120	<	20	20	20	40	80	20	40	40
B/Georgia/559/2017	1A	2017-04-24	C1/MDCK1	5120	80	20	20	20	40	80	40	40	40
B/St Petersburg/287/2017	1A	2017-04-26	MDCK1/MDCK1	5120	<	20	20	20	40	80	20	40	40
B/Georgia/564/2017	1A	2017-04-28	C1/MDCK1	5120	40	40	20	20	40	80	40	40	40
B/Georgia/566/2017	1A	2017-05-01	C1/MDCK1	5120	<	20	10	<	40	80	40	80	40
B/St Petersburg/289/2017	1A	2017-05-03	MDCK1/MDCK1	5120	<	20	10	<	40	40	20	40	40
B/St Petersburg/297/2017	1A	2017-05-17	MDCK1/MDCK1	5120	80	320	80	80	160	320	40	40	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-11. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-07-25

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574* ^{1,3}	F41/14 ²	NIB F52/16 ²	F29/13 ²	F01/13 ⁴	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²
	Passage history												
	Ferret number												
	Genetic group			1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	5120	320	80	40	80	20	80	20	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	5120	80	320	160	80	160	1280	40	20	40
B/Malta/636714/2011	1A	2011-03-07	E4/E1	5120	80	320	160	80	80	640	40	20	40
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	640	640	640	1280	160	80	320
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	160	80	80	80	320	40	20	40
B/South Australia/81/2012	1A	2012-11-28	E4/E1	640	80	160	80	80	80	320	20	20	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	640	<	20	20	<	40	80	40	40	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	640	<	20	20	<	40	80	20	40	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	640	<	20	20	<	40	80	20	40	40
TEST VIRUSES													
B/Vladivostok/9/2017	1A	2017-01-26	MDCK1/MDCK1	2560	<	20	10	<	40	40	20	40	80
B/Vladivostok/13/2017	1A	2017-01-26	MDCK1/MDCK1	2560	<	20	10	<	40	40	20	40	80
B/Yaroslavl/2/2017	1A	2017-02-06	MDCK1/MDCK1	2560	<	20	10	<	40	40	20	40	80
B/Moscow/65/2017	1A	2017-02-15	MDCK1/MDCK1	1280	<	20	10	<	40	40	20	40	80
B/Yaroslavl/88/2017	1A	2017-02-15	MDCK1/MDCK1	2560	<	20	10	<	40	40	20	40	80
B/Moscow/77/2017	1A	2017-02-18	MDCK1/MDCK1	1280	<	20	10	<	40	40	20	40	80
B/Tomsk/5/2017	1A	2017-02-19	MDCK1/MDCK1	1280	<	20	10	<	40	40	20	40	40
B/Tomsk/3/2017	1A	2017-02-28	MDCK1/MDCK1	2560	<	20	10	<	40	40	20	40	80
B/Moscow/84/2017	1A	2017-03-01	MDCK1/MDCK1	640	<	20	20	<	40	40	20	40	40
B/Moscow/102/2017	1A	2017-03-02	MDCK1/MDCK1	640	<	20	20	<	40	40	20	40	40
B/Yaroslavl/3/2017	1A	2017-03-03	MDCK1/MDCK1	1280	<	20	20	<	40	40	20	40	40
B/Cheboksary/120/2017	1A	2017-03-07	MDCK1/MDCK1	1280	<	20	<	<	40	40	20	40	80
B/Cheboksary/104/2017	1A	2017-03-10	MDCK1/MDCK1	1280	<	20	20	<	40	40	20	40	80
B/Tomsk/20/2017	1A	2017-03-21	MDCK2/MDCK1	1280	<	20	20	<	40	40	20	40	80
B/Moscow/92/2017	1A	2017-03-22	MDCK1/MDCK1	1280	<	20	10	<	20	20	10	40	40
B/Tomsk/13/2017	1A	2017-03-22	MDCK1/MDCK1	1280	<	20	20	<	20	40	20	40	40
B/Orenburg/17/2017	1A	2017-03-23	MDCK1/MDCK2	1280	<	20	20	<	40	40	20	40	40
B/Tomsk/30/2017	1A	2017-03-24	MDCK1/MDCK1	1280	<	20	20	<	40	40	20	40	40
B/Orenburg/18/2017	1A	2017-03-29	MDCK1/MDCK1	1280	<	10	10	<	20	10	10	20	40
B/Moscow/98/2017	1A	2017-03-31	MDCK1/MDCK1	2560	<	20	20	<	40	40	20	40	40
B/Nosy Be/1638/2017	1A	2017-04-04	MDCK1	1280	<	20	10	<	40	40	20	40	40
B/Moscow/107/2017	1A	2017-04-05	MDCK1/MDCK1	1280	<	20	10	<	40	40	20	40	40
B/Moscow/110/2017	1A	2017-04-06	MDCK1/MDCK1	640	<	20	20	<	40	80	20	40	80
B/Moscow/114/2017	1A	2017-04-19	MDCK1/MDCK1	640	<	20	20	<	40	80	20	40	80
B/Moscow/118/2017	1A	2017-04-27	MDCK1/MDCK1	640	<	20	20	<	40	80	20	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-12. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-08-01

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	F41/14 ^{*2}	NIB F52/16 ^{*2}	F29/13 ^{*2}	F01/13 ^{*4}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}
				1A		1A	1A	1A	1A	1A	1B	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	320	160	40	80	40	80	10	<	<
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	160	640	320	160	160	1280	80	40	20
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	160	640	320	160	80	640	40	20	20
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	1280	1280	1280	1280	320	160	160
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	320	160	160	160	320	40	40	40
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	160	320	160	160	160	640	40	20	20
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	20	20	20	80	80	40	80	40
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK3	2560	10	20	20	<	80	40	40	80	80
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	2560	<	40	20	20	80	80	40	80	80
TEST VIRUSES													
B/Toamasina/0289/2017	1A	2017-02-01	MDCK1/MDCK1	2560	<	40	20	20	40	80	40	80	40
B/Novgorog Velikiy/85/2017	1A	2017-02-15	MDCK2/MDCK2	2560	20	40	20	40	80	80	40	80	80
B/Toamasina/0740/2017	1A	2017-03-06	MDCK1/MDCK1	2560	10	40	20	20	80	80	40	80	40
B/Nosy Be/1095/2017	1A	2017-03-09	MDCK1/MDCK1	2560	10	40	20	20	80	80	40	80	40
B/England/36/2017	1A	2017-03-17	SIAT1/MDCK1	2560	<	40	20	20	80	80	40	80	40
B/Toamasina/1433/2017	1A	2017-04-05	MDCK1/MDCK1	2560	<	40	20	20	80	80	40	80	40
B/England/46/2017	1A	2017-04-26	SIAT1/MDCK1	2560	<	20	20	20	80	80	40	80	40
B/Fianarantsoa/2440/2017	1A	2017-05-19	MDCK1/MDCK1	2560	<	40	20	40	80	80	40	80	80
B/Hong Kong/269/2017	△ 162-164	2017-05-24	Cx/MDCK1	160	<	<	<	<	20	<	<	<	<
B/Hong Kong/286/2017	△ 162-164	2017-06-06	Cx/MDCK1	80	<	<	<	<	10	<	<	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-13. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-08-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										NEW	NEW	NEW	
				Post-infection ferret antisera													
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Maryland 15/16 MDCK	B/Maryland 15/16 Egg	B/Nor 2409/17 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	F41/14 ^{*2}	NIB F52/16 ^{*2}	F29/13 ^{*2}	F01/13 ^{*4}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}	CDC F2017-045 ^{*2}	CDC F2017-041 ^{*2}	F26/17 ^{*2}	
				Genetic group	1A	1A	1A	1A	1A	1A	1B	1A	1A	1A	1A	1A	
REFERENCE VIRUSES																	
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	320	640	40	40	40	160	10	<	<	<	320	20	
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	160	1280	320	160	160	1280	40	20	20	<	320	20	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	1280	320	160	160	1280	40	40	40	<	320	10	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	1280	1280	1280	320	160	320	80	1280	80		
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	40	1280	160	160	320	640	40	40	40	<	320	10	
B/South Australia/81/2012	1A	2012-11-28	E4/E1	2560	160	1280	160	160	320	1280	40	40	40	<	320	40	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	20	20	80	160	40	80	80	<	20	20	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK3	5120	<	80	20	<	80	80	20	80	80	<	40	20	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	5120	<	80	20	20	80	160	40	80	80	<	40	20	
B/Maryland/15/2016	Δ 162-163	1A	2016-12-27	C3/MDCK1	80	<	20	<	20	<	<	<	<	1280	160	320	
B/Maryland/15/2016	Δ 162-163	1A	2016-12-27	E4/E1	1280	160	640	40	40	80	320	<	<	<	1280	640	
B/Norway/2409/2017	Δ 162-163	1A	2017-04-27	MDCK1/MDCK1	80	<	10	<	<	10	<	<	<	1280	160	160	
TEST VIRUSES																	
B/Almatinskay/237/2017	1A	2017-02-06	MDCKx/MDCK1	2560	<	160	20	<	40	80	40	40	40	<	20	80	
B/Almatinskay/270/2017	1A	2017-02-09	MDCKx/MDCK1	2560	<	160	20	20	80	160	40	80	80	<	80	80	
B/Karaganda/190/2017	1A	2017-02-13	MDCKx/MDCK1	2560	<	160	20	20	80	160	40	80	80	<	40	80	
B/Irkutsk/579/2017	1A	2017-02-18	MDCK2/MDCK1	5120	<	160	40	40	80	320	40	80	80	10	80	80	
B/Karaganda/169/2017	1A	2017-02-20	MDCKx/MDCK1	2560	<	80	20	20	40	160	20	40	40	<	40	40	
B/Khabarovsk/344/2017	1A	2017-02-22	MDCK2/MDCK1	2560	<	160	20	40	80	160	40	80	80	<	40	80	
B/Khakassia/A1/2017	1A	2017-03-06	MDCK1/MDCK1	2560	160	1280	160	80	160	640	20	20	20	<	320	<	
B/Krasnoyarsk/5450/2017	1A	2017-03-21	MDCK2/MDCK1	2560	<	80	20	20	40	160	40	40	40	<	40	10	
B/Novosibirsk/75/2017	1A	2017-04-10	MDCK1/MDCK1	2560	<	80	10	<	80	80	40	40	40	<	40	80	
B/Yekaterinburg/29/2017	1A	2017-04-19	MDCK1/MDCK1	2560	<	80	<	<	40	80	20	40	40	<	20	10	
B/Analavory/2827/2017	1A	2017-06-20	MDCK1/MDCK1	2560	<	320	10	<	40	80	20	40	40	<	40	10	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used);

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

Sequences in phylogenetic trees

Vaccine

Table 11-14. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-08-15

Viruses	Other information	Passage history	Collection date	Haemagglutination inhibition titre										NEW	NEW	NEW	NEW	
				Post-infection ferret antisera										B/Maryland 15/16 MDCK CDC F2017-045 ²	B/Maryland 15/16 Egg CDC F2017-041 ²	B/Nor 2409/16 MDCK F20/17 ²	B/Nor 2409/17 MDCK F26/17 ²	
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK					
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F41/14 ²	NIB F52/16 ²	F29/13 ²	F01/13 ⁴	F04/13 ²	F41/13 ²	F09/13 ²	F15/16 ²	F16/16 ²					
				Genetic group	1A	1A	1A	1A	1A	1A	1B	1A	1A					1A
REFERENCE VIRUSES																		
B/Malaysia/2506/2004			2004-12-06	E3/E7	2560	640	320	80	80	80	160	20	<	<	<	320	10	20
B/Brisbane/60/2008	1A		2008-08-04	E4/E3	2560	160	1280	320	160	160	640	40	40	20	<	320	10	20
B/Malta/636714/2011	1A		2011-03-07	E4/E1	1280	80	640	160	160	160	320	40	40	20	<	320	10	10
B/Johannesburg/3964/2012	1A		2012-08-03	E1/E2	5120	640	1280	1280	1280	1280	320	320	320	80	1280	80	80	80
B/Formosa/V2367/2012	1A		2012-08-06	MDCK1/MDCK3	5120	80	640	160	160	320	640	40	80	80	<	320	10	20
B/South Australia/81/2012	1A		2012-11-28	E4/E1	1280	80	640	160	80	160	320	40	40	20	<	320	10	20
B/Hong Kong/514/2009	1B		2009-10-11	MDCK3	2560	<	80	20	20	80	80	40	80	80	<	20	<	20
B/Ireland/3154/2016	1A		2016-01-14	MDCK1/MDCK3	5120	<	80	20	20	80	80	40	160	80	<	40	10	20
B/Nordrhein-Westfalen/1/2016	1A		2016-01-04	C2/MDCK2	5120	10	80	40	20	80	80	40	160	160	<	80	10	20
B/Maryland/15/2016	△ 162-163	1A	2016-12-27	C3/MDCK1	160	<	10	<	<	20	<	<	10	<	640	320	640	640
B/Maryland/15/2016	△ 162-163	1A	2016-12-27	E4/E1	640	160	160	40	40	40	80	<	<	<	320	640	320	640
B/Norway/2409/2017	△ 162-163	1A	2017-04-27	MDCK1/MDCK1	80	<	<	<	<	10	<	<	<	<	320	320	640	320
TEST VIRUSES																		
B/Kazakhstan/7420/2017	1A			MDCK1	2560	<	80	40	<	80	160	40	160	80	<	<	<	20
B/Kazakhstan/7436/2017	1A		2017-02-22	MDCK1	2560	20	80	40	20	80	80	40	80	80	<	<	<	20
B/Kazakhstan/7440/2017	1A		2017-03-09	MDCK1	5120	10	160	40	40	80	160	80	160	160	10	10	10	40
B/Bucuresti/209398/2017	1A		2017-03-10	MDCK1/MDCK1	5120	10	80	20	20	80	80	40	160	80	ND	ND	10	40
B/Bucuresti/209401/2017	1A		2017-03-12	MDCK1/MDCK1	2560	<	80	20	<	40	80	40	80	80	ND	ND	10	40
B/Bucuresti/209562/2017	1A		2017-03-15	MDCK1/MDCK1	5120	<	80	20	20	80	80	40	160	40	ND	ND	ND	ND
B/Buzau/210450/2017	1A		2017-03-25	MDCK1/MDCK1	2560	20	160	40	40	160	160	40	80	80	<	10	10	20
B/Olt/210464/2017	1A		2017-03-30	MDCK1/MDCK1	2560	<	80	20	20	80	80	40	160	80	<	10	10	20
B/Bucuresti/210584/2017	1A		2017-04-03	MDCK1/MDCK1	2560	<	80	20	<	40	80	40	80	80	<	<	<	20
B/St Petersburg/282/2017	1A		2017-04-20	E1/E1	1280	320	320	160	80	80	320	40	80	20	<	320	<	10
B/Hong Kong/269/2017	△ 162-164	1A	2017-05-24	Cx/MDCK1	40	<	<	<	<	<	<	<	<	<	10	20	20	40
B/Hong Kong/286/2017	△ 162-164	1A	2017-06-06	Cx/MDCK1	80	<	<	<	<	<	10	<	<	<	20	40	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

ND = Not Done

Sequences in phylogenetic trees

Vaccine

Table 11-15. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-08-30

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Nor 2409/17 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574* ^{1,3}	F41/14 ^{*2}	NIB F52/16 ^{*2}	F29/13 ^{*2}	F01/13 ^{*4}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}	F20/17 ^{*2}	
				1A	1A	1A	1A	1A	1A	1A	1B	1A	1A	1A	
REFERENCE VIRUSES															
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	640	160	40	80	40	80	20	<	<	ND	
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	160	640	160	160	160	640	80	40	20	ND	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	160	640	160	160	160	640	80	40	20	ND	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	320	1280	640	640	640	1280	160	160	80	ND	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	2560	80	640	160	160	320	320	80	80	40	ND	
B/South Australia/81/2012	1A	2012-11-28	E4/E1	2560	160	640	160	80	160	320	80	40	20	ND	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	80	20	<	40	80	160	80	80	ND	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	<	80	10	<	40	40	80	80	80	ND	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	5120	<	80	20	<	80	80	80	80	80	ND	
B/Norway/2409/2017	1A	2017-04-27	MDCK1/MDCK2	160	<	<	<	<	10	<	<	<	<	320	
TEST VIRUSES															
B/Pennsylvania/19/2016		2016-03-16	E4/E2	1280	160	640	320	160	160	640	80	40	40	<	
B/St Petersburg/293/2016		2016-04-20	E1/E3/E1	1280	320	320	160	80	160	320	80	40	20	10	
B/Singapore/INFKK-16-0575/2016		2016-06-02	E5/E1	1280	320	640	160	80	80	640	80	40	20	<	
B/Kazakhstan/127/2017	1A	2017-02-06	MDCK1	2560	20	80	20	20	80	80	80	160	80	ND	
B/Kazakhstan/7410/2017	1A	2017-02-09	MDCK1	2560	<	80	20	<	80	80	160	160	80	ND	
B/Saudi Arabia/1057866160/2017	1A	2017-02-15	MDCK1	5120	10	80	20	20	80	80	160	160	80	ND	
B/Kazakhstan/7416/2017	1A	2017-02-18	MDCK1	2560	10	80	20	20	80	80	80	160	80	ND	
B/Ireland/02932/2017		2017-02-24	C1/MDCK1	2560	<	40	20	<	40	80	80	80	80	ND	
B/Colorado/06/2017	△ 162-163	2017-02-25	C1/MDCK1	160	10	10	<	<	20	<	<	<	<	320	
B/Colorado/06/2017	△ 162-163	2017-02-25	E5/E1	640	40	640	20	20	40	160	10	<	<	160	
B/Saudi Arabia/247224/2017	1A	2017-02-27	MDCK1	5120	160	640	160	160	320	640	80	80	40	ND	
B/Bucuresti/210480/2017	1A	2017-04-04	MDCK1	2560	10	80	20	20	80	80	80	80	80	ND	
B/Bucuresti/931-C8857/2017	1A	2017-04-05	MDCK1	5120	80	640	80	80	160	640	160	160	80	ND	
B/St Petersburg/238/2017	1A	2017-04-05	E1/E2	2560	160	1280	320	160	320	640	160	80	40	<	
B/Olt/210799/2017	1A	2017-04-08	MDCK1	2560	10	80	20	20	80	80	80	80	80	ND	
B/Bucuresti/210885/2017	1A	2017-04-09	MDCK1	2560	10	80	20	<	80	80	80	80	80	ND	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

ND = Not Done

Sequences in phylogenetic trees

Vaccine

Table 11-16. Antigenic analyses of influenza B viruses (Victoria lineage) 2017-09-05

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre																NEW	NEW	NEW	NEW
				Post-infection ferret antisera																B/Nor 2409/17 MDCK	B/HK 269/17 Egg	B/Nor 2409/17 MDCK	B/Nor 2409/17 Egg
				B/Bris 60/08 Egg	B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Nor 2409/17 MDCK	B/HK 269/17 MDCK	B/HK 269/17 Egg	B/Nor 2409/17 MDCK	B/Nor 2409/17 Egg					
				Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	F41/14 ^{*2}	NIB F52/16 ^{*2}	F29/13 ^{*2}	F01/13 ^{*4}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}	F20/17 ^{*2}	F33/17 ^{*2}	F34/17 ^{*2}	F35/17 ^{*2}	F36/17 ^{*2}					
				Genetic group	1A		1A	1A	1A	1A	1A	1B	1A	1A									
REFERENCE VIRUSES																							
B/Malaysia/2506/2004		2004-12-06	E3/E7	2560	640	160	40	80	40	160	20	<	<	<	<	160	<	160					
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	320	640	160	160	160	640	80	40	40	10	<	80	10	160					
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	160	640	320	160	160	640	80	40	20	<	<	80	<	80					
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	640	640	640	1280	320	160	160	40	<	320	40	320					
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	80	640	160	160	320	640	80	80	40	<	<	80	<	160					
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	160	640	80	80	160	320	80	40	20	10	<	40	10	80					
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	10	40	20	10	40	80	80	80	80	<	<	<	<	<					
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	10	80	20	20	80	80	80	160	80	10	<	<	<	20					
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	5120	<	80	20	20	80	80	80	80	80	<	<	<	<	<					
B/Hong Kong/269/2017	Δ 162-164	1A	Cx/MDCK2	160	<	<	<	<	10	10	<	<	<	40	640	640	10	40					
B/Hong Kong/269/2017	clone 75	1A	E4	640	160	160	20	40	40	80	10	<	<	20	640	1280	<	320					
B/Norway/2409/2017	Δ 162-163	1A	MDCK1/MDCK2	160	<	<	<	<	20	<	<	20	<	640	<	20	160	160					
B/Norway/2409/2017	clone 1	1A	E7	1280	80	80	20	<	80	80	10	<	<	160	<	80	20	640					
TEST VIRUSES																							
B/Shanghai-Songjiang/1173/2017	1A	2017-03-27	E3/E1	2560	40	80	10	<	80	80	80	80	80	<	<	<	<	<					
B/Shanghai-Songjiang/1173/2017	1A	2017-03-27	C2/MDCK1	1280	40	320	80	80	160	320	80	40	40	<	<	20	<	40					
B/Shanghai-Baoshan/1325/2017	1A	2017-04-24	E3/E2	1280	80	320	160	80	80	320	80	40	20	ND	ND	ND	ND	ND					

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

ND = Not Done

Sequences in phylogenetic trees

Vaccine

Table 11-17. Antigenic analyses of influenza B viruses (Victoria lineage) with collection dates after 2017-01-31 - Summary

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										NEW	NEW	NEW	NEW
				Post-infection ferret antisera													
				B/Mal 2506/04 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Jhb 3964/12 Egg	B/For V2367/12 MDCK	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Maryland 15/16 MDCK	B/Maryland 15/16 Egg	B/Nor 2409/16 MDCK	B/Nor 2409/17 MDCK	
				F41/14 ^{*2}	NIB F52/16 ^{*2}	F29/13 ^{*2}	F01/13 ^{*4}	F04/13 ^{*2}	F41/13 ^{*2}	F09/13 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}	CDC F2017-045 ^{*2}	CDC F2017-041 ^{*2}	F20/17 ^{*2}	F26/17 ^{*2}	
				Genetic group		1A	1A	1A	1A	1A	1B	1A	1A	1A	1A	1A	1A
REFERENCE VIRUSES																	
B/Malaysia/2506/2004		2004-12-06	E3/E7	640	320	80	80	80	160	20	<	<	<	320	10	20	
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	160	1280	320	160	160	640	40	40	20	<	320	10	20	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	80	640	160	160	160	320	40	40	20	<	320	10	10	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	640	1280	1280	1280	1280	1280	320	320	320	80	1280	80	80	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	80	640	160	160	320	640	40	80	80	<	320	10	20	
B/South Australia/81/2012	1A	2012-11-28	E4/E1	80	640	160	80	160	320	40	40	20	<	320	10	20	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	<	80	20	20	80	80	40	80	80	<	20	<	20	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK3	<	80	20	20	80	80	40	160	160	<	40	10	20	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	10	80	40	20	80	80	40	160	160	<	80	10	20	
B/Maryland/15/2016	△ 162-163	2016-12-27	C3/MDCK1	<	10	<	<	20	<	<	10	<	640	320	640	640	
B/Maryland/15/2016	△ 162-163	2016-12-27	E4/E1	160	160	40	40	40	80	<	<	<	320	640	320	640	
B/Norway/2409/2017	△ 162-163	2017-04-27	MDCK1/MDCK1	<	<	<	<	10	<	<	<	<	320	320	640	320	
TEST VIRUSES																	
Number of viruses tested*				174	174	174	174	174	174	174	174	174	19	19	17	21	
No with titre reduction ≤2-fold				3	12	10		101	24	167	165	163		1	2	6	
%				1.7	6.9	5.7		58.0	13.8	96.0	94.8	93.7		5.3	11.8	28.6	
No with titre reduction =4-fold				5	2	7	3	58	97	2	6	7		1		2	
%				2.9	1.1	4.1	1.7	33.3	55.7	1.1	3.4	4.0		5.3		9.5	
No with titre reduction ≥8-fold				166	160	157	171	15	53	5	3	4	19	17	15	13	
%				95.4	92.0	90.2	98.3	8.7	30.5	2.9	1.8	2.3	100	89.4	88.2	61.9	

* All viruses sequenced carried HA genes that fell in the B/Brisbane/60/2008 clade (clade 1A)

Vaccine

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 11. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine virus

Reference viruses

Collection date

Mar 2017

Apr 2017

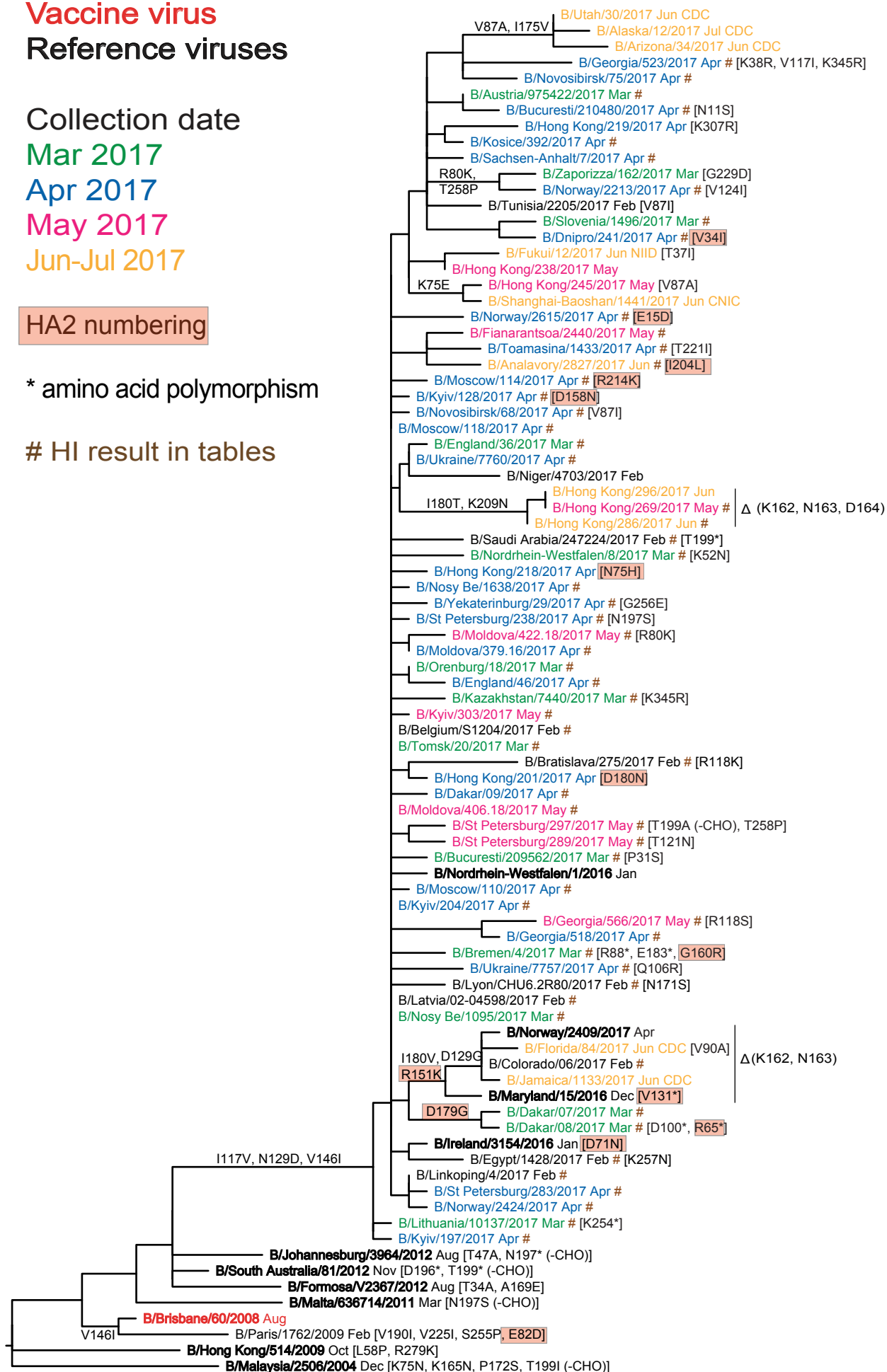
May 2017

Jun-Jul 2017

HA2 numbering

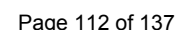
* amino acid polymorphism

HI result in tables



0.001

Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



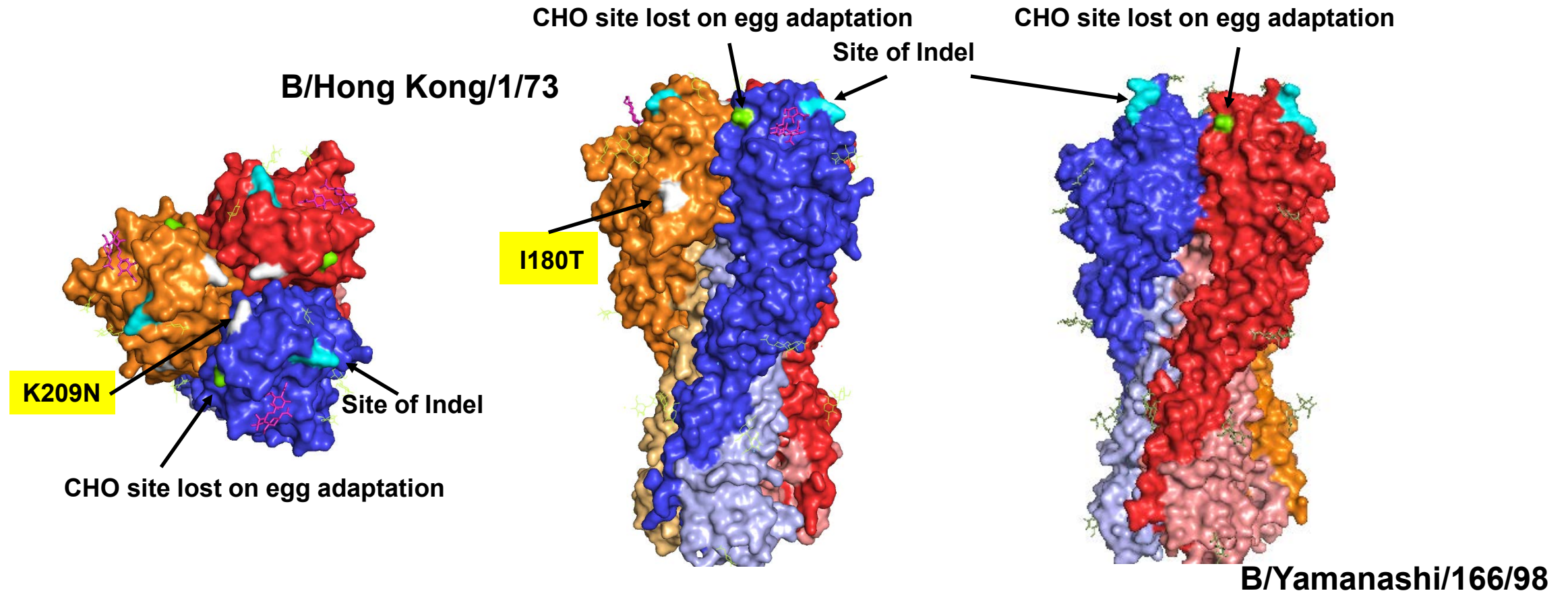
	310	320	330	340	3	13	23	33	43	53
B/Malaysia/2506/2004	G	G	L	N	K	S	K	P	Y	T
B/Hong Kong/514/2009	G	G	L	N	K	S	K	P	Y	T
B/Brisbane/60/2008	G	G	L	N	K	S	K	P	Y	T
B/Malta/636714/2011	G	G	L	N	K	S	K	P	Y	T
B/Formosa/V2367/2012	G	G	L	N	K	S	K	P	Y	T
B/South Australia/81/2012	G	G	L	N	K	S	K	P	Y	T
B/Johannesburg/3964/2012	G	G	L	N	K	S	K	P	Y	T
B/Kyiv/197/2017	G	G	L	N	K	S	K	P	Y	T
B/Ireland/3154/2016	G	G	L	N	K	S	K	P	Y	T
B/Dakar/07/2017	G	G	L	N	K	S	K	P	Y	T
B/Maryland/15/2016	G	G	L	N	K	S	K	P	Y	T
B/Norway/2409/2017	G	G	L	N	K	S	K	P	Y	T
B/Ukraine/7757/2017	G	G	L	N	K	S	K	P	Y	T
B/Georgia/566/2017	G	G	L	N	K	S	K	P	Y	T
B/Nordrhein-Westfalen/1/2016	G	G	L	N	K	S	K	P	Y	T
B/St Petersburg/289/2017	G	G	L	N	K	S	K	P	Y	T
B/Kazakhstan/7440/2017	G	G	L	N	K	S	K	P	Y	T
B/Yekaterinburg/29/2017	G	G	L	N	K	S	K	P	Y	T
B/Nordrhein-Westfalen/8/2017	G	G	L	N	K	S	K	P	Y	T
B/Hong Kong/286/2017	G	G	L	N	K	S	K	P	Y	T
B/Hong Kong/269/2017	G	G	L	N	K	S	K	P	Y	T
B/Novosibirsk/68/2017	G	G	L	N	K	S	K	P	Y	T
B/Analavory/2827/2017	G	G	L	N	K	S	K	P	Y	T
B/Hong Kong/245/2017	G	G	L	N	K	S	K	P	Y	T
B/Norway/2213/2017	G	G	L	N	K	S	K	P	Y	T
B/Zaporizha/162/2017	G	G	L	N	K	S	K	P	Y	T
B/Hong Kong/219/2017	G	G	L	N	K	S	K	P	Y	T
B/Bucuresti/210480/2017	G	G	L	N	K	S	K	P	Y	T
B/Georgia/523/2017	G	G	L	N	K	S	K	P	Y	T
B/Alaska/12/2017	G	G	L	N	K	S	K	P	Y	T

	63	73	83	93	103	113	123	133	143	153
B/Malaysia/2506/2004	S	L	S	E	L	V	K	N	L	Q
B/Hong Kong/514/2009	S	L	S	E	L	V	K	N	L	Q
B/Brisbane/60/2008	S	L	S	E	L	V	K	N	L	Q
B/Malta/636714/2011	S	L	S	E	L	V	K	N	L	Q
B/Formosa/V2367/2012	S	L	S	E	L	V	K	N	L	Q
B/South Australia/81/2012	S	L	S	E	L	V	K	N	L	Q
B/Johannesburg/3964/2012	S	L	S	E	L	V	K	N	L	Q
B/Kyiv/197/2017	S	L	S	E	L	V	K	N	L	Q
B/Ireland/3154/2016	S	L	S	E	L	V	K	N	L	Q
B/Dakar/07/2017	S	L	S	E	L	V	K	N	L	Q
B/Maryland/15/2016	S	L	S	E	L	V	K	N	L	Q
B/Norway/2409/2017	S	L	S	E	L	V	K	N	L	Q
B/Ukraine/7757/2017	S	L	S	E	L	V	K	N	L	Q
B/Georgia/566/2017	S	L	S	E	L	V	K	N	L	Q
B/Nordrhein-Westfalen/1/2016	S	L	S	E	L	V	K	N	L	Q
B/St Petersburg/289/2017	S	L	S	E	L	V	K	N	L	Q
B/Kazakhstan/7440/2017	S	L	S	E	L	V	K	N	L	Q
B/Yekaterinburg/29/2017	S	L	S	E	L	V	K	N	L	Q
B/Nordrhein-Westfalen/8/2017	S	L	S	E	L	V	K	N	L	Q
B/Hong Kong/286/2017	S	L	S	E	L	V	K	N	L	Q
B/Hong Kong/269/2017	S	L	S	E	L	V	K	N	L	Q
B/Novosibirsk/68/2017	S	L	S	E	L	V	K	N	L	Q
B/Analavory/2827/2017	S	L	S	E	L	V	K	N	L	Q
B/Hong Kong/245/2017	S	L	S	E	L	V	K	N	L	Q
B/Norway/2213/2017	S	L	S	E	L	V	K	N	L	Q
B/Zaporizha/162/2017	S	L	S	E	L	V	K	N	L	Q
B/Hong Kong/219/2017	S	L	S	E	L	V	K	N	L	Q
B/Bucuresti/210480/2017	S	L	S	E	L	V	K	N	L	Q
B/Georgia/523/2017	S	L	S	E	L	V	K	N	L	Q
B/Alaska/12/2017	S	L	S	E	L	V	K	N	L	Q

	163	173	183	193	203	213	223
B/Malaysia/2506/2004	A	G	T	F	D	A	G
B/Hong Kong/514/2009	A	G	T	F	D	A	G
B/Brisbane/60/2008	A	G	T	F	D	A	G
B/Malta/636714/2011	A	G	T	F	D	A	G
B/Formosa/V2367/2012	A	G	T	F	D	A	G
B/South Australia/81/2012	A	G	T	F	D	A	G
B/Johannesburg/3964/2012	A	G	T	F	D	A	G
B/Kyiv/197/2017	A	G	T	F	D	A	G
B/Ireland/3154/2016	A	G	T	F	D	A	G
B/Dakar/07/2017	A	G	T	F	D	A	G
B/Maryland/15/2016	A	G	T	F	D	A	G
B/Norway/2409/2017	A	G	T	F	D	A	G
B/Ukraine/7757/2017	A	G	T	F	D	A	G
B/Georgia/566/2017	A	G	T	F	D	A	G
B/Nordrhein-Westfalen/1/2016	A	G	T	F	D	A	G
B/St Petersburg/289/2017	A	G	T	F	D	A	G
B/Kazakhstan/7440/2017	A	G	T	F	D	A	G
B/Yekaterinburg/29/2017	A	G	T	F	D	A	G
B/Nordrhein-Westfalen/8/2017	A	G	T	F	D	A	G
B/Hong Kong/286/2017	A	G	T	F	D	A	G
B/Hong Kong/269/2017	A	G	T	F	D	A	G
B/Novosibirsk/68/2017	A	G	T	F	D	A	G
B/Analavory/2827/2017	A	G	T	F	D	A	G
B/Hong Kong/245/2017	A	G	T	F	D	A	G
B/Norway/2213/2017	A	G	T	F	D	A	G
B/Zaporizha/162/2017	A	G	T	F	D	A	G
B/Hong Kong/219/2017	A	G	T	F	D	A	G
B/Bucuresti/210480/2017	A	G	T	F	D	A	G
B/Georgia/523/2017	A	G	T	F	D	A	G
B/Alaska/12/2017	A	G	T	F	D	A	G

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 13. Influenza B (Victoria-lineage) HA - HA1 residues defining deletion variants



Location of key sites on B/Hong Kong/1/73 (triple deletion) and BN/Yamanashi/166/98 (B/Yamagata-lineage)

Double deletion variants are defined by HA1 I180V and HA2 R151K substitutions, with many also having HA1 D129G substitution

Figure 14. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine virus

Reference viruses

Collection date

Mar 2017

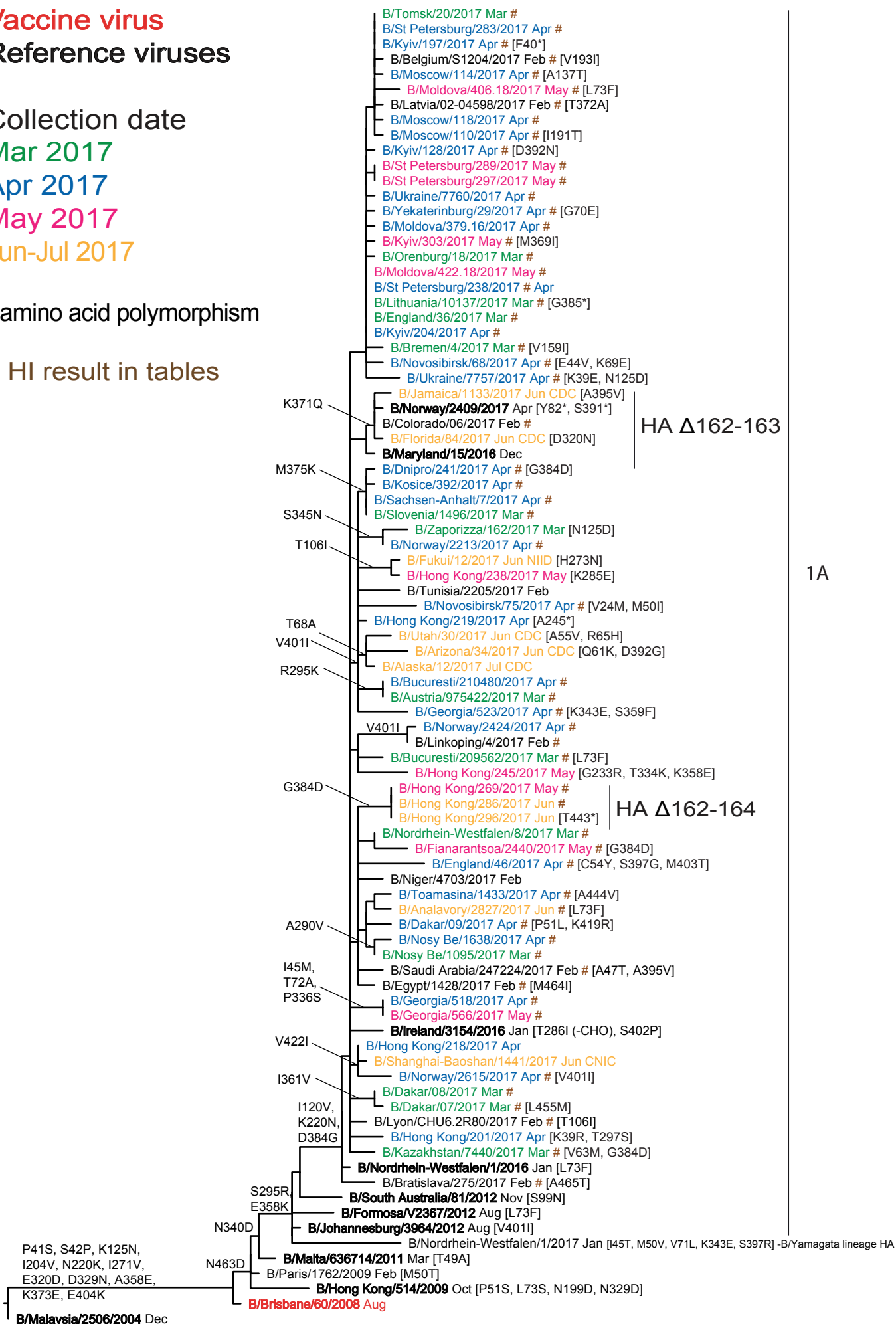
Apr 2017

May 2017

Jun-Jul 2017

* amino acid polymorphism

HI result in tables



	310	320	330	340	350	360	370	380	390	400
B/Malaysia/2506/2004	P	F	V	K	L	N	V	E	T	D
B/Hong Kong/514/2009	A	E	I	R	L	M	C	T	E	T
B/Brisbane/60/2008	Y	L	D	T	P	R	P	D	D	G
B/Malta/636714/2011	S	I	T	G	P	C	E	S	N	G
B/Formosa/V2367/2012	D	K	G	S	G	G	I	K	G	G
B/South Australia/81/2012	F	V	H	Q	R	M	A	S	K	I
B/Johannesburg/3964/2012	G	R	W	Y	S	R	T	M	S	K
B/Kyiv/197/2017	T	K	R	M	G	M	G	L	Y	V
B/Ireland/3154/2016	K	Y	D	G	P	W	A	D	S	D
B/Dakar/07/2017	A	L	A	F	S	G	V	M		
B/Maryland/15/2016										
B/Norway/2409/2017										
B/Ukraine/7757/2017										
B/Georgia/566/2017										
B/Nordrhein-Westfalen/1/2016										
B/St Petersburg/289/2017										
B/Kazakhstan/7440/2017										
B/Yekaterinburg/29/2017										
B/Nordrhein-Westfalen/8/2017										
B/Hong Kong/286/2017										
B/Hong Kong/269/2017										
B/Novosibirsk/68/2017										
B/Analavory/2827/2017										
B/Hong Kong/245/2017										
B/Norway/2213/2017										
B/Zaporizka/162/2017										
B/Hong Kong/219/2017										
B/Bucuresti/210480/2017										
B/Georgia/523/2017										
B/Alaska/12/2017										
B/Malaysia/2506/2004										
B/Hong Kong/514/2009										
B/Brisbane/60/2008										
B/Malta/636714/2011										
B/Formosa/V2367/2012										
B/South Australia/81/2012										
B/Johannesburg/3964/2012										
B/Kyiv/197/2017										
B/Ireland/3154/2016										
B/Dakar/07/2017										
B/Maryland/15/2016										
B/Norway/2409/2017										
B/Ukraine/7757/2017										
B/Georgia/566/2017										
B/Nordrhein-Westfalen/1/2016										
B/St Petersburg/289/2017										
B/Kazakhstan/7440/2017										
B/Yekaterinburg/29/2017										
B/Nordrhein-Westfalen/8/2017										
B/Hong Kong/286/2017										
B/Hong Kong/269/2017										
B/Novosibirsk/68/2017										
B/Analavory/2827/2017										
B/Hong Kong/245/2017										
B/Norway/2213/2017										
B/Zaporizka/162/2017										
B/Hong Kong/219/2017										
B/Bucuresti/210480/2017										
B/Georgia/523/2017										
B/Alaska/12/2017										

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 12. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2017-01-31

MONTH	B Yamagata lineage				
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³
2017 FEBRUARY					
Austria	6	6	5	1	
Belgium	1	1			1
France	1	1		1	
Germany	4	4		1	3
Ghana	1	1	1		
Iceland	3	3	1	1	1
Latvia	3	3	1		2
Niger	2	in process			
Norway	7	5		1	4
Russia	1	1		1	
Saudi Arabia	1	1	1		
Serbia	1	1			1
Slovakia	1	1			1
Slovenia	2	2		1	1
2017 MARCH					
Austria	1	1		1	
France	1	1		1	
Germany	2	2		2	
Iceland	2	2	1		1
Ireland	1	1	1		
Italy	3	2		1	1
Jordan	1	1		1	
Latvia	1	1			1
Lithuania	4	4	1	1	2
Niger	1	in process			
Norway	4	3		1	2
Saudi Arabia	2	2	2		
Serbia	1	1		1	
Slovakia	1	1		1	
Slovenia	4	4		1	3
2017 APRIL					
Austria	2	2	2		
Germany	1	1	1		
Ireland	2	2	1	1	
Latvia	1	1			1
Norway	8	8	5	3	
Romania	1	1	1		
Slovenia	1	1			1
Ukraine	1	1	1		
United Kingdom	1	1			1
2017 MAY					
Hong Kong	24	sequenced		3	
Norway	2	2	1	1	
United Kingdom	1	1	1		
2017 JUNE					
Ghana	1	0			
Niger	1	in process			
2017 JULY					
South Africa	3	3	3		
	113	80	30	26	27
23 Countries			36.1%	31.3%	32.5%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay

3. Reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 13-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-02-28 + 2017-03-08

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre												B/HK 3417/14 Egg St Jude F715/14 ^{1,4}
					Post-infection ferret antisera												
					NEW												
					B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg		
					SH614 ^{1,3}	F17/13 ¹	F38/14 ²	F27/13 ¹	F05/15 ¹	F42/14 ²	F06/17 ²	F10/13 ²	F06/15 ²	F35/14 ²	F37/14 ²		
Genetic Group	3	1	2	2	2	2	2	3	3	3	3	3					
REFERENCE VIRUSES																	
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	320	160	640	ND	160	160	20	160	160		
B/Brisbane/3/2007	2	2007-09-03	E2/E2	2560	1280	1280	320	160	1280	ND	320	320	20	320	320		
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	640	160	40	ND	40	20	40	40	160		
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	640	320	320	ND	160	80	40	160	320		
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	640	320	320	160	80	320	640	40	80	<	80	160		
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	5120	640	320	320	40	640	ND	640	320	80	640	640		
B/Stockholm/12/2011	3	2011-03-28	Ex/E2	1280	160	80	40	<	80	ND	80	160	20	80	80		
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	320	320	320	640	320	ND	320	160	640	320	320		
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	160	160	40	10	160	160	160	80	20	160	160		
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	40	40	10	80	ND	80	40	10	80	160		
TEST VIRUSES																	
B/Ghana/DARI-16-0165/2016	3	2016-08-10	MDCK1	1280	40	20	40	40	20	40	40	20	40	40	80		
B/Guangxi-Xixiangtang/11019/2016	3	2016-09-19	C2/MDCK1	1280	40	20	80	20	10	ND	20	10	40	20	80		
B/Ghana/FS-16-1501/2016	3	2016-10-11	Cx/MDCK1	2560	40	20	80	40	40	80	80	20	80	40	80		
B/Ghana/DILI-16-1031/2016	3	2016-10-12	Cx/MDCK1	1280	40	20	40	20	20	<	40	20	40	20	80		
B/Guangxi-Jiangzhou/1917/2016	3	2016-11-18	C4/MDCK1	1280	40	10	40	10	10	ND	20	10	10	20	80		
B/Sachsen/74/2016	3	2016-11-21	C1/MDCK1	1280	40	20	80	40	20	<	40	20	80	40	160		
B/Yunnan-Hongta/11394/2016	3	2016-11-25	C4/MDCK1	1280	40	20	40	40	20	ND	40	20	40	20	80		
B/Luxembourg/63550/2016	3	2016-12-15	MDCK1	1280	40	20	40	20	20	<	40	20	40	40	80		
B/Diyarbakir/1988/2016	3	2016-12-20	SIAT1/MDCK1	1280	40	20	40	20	20	ND	40	20	20	40	80		
B/Croatia/4008/2016	3	2016-12-24	MDCKx/M3	1280	40	20	40	20	20	ND	40	10	40	40	80		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

ND = Not Done

B/Yamagata-lineage virus recommended for use in quadrivalent vaccines

Sequences in phylogenetic trees

Vaccine[#]

Table 13-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-03-15 + 2017-03-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Phuket 3073/13	B/FI 4/06	B/Bris 3/07	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/HK 3417/14	
				Egg	Egg	Egg	MDCK	MDCK	Egg	Egg	Egg	Egg	MDCK	Egg	
				Ferret number	SH614 ^{*1,3}	F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*2}	F05/15 ^{*2}	F42/14 ^{*2}	F10/13 ^{*2}	F06/15 ^{*2}	F35/14 ^{*2}	F37/14 ^{*2}	St Jude F715/14 ^{*2,4}
Genetic Group	3	1	2	2	2	2	3	3	3	3	3				
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	320	160	640	160	160	20	160	160	
B/Brisbane/3/2007	2	2007-09-03	E2/E2	2560	1280	1280	320	160	1280	320	320	20	320	320	
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	640	160	40	40	20	40	40	160	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	640	320	320	160	80	40	160	320	
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	640	320	320	160	40	640	40	80	<	160	160	
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	5120	640	320	320	40	640	640	320	80	640	640	
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	40	20	<	80	40	80	<	40	80	
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	320	320	320	640	320	320	160	640	320	320	
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	160	160	40	10	160	160	80	20	160	160	
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	40	40	10	80	80	40	10	80	160	
TEST VIRUSES															
B/New Hampshire/01/2016	3	2016-01-07	MDCK2/MDCK1	5120	320	320	320	640	320	320	160	640	320	320	
B/Texas/81/2016	3	2016-04-19	MDCK2/MDCK1	2560	80	80	80	20	80	80	40	80	80	160	
B/Cameroon/7497/2016	3	2016-09-02	MDCK3/MDCK1	2560	80	160	80	80	80	160	40	80	80	160	
B/Cameroon/8244/2016	3	2016-09-14	MDCK2/MDCK1	1280	80	80	80	40	40	80	20	80	40	160	
B/Cameroon/8223/2016	3	2016-09-29	MDCK1/MDCK1	2560	80	160	160	160	160	160	40	160	80	160	
B/Cameroon/8242/2016	3	2016-09-30	MDCK3/MDCK1	2560	40	80	80	40	80	80	20	160	80	80	
B/Cameroon/8224/2016	3	2016-10-01	MDCK3/MDCK1	2560	80	80	80	80	40	80	20	80	80	160	
B/Cameroon/8227/2016	3	2016-10-01	MDCK1/MDCK1	640	40	80	40	20	10	40	<	40	10	80	
A/Denmark/48/2016	3	2016-10-06	SIAT2/MDCK1/SIAT1	2560	80	160	80	80	80	160	40	80	80	160	
B/Ghana/DARI-16-0225/2016	3	2016-10-25	MDCK2	2560	40	80	80	40	40	80	20	80	40	160	
B/Yokohama/112/2016	3	2016-11-15	MDCK2/MDCK2/MDCK1	1280	40	80	40	40	40	80	<	40	40	80	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-04-05 + 2017-05-03

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg
				SH614 ^{*1,3}	F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*5}	F05/15 ^{*2}	F42/14 ^{*2}	F10/13 ^{*2}	F06/15 ^{*2}	F35/14 ^{*2}	F51/16 ^{*2}	F715/14 ^{*2,4}
	Passage history			3	1	2	2	2	2	3	3	3	3	3
	Ferret number													
	Genetic Group													
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	640	640	160	80	640	160	160	10	160	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	1280	640	640	160	160	640	160	160	<	320	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	40	40	320	160	20	10	<	<	40	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	320	160	320	80	80	10	80	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	320	640	160	80	640	80	80	<	320	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	320	320	80	<	320	320	160	40	320	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	80	<	<	80	40	80	<	40	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	320	160	320	320	320	80	320	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	320	160	40	10	160	160	80	20	320	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	40	40	<	<	40	40	20	<	40	160
TEST VIRUSES														
B/Dakar/03/2016	3	2016-11-09	P1/MDCK1	2560	40	40	40	40	40	80	20	40	40	80
B/Catalonia/2134258NS/2017	3	2017-01-04	C0/MDCK1	2560	160	160	80	320	160	160	40	160	160	160
B/Norway/877/2017	3	2017-01-25	MDCK1/MDCK1	1280	40	40	40	20	40	40	20	40	40	80
B/Rheinland-Pfalz/1/2017	3	2017-01-31	C1/MDCK1	1280	40	80	40	20	40	80	20	40	40	80
B/Niedersachsen/3/2017	3	2017-02-06	C1/MDCK1	1280	40	40	40	20	40	80	20	40	40	80
B/Iceland/09/2017	3	2017-02-07	MDCK1/MDCK1	1280	80	80	80	40	80	80	20	80	40	80
B/Norway/1073/2017	3	2017-02-13	MDCK1/MDCK1	2560	40	80	80	40	80	80	20	80	40	160
B/Norway/1014/2017	3	2017-02-14	MDCK1/MDCK1	2560	40	40	80	40	40	80	20	40	40	80
B/Nordrhein-Westfalen/5/2017	3	2017-02-14	C1/MDCK1	1280	80	80	40	20	40	80	20	40	40	160
B/Norway/1152/2017	3	2017-02-15	MDCK1/MDCK1	2560	40	80	80	40	80	80	20	40	40	160
B/Bremen/2/2017	3	2017-02-15	C1/MDCK1	1280	40	40	40	20	40	40	20	40	40	80
B/Iceland/46/2017	3	2017-02-24	MDCK1/MDCK1	2560	160	160	160	320	160	160	80	640	160	160
B/Iceland/45/2017	3	2017-02-24	MDCK1/MDCK1	1280	40	80	40	20	80	80	20	40	80	80
B/Iceland/58/2017	3	2017-03-03	MDCK1/MDCK1	1280	40	80	80	40	80	40	20	40	40	80
B/Iceland/62/2017	3	2017-03-06	MDCKx/MDCK1	2560	160	160	320	640	320	160	80	640	320	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC; 5 < = <20

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-05-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg
				SH614 ^{*1,3}	F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*4}	F05/15 ^{*2}	F42/14 ^{*2}	F10/13 ^{*2}	F06/15 ^{*2}	F35/14 ^{*2}	NIB F51/16 ^{*2}	St Jude's F715/14 ^{*2,4}
				3	1	2	2	2	2	3	3	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	640	640	160	80	640	160	160	10	160	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	1280	640	640	160	160	640	160	160	<	320	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	40	40	320	160	20	10	<	<	40	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	320	160	320	80	80	10	80	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	640	160	160	80	640	80	80	<	320	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	320	320	80	<	320	320	160	40	320	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	80	<	<	80	40	80	<	40	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	320	160	320	320	320	80	320	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	5120	320	80	40	20	320	320	160	20	640	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	40	40	<	<	40	40	20	<	40	160
TEST VIRUSES														
B/Latvia/02-018885/2017	3	2017-02-07	C2/MDCK1	5120	320	320	320	640	320	320	160	640	320	320
B/Latvia/02-031654/2017	3	2017-02-13	C2/MDCK1	2560	80	160	160	160	160	160	40	160	80	320
B/Latvia/02-058113/2017	3	2017-02-21	C2/MDCK1	2560	80	80	80	80	80	80	20	80	80	160
B/Lithuania/8622/2017	3	2017-03-10	MDCK2/MDCK1	5120	160	160	160	320	320	160	80	160	160	160
B/Lithuania/8865/2017	3	2017-03-14	MDCK2/MDCK1	2560	40	80	80	40	40	80	20	40	40	160
B/Lithuania/9464/2017	3	2017-03-20	MDCK2/MDCK1	5120	320	320	640	1280	320	320	160	1280	320	320
B/Lithuania/10115/2017	3	2017-03-23	MDCK2/MDCK1	2560	80	80	80	40	80	80	40	80	80	160
B/Latvia/03-082708/2017	3	2017-03-30	C2/MDCK1	2560	80	80	160	160	80	80	40	160	80	160
B/Latvia/04-014585/2017	3	2017-04-05	C1/MDCK1	1280	40	80	40	80	40	40	20	40	40	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-5. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-05-31 + 2017-06-06

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg
Passage history		Passage history			SH614 ^{*1,3}	F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*2}	F05/15 ^{*2}	F42/14 ^{*2}	F10/13 ^{*2}	F06/15 ^{*2}	F35/14 ^{*2}	NIB F51/16 ^{*2}	St Jude's F715/14 ^{*2,4}
Ferret number		Ferret number			3	1	2	2	2	2	3	3	3	3	3
Genetic Group		Genetic Group													
REFERENCE VIRUSES															
B/Florida/4/2006		1	2006-12-15	E7/E1	2560	640	640	160	80	640	160	160	20	320	160
B/Brisbane/3/2007		2	2007-09-03	E2/E2	2560	640	640	160	80	640	160	160	20	320	160
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	1280	80	160	640	160	80	40	20	20	40	160
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	640	320	640	160	80	40	320	320
B/Massachusetts/02/2012		2	2012-03-13	E3/E3	1280	640	320	160	80	640	80	160	40	320	160
B/Wisconsin/1/2010		3	2010-02-20	E3/E2	5120	320	640	80	20	320	320	160	40	320	320
B/Stockholm/12/2011		3	2011-03-28	EX/E2	1280	160	160	40	10	80	40	80	20	160	80
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK2	5120	320	320	160	320	320	320	160	320	320	320
B/Phuket/3073/2013		3	2013-11-21	E4/E3	5120	320	160	40	20	320	320	160	40	640	160
B/Hong Kong/3417/2014		3	2014-06-04	E4/E1	1280	80	80	20	10	40	80	20	10	80	160
TEST VIRUSES															
B/Jordan/20160020680/2016		3	2016-11-08	MDCK1	5120	160	160	80	80	160	160	40	80	160	160
B/Slovenia/741/2017		3	2017-01-30	MDCKx/MDCK1	2560	80	160	80	80	80	160	40	80	80	160
B/Slovenia/739/2017		3	2017-01-30	MDCK1/MDCK1	5120	320	320	160	160	320	160	80	320	160	320
B/Slovenia/816/2017		3	2017-02-01	MDCKx/MDCK1	5120	160	160	160	80	160	160	40	160	160	160
B/Serbia/2720/2017		3	2017-02-14	MDCK1	2560	80	160	80	80	80	160	40	80	80	160
B/Norway/1473/2017		3	2017-02-25	MDCK2	5120	160	160	80	80	160	160	40	160	160	160
B/Slovenia/1321/2017		3	2017-02-27	MDCK1/MDCK1	5120	80	160	80	80	160	160	40	160	80	320
B/Norway/1484/2017		3	2017-02-28	MDCK1	5120	80	80	80	80	80	160	40	80	80	160
B/Serbia/2771/2017		3	2017-03-01	MDCK1	5120	80	160	80	80	160	160	40	160	160	160
B/Norway/1471/2017		3	2017-03-03	MDCK1	2560	80	80	80	80	40	80	40	80	80	160
B/Slovenia/1427/2017		3	2017-03-14	MDCK1/MDCK1	2560	80	80	80	40	40	80	20	40	80	160
B/Slovenia/1467/2017		3	2017-03-20	SIATx/MDCK1	2560	40	80	80	40	40	80	20	80	40	160
B/Slovenia/1560/2017		3	2017-03-27	MDCKx/MDCK1	2560	40	80	40	40	40	80	20	40	40	160
B/Slovenia/1666/2017		3	2017-04-11	MDCKx/MDCK1	2560	40	80	80	40	40	80	20	40	40	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-6. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-06-20 + 2017-07-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg
				SH614 ^{*1,3}	F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*2}	F05/15 ^{*2}	F42/14 ^{*2}	F36/15 ^{*2}	F06/15 ^{*2}	F35/14 ^{*2}	F51/16 ^{*2}	F715/14 ^{*2,4}
	Passage history			3	1	2	2	2	2	3	3	3	3	3
	Ferret number													
	Genetic Group													
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	160	80	640	160	160	20	320	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	2560	640	640	160	80	640	160	160	20	320	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	160	640	160	80	40	20	20	40	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	640	320	640	160	80	40	320	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	640	640	160	80	640	80	80	<	320	80
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	5120	320	640	80	20	320	320	160	40	320	320
B/Stockholm/12/2011	3	2011-03-28	EX/E2	1280	160	160	40	10	80	40	80	20	160	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	320	320	160	320	320	320	160	320	320	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	160	160	40	20	160	80	80	20	320	80
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	80	20	10	40	80	20	10	80	160
TEST VIRUSES														
B/Trnava/365/2017	3		MDCKx/MDCK1	2560	80	80	80	40	40	40	20	80	80	160
B/Kosice/231/2017	3	2017-01-12	MDCKx/MDCK1	2560	160	160	80	40	160	80	40	80	80	160
B/Slovenia/468/2017	3	2017-01-19	MDCKx/MDCK1	2560	80	40	40	40	20	40	20	40	40	80
B/Perugia/1/2017	3	2017-01-23	MDCK2/MDCK1	1280	40	40	40	20	20	20	10	40	20	80
B/Thuringen/2/2017	3	2017-02-15	C1/MDCK1	1280	40	40	40	10	40	40	20	40	80	80
B/Banska Bystrica/360/2017	3	2017-02-20	MDCKx/MDCK1	1280	40	40	40	20	20	20	20	20	20	80
B/Norway/1453/2017	3	2017-03-03	MDCK3	1280	40	40	40	20	20	20	10	20	20	80
B/Friuli-Venezia Giulia/18/2017	3	2017-03-06	MDCK1/MDCK1	2560	80	40	40	40	40	40	20	40	40	80
B/Sassari/1/2017	3	2017-03-10	MDCK3/MDCK1	2560	80	160	80	80	40	40	40	80	80	160
B/Komarno/383/2017	3	2017-03-23	MDCKx/MDCK1	2560	80	80	80	40	40	80	20	80	80	160
B/Norway/2431/2017	3	2017-04-21	MDCK2	2560	80	80	40	10	40	40	20	80	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-7. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-06-28

				Haemagglutination inhibition titre										
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg
				SH614 ^{*1,3}	F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*2}	F05/15 ^{*2}	F42/14 ^{*2}	F36/15 ^{*2}	F06/15 ^{*2}	F35/14 ^{*2}	NIB F51/16 ^{*2}	St Jude's F715/14 ^{*2,4}
Passage history	Passage history			3	1	2	2	2	2	3	3	3	3	3
Ferret number	Ferret number													
Genetic Group	Genetic Group													
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	640	1280	160	80	640	80	160	20	640	160
B/Brisbane/3/2007	2	2007-09-03	E2/E2	2560	1280	1280	320	160	1280	160	320	40	640	320
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	2560	160	160	1280	320	320	80	40	160	160	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	640	320	640	80	80	80	640	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	80	40	640	160	160	40	640	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	5120	320	320	80	20	640	160	320	40	640	160
B/Stockholm/12/2011	3	2011-03-28	EX/E2	1280	160	160	20	<	80	40	80	20	320	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	320	320	320	640	320	320	160	320	640	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	160	160	40	10	160	80	80	20	640	40
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	80	20	10	80	40	20	10	160	80
TEST VIRUSES														
B/Ukraine/7576/2017	3	2017-01-06	C1/MDCK1	2560	80	160	80	80	160	80	40	160	160	160
B/Jordan/20221/2017	3	2017-03-05	MDCK1	2560	160	80	80	40	80	80	40	160	160	160
B/Mecklenburg-Vorpommern/1/2017	3	2017-03-13	C1/MDCK1	2560	80	80	80	80	80	80	40	160	160	160
B/Rheinland-Pfalz/3/2017	3	2017-03-22	C1/MDCK1	2560	80	80	80	40	160	80	40	80	160	160
B/Norway/2133/2017	3	2017-03-28	MDCK1/MDCK1	2560	80	160	80	40	80	80	40	40	160	160
B/Norway/2162/2017	3	2017-04-03	MDCK1/MDCK1	2560	160	160	80	80	160	80	80	80	320	160
B/Norway/2155/2017	3	2017-04-04	MDCK1	2560	80	160	80	40	80	80	40	40	160	160
B/Norway/2169/2017	3	2017-04-04	MDCK1/MDCK1	5120	160	160	80	80	160	160	80	80	320	320
B/Saarland/3/2017	3	2017-04-06	C1/MDCK1	2560	160	160	80	80	160	80	40	160	320	160
B/Ukraine/7758/2017	3	2017-04-20	C1/MDCK1	5120	160	160	320	640	320	320	160	640	640	160
B/Norway/2363/2017	3	2017-04-21	MDCK1	2560	160	160	80	40	80	80	40	40	160	160
B/Norway/2387/2017	3	2017-04-24	MDCK1	2560	160	160	80	80	160	80	40	80	320	160
B/Norway/2410/2017	3	2017-04-26	MDCK1	2560	80	160	80	80	160	80	40	80	320	160
B/Norway/2421/2017	3	2017-04-28	MDCK2	2560	80	160	80	40	80	80	40	80	160	160
B/Norway/2613/2017	3	2017-05-02	MDCK1	5120	160	160	160	80	320	160	160	320	320	160
B/Norway/2600/2017	3	2017-05-02	MDCK1	2560	80	80	80	20	80	80	20	80	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 <= <40; 2 <= <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-8. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-07-19

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											B/Phuket 3073/13 Egg NIB F51/16*2	B/HK 3417/14 Egg St Jude's F715/14*2,4
				Post-infection ferret antisera												
				B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg		
				SH614*1,3	F17/13*1	F38/14*2	F27/13*2	F05/15*2	F42/14*2	F36/15*2	F06/15*2	F27/15*2	F51/16*2	F715/14*2,4		
				Genetic Group	3	1	2	2	2	2	3	3	3	3		
REFERENCE VIRUSES																
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	640	640	160	80	640	160	160	<	640	160		
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	640	640	160	80	640	160	160	<	640	160		
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	2560	160	160	640	640	160	320	20	80	160	160		
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	320	320	320	160	40	<	320	320		
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	640	640	160	80	640	80	80	<	320	160		
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	160	160	40	20	320	160	160	<	320	160		
B/Stockholm/12/2011	3	2011-03-28	EX/E2	1280	160	80	20	<	80	80	80	<	160	80		
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	160	160	160	320	80	160	320	160		
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	160	160	20	10	80	80	80	<	320	80		
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	640	40	40	10	<	40	40	20	<	80	80		
TEST VIRUSES																
B/Helsingborg/1/2016	3	2016-12-08	MDCK0/MDCK1	2560	80	80	80	80	40	80	20	40	80	80		
B/Lund/4/2016	3	2016-12-10	MDCK0/MDCK1	2560	80	80	40	40	40	80	20	40	80	80		
B/Belgium/S918/2017	3	2017-02-11	SIATx/MDCK1	1280	40	40	40	20	20	40	10	<	40	80		
B/St Petersburg/118/2017	3	2017-02-11	MDCK1/MDCK1	1280	80	80	40	40	40	80	20	<	80	80		
B/Austria/973511/2017	3	2017-02-27	C1/MDCK1	5120	320	160	160	320	320	320	160	160	320	160		
B/Austria/979756/2017	3	2017-03-27	C2/MDCK1	2560	80	80	80	40	20	80	20	<	80	160		
B/Austria/986609/2017	3	2017-04-24	C1/MDCK1	5120	160	160	160	320	160	160	80	80	320	160		
B/Austria/986875/2017	3	2017-04-24	C1/MDCK1	5120	320	160	160	320	320	320	80	160	320	160		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-9. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-08-01 + 2017-08-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg
				SH614 ^{*1,3}	F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*2}	F05/15 ^{*2}	F42/14 ^{*2}	F36/15 ^{*2}	F06/15 ^{*2}	F27/15 ^{*4}	F51/16 ^{*2}	F715/14 ^{*2,4}
	Passage history			3	1	2	2	2	2	3	3	3	3	3
	Ferret number													
	Genetic Group													
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7	5120	1280	1280	320	160	1280	320	320	40	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	2560	1280	1280	160	160	1280	160	320	20	1280	320
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	2560	80	160	640	640	160	160	20	10	80	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	2560	640	320	640	640	640	320	160	80	640	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	640	640	80	80	640	80	80	10	320	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	160	160	20	20	160	80	80	20	640	160
B/Stockholm/12/2011	3	2011-03-28	EX/E2	1280	160	80	20	<	80	80	80	10	160	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	160	160	320	160	80	160	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	160	160	10	<	80	80	40	10	320	80
B/Hong Kong/3417/2014	3	2014-06-04	E4/E3	1280	80	40	20	<	40	40	20	<	80	160
TEST VIRUSES														
B/Lisboa/2/2017	3	2017-01-24	SIAT1/MDCK1	5120	160	320	320	640	320	320	160	320	640	320
B/Austria/971742/2017	3	2017-02-17	C1/MDCK1	5120	320	320	320	1280	320	320	160	160	640	160
B/Austria/971746/2017	3	2017-02-18	C1/MDCK1	5120	320	640	640	1280	640	640	320	320	640	320
B/Austria/972293/2017	3	2017-02-20	C1/MDCK2	5120	160	160	160	320	160	320	80	160	320	320
B/Austria/972415/2017	3	2017-02-20	C1/MDCK1	2560	80	80	80	40	80	80	20	40	160	160
B/Austria/972769/2017	3	2017-02-20	C1/MDCK1	2560	80	80	40	20	40	40	20	40	80	160
B/Lorraine/1080/2017	3	2017-02-22	MDCK1/MDCK1	1280	40	40	80	20	40	40	20	20	80	160
B/Brest/1191/2017	3	2017-03-05	MDCK1/MDCK1	2560	40	40	40	20	40	40	20	20	80	160
B/England/42/2017	3	2017-04-01	SIAT1/MDCK1	1280	40	40	20	10	20	40	10	40	40	80
B/Hong Kong/251/2017	3	2017-05-12	Cx/MDCK1	1280	40	40	40	20	40	40	20	20	80	160
B/Hong Kong/253/2017	3	2017-05-13	Cx/MDCK1	1280	40	40	20	10	20	40	10	<	80	80
B/Hong Kong/257/2017	3	2017-05-13	Cx/MDCK1	1280	40	40	40	10	20	40	10	20	80	80
B/England/53/2017	3	2017-05-31	MDCK1/MDCK1	2560	160	160	80	80	160	160	40	80	320	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-10. Antigenic analyses of influenza B viruses (Yamagata lineage) 2017-08-15 + 2017-08-30 + 7017-09-05 + 2017-09-14

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											B/Phuket 3073/13 Egg	B/Phuket 3073/13 MDCK	B/HK 3417/14 Egg
				Post-infection ferret antisera													
				B/Phuket 3073/13 Egg	B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg			
				SH614 ^{*1,3}	F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*2}	F05/15 ^{*2}	F42/14 ^{*2}	F36/15 ^{*2}	F06/15 ^{*2}	F27/15 ^{*4}	NIB F51/16 ^{*2}	St Jude's F715/14 ^{*2,4}			
				Genetic Group	3	1	2	2	2	2	3	3	3	3			
REFERENCE VIRUSES																	
B/Florida/4/2006	1	2006-12-15	E7	2560	1280	1280	320	320	1280	320	320	80	1280	320			
B/Brisbane/3/2007	2	2007-09-03	E2/E2	2560	1280	1280	160	160	1280	320	320	80	1280	320			
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	2560	80	160	640	640	160	80	20	160	80	160			
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	2560	320	640	640	640	640	320	160	160	640	320			
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	1280	640	640	80	80	640	160	160	40	640	160			
B/Wisconsin/1/2010	3	2010-02-20	E3/E1	5120	320	320	40	40	320	160	160	1280	320	320			
B/Stockholm/12/2011	3	2011-03-28	EX/E2	1280	160	80	20	<	80	80	80	20	160	80			
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	80	80	320	320	80	320	320	160			
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	80	80	10	10	80	80	40	40	320	80			
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	40	10	10	40	40	20	20	80	160			
TEST VIRUSES																	
B/New Hampshire/01/2016	3	2016-01-07	E4/E1	2560	40	40	40	40	10	40	20	80	40	160			
B/Singapore/INFTT-16-0610/2016	3	2016-06-02	E3/E1	1280	80	80	20	10	80	80	20	40	160	160			
B/Fujian-Tongan/1907/2016	3	2016-11-14	E4/E3	640	80	80	10	<	40	40	20	20	160	40			
B/Fujian-Tongan/1907/2016	3	2016-11-14	C3/MDCK1	5120	160	320	80	160	80	160	80	320	320	160			
B/Ireland/94721/2017	3	2017-01-31	C1/MDCK1	5120	160	160	80	80	160	160	40	160	320	160			
B/Maine/04/2014	3	2017-02-02	C1/MDCK1	2560	80	80	80	80	80	80	40	160	160	160			
B/Saudi Arabia/246315/2017	3	2017-02-24	MDCK1	5120	160	160	80	80	160	160	40	160	160	320			
B/Ghana/532/2017	3	2017-02-28	C2/MDCK1	2560	80	80	80	160	160	160	40	160	160	160			
B/Saudi Arabia/1123799031/2017	3	2017-03-09	MDCK1	2560	80	80	40	40	80	80	40	80	160	160			
B/Ireland/06651/2017	3	2017-03-13	C1/MDCK1	5120	80	80	80	40	80	80	40	160	160	160			
B/Saudi Arabia/1160382303/2017	3	2017-03-14	MDCK1	2560	160	80	80	40	80	80	40	160	160	160			
B/Ireland/11620/2017	3	2017-04-02	C1/MDCK1	2560	80	80	40	40	40	80	20	80	80	160			
B/Bucuresti/210437/2017	3	2017-04-04	MDCK1/MDCK1	5120	160	160	80	80	160	160	40	160	160	160			
B/Ireland/12497/2017	3	2017-04-06	C1/MDCK1	2560	80	80	80	80	160	160	40	160	160	160			
B/Shanghai-Huangpu/1633/2017	3	2017-04-06	C3/MDCK1	5120	320	320	320	640	320	640	160	640	640	320			
B/South Africa/R07204/2017	3	2017-07-21	MDCK1	2560	80	80	40	40	80	80	40	80	160	160			
B/South Africa/R07200/2017	3	2017-07-24	MDCK1	2560	80	80	80	40	160	160	40	160	160	160			
B/South Africa/R07336/2017	3	2017-07-31	MDCK1	2560	80	160	80	80	160	160	40	160	160	160			

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 RDE serum pre-adsorbed with TRBC

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 13-11. Antigenic analyses of influenza B viruses (Yamagata lineage) with collection dates after 2017-01-31 - Summary

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/FI 4/06 Egg	B/Bris 3/07 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/HK 3417/14 Egg
				F17/13 ^{*1}	F38/14 ^{*2}	F27/13 ^{*2}	F05/15 ^{*2}	F42/14 ^{*2}	F36/15 ^{*2}	F06/15 ^{*2}	F27/15 ^{*4}	F51/16 ^{*2}	F715/14 ^{*2,4}
	Passage history			1	2	2	2	2	3	3	3	3	3
	Ferret number												
	Genetic Group												
REFERENCE VIRUSES													
B/Florida/4/2006	1	2006-12-15	E7	1280	1280	320	160	1280	320	320	40	1280	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	1280	160	160	1280	160	320	20	1280	320
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	80	160	640	640	160	160	20	10	80	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	640	320	640	640	640	320	160	80	640	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	640	640	80	80	640	160	160	40	640	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	160	160	20	20	160	80	80	20	640	160
B/Stockholm/12/2011	3	2011-03-28	EX/E2	160	80	20	<	80	80	80	10	160	80
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	160	160	160	160	320	160	80	160	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	80	80	10	10	80	80	40	40	320	80
B/Hong Kong/3417/2014	3	2014-06-04	E4/E3	80	40	20	<	40	40	20	<	80	160
TEST VIRUSES													
Number of viruses tested*				83	83	83	83	83	83	83	83	83	83
No with titre reduction ≤2-fold				4	3	8	16	10	56	48	32	31	83
%				4.8	3.6	9.7	19.3	12.0	67.5	57.8	38.6	37.3	100
No with titre reduction =4-fold				16	15	10	23	21	18	30	24	25	
%				19.3	18.1	12.0	27.7	25.3	21.7	36.1	28.9	30.1	
No with titre reduction ≥8-fold				63	65	65	44	52	9	5	27	27	
%				75.9	78.3	78.3	53.0	62.7	10.8	6.1	32.5	32.6	

* All viruses sequenced in time for the VCM fell in clade 3

Vaccine[#]

[#] B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Vaccine virus
Reference viruses

Jul 2017

* amino acid polymorphism

HI result in tables

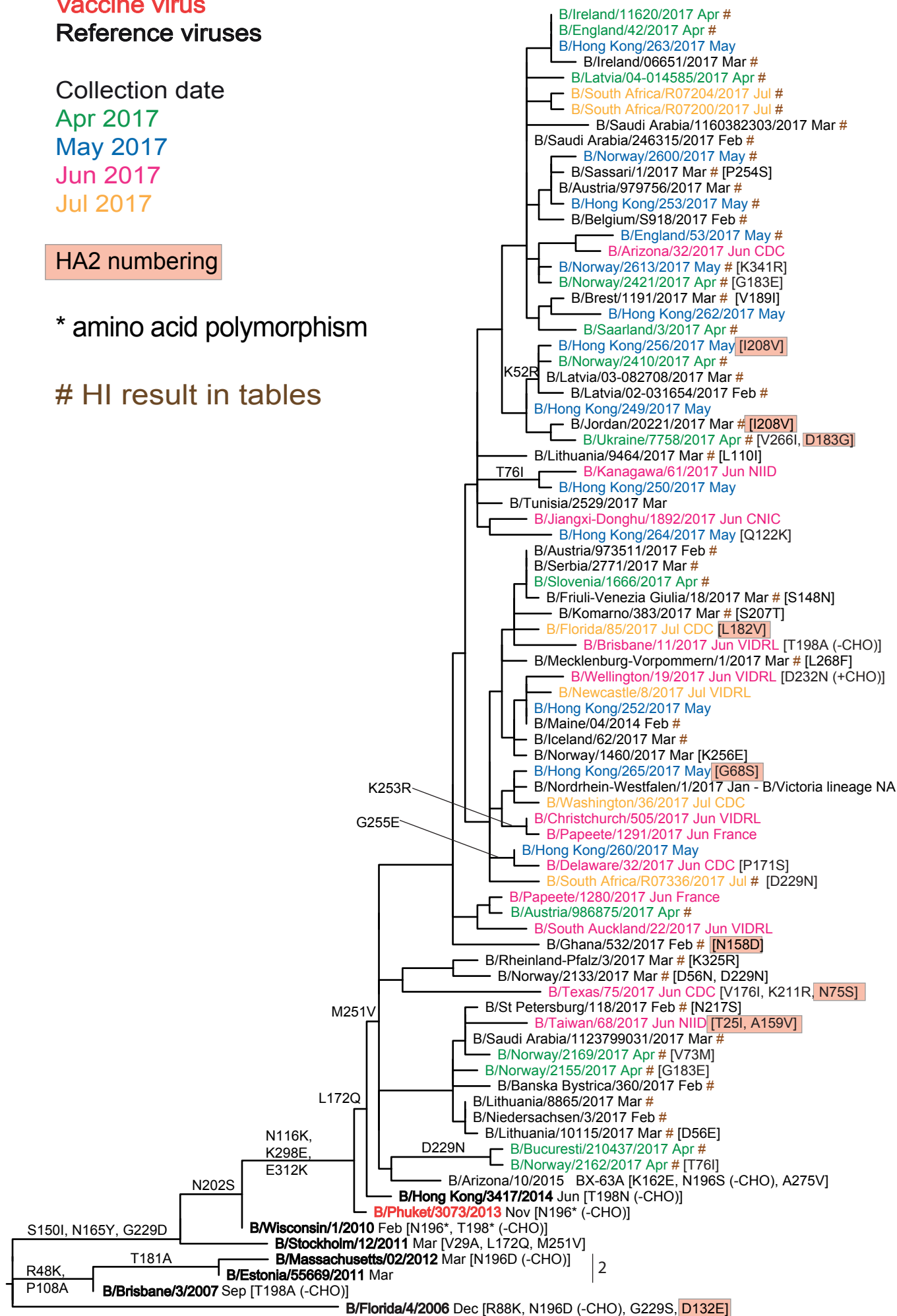


Figure 17. HA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100	
B/Florida/4/2006	DR	I	C	T	G	I	T	S	S	N	1
B/Estonia/55669/2011											2
B/Massachusetts/02/2012											2
B/Brisbane/3/2007											2
B/Stockholm/12/2011											3
B/Wisconsin/1/2010											3
B/Phuket/3073/2013											3
B/Hong Kong/3417/2014											3
B/Norway/2162/2017											3
B/Norway/2169/2017											3
B/Taiwan/68/2017											3
B/Texas/75/2017											3
B/South Africa/R07336/2017											3
B/Delaware/32/2017											3
B/Hong Kong/260/2017											3
B/Christchurch/505/2017											3
B/Hong Kong/265/2017											3
B/Wellington/19/2017											3
B/Friuli-Venezia Giulia/18/201											3
B/Hong Kong/264/2017											3
B/Hong Kong/250/2017											3
B/Kanagawa/61/2017											3
B/Ukraine/7758/2017											3
B/Norway/2410/2017											3
B/Hong Kong/256/2017											3
B/Hong Kong/262/2017											3
B/Norway/2421/2017											3
B/England/53/2017											3
B/Norway/2600/2017											3
B/South Africa/R07204/2017											3

	110	120	130	140	150	160	170	180	190	200	
B/Florida/4/2006	RT	K	I	R	Q	L	P	N	L	R	1
B/Estonia/55669/2011											2
B/Massachusetts/02/2012											2
B/Brisbane/3/2007											2
B/Stockholm/12/2011											3
B/Wisconsin/1/2010											3
B/Phuket/3073/2013											3
B/Hong Kong/3417/2014											3
B/Norway/2162/2017											3
B/Norway/2169/2017											3
B/Taiwan/68/2017											3
B/Texas/75/2017											3
B/South Africa/R07336/2017											3
B/Delaware/32/2017											3
B/Hong Kong/260/2017											3
B/Christchurch/505/2017											3
B/Hong Kong/265/2017											3
B/Wellington/19/2017											3
B/Friuli-Venezia Giulia/18/201											3
B/Hong Kong/264/2017											3
B/Hong Kong/250/2017											3
B/Kanagawa/61/2017											3
B/Ukraine/7758/2017											3
B/Norway/2410/2017											3
B/Hong Kong/256/2017											3
B/Hong Kong/262/2017											3
B/Norway/2421/2017											3
B/England/53/2017											3
B/Norway/2600/2017											3
B/South Africa/R07204/2017											3

	210	220	230	240	250	260	270	280	290	300	
B/Florida/4/2006	K	N	L	Y	G	D	S	N	P	K	1
B/Estonia/55669/2011											2
B/Massachusetts/02/2012											2
B/Brisbane/3/2007											2
B/Stockholm/12/2011											3
B/Wisconsin/1/2010											3
B/Phuket/3073/2013											3
B/Hong Kong/3417/2014											3
B/Norway/2162/2017											3
B/Norway/2169/2017											3
B/Taiwan/68/2017											3
B/Texas/75/2017											3
B/South Africa/R07336/2017											3
B/Delaware/32/2017											3
B/Hong Kong/260/2017											3
B/Christchurch/505/2017											3
B/Hong Kong/265/2017											3
B/Wellington/19/2017											3
B/Friuli-Venezia Giulia/18/201											3
B/Hong Kong/264/2017											3
B/Hong Kong/250/2017											3
B/Kanagawa/61/2017											3
B/Ukraine/7758/2017											3
B/Norway/2410/2017											3
B/Hong Kong/256/2017											3
B/Hong Kong/262/2017											3
B/Norway/2421/2017											3
B/England/53/2017											3
B/Norway/2600/2017											3
B/South Africa/R07204/2017											3

[illegible]

Figure 18. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

Vaccine virus

Reference viruses

Collection date

Apr 2017

May 2017

Jun 2017

Jul 2017

* amino acid polymorphism

HI result in tables

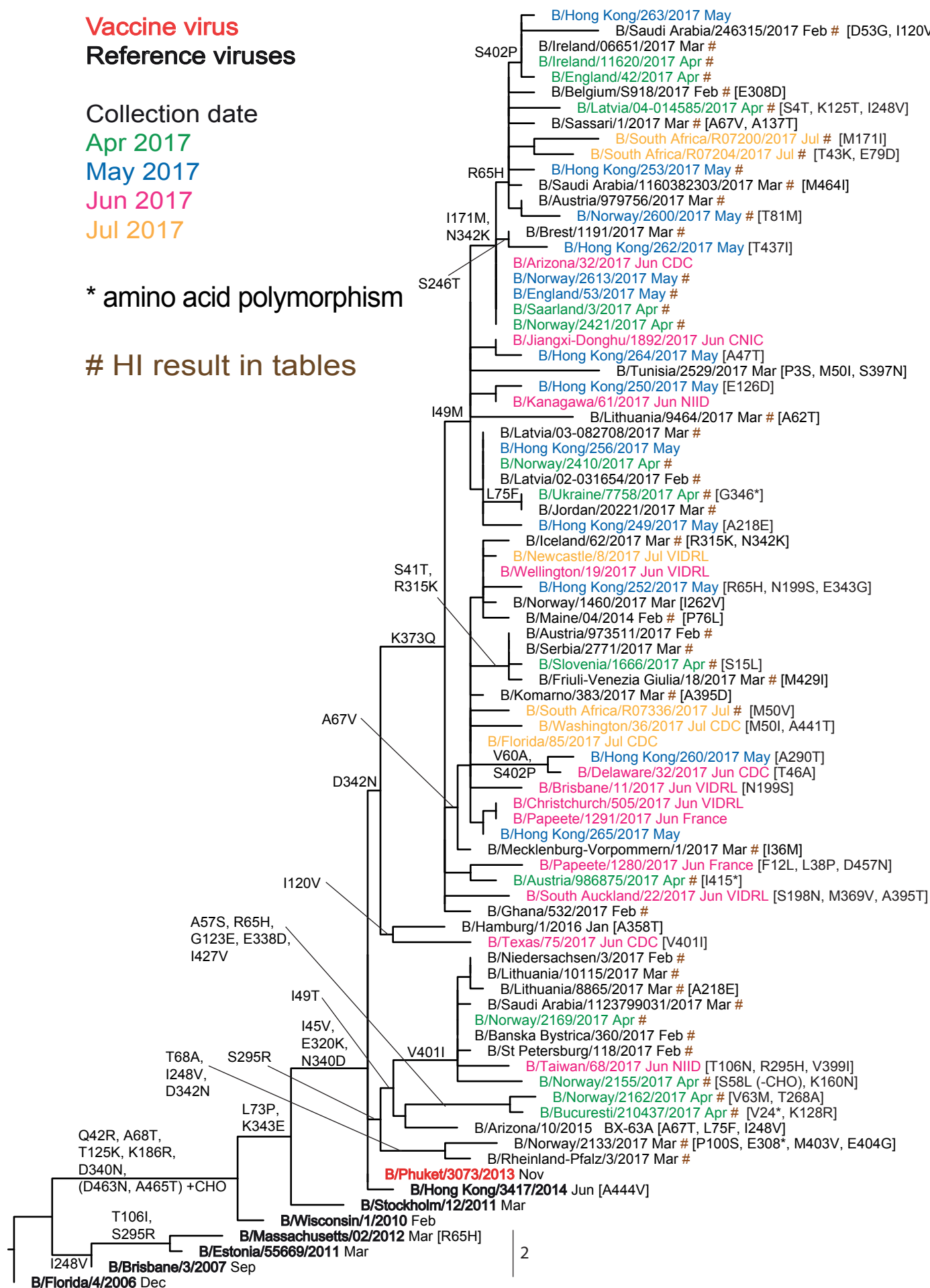


Figure 19. NA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100		
B/Florida/4/2006	M	L	P	S	T	I	Q	T	L	F	L	1
B/Estonia/55669/2011	M	L	P	S	T	I	Q	T	L	F	L	2
B/Massachusetts/02/2012	M	L	P	S	T	I	Q	T	L	F	L	2
B/Brisbane/3/2007	M	L	P	S	T	I	Q	T	L	F	L	2
B/Stockholm/12/2011	M	L	P	S	T	I	Q	T	L	F	L	3
B/Wisconsin/1/2010	M	L	P	S	T	I	Q	T	L	F	L	3
B/Phuket/3073/2013	M	L	P	S	T	I	Q	T	L	F	L	3
B/Hong Kong/3417/2014	M	L	P	S	T	I	Q	T	L	F	L	3
B/Norway/2162/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Norway/2169/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Taiwan/68/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Texas/75/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/South Africa/R07336/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Delaware/32/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Hong Kong/260/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Christchurch/505/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Hong Kong/265/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Wellington/19/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Friuli-Venezia Giulia/18/201	M	L	P	S	T	I	Q	T	L	F	L	3
B/Hong Kong/264/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Hong Kong/250/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Kanagawa/61/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Ukraine/7758/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Norway/2410/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Hong Kong/256/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Hong Kong/262/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Norway/2421/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/England/53/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/Norway/2600/2017	M	L	P	S	T	I	Q	T	L	F	L	3
B/South Africa/R07204/2017	M	L	P	S	T	I	Q	T	L	F	L	3

	110	120	130	140	150	160	170	180	190	200		
B/Florida/4/2006	H	R	F	G	E	T	K	G	N	S	A	1
B/Estonia/55669/2011	I											2
B/Massachusetts/02/2012	I											2
B/Brisbane/3/2007												2
B/Stockholm/12/2011			K							R		3
B/Wisconsin/1/2010			K							R		3
B/Phuket/3073/2013			K							R		3
B/Hong Kong/3417/2014			K							R		3
B/Norway/2162/2017		E	K							R		3
B/Norway/2169/2017			K							R		3
B/Taiwan/68/2017	N		K							R		3
B/Texas/75/2017		V	K							R		3
B/South Africa/R07336/2017			K							R		3
B/Delaware/32/2017			K							R		3
B/Hong Kong/260/2017			K							R		3
B/Christchurch/505/2017			K							R		3
B/Hong Kong/265/2017			K							R		3
B/Wellington/19/2017			K							R		3
B/Friuli-Venezia Giulia/18/201			K							R		3
B/Hong Kong/264/2017			K							R		3
B/Hong Kong/250/2017			KD							R		3
B/Kanagawa/61/2017			K							R		3
B/Ukraine/7758/2017			K							R		3
B/Norway/2410/2017			K							R		3
B/Hong Kong/256/2017			K							R		3
B/Hong Kong/262/2017			K				M			R		3
B/Norway/2421/2017			K				M			R		3
B/England/53/2017			K				M			R		3
B/Norway/2600/2017			K				M			R		3
B/South Africa/R07204/2017			K				M			R		3

	210	220	230	240	250	260	270	280	290	300		
B/Florida/4/2006	L	L	K	I	K	Y	G	E	A	Y	T	1
B/Estonia/55669/2011						V						2
B/Massachusetts/02/2012						V				R		2
B/Brisbane/3/2007						V						2
B/Stockholm/12/2011												3
B/Wisconsin/1/2010												3
B/Phuket/3073/2013												3
B/Hong Kong/3417/2014												3
B/Norway/2162/2017							A			R		3
B/Norway/2169/2017										R		3
B/Taiwan/68/2017										H		3
B/Texas/75/2017												3
B/South Africa/R07336/2017												3
B/Delaware/32/2017												3
B/Hong Kong/260/2017										T		3
B/Christchurch/505/2017												3
B/Hong Kong/265/2017												3
B/Wellington/19/2017												3
B/Friuli-Venezia Giulia/18/201												3
B/Hong Kong/264/2017												3
B/Hong Kong/250/2017												3
B/Kanagawa/61/2017												3
B/Ukraine/7758/2017												3
B/Norway/2410/2017												3
B/Hong Kong/256/2017												3
B/Hong Kong/262/2017					T							3
B/Norway/2421/2017												3
B/England/53/2017												3
B/Norway/2600/2017												3
B/South Africa/R07204/2017												3

Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**

Table 14. Antiviral (NAI) testing of isolates with collection dates after 2017-01-31*

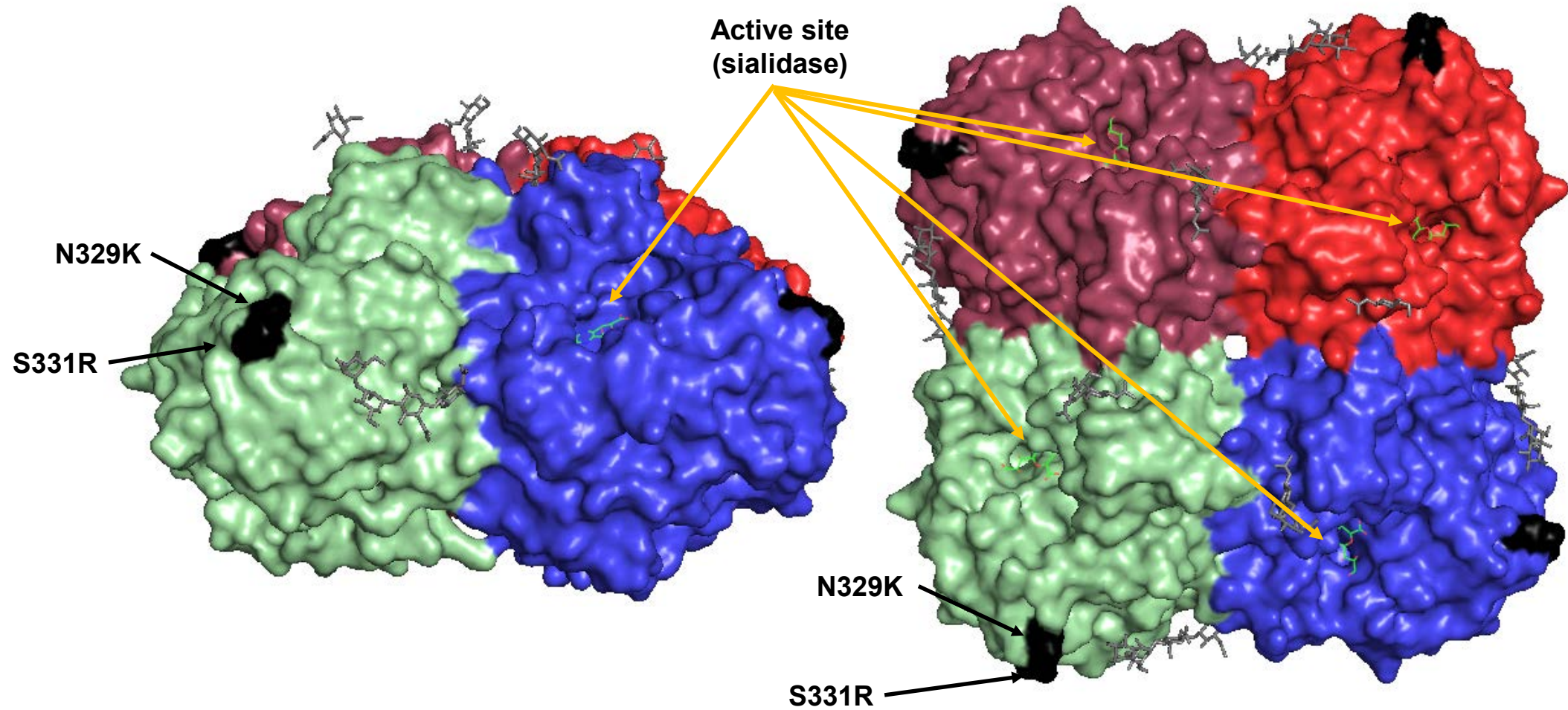
Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2017									
February	17		109	5	38		29		169
March	14		37	3	38		23		92
April	9		9	1	50	1	13		70
May	2	1	27		8		5		38
June	4		3		2		0		9
July	6		6						
Totals	46	1	185	9	136	1	70	0	378

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2017-09-12

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Paris/1227/2017	H1N1pdm09	2017-05-03	834.7 (HRI)	3.3	H275Y
A/Austria/971753/2017	H3N2	2017-02-18	32.7 (RI)	24.8 (RI)	S331R (-CHO)
A/Belgium/G448/2017	H3N2	2017-02-16	26.5 (RI)	15.1 (RI)	S331R (-CHO)
A/Latvia/02-062187/2017	H3N2	2017-02-22	14.3 (RI)	10.9 (RI)	N329K (-CHO)
A/St Petersburg/301/2017	H3N2	2017-02-17	24.9 (RI)	18.9 (RI)	S331R (-CHO)
A/St Petersburg/224/2017	H3N2	2017-02-16	77.0 (RI)	22.9 (RI)	S331R (-CHO)
A/Khabarovsk/18/2017	H3N2	2017-03-02	20.3 (RI)	15.7 (RI)	S331R (-CHO)
A/Poland/17C/2017	H3N2	2017-03-01	14.0 (RI)	8.1	N329K (-CHO)
A/Novosibirsk/47/2017	H3N2	2017-03-07	17.4 (RI)	12.7 (RI)	S331R (-CHO)
A/Bremen/27/2017	H3N2	2017-04-19	10.1 (RI)	6.9	N329K (-CHO)
B/Moscow/107/2017	Victoria	2017-04-05	4.9 (RI)	2.4	H273Y

Figure 20. N2 neuraminidase - location of amino acid substitutions associated with RI by NAIs



Substitutions introduce + charge and result in loss of glycosylation at position 329